

Chapterwise DPP for JEE Main

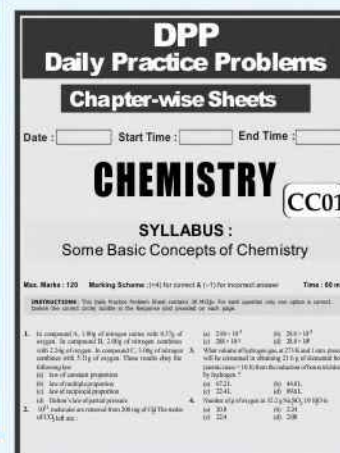


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Scoring Grid	DAILY PRACTICE PROBLEM DPP CHAPTERWISE CC01 - CHEMISTRY			
	Total Questions	30	Total Marks	120
	Attempted		Correct	
	Incorrect		Net Score	
	Cut-off Score	37	Qualifying Score	52
	Success Gap = Net Score – Qualifying Score			
	Net Score = (Correct × 4) – (Incorrect × 1)			

Chapterwise DPP for JEE Main

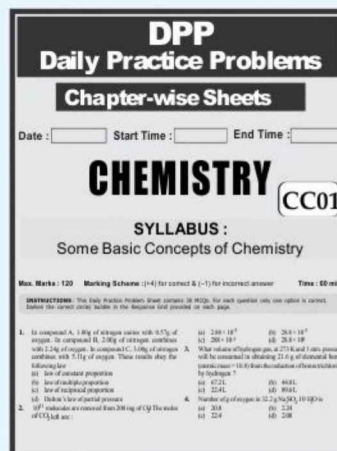


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	Cut-off Score	37	Qualifying Score	52
	Success Gap = Net Score – Qualifying Score			
	Net Score = (Correct × 4) – (Incorrect × 1)			

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DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC01

SYLLABUS : Some Basic Concepts of Chemistry

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- In compound A, 1.00g of nitrogen unites with 0.57g of oxygen. In compound B, 2.00g of nitrogen combines with 2.24g of oxygen. In compound C, 3.00g of nitrogen combines with 5.11g of oxygen. These results obey the following law
 - law of constant proportion
 - law of multiple proportion
 - law of reciprocal proportion
 - Dalton's law of partial pressure
- 10^{21} molecules are removed from 200 mg of CO_2 . The moles of CO_2 left are :
 - 2.88×10^{-3}
 - 28.8×10^{-3}
 - 288×10^{-3}
 - 28.8×10^3
- What volume of hydrogen gas, at 273 K and 1 atm. pressure will be consumed in obtaining 21.6 g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ?
 - 67.2L
 - 44.8L
 - 22.4L
 - 89.6L
- Number of g of oxygen in 32.2 g $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ is
 - 20.8
 - 2.24
 - 22.4
 - 2.08

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. 6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is
 (a) 0.02 M (b) 0.01 M
 (c) 0.001 M (d) 0.1 M
 (Avogadro constant, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)
6. If we consider that $1/6$, in place of $1/12$, mass of carbon atom is taken to be the relative atomic mass unit, the mass of one mole of the substance will
 (a) be a function of the molecular mass of the substance
 (b) remain unchanged
 (c) increase two fold
 (d) decrease twice
7. The weight of NaCl decomposed by 4.9 g of H_2SO_4 , if 6 g of sodium hydrogen sulphate and 1.825 g of HCl, were produced in the reaction is:
 (a) 6.921 g (b) 4.65 g
 (c) 2.925 g (d) 1.4 g
8. Which one of the following is the lightest?
 (a) 0.2 mole of hydrogen gas
 (b) 6.023×10^{22} molecules of nitrogen
 (c) 0.1 g of silver
 (d) 0.1 mole of oxygen gas
9. How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms?
 (a) 1.25×10^{-2} (b) 2.5×10^{-2}
 (c) 0.02 (d) 3.125×10^{-2}
10. The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% H_2SO_4 (molar mass = 98 g mol^{-1}) by mass will be
 (a) 1.45 (b) 1.64
 (c) 1.88 (d) 1.22
11. A gas occupies a volume of 300 cc at 27°C and 620 mm pressure. The volume of gas at 47°C and 640 mm pressure is:
 (a) 260 cc (b) 310 cc
 (c) 390 cc (d) 450 cc
12. Haemoglobin contains 0.33% of iron by weight. The molecular weight of haemoglobin is approximately 67200. The number of iron atoms (at. wt. of Fe = 56) present in one molecule of haemoglobin is
 (a) 6 (b) 1
 (c) 2 (d) 4
13. The volume of 20 volume H_2O_2 required to get 5 litres of O_2 at STP is
 (a) 250 ml (b) 125 ml
 (c) 100 ml (d) 50 ml.
14. In the reaction,

$$2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{Al}^{3+}(\text{aq}) + 6\text{Cl}^{-}(\text{aq}) + 3\text{H}_2(\text{g})$$

 (a) 11.2 L $\text{H}_2(\text{g})$ at STP is produced for every mole HCl(aq) consumed
 (b) 6 L HCl(aq) is consumed for every 3 L $\text{H}_2(\text{g})$ produced
 (c) 33.6 L $\text{H}_2(\text{g})$ is produced regardless of temperature and pressure for every mole Al that reacts
 (d) 67.2 $\text{H}_2(\text{g})$ at STP is produced for every mole Al that reacts.

**RESPONSE
GRID**

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. The concentrated sulphuric acid that is peddled commercial is 95% H_2SO_4 by weight. If the density of this commercial acid is 1.834 g cm^{-3} , the molarity of this solution is
 (a) 17.8 M (b) 12.0 M
 (c) 10.5 M (d) 15.7 M
16. What is the mass of precipitate formed when 50 mL of 16.9% solution of AgNO_3 is mixed with 50 mL of 5.8% NaCl solution ?
 (Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)
 (a) 28 g (b) 3.5 g
 (c) 7 g (d) 14 g
17. Number of valence electrons in 4.2 gram of N_3^- ion is
 (a) $4.2 N_A$ (b) $0.1 N_A$
 (c) $1.6 N_A$ (d) $3.2 N_A$
18. A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be
 (a) MCl_3 (b) MCl_2
 (c) MCl_4 (d) MCl_5
19. A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g. of CO_2 . The empirical formula of the hydrocarbon is :
 (a) C_2H_4 (b) C_3H_4
 (c) C_6H_5 (d) C_7H_8
20. Following is the composition of a washing soda sample :

Substance	Molecular Wt.	Mass percent
Na_2CO_3	106.0	84.8
NaHCO_3	84.0	8.4
NaCl	58.5	6.8

On complete reaction with excess HCl , one kilogram of the washing soda will evolve:

- (a) 9 mol of CO_2 (b) 16 mol of CO_2
 (c) 17 mol of CO_2 (d) 18 mol of CO_2
21. Arrange the numbers in increasing no. of significant figures.
 0.002600, 2.6000, 2.6, 0.260
 (a) $2.6 < 0.260 < 0.002600 < 2.6000$
 (b) $2.6000 < 2.6 < 0.002600 < 0.260$
 (c) $0.260 < 2.6 < 0.002600 < 2.6000$
 (d) $0.002600 < 0.260 < 2.6 < 2.6000$
22. Dissolving 120 g of a compound (mol. wt. 60) in 1000 g of water gave a solution of density 1.12 g/mL. The molarity of the solution is:
 (a) 1.00 M (b) 2.00 M
 (c) 2.50 M (d) 4.00 M
23. A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is:
 (a) NH_2 (b) N_3H
 (c) NH_3 (d) N_2H_4
24. The amount of BaSO_4 formed upon mixing 100 mL of 20.8% BaCl_2 solution with 50 mL of 9.8% H_2SO_4 solution with 50 mL of 9.8% H_2SO_4 solution will be:
 (Ba = 137, Cl = 35.5, S = 32, H = 1 and O = 16)
 (a) 23.3 g (b) 11.65 g
 (c) 30.6 g (d) 33.2 g

RESPONSE
GRID

15. (a) (b) (c) (d) 16. (a) (b) (c) (d) 17. (a) (b) (c) (d) 18. (a) (b) (c) (d) 19. (a) (b) (c) (d)
 20. (a) (b) (c) (d) 21. (a) (b) (c) (d) 22. (a) (b) (c) (d) 23. (a) (b) (c) (d) 24. (a) (b) (c) (d)

Space for Rough Work

C-4

DPP/ CC01

25. 2 g of a mixture of CO and CO₂ on reaction with excess I₂O₅ produced 2.54 g of I₂. What will be the mass % of CO₂ in the original mixture ?
 (a) 35 (b) 70
 (c) 30 (d) 60
26. 7.5 grams of a gas occupy 5.6 litres of volume at STP. The gas is
 (a) N₂O (b) NO
 (c) CO (d) CO₂
27. Number of moles of KMnO₄ required to oxidize one mole of Fe(C₂O₄) in acidic medium is
 (a) 0.167 (b) 0.6
 (c) 0.2 (d) 0.4
28. What is the weight of oxygen required for the complete combustion of 2.8 kg of ethylene ?
 (a) 2.8 kg (b) 6.4 kg
 (c) 9.6 kg (d) 96 kg
29. A gas mixture of 3 litres of propane (C₃H₈) and butane (C₄H₁₀) on complete combustion at 25° C produced 10 litre CO₂. Find out the composition of gas mixture (Propane : Butane)
 (a) 2 : 1 (b) 1 : 2
 (c) 1.5 : 1.5 (d) 0.5 : 2.5
30. An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of the compound is :
 (a) C₃H₅O₂ (b) C₄H₁₀O₂
 (c) C₆H₁₀O₄ (d) C₃H₁₀O₂

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 1 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC02

SYLLABUS : Structure of Atom

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

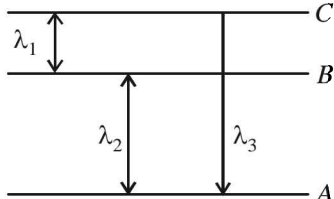
INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Among the following groupings which represents the collection of isoelectronic species?
(a) NO^+ , C_2^{2-} , O_2^- , CO (b) N_2 , C_2^{2-} , CO, NO
(c) CO, NO^+ , CN^- , C_2^{2-} (d) NO, CN^- , N_2 , O_2^-
- The compound in which cation is isoelectronic with anion is :
(a) NaCl (b) CsF
(c) NaI (d) K_2S
- The de-Broglie wavelength of an electron in the ground state of hydrogen atom is : [K.E. = 13.6 eV; $1\text{eV} = 1.602 \times 10^{-19}\text{J}$]
(a) 33.28 nm (b) 3.328 nm
(c) 0.3328 nm (d) 0.0332 nm
- The frequency of light emitted for the transition $n = 4$ to $n = 2$ of the He^+ is equal to the transition in H atom corresponding to which of the following ?
(a) $n = 2$ to $n = 1$ (b) $n = 3$ to $n = 2$
(c) $n = 4$ to $n = 3$ (d) $n = 3$ to $n = 1$

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. The first emission line in the atomic spectrum of hydrogen in the Balmer series appears at
- (a) $\frac{9R}{400}\text{cm}^{-1}$ (b) $\frac{7R}{144}\text{cm}^{-1}$
 (c) $\frac{3R}{4}\text{cm}^{-1}$ (d) $\frac{5R}{36}\text{cm}^{-1}$
6. In hydrogen atomic spectrum, a series limit is found at 12186.3 cm^{-1} . Then it belong to
- (a) Lyman series (b) Balmer series
 (c) Paschen series (d) Brackett series
7. Two fast moving particles X and Y are associated with de Broglie wavelengths 1 nm and 4 nm respectively. If mass of X is nine times the mass of Y, the ratio of kinetic energies of X and Y would be
- (a) 3 : 1 (b) 9 : 1
 (c) 5 : 12 (d) 16 : 9
8. The ratio of magnetic moments of Fe(III) and Co(II) is
- (a) 7 : 3 (b) 3 : 7
 (c) $\sqrt{7} : \sqrt{3}$ (d) $\sqrt{3} : \sqrt{7}$
9. The values of Planck's constant is $6.63 \times 10^{-34}\text{ Js}$. The velocity of light is $3.0 \times 10^8\text{ m s}^{-1}$. Which value is closest to the wavelength in nanometres of a quantum of light with frequency of $8 \times 10^{15}\text{ s}^{-1}$?
- (a) 5×10^{-18} (b) 4×10^1
 (c) 3×10^7 (d) 2×10^{-25}
10. Li and a proton are accelerated by the same potential, their de Broglie wavelengths λ_{Li} and λ_p have the ratio (assume $m_{\text{Li}} = 9m_p$)
- (a) 1 : 2 (b) 1 : 4
 (c) 1 : 1 (d) $1 : 3\sqrt{3}$
11. Energy levels, A, B, C, of a certain atom correspond to increasing values of energy i.e., $E_A < E_B < E_C$. If $\lambda_1, \lambda_2, \lambda_3$ are the wave lengths of radiations corresponding to the transition from C to B, B to A and C to A respectively, which of the following statements is correct ?
- 
- (a) $\lambda_3 = \lambda_1 + \lambda_2$ (b) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
 (c) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (d) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$
12. If uncertainty in position and momentum are equal, then uncertainty in velocity is :
- (a) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (b) $\sqrt{\frac{h}{2\pi}}$
 (c) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ (d) $\sqrt{\frac{h}{\pi}}$
13. The electrons, identified by quantum numbers n and l (i) n = 4, l = 1 (ii) n = 4, l = 0 (iii) n = 3, l = 2 (iv) n = 3, l = 1 can be placed in order of increasing energy, from the lowest to highest, as
- (a) (iv) < (ii) < (iii) < (i) (b) (ii) < (iv) < (i) < (iii)
 (c) (i) < (iii) < (ii) < (iv) (d) (iii) < (i) < (iv) < (ii)
14. Ionisation energy of He^+ is $19.6 \times 10^{-18}\text{ J atom}^{-1}$. The energy of the first stationary state (n = 1) of Li^{2+} is
- (a) $4.41 \times 10^{-16}\text{ J atom}^{-1}$ (b) $-4.41 \times 10^{-17}\text{ J atom}^{-1}$
 (c) $-2.2 \times 10^{-15}\text{ J atom}^{-1}$ (d) $8.82 \times 10^{-17}\text{ J atom}^{-1}$

RESPONSE
GRID

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

11. (a)(b)(c)(d)

12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

Space for Rough Work

15. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] :
- (a) $\frac{h^2}{4\pi^2 m a_0^2}$ (b) $\frac{h^2}{16\pi^2 m a_0^2}$
 (c) $\frac{h^2}{32\pi^2 m a_0^2}$ (d) $\frac{h^2}{64\pi^2 m a_0^2}$
16. Energy of an electron is given by $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$.
 Wavelength of light required to excite an electron in an hydrogen atom from level $n = 1$ to $n = 2$ will be :
 ($h = 6.62 \times 10^{-34} \text{ Js}$ and $c = 3.0 \times 10^8 \text{ ms}^{-1}$)
 (a) $1.214 \times 10^{-7} \text{ m}$ (b) $2.816 \times 10^{-7} \text{ m}$
 (c) $6.500 \times 10^{-7} \text{ m}$ (d) $8.500 \times 10^{-7} \text{ m}$
17. If the nitrogen atom had electronic configuration $1s^7$ it would have energy lower than that of the normal ground state configuration $1s^2 2s^2 2p^3$ because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed. It violates
 (a) Heisenberg's uncertainty principle
 (b) Hund's rule
 (c) Pauli exclusion principle
 (d) Bohr postulate of stationary orbits
18. In a hydrogen atom, if energy of an electron in ground state is 13.6 eV, then that in the 2nd excited state is
 (a) 1.51 eV (b) 3.4 eV
 (c) 6.04 eV (d) 13.6 eV
19. Of the following sets which one does NOT contain isoelectronic species?
 (a) BO_3^{3-} , CO_3^{2-} , NO_3^-
 (b) SO_3^{2-} , CO_3^{2-} , NO_3^-
 (c) CN^- , N_2 , C_2^{2-}
 (d) PO_4^{3-} , SO_4^{2-} , ClO_4^-
20. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite the electron in the atom from $n = 1$ to $n = 2$ is
 (a) $8.51 \times 10^5 \text{ J mol}^{-1}$ (b) $6.56 \times 10^5 \text{ J mol}^{-1}$
 (c) $7.56 \times 10^5 \text{ J mol}^{-1}$ (d) $9.84 \times 10^5 \text{ J mol}^{-1}$
21. The limiting line in Balmer series will have a frequency of (Rydberg constant, $R_\infty = 3.29 \times 10^{15} \text{ cycles/s}$)
 (a) $8.22 \times 10^{14} \text{ s}^{-1}$ (b) $3.29 \times 10^{15} \text{ s}^{-1}$
 (c) $3.65 \times 10^{14} \text{ s}^{-1}$ (d) $5.26 \times 10^{13} \text{ s}^{-1}$
22. The energy required to break one mole of Cl – Cl bonds in Cl_2 is 242 kJ mol^{-1} . The longest wavelength of light capable of breaking a single Cl – Cl bond is
 ($c = 3 \times 10^8 \text{ ms}^{-1}$ and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$).
 (a) 594 nm (b) 640 nm
 (c) 700 nm (d) 494 nm
23. The de Broglie wavelength of a car of mass 1000 kg and velocity 36 km/hr is :
 (a) $6.626 \times 10^{-34} \text{ m}$ (b) $6.626 \times 10^{-38} \text{ m}$
 (c) $6.626 \times 10^{-31} \text{ m}$ (d) $6.626 \times 10^{-30} \text{ m}$
24. If the radius of first orbit of H atom is a_0 , the de-Broglie wavelength of an electron in the third orbit is
 (a) $4\pi a_0$ (b) $8\pi a_0$
 (c) $6\pi a_0$ (d) $2\pi a_0$

RESPONSE
GRID

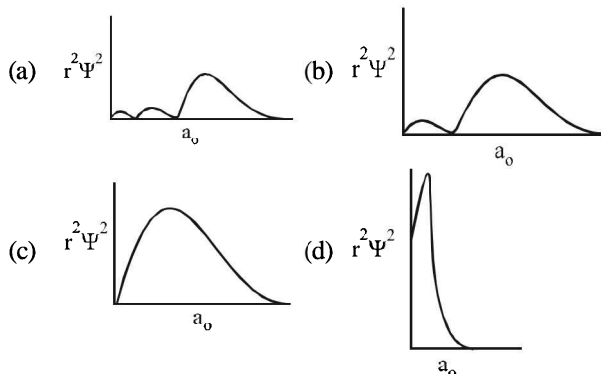
15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
 20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

C-8

DPP/ CC02

25. Which of the following radial distribution graphs correspond to $\ell = 2$ for the H atom ?



26. If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become
 (a) one fourth (b) half
 (c) four times (d) two times
27. The correct set of four quantum numbers for the valence electrons of rubidium atom ($Z = 37$) is:
 (a) $5, 0, 0, +\frac{1}{2}$ (b) $5, 1, 0, +\frac{1}{2}$
 (c) $5, 1, 1, +\frac{1}{2}$ (d) $5, 0, 1, +\frac{1}{2}$

28. If λ_0 and λ be threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is:

(a) $\sqrt{\frac{2h}{m}(\lambda_0 - \lambda)}$ (b) $\sqrt{\frac{2hc}{m}(\lambda_0 - \lambda)}$
 (c) $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$ (d) $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda}\right)}$

29. If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:

(a) $\frac{1}{2} \frac{e^2}{r}$ (b) $-\frac{e^2}{r}$
 (c) $\frac{me^2}{r}$ (d) $-\frac{1}{2} \frac{e^2}{r}$

30. The dissociation energy of H_2 is $430.53 \text{ KJ mol}^{-1}$. If hydrogen is dissociated by illumination with radiation of wavelength 253.7 nm the fraction of the radiant energy which will be converted into kinetic energy is given by
 (a) 100% (b) 8.76%
 (c) 2.22% (d) 1.22%

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 2 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	51
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC03

SYLLABUS : Classification of Elements and Periodicity in Properties

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

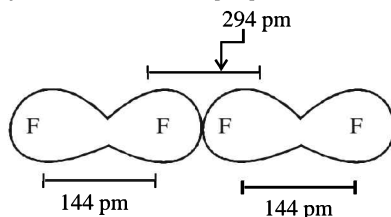
Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. The correct sequence which shows decreasing order of the ionic radii of the elements is

- (a) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$
(b) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$
(c) $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$
(d) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

2. The van der Waal and covalent radii of fluorine atom respectively from the following figure are.



- (a) 219pm, 72pm (b) 75pm, 72pm
(c) 147pm, 72pm (d) 147pm, 144pm

3. Arrange the following in increasing order of ionic radii?
 $\text{C}^{4-}, \text{N}^{3-}, \text{F}^-, \text{O}^{2-}$

- (a) $\text{C}^{4-} < \text{N}^{3-} < \text{O}^{2-} < \text{F}^-$
(b) $\text{N}^{3-} < \text{C}^{4-} < \text{O}^{2-} < \text{F}^-$
(c) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-} < \text{C}^{4-}$
(d) $\text{O}^{2-} < \text{F}^- < \text{N}^{3-} < \text{C}^{4-}$

4. Which is not the correct order for the stated property.

- (a) $\text{Ba} > \text{Sr} > \text{Mg}$; atomic radius
(b) $\text{F} > \text{O} > \text{N}$; first ionization enthalpy
(c) $\text{Cl} > \text{F} > \text{I}$; electron affinity
(d) $\text{O} > \text{Se} > \text{Te}$; electronegativity

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

C-10

DPP/ CC03

5. In which of the following arrangements, the order is NOT according to the property indicated against it?
- $\text{Li} < \text{Na} < \text{K} < \text{Rb}$:
Increasing metallic radius
 - $\text{I} < \text{Br} < \text{F} < \text{Cl}$:
Increasing electron gain enthalpy (with negative sign)
 - $\text{B} < \text{C} < \text{N} < \text{O}$
Increasing first ionization enthalpy
 - $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$
Increasing ionic size
6. The symbol and IUPAC name for the element with atomic number 120, respectively are
- Ubn and unbnilium
 - Ubn and unbiunium
 - Ubn and unnibium
 - Ubn and unnilium
7. Sequence of acidic character is
- $\text{N}_2\text{O}_5 > \text{SO}_2 > \text{CO} > \text{CO}_2$
 - $\text{N}_2\text{O}_5 > \text{SO}_2 > \text{CO}_2 > \text{CO}$
 - $\text{SO}_2 > \text{CO}_2 > \text{CO} > \text{N}_2\text{O}_5$
 - $\text{SO}_2 > \text{N}_2\text{O}_5 > \text{CO} > \text{CO}_2$
8. The correct order of ionization energy for carbon, nitrogen and oxygen atoms is:
- $\text{C} > \text{N} > \text{O}$
 - $\text{C} > \text{N} < \text{O}$
 - $\text{C} < \text{N} > \text{O}$
 - $\text{C} < \text{N} < \text{O}$
9. Which of the following order is wrong?
- $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ – Acidic
 - $\text{Li} < \text{Be} < \text{B} < \text{C}$ – First IP
 - $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$ – Basic
 - $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Cs}^+$ – Ionic radius
10. The radii of F , F^- , O and O^{2-} are in the order
- $\text{O}^{2-} > \text{F}^- > \text{F} > \text{O}$
 - $\text{F}^- > \text{O}^{2-} > \text{F} > \text{O}$
 - $\text{O}^{2-} > \text{O} > \text{F}^- > \text{F}$
 - $\text{O}^{2-} > \text{F}^- > \text{O} > \text{F}$
11. Which of the following has the maximum number of unpaired electrons?
- Mg^{2+}
 - Ti^{3+}
 - V^{3+}
 - Fe^{2+}
12. The incorrect statement among the following is
- The first ionization potential of Al is less than the first ionization potential of Mg
 - The second ionization potential of Mg is greater than the second ionization potential of Na
 - The first ionization potential of Na is less than the first ionization potential of Mg
 - The third ionization potential of Mg is greater than the third ionization potential of Al.
13. According to the Periodic Law of elements, the variation in properties of elements is related to their
- nuclear masses
 - atomic numbers
 - nuclear neutron-proton number ratios
 - atomic masses
14. Identify the correct order of the size of the following:
- $\text{Ca}^{2+} < \text{K}^+ < \text{Ar} < \text{Cl}^- < \text{S}^{2-}$
 - $\text{Ar} < \text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$
 - $\text{Ca}^{2+} < \text{Ar} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$
 - $\text{Ca}^{2+} < \text{K}^+ < \text{Ar} < \text{S}^{2-} < \text{Cl}^-$

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. Following statements regarding the periodic trends of chemical reactivity of the alkali metals and the halogens are given. Which of these statements gives the correct picture?
- Chemical reactivity increases with increase in atomic number down the group in both the alkali metals and halogens
 - In alkali metals the reactivity increases but in the halogens it decreases with increase in atomic number down the group
 - The reactivity decreases in the alkali metals but increases in the halogens with increase in atomic number down the group
 - In both the alkali metals and the halogens the chemical reactivity decreases with increase in atomic number down the group
16. In which of the following arrangements, the sequence is *not* strictly according to the property written against it?
- $\text{HF} < \text{HCl} < \text{HBr}, \text{HI}$: increasing acid strength
 - $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: increasing basic strength
 - $\text{B} < \text{C} < \text{O} < \text{N}$: increasing first ionization enthalpy
 - $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: increasing oxidising power
17. Which of the following order is wrong?
- $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ — Acidic
 - $\text{Li} < \text{Be} < \text{B} < \text{C}$ — IE_1
 - $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$ — Basic
 - $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Cs}^+$ — Ionic radius
18. The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I, having atomic number 9, 17, 35 and 53 respectively, is :
- $\text{F} > \text{Cl} > \text{Br} > \text{I}$
 - $\text{Cl} > \text{F} > \text{Br} > \text{I}$
 - $\text{Br} > \text{Cl} > \text{I} > \text{F}$
 - $\text{I} > \text{Br} > \text{Cl} > \text{F}$
19. Which one of the following has largest ionic radius?
- Li^+
 - O_2^{2-}
 - B^{3+}
 - F^-
20. Which one of the following arrangements represents the correct order of least negative to most negative electron gain enthalpy for C, Ca, Al, F and O?
- $\text{Ca} < \text{Al} < \text{C} < \text{O} < \text{F}$
 - $\text{Al} < \text{Ca} < \text{O} < \text{C} < \text{F}$
 - $\text{Al} < \text{O} < \text{C} < \text{Ca} < \text{F}$
 - $\text{C} < \text{F} < \text{O} < \text{Al} < \text{Ca}$
21. Which of the following elements represents highly electropositive as well as highly electronegative character in its period?
- Hydrogen
 - Nitrogen
 - Fluorine
 - None of these
22. Which one of the following ions has the highest value of ionic radius ?
- O^{2-}
 - B^{3+}
 - Li^+
 - F^-
23. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 the correct order of acid strength is
- $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_2 < \text{P}_2\text{O}_3$
 - $\text{SiO}_2 < \text{SO}_2 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_3$
 - $\text{SO}_2 < \text{P}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$
 - $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$
24. Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species O^{2-} , S^{2-} , N^{3-} , P^{3-} ?
- $\text{O}^{2-} < \text{N}^{3-} < \text{S}^{2-} < \text{P}^{3-}$
 - $\text{O}^{2-} < \text{P}^{3-} < \text{N}^{3-} < \text{S}^{2-}$
 - $\text{N}^{3-} < \text{O}^{2-} < \text{P}^{3-} < \text{S}^{2-}$
 - $\text{N}^{3-} < \text{S}^{2-} < \text{O}^{2-} < \text{P}^{3-}$

RESPONSE
GRID

15. (a) (b) (c) (d) 16. (a) (b) (c) (d) 17. (a) (b) (c) (d) 18. (a) (b) (c) (d) 19. (a) (b) (c) (d)
20. (a) (b) (c) (d) 21. (a) (b) (c) (d) 22. (a) (b) (c) (d) 23. (a) (b) (c) (d) 24. (a) (b) (c) (d)

Space for Rough Work

C-12

DPP/ CC03

25. Which of the following oxides is amphoteric in character?
 (a) SnO_2 (b) SiO_2
 (c) CO_2 (d) CaO
26. The formation of the oxide ion $\text{O}^{2-}(\text{g})$, from oxygen atom requires first an exothermic and then an endothermic step as shown below :
 $\text{O}(\text{g}) + \text{e}^- \rightarrow \text{O}^-(\text{g}); \Delta_f H^\ominus = -141 \text{ kJ mol}^{-1}$
 $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g}); \Delta_f H^\ominus = +780 \text{ kJ mol}^{-1}$
 Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that
 (a) Electron repulsion outweighs the stability gained by achieving noble gas configuration
 (b) O^- ion has comparatively smaller size than oxygen atom
 (c) Oxygen is more electronegative
 (d) Addition of electron in oxygen results in larger size of the ion.
27. Which of the following statements is wrong ?
 (a) van der Waal's radius of iodine is more than its covalent radius
 (b) All isoelectronic ions belong to same period of the periodic table
 (c) I.E._1 of N is higher than that of O while I.E._2 of O is higher than that of N
 (d) The electron gain enthalpy of N is almost zero while that of P is 74.3 kJ mol^{-1}
28. The first ($\Delta_i H_1$) and second ($\Delta_i H_2$) ionization enthalpies (in kJ mol^{-1}) and the electron gain enthalpy ($\Delta_{\text{eg}} H$) (in kJ mol^{-1}) of the elements I, II, III, IV and V are given below

Element	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{\text{eg}} H$
I	520	7300	-60
II	419	3051	-48
III	1681	3374	-328
IV	1008	1846	-295
V	2372	5251	+48

The most reactive metal and the least reactive non-metal of these are respectively

- (a) I and V (b) V and II
 (c) II and V (d) IV and V
29. Consider the following statements:
 (i) Atomic radii decreases across a row of the periodic table when we move from left to right.
 (ii) Atomic radii increases down the column as we move from top to bottom.
 (iii) Although the order of elements is based on atomic numbers, vertical families share similar chemical properties.
 Which of the statement(s) given above is/are correct?
 (a) (i) and (ii) (b) (i) and (iii)
 (c) (ii) and (iii) (d) (i), (ii) and (iii)
30. Match Column-I with Column-II and select the correct answer by the given codes.
- | Column-I
(Atoms) | Column-II
(Properties) |
|--|-------------------------------|
| (A) He | (p) High electronegative |
| (B) F | (q) Most electropositive |
| (C) Rb | (r) Strongest reducing agent |
| (D) Li | (s) Highest ionisation energy |
| (a) A - (s), B - (q), C - (r), D - (p) | |
| (b) A - (p), B - (s), C - (q), D - (r) | |
| (c) A - (s), B - (p), C - (r), D - (q) | |
| (d) A - (s), B - (p), C - (q), D - (r) | |

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 3 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	55
Success Gap = Net Score - Qualifying Score			
Net Score = (Correct $\times$ 4) - (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC04

SYLLABUS : Chemical Bonding and Molecular Structure

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

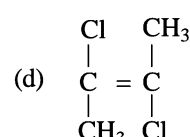
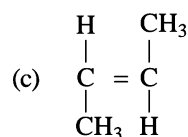
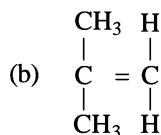
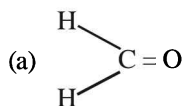
Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

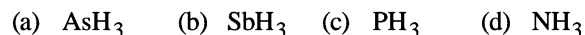
1. Which of the following does not contain coordinate bond ?



2. Which of the following has the highest dipole moment?



3. Of the following hydrides which one has the lowest boiling point ?



4. In compounds of type ECl_3 , where E = B, P, As or Bi, the angles Cl - E - Cl for different E are in the order.



RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5 D) is larger than that of NF_3 (0.2D). This is because
- in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions
 - in NH_3 as well as NF_3 the atomic dipole and bond dipole are in opposite directions
 - in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction
 - in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction
6. KF combines with HF to form KHF_2 . The compound contains the species
- K^+ , F^- and H^+
 - K^+ , F^- and HF
 - K^+ , and $[\text{HF}_2]^-$
 - $[\text{KHF}]^+$ and F_2
7. On changing N_2 to N_2^+ , the dissociation energy of N-N bond and on changing O_2 to O_2^+ the dissociation energy of O-O bond....
- increases, decreases
 - decreases, increases
 - decreases in both cases
 - increases in both cases
8. According to MO theory which of the following lists ranks the nitrogen species in terms of increasing bond order?
- $\text{N}_2^{2-} < \text{N}_2^- < \text{N}_2$
 - $\text{N}_2 < \text{N}_2^{2-} < \text{N}_2^-$
 - $\text{N}_2^- < \text{N}_2^{2-} < \text{N}_2$
 - $\text{N}_2^- < \text{N}_2 < \text{N}_2^{2-}$
9. Which of the two ions from the list given below have the geometry that is explained by the same hybridization of orbitals, NO_2^- , NO_3^- , NH_2^- , NH_4^+ , SCN^- ?
- NO_2^- and NO_3^-
 - NH_2^- and NO_3^-
 - SCN^- and NH_2^-
 - NO_2^- and NH_2^-
10. The values of electronegativity of atoms A and B are 1.20 and 4.0 respectively. The percentage of ionic character of A - B bond is
- 50%
 - 72.24%
 - 55.3%
 - 43%
11. In PO_4^{3-} ion the formal charge on the oxygen atom of P-O bond is
- +1
 - 1
 - 0.75
 - +0.75
12. Which molecule/ion out of the following does not contain unpaired electrons?
- N_2^+
 - O_2
 - O_2^{2-}
 - B_2
13. Which of the following pairs of ions are isoelectronic and isostructural ?
- ClO_3^- , CO_3^{2-}
 - SO_3^{2-} , NO_3^-
 - ClO_3^- , SO_3^{2-}
 - CO_3^{2-} , SO_3^{2-}
14. Which of the following is correct increasing order of lone pair of electrons on the central atom?
- $\text{IF}_7 < \text{IF}_5 < \text{ClF}_3 < \text{XeF}_2$
 - $\text{IF}_7 < \text{XeF}_2 < \text{ClF}_2 < \text{IF}_5$
 - $\text{IF}_7 < \text{ClF}_3 < \text{XeF}_2 < \text{IF}_5$
 - $\text{IF}_7 < \text{XeF}_2 < \text{IF}_5 < \text{ClF}_3$

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false?
- The H–C–H bond angle in CH_4 , the H–N–H bond angle in NH_3 , and the H–O–H bond angle in H_2O are all greater than 90°
 - The H–O–H bond angle in H_2O is larger than the H–C–H bond angle in CH_4 .
 - The H–O–H bond angle in H_2O is smaller than the H–N–H bond angle in NH_3 .
 - The H–C–H bond angle in CH_4 is larger than the H–N–H bond angle in NH_3 .
16. Predict the correct order among the following :
- lone pair - lone pair > lone pair - bond pair > bond pair - bond pair
 - lone pair - lone pair > bond pair - bond pair > lone pair - bond pair
 - bond pair - bond pair > lone pair - bond pair > lone pair - lone pair
 - lone pair - bond pair > bond pair - bond pair > lone pair - lone pair
17. In which of the following species the interatomic bond angle is $109^\circ 28'$?
- NH_3 , BF_4^-
 - NH_4^+ , BF_3
 - NH_3 , BF_4
 - NH_2^- , BF_3 .
18. In XeF_2 , XeF_4 and XeF_6 , the number of lone pairs on Xe are respectively
- 2, 3, 1
 - 1, 2, 3
 - 4, 1, 2
 - 3, 2, 1
19. Hybridisation of the underline atom changes in:
- $\underline{\text{Al}}\text{H}_3$ changes to $\text{Al}\underline{\text{H}}_4^-$
 - $\text{H}_2\underline{\text{O}}$ changes to $\text{H}_3\underline{\text{O}}^+$
 - $\underline{\text{N}}\text{H}_3$ changes to NH_4^+
 - in all cases
20. The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively
- sp^3 and sp^2
 - sp^2 and sp^3
 - sp^2 and sp^2
 - sp^3 and sp^3
21. Among the following ions, the $p\pi - d\pi$ overlap could be present in
- NO_3^-
 - PO_4^{3-}
 - CO_3^{2-}
 - NO_2^-
22. The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down group-15 of the periodic table is due to
- decreasing lp-bp repulsion
 - decreasing electronegativity
 - increasing bp-bp repulsion
 - increasing p-orbital character in sp^3
23. Which one of the following pairs of species have the same bond order?
- CN^- and NO^+
 - CN^- and CN^+
 - O_2^- and CN^-
 - NO^+ and CN^+
24. N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which of the following statements is wrong?
- In N_2^+ , N - N bond weakens
 - In O_2^+ , the O - O bond order increases
 - In O_2^+ , paramagnetism decreases
 - N_2^+ becomes diamagnetic

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

Space for Rough Work

C-16

DPP/ CC04

25. The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} whereas that of C – F in CF_4 is 515 kJ mol^{-1} . The correct reason for higher B – F bond dissociation energy as compared to that of C – F is
- stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4 .
 - significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4 .
 - lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4 .
 - smaller size of B-atom as compared to that of C-atom.
26. Ortho-Nitrophenol is less soluble in water than *p*- and *m*-Nitrophenols because :
- o*-Nitrophenol is more volatile steam than those of *m*- and *p*-isomers.
 - o*-Nitrophenol shows intramolecular H-bonding
 - o*-Nitrophenol shows intermolecular H-bonding
 - Melting point of *o*-Nitrophenol is lower than those of *m*- and *p*-isomers.
27. Amongst LiCl , RbCl , BeCl_2 and MgCl_2 the compounds with the greatest and the least ionic character, respectively are:
- LiCl and RbCl
 - RbCl and BeCl_2
 - MgCl_2 and BeCl_2
 - RbCl and MgCl_2
28. Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2 , Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is
- $\text{H}_2 > \text{B}_2 > \text{Li}_2$
 - $\text{Li}_2 > \text{H}_2 > \text{B}_2$
 - $\text{Li}_2 > \text{B}_2 > \text{H}_2$
 - $\text{B}_2 > \text{H}_2 > \text{Li}_2$
29. The shape of IF_6^- is :
- Trigonally distorted octahedron
 - Pyramidal
 - Octahedral
 - Square antiprism
30. Which one of the following molecules is paramagnetic?
- N_2
 - NO
 - CO
 - O_3

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 4 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	50
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC05

SYLLABUS : States of Matter

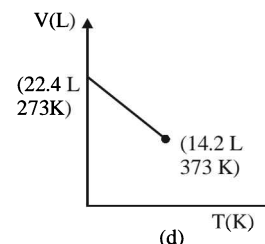
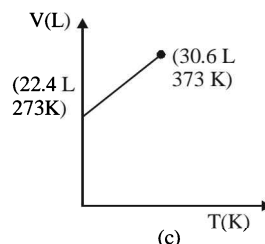
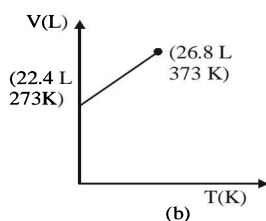
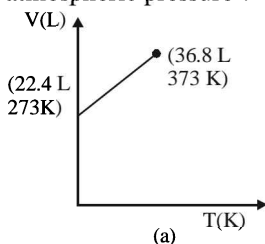
Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. Which of the following volume (V) - temperature (T) plots represents the behaviour of one mole of an ideal gas at one atmospheric pressure ?



RESPONSE GRID

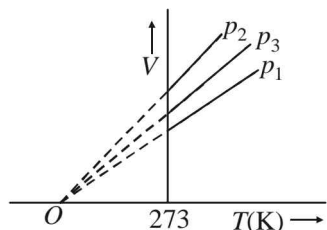
1. (a) (b) (c) (d)

Space for Rough Work

C-18

DPP/ CC05

2. The volume-temperature graphs of a given mass of an ideal gas at constant pressure are shown below.



What is the correct order of pressures ?

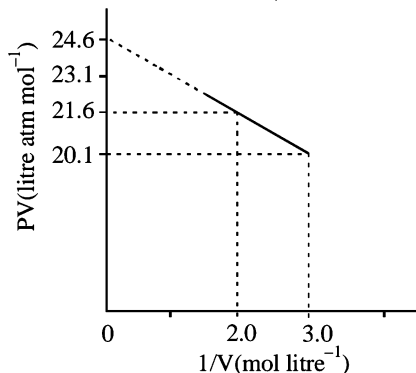
- (a) $p_1 > p_3 > p_2$ (b) $p_1 > p_2 > p_3$
 (c) $p_2 > p_3 > p_1$ (d) $p_2 > p_1 > p_3$
3. The inversion temperature T_i (K) of hydrogen is (given van der Waal's constants a and b are $0.244 \text{ atm L}^2 \text{ mol}^{-2}$ and 0.027 L mol^{-1} respectively)
 (a) 440 (b) 220 (c) 110 (d) 330
4. The temperature of the gas is raised from 27°C to 927°C , the root mean square velocity is
 (a) $\sqrt{927/27}$ time the earlier value
 (b) same as before
 (c) halved
 (d) doubled
5. Equal masses of methane and oxygen are mixed in an empty container at 25°C . The fraction of the total pressure exerted by oxygen is
 (a) $1/2$ (b) $2/3$
 (c) $\frac{1}{3} \times \frac{273}{298}$ (d) $1/3$
6. The ratio between the root mean square speed of H_2 at 50 K and that of O_2 at 800 K is,
 (a) 4 (b) 2
 (c) 1 (d) $1/4$
7. Calculate the total pressure in a 10.0 L cylinder which contains 0.4g helium, 1.6 g oxygen and 1.4 g nitrogen at 27°C .
 (a) 0.492 atm (b) 49.2 atm
 (c) 4.52 atm (d) 0.0492 atm
8. A bottle of dry ammonia and a bottle of dry hydrogen chloride connected through a long tube are opened simultaneously at both ends the white ammonium chloride ring first formed will be
 (a) at the centre of the tube.
 (b) near the hydrogen chloride bottle.
 (c) near the ammonia bottle.
 (d) throughout the length of the tube.
9. The r.m.s velocity of CO_2 at temperature T (in kelvin) is $x \text{ cms}^{-1}$. At what temperature (in kelvin) the r.m.s. velocity of nitrous oxide would be $4x \text{ cms}^{-1}$?
 (a) $16T$ (b) $2T$
 (c) $4T$ (d) $32T$
10. A mixture of two gases A and B in the mole ratio 2 : 3 is kept in a 2 litre vessel. A second 3L vessel has the same two gases in the mole ratio 3 : 5. Both gas mixtures have the same temperature and same total pressure. They are allowed to inter-mix and the final temperature and the total pressure are the same as the initial values, the final volume being 5 litres. Given that the molar masses are M_A and M_B . What is the mean molar mass of the final mixture?
 (a) $\frac{5M_A + 8M_B}{13}$ (b) $\frac{77M_A + 123M_B}{200}$
 (c) $\frac{123M_A + 77M_B}{250}$ (d) $\frac{123M_A + 77M_B}{150}$
11. At constant volume and temperature conditions, the rate of diffusion D_A and D_B of gases A and B having densities ρ_A and ρ_B are related by the expression.
 (a) $D_A = \left[D_B \frac{\rho_A}{\rho_B} \right]^{1/2}$ (b) $D_A = \left[D_B \frac{\rho_B}{\rho_A} \right]^{1/2}$
 (c) $D_A = D_B \left(\frac{\rho_A}{\rho_B} \right)^{1/2}$ (d) $D_A = D_B \left(\frac{\rho_B}{\rho_A} \right)^{1/2}$

RESPONSE
GRID

2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d) 5. (a)(b)(c)(d) 6. (a)(b)(c)(d)
 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d) 10. (a)(b)(c)(d) 11. (a)(b)(c)(d)

Space for Rough Work

12. For one mole of a van der Waals gas when $b = 0$ and $T = 300\text{ K}$, the PV vs. $1/V$ plot is shown below. The value of the van der Waals constant a ($\text{atm. liter}^2 \text{ mol}^{-2}$) is :



- (a) 1.0 (b) 4.5 (c) 1.5 (d) 3.0
13. As the temperature is raised from 20°C to 40°C , the average kinetic energy of neon atoms changes by a factor of which of the following ?
 (a) $313/293$ (b) $\sqrt{313/293}$
 (c) $1/2$ (d) 2
14. Maximum deviation from ideal gas is expected from :
 (a) $\text{N}_2(\text{g})$ (b) $\text{CH}_4(\text{g})$
 (c) $\text{NH}_3(\text{g})$ (d) $\text{H}_2(\text{g})$
15. Equal masses of H_2 , O_2 and methane have been taken in a container of volume $\bar{V}$ at temperature 27°C in identical conditions. The ratio of the volumes of gases $\text{H}_2 : \text{O}_2 : \text{methane}$ would be :
 (a) $8:16:1$ (b) $16:8:1$
 (c) $16:1:2$ (d) $8:1:2$
16. Equal moles of hydrogen and oxygen gases are placed in a container with a pin-hole through which both can escape. What fraction of the oxygen escapes in the time required for one-half of the hydrogen to escape ?
 (a) $1/8$ (b) $1/4$
 (c) $3/8$ (d) $1/2$
17. Which one of the following statements is NOT true about the effect of an increase in temperature on the distribution of molecular speeds in a gas?
 (a) The area under the distribution curve remains the same as under the lower temperature
 (b) The distribution becomes broader
 (c) The fraction of the molecules with the most probable speed increases
 (d) The most probable speed increases
18. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels
 (a) in a wavy path
 (b) in a straight line path
 (c) with an accelerated velocity
 (d) in a circular path
19. For 1 mol of an ideal gas at a constant temperature T , the plot of $(\log P)$ against $(\log V)$ is a (P : Pressure, V : Volume)
 (a) Straight line parallel to x -axis.
 (b) Straight line with a negative slope.
 (c) Curve starting at origin.
 (d) Straight line passing through origin.
20. When $\text{CO}_2(\text{g})$ is passed over red hot coke it partially gets reduced to $\text{CO}(\text{g})$. Upon passing 0.5 L of $\text{CO}_2(\text{g})$ over red hot coke, the total volume of the gases increased to 700 mL . The composition of the gaseous mixture at STP is
 (a) $\text{CO}_2 = 300\text{ mL}$; $\text{CO} = 400\text{ mL}$
 (b) $\text{CO}_2 = 0.0\text{ mL}$; $\text{CO} = 700\text{ mL}$
 (c) $\text{CO}_2 = 200\text{ mL}$; $\text{CO} = 500\text{ mL}$
 (d) $\text{CO}_2 = 350\text{ mL}$; $\text{CO} = 350\text{ mL}$
21. For gaseous state, if most probable speed is denoted by C^* , average speed by $\bar{C}$ and mean square speed by C , then for a large number of molecules the ratios of these speeds are :
 (a) $C^* : \bar{C} : C = 1.225 : 1.128 : 1$
 (b) $C^* : \bar{C} : C = 1.128 : 1.225 : 1$
 (c) $C^* : \bar{C} : C = 1 : 1.128 : 1.225$
 (d) $C^* : \bar{C} : C = 1 : 1.225 : 1.128$

RESPONSE
GRID

12. (a) (b) (c) (d)

13. (a) (b) (c) (d)

14. (a) (b) (c) (d)

15. (a) (b) (c) (d)

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

Space for Rough Work

C-20

DPP/ CC05

22. Sulphur dioxide and oxygen were allowed to diffuse through a porous partition. 20 dm^3 of SO_2 diffuses through the porous partition in 60 seconds. The volume of O_2 in dm^3 which diffuses under the similar condition in 30 seconds will be (atomic mass of sulphur = 32 u):
 (a) 7.09 (b) 14.1
 (c) 10.0 (d) 28.2
23. By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from $5 \times 10^4 \text{ cm/s}$ to $10 \times 10^4 \text{ cm/s}$?
 (a) Two (b) Three
 (c) Six (d) Four
24. The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is: Atomic masses: He = 4 u, O = 16 u
 (a) 300 K (b) 600 K
 (c) 1200 K (d) 2400 K
25. If Z is a compressibility factor, van der Waals equation at low pressure can be written as:
 (a) $Z = 1 + \frac{RT}{Pb}$ (b) $Z = 1 - \frac{a}{VRT}$
 (c) $Z = 1 - \frac{Pb}{RT}$ (d) $Z = 1 + \frac{Pb}{RT}$
26. The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, the final volume the gas will be:
 (a) 1.750 L (b) 3.60 L
 (c) 4.032 L (d) 7.50 L
27. 56 g of nitrogen and 96 g of oxygen are mixed isothermally and at a total pressure of 10 atm. The partial pressures of oxygen and nitrogen (in atm) are respectively
 (a) 4, 6 (b) 5, 5
 (c) 2, 8 (d) 6, 4
28. Two vessels containing gases A and B are interconnected as shown in the figure. The stopper is opened, the gases are allowed to mix homogeneously. The partial pressures of A and B in the mixture will be, respectively
- Gas A
 12 L
 8 atm

Gas B
 8 L
 5 atm
- (a) 8 and 5 atm (b) 9.6 and 4 atm
 (c) 4.8 and 2 atm (d) 6.4 and 4 atm
29. A neon-dioxygen mixture contains 70.6 g O_2 and 167.5 g neon. If pressure of the mixture of gases in the cylinder is 25 bar. What is the partial pressure of O_2 and Ne in the mixture respectively?
 (a) 5.25 bar, 10 bar (b) 19.75 bar, 5.25 bar
 (c) 19.75 bar, 10 bar (d) 5.75 bar, 19.75 bar
30. Let the most probable velocity of hydrogen molecules at a temperature of $t^\circ \text{C}$ be V_0 . When the temperature is raised to $(2t + 273)^\circ \text{C}$ the new rms velocity is (suppose all the molecules dissociate into atoms at latter temperature)
 (a) $2\sqrt{3}V_0$ (b) $\sqrt{6}V_0$
 (c) $\sqrt{3\left(2 + \frac{273}{t}\right)}V_0$ (d) $\sqrt{\frac{2}{3}}V_0$

**RESPONSE
GRID**

 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d) 25. (a)(b)(c)(d) 26. (a)(b)(c)(d)
 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d) 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 5 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	53
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC06

SYLLABUS : Thermodynamics

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

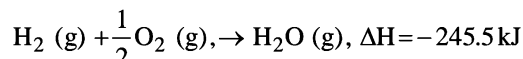
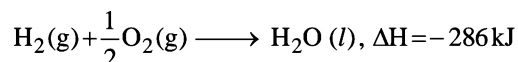
- Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and 100°C , (given : molar enthalpy of vapourisation of water at 1 bar and 373 K = 41 kJ mol^{-1} and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$) will be
 - $41.00 \text{ kJ mol}^{-1}$
 - $4.100 \text{ kJ mol}^{-1}$
 - $3.7904 \text{ kJ mol}^{-1}$
 - $37.904 \text{ kJ mol}^{-1}$
- In a closed insulated container, a liquid is stirred with a paddle to increase the temperature, which of the following is true?
 - $\Delta E = W \neq 0, q = 0$
 - $\Delta E = W = q \neq 0$
 - $\Delta E = 0, W = q \neq 0$
 - $W = 0, \Delta E = q \neq 0$
- The maximum work which a system can perform at constant temperature and pressure equals
 - $\Delta H - \Delta E$
 - $(-\Delta G)$
 - $q - \Delta E$
 - $P\Delta V - \Delta G$
- The entropy change in the fusion of one mole of a solid melting at 27°C (Latent heat of fusion, 2930 J mol^{-1}) is :
 - $9.77 \text{ J K}^{-1} \text{ mol}^{-1}$
 - $10.73 \text{ J K}^{-1} \text{ mol}^{-1}$
 - $2930 \text{ J K}^{-1} \text{ mol}^{-1}$
 - $108.5 \text{ J K}^{-1} \text{ mol}^{-1}$

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

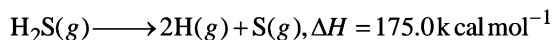
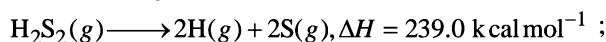
Space for Rough Work

5. The enthalpy of vapourisation of water from the following two equations is.



- (a) 6.02 kJ (b) 40.5 kJ
(c) 62.3 kJ (d) 1.25 kJ
6. The standard enthalpy of formation of NH_3 is $-46.0 \text{ kJ mol}^{-1}$. If the enthalpy of formation of H_2 from its atoms is -436 kJ mol^{-1} and that of N_2 is -712 kJ mol^{-1} , the average bond enthalpy of N – H bond in NH_3 is
- (a) -964 kJ mol^{-1} (b) $+352 \text{ kJ mol}^{-1}$
(c) $+1056 \text{ kJ mol}^{-1}$ (d) $-1102 \text{ kJ mol}^{-1}$

7. Calculate enthalpy change for the change $8\text{S}(\text{g}) \rightarrow 4\text{S}_8(\text{g})$, given that



- (a) $+512.0 \text{ kcal}$ (b) -512.0 kcal
(c) 508.0 kcal (d) -508.0 kcal

8. From the following bond energies:

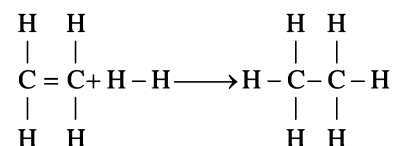
H – H bond energy: $431.37 \text{ kJ mol}^{-1}$

C = C bond energy: $606.10 \text{ kJ mol}^{-1}$

C – C bond energy: $336.49 \text{ kJ mol}^{-1}$

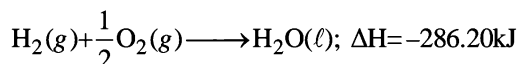
C – H bond energy: $410.50 \text{ kJ mol}^{-1}$

Enthalpy for the reaction,



will be:

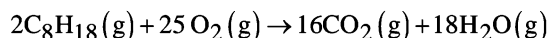
- (a) $-243.6 \text{ kJ mol}^{-1}$ (b) $-120.0 \text{ kJ mol}^{-1}$
(c) $553.0 \text{ kJ mol}^{-1}$ (d) $1523.6 \text{ kJ mol}^{-1}$
9. On the basis of the following thermochemical data :
[$\Delta_f H^\circ(\text{H}^+_{\text{aq}}) = 0$]



The value of enthalpy of formation of OH^- ion at 25°C is:

- (a) -228.88 kJ (b) $+228.88 \text{ kJ}$
(c) -343.52 kJ (d) -22.88 kJ
10. For a particular reversible reaction at temperature T , ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, the reaction would be spontaneous when
- (a) $T_e > T$ (b) $T > T_e$
(c) T_e is 5 times T (d) $T = T_e$

11. Consider the following spontaneous reaction

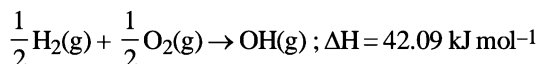


The sign of ΔH , ΔS , ΔG would be respectively

- (a) $+, -, +$ (b) $-, +, -$
(c) $+, +, -$ (d) $-, -, -$
12. The favourable conditions for a spontaneous reaction are

- (a) $T\Delta S > \Delta H$, $\Delta H = +\text{ve}$, $\Delta S = +\text{ve}$
(b) $T\Delta S > \Delta H$, $\Delta H = +\text{ve}$, $\Delta S = -\text{ve}$
(c) $T\Delta S = \Delta H$, $\Delta H = -\text{ve}$, $\Delta S = -\text{ve}$
(d) $T\Delta S = \Delta H$, $\Delta H = +\text{ve}$, $\Delta S = +\text{ve}$

13. The enthalpies of the following reactions are shown below.



Calculate the O – H bond energy for the hydroxyl radical.

- (a) $223.18 \text{ kJ mol}^{-1}$ (b) $423.38 \text{ kJ mol}^{-1}$
(c) $513.28 \text{ kJ mol}^{-1}$ (d) $113.38 \text{ kJ mol}^{-1}$
14. When 5 litres of a gas mixture of methane and propane is perfectly combusted at 0°C and 1 atmosphere, 16 litre of oxygen at the same temperature and pressure is consumed. The amount of heat released from this combustion in kJ ($\Delta H_{\text{comb}}(\text{CH}_4) = 890 \text{ kJ mol}^{-1}$, $\Delta H_{\text{comb}}(\text{C}_3\text{H}_8) = 2220 \text{ kJ mol}^{-1}$) is
- (a) 32 (b) 38
(c) 317 (d) 477

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. The heat of combustion of carbon to CO_2 is -393.5 kJ/mol . The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is
 (a) -315 kJ (b) $+315 \text{ kJ}$
 (c) -630 kJ (d) -3.15 kJ
16. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a $\text{C} \equiv \text{C}$ bond in C_2H_2 . That energy is (take the bond energy of a $\text{C} - \text{H}$ bond as 350 kJ mol^{-1})
 $2\text{C}(s) + \text{H}_2(g) \longrightarrow \text{HC} \equiv \text{CH}(g); \Delta H = 225 \text{ kJ mol}^{-1}$
 $2\text{C}(s) \longrightarrow 2\text{C}(g); \Delta H = 1410 \text{ kJ mol}^{-1}$
 $\text{H}_2(g) \longrightarrow 2\text{H}(g); \Delta H = 330 \text{ kJ mol}^{-1}$
 (a) 1165 (b) 837
 (c) 865 (d) 815
17. For a given reaction, $\Delta H = 35.5 \text{ kJ mol}^{-1}$ and $\Delta S = 83.6 \text{ JK}^{-1} \text{ mol}^{-1}$. The reaction is spontaneous at : (Assume that ΔH and ΔS do not vary with temperature)
 (a) $T > 425 \text{ K}$ (b) All temperatures
 (c) $T > 298 \text{ K}$ (d) $T < 425 \text{ K}$
18. If an endothermic reaction is non-spontaneous at freezing point of water and becomes feasible at its boiling point, then
 (a) ΔH is $-ve$, ΔS is $+ve$
 (b) ΔH and ΔS both are $+ve$
 (c) ΔH and ΔS both are $-ve$
 (d) ΔH is $+ve$, ΔS is $-ve$
19. The standard enthalpies of formation of $\text{CO}_2(g)$, $\text{H}_2\text{O}(l)$ and glucose(s) at 25°C are -400 kJ/mol , -300 kJ/mol and -1300 kJ/mol , respectively. The standard enthalpy of combustion per gram of glucose at 25°C is
 (a) $+2900 \text{ kJ}$ (b) -2900 kJ
 (c) -16.11 kJ (d) $+16.11 \text{ kJ}$
20. If at 298 K the bond energies of $\text{C} - \text{H}$, $\text{C} - \text{C}$, $\text{C} = \text{C}$ and $\text{H} - \text{H}$ bonds are respectively 414 , 347 , 615 and 435 kJ mol^{-1} , the value of enthalpy change for the reaction
 $\text{H}_2\text{C} = \text{CH}_2(g) + \text{H}_2(g) \rightarrow \text{H}_3\text{C} - \text{CH}_3(g)$ at 298 K will be
 (a) -250 kJ (b) $+125 \text{ kJ}$
 (c) -125 kJ (d) $+250 \text{ kJ}$
21. Standard entropy of X_2 , Y_2 and X Y_3 are 60 , 40 and $50 \text{ J K}^{-1} \text{ mol}^{-1}$, respectively. For the reaction,
 $\frac{1}{2}\text{X}_2 + \frac{3}{2}\text{Y}_2 \rightarrow \text{X Y}_3, \Delta H = -30 \text{ kJ}$, to be at equilibrium, the temperature will be
 (a) 1250 K (b) 500 K
 (c) 750 K (d) 1000 K
22. The enthalpy changes for the following processes are listed below :
 $\text{Cl}_2(g) \rightarrow 2\text{Cl}(g), 242.3 \text{ kJ mol}^{-1}$
 $\text{I}_2(g) \rightarrow 2\text{I}(g), 151.0 \text{ kJ mol}^{-1}$
 $\text{ICl}(g) \rightarrow \text{I}(g) + \text{Cl}(g), 211.3 \text{ kJ mol}^{-1}$
 $\text{I}_2(s) \rightarrow \text{I}_2(g), 62.76 \text{ kJ mol}^{-1}$
 Given that the standard states for iodine and chlorine are $\text{I}_2(s)$ and $\text{Cl}_2(g)$, the standard enthalpy of formation for $\text{ICl}(g)$ is :
 (a) $+16.8 \text{ kJ mol}^{-1}$
 (b) $+244.8 \text{ kJ mol}^{-1}$
 (c) $-14.6 \text{ kJ mol}^{-1}$
 (d) $-16.8 \text{ kJ mol}^{-1}$
23. Bond enthalpies of H_2 , X_2 and HX are in the ratio $2 : 1 : 2$. If enthalpy of formation of HX is -50 kJ mol^{-1} , the bond enthalpy of X_2 is
 (a) 100 kJ mol^{-1} (b) 300 kJ mol^{-1}
 (c) 200 kJ mol^{-1} (d) 400 kJ mol^{-1}
24. Given that:
 (i) $\Delta_f H^\circ$ of N_2O is 82 kJ mol^{-1}
 (ii) Bond energies of $\text{N} \equiv \text{N}$, $\text{N} = \text{N}$, $\text{O} = \text{O}$ and $\text{N} = \text{O}$ are 946 , 418 , 498 and 607 kJ mol^{-1} respectively,
 The resonance energy of N_2O is :
 (a) -88 kJ (b) -66 kJ
 (c) -62 kJ (d) -44 kJ

RESPONSE
GRID

15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
 20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

C-24

DPP/ CC06

25. Given that bond energies of H–H and Cl–Cl are 430 kJ mol^{-1} and 240 kJ mol^{-1} respectively and ΔH_f for HCl is -90 kJ mol^{-1} , bond enthalpy of HCl is
 (a) 380 kJ mol^{-1} (b) 425 kJ mol^{-1}
 (c) 245 kJ mol^{-1} (d) 290 kJ mol^{-1}
26. Under isothermal condition for one mole of ideal gas what is the ratio of work done under reversible to irreversible process, initially held at 20 atm undergoes expansion from 1L to 2L, at 298K, under external pressure of 10 atm?
 (a) 1.7 (b) 2.0
 (c) 1.4 (d) 1.0
27. Temperature of 5 moles of a gas is decreased by 2K at constant pressure. Indicate the correct statement
 (a) Work done by gas is $= 5 R$
 (b) Work done over the gas is $= 10 R$
 (c) Work done by the gas $= 10 R$
 (d) Work done $= 0$
28. For complete combustion of ethanol,
 $\text{C}_2\text{H}_5\text{OH}(l) + 3\text{O}_2(g) \longrightarrow 2\text{CO}_2(g) + 3\text{H}_2\text{O}(l)$,
 the amount of heat produced as measured in bomb calorimeter, is $1364.47 \text{ kJ mol}^{-1}$ at 25°C . Assuming ideality the enthalpy of combustion, $\Delta_c H$, for the reaction will be:
 ($R = 8.314 \text{ kJ mol}^{-1}$)
 (a) $-1366.95 \text{ kJ mol}^{-1}$ (b) $-1361.95 \text{ kJ mol}^{-1}$
 (c) $-1460.95 \text{ kJ mol}^{-1}$ (d) $-1350.50 \text{ kJ mol}^{-1}$
29. Read the following statements carefully and choose the correct answer
 (i) Expansion of a gas in vacuum ($p_{\text{ex}} = 0$) is called free expansion.
 (ii) Work is done during free expansion of an ideal gas whether the process is reversible or irreversible
 (iii) No work is done during free expansion of an ideal gas whether the process is reversible or irreversible
 (iv) No work is done during free expansion of an ideal gas when the process is reversible
 (a) Only statement (iii) is correct
 (b) Statements (i) and (iii) are correct
 (c) Statements (ii) and (iv) are correct
 (d) Statements (i) and (iv) are correct
30. The (S°) of the following substances are:
 $\text{CH}_4(g)$ $186.2 \text{ JK}^{-1} \text{ mol}^{-1}$
 $\text{O}_2(g)$ $205.2 \text{ JK}^{-1} \text{ mol}^{-1}$
 $\text{CO}_2(g)$ $213.6 \text{ JK}^{-1} \text{ mol}^{-1}$
 $\text{H}_2\text{O}(g)$ $69.9 \text{ JK}^{-1} \text{ mol}^{-1}$
 The entropy change (ΔS°) for the reaction
 $\text{CH}_4(g) + 2\text{O}_2(g) \rightarrow \text{CO}_2(g) + 2\text{H}_2\text{O}(l)$ is:
 (a) $-312.5 \text{ JK}^{-1} \text{ mol}^{-1}$
 (b) $-242.8 \text{ JK}^{-1} \text{ mol}^{-1}$
 (c) $-108.1 \text{ JK}^{-1} \text{ mol}^{-1}$
 (d) $-37.6 \text{ JK}^{-1} \text{ mol}^{-1}$

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 6 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	36	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC07

SYLLABUS : Equilibrium

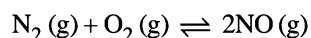
Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. K_c for the reaction



at 300 K is 4.0×10^{-6} . K_p for the above reaction will be ($R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$)

- (a) 2.4×10^{-3} (b) 4×10^{-6}
(c) $4 \times 10^{-6} (RT)^2$ (d) 16×10^{-12}

2. At 100°C the K_w of water is 55 times its value at 25°C. What will be the pH of neutral solution? ($\log 55 = 1.74$)

- (a) 6.13 (b) 7.00
(c) 7.87 (d) 5.13

3. The dissociation constant of a weak acid is 1×10^{-4} . In order to prepare a buffer solution with a pH = 5 the $[\text{Salt}]/[\text{Acid}]$ ratio should be

- (a) 1 : 10 (b) 4 : 5
(c) 10 : 1 (d) 5 : 4

4. The exothermic formation of ClF_3 is represented by the equation :



Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2 , F_2 and ClF_3 ?

- (a) Adding F_2
(b) Increasing the volume of the container
(c) Removing Cl_2
(d) Increasing the temperature

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

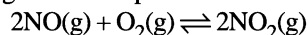
5. The K_{sp} of Ag_2CrO_4 , $AgCl$, $AgBr$ and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} , 8.3×10^{-17} . Which one of the following salts will precipitate last if $AgNO_3$ solution is added to the solution containing equal moles of $NaCl$, $NaBr$, NaI and Na_2CrO_4 ?

(a) $AgCl$ (b) $AgBr$
(c) Ag_2CrO_4 (d) AgI

6. MY and NY_3 , two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be true in regard to MY and NY_3 ?

(a) The molar solubilities of MY and NY_3 in water are identical.
(b) The molar solubility of MY in water is less than that of NY_3
(c) The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water.
(d) The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities.

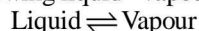
7. The following reaction is performed at 298 K.



The standard free energy of formation of $NO(g)$ is 86.6 kJ/mol at 298 K. What is the standard free energy of formation of $NO_2(g)$ at 298 K? ($K_p = 1.6 \times 10^{12}$)

(a) $86600 - \frac{\ln(1.6 \times 10^{12})}{R(298)}$
(b) $0.5[2 \times 86,600 - R(298) \ln(1.6 \times 10^{12})]$
(c) $R(298) \ln(1.6 \times 10^{12}) - 86600$
(d) $86600 + R(298) \ln(1.6 \times 10^{12})$

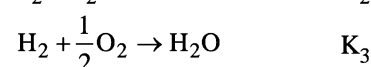
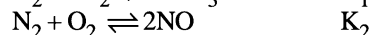
8. Consider the following liquid - vapour equilibrium.



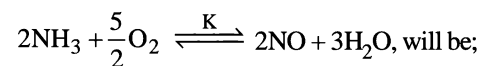
Which of the following relations is correct?

(a) $\frac{d \ln G}{dT^2} = \frac{\Delta H_v}{RT^2}$ (b) $\frac{d \ln P}{dT} = \frac{\Delta H_v}{RT}$
(c) $\frac{d \ln P}{dT^2} = \frac{-\Delta H_v}{T^2}$ (d) $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT^2}$

9. The equilibrium constant of the following are:

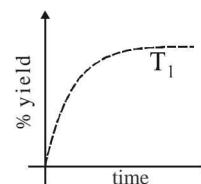


The equilibrium constant (K) of the reaction:

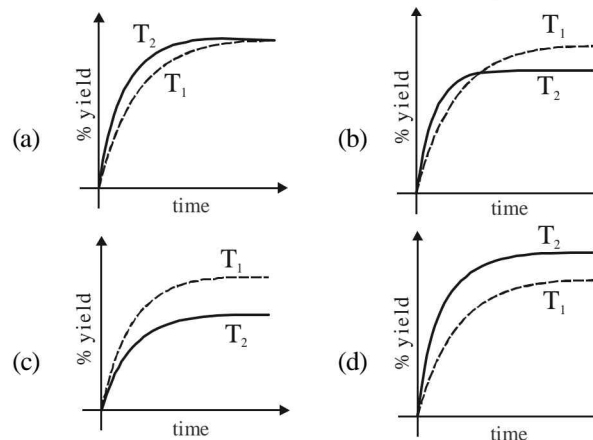


(a) $K_2K_3^3/K_1$ (b) K_2K_3/K_1
(c) $K_2^3K_3/K_1$ (d) $K_1K_3^3/K_2$

10. The % yield of ammonia as a function of time in the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H < 0$ at (P, T_1) is given below



If this reaction is conducted at (P, T_2) , with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by



RESPONSE
GRID

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

Space for Rough Work

11. Concentration of the Ag^+ ions in a saturated solution of $\text{Ag}_2\text{C}_2\text{O}_4$ is $2.2 \times 10^{-4} \text{ mol L}^{-1}$. Solubility product of $\text{Ag}_2\text{C}_2\text{O}_4$ is :-
 (a) 2.66×10^{-12} (b) 4.5×10^{-11}
 (c) 5.3×10^{-12} (d) 2.42×10^{-8}
12. For the following reaction in gaseous phase
 $\text{CO(g)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}), K_p / K_c$ is
 (a) $(RT)^{1/2}$ (b) $(RT)^{-1/2}$
 (c) (RT) (d) $(RT)^{-1}$
13. The degree of dissociation of $\text{PCl}_5(\alpha)$ obeying the equilibrium $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ is related to the equilibrium pressure by
 (a) $\alpha \propto \frac{1}{P^4}$ (b) $\alpha \propto \frac{1}{\sqrt{P}}$
 (c) $\alpha \propto \frac{1}{P^2}$ (d) $\alpha \propto P$
14. A 20 litre container at 400 K contains $\text{CO}_2(\text{g})$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :-
 (Given that : $\text{SrCO}_3(\text{s}) \rightleftharpoons \text{SrO}(\text{s}) + \text{CO}_2(\text{g}), K_p = 1.6 \text{ atm}$)
 (a) 10 litre (b) 4 litre
 (c) 2 litre (d) 5 litre
15. For the reaction $\text{C}_{(\text{s})} + \text{CO}_{2(\text{g})} \rightarrow 2\text{CO}_{(\text{g})}, K_p = 63 \text{ atm}$ at 1000 K. If at equilibrium : $P_{\text{CO}} = 10 P_{\text{CO}_2}$, then the total pressure of the gases at equilibrium is
 (a) 6.3 atm (b) 6.93 atm
 (c) 0.63 atm (d) 0.693 atm
16. The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be
 (a) 0.2×10^5 (b) 5.0×10^{-5}
 (c) 5.0×10^{15} (d) 5.0×10^{-15}
17. 2.5 ml of (2/5) M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°) is titrated with (2/15) M HCl in water at 25°C . The concentration of H^+ at equivalence point is ($K_w = 1 \times 10^{-14}$ at 25°C)
 (a) $3.7 \times 10^{-14} \text{ M}$ (b) $3.2 \times 10^{-7} \text{ M}$
 (c) $3.2 \times 10^{-2} \text{ M}$ (d) $2.7 \times 10^{-2} \text{ M}$
18. Calculate the pH of a solution obtained by diluting 1 mL of 0.10 M weak monoacidic base to 100 mL at constant temperature if K_b of the base is 1×10^{-5} ?
 (a) 8 (b) 9
 (c) 10 (d) 11
19. In a saturated solution of the sparingly soluble strong electrolyte AgIO_3 (molecular mass = 283) the equilibrium which sets is $\text{AgIO}_3(\text{s}) \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{IO}_3^-(\text{aq})$. If the solubility product constant K_{sp} of AgIO_3 at a given temperature is 1.0×10^{-8} , what is the mass of AgIO_3 contained in 100 ml of its saturated solution?
 (a) $1.0 \times 10^{-4} \text{ g}$ (b) $28.3 \times 10^{-2} \text{ g}$
 (c) $2.83 \times 10^{-3} \text{ g}$ (d) $1.0 \times 10^{-7} \text{ g}$
20. Species acting as both Bronsted acid and base is
 (a) $(\text{HSO}_4)^{-1}$ (b) Na_2CO_3
 (c) NH_3 (d) OH^{-1}

RESPONSE
GRID

11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d) 15. (a)(b)(c)(d)
 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d) 20. (a)(b)(c)(d)

Space for Rough Work

21. Let the solubility of an aqueous solution of $\text{Mg}(\text{OH})_2$ be x then its K_{sp} is
 (a) $4x^3$ (b) $108x^5$
 (c) $27x^4$ (d) $9x$
22. The K_{sp} for $\text{Cr}(\text{OH})_3$ is 1.6×10^{-30} . The solubility of this compound in water is :
 (a) $\sqrt[4]{1.6 \times 10^{-30}}$ (b) $\sqrt[4]{1.6 \times 10^{-30}} / 27$
 (c) $1.6 \times 10^{-30/27}$ (d) $\sqrt{1.6 \times 10^{-30}}$
23. Which equilibrium can be described as an acid-base reaction using the Lewis acid-base definition but not using the Bronsted-Lowry definition?
 (a) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightleftharpoons 2\text{NH}_4^+ + \text{SO}_4^{2-}$
 (b) $\text{NH}_3 + \text{CH}_3\text{COOH} \rightleftharpoons \text{NH}_4^+ + \text{CH}_3\text{COO}^-$
 (c) $\text{H}_2\text{O} + \text{CH}_3\text{COOH} \rightleftharpoons \text{H}_3\text{O}^+ + \text{CH}_3\text{COO}^-$
 (d) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$
24. Which one of the following statements is not true?
 (a) $\text{pH} + \text{pOH} = 14$ for all aqueous solutions
 (b) The pH of $1 \times 10^{-8} \text{ M HCl}$ is 8
 (c) 96,500 coulombs of electricity when passed through a CuSO_4 solution deposits 1 gram equivalent of copper at the cathode
 (d) The conjugate base of H_2PO_4^- is HPO_4^{2-}
25. How many litres of water must be added to 1 litre an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ?
 (a) 0.1 L (b) 0.9 L (c) 2.0 L (d) 9.0 L
26. For the reaction :
 $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g}),$
 $(K_c = 1.8 \times 10^{-6} \text{ at } 184^\circ\text{C}) (R = 0.0831 \text{ kJ / (mol. K)})$
- When K_p and K_c are compared at 184°C , it is found that
 (a) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure
 (b) $K_p = K_c$
 (c) K_p is less than K_c
 (d) K_p is greater than K_c
27. Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge + 4 and four phosphate anions of charge - 3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct?
 (a) $S = \{K_{sp} / (6912)^{1/7}\}$ (b) $S = \{K_{sp} / 144\}^{1/7}$
 (c) $S = \{K_{sp} / 6912\}^{1/7}$ (d) $S = \{K_{sp} / 6912\}^7$
28. The pK_a of a weak acid, HA , is 4.80. The pK_b of a weak base, BOH , is 4.78. The pH of an aqueous solution of the corresponding salt, BA , will be
 (a) 9.58 (b) 4.79
 (c) 7.01 (d) 9.22
29. Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts Hg_2Cl_2 , $\text{Cr}_2(\text{SO}_4)_3$, BaSO_4 and CrCl_3 respectively ?
 (a) $\text{BaSO}_4 > \text{Hg}_2\text{Cl}_2 > \text{Cr}_2(\text{SO}_4)_3 > \text{CrCl}_3$
 (b) $\text{BaSO}_4 > \text{Hg}_2\text{Cl}_2 > \text{CrCl}_3 > \text{Cr}_2(\text{SO}_4)_3$
 (c) $\text{BaSO}_4 > \text{CrCl}_3 > \text{Hg}_2\text{Cl}_2 > \text{Cr}_2(\text{SO}_4)_3$
 (d) $\text{Hg}_2\text{Cl}_2 > \text{BaSO}_4 > \text{CrCl}_3 > \text{Cr}_2(\text{SO}_4)_3$
30. Calculate the pH of 0.5 M aqueous solution of NaCN , the pK_b of CN^- is 4.70
 (a) 4.70 (b) 11.5
 (c) 7 (d) 6.5

RESPONSE GRID

21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d) 25. (a)(b)(c)(d)
 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d) 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 7 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	50
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC08

SYLLABUS : Redox Reaction

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

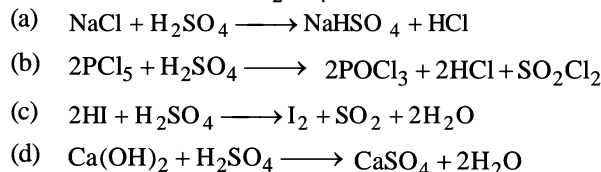
- In which of the following pairs, there is greatest difference in the oxidation number of the underlined elements ?
(a) $\underline{\text{N}}\text{O}_2$ and $\underline{\text{N}}_2\text{O}_4$ (b) $\underline{\text{P}}_2\text{O}_5$ and $\underline{\text{P}}_4\text{O}_{10}$
(c) $\underline{\text{N}}_2\text{O}$ and $\underline{\text{N}}\text{O}$ (d) $\underline{\text{S}}\text{O}_2$ and $\underline{\text{S}}\text{O}_3$
- Which of the following is a redox reaction?
(a) $\text{NaCl} + \text{KNO}_3 \rightarrow \text{NaNO}_3 + \text{KCl}$
(b) $\text{CaC}_2\text{O}_4 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$
(c) $\text{Mg}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{MgCl}_2 + 2\text{NH}_4\text{OH}$
(d) $\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn}(\text{CN})_2$
- Several blocks of magnesium are fixed to the bottom of a ship to
(a) make the ship lighter
(b) prevent action of water and salt
(c) prevent puncturing by under-sea rocks
(d) keep away the sharks
- When KMnO_4 reacts with acidified FeSO_4
(a) FeSO_4 is oxidised and KMnO_4 is reduced
(b) only KMnO_4 is oxidised
(c) only FeSO_4 is oxidised
(d) None of these

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

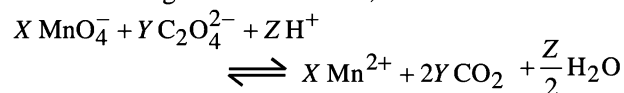
5. Which of the following chemical reactions depict the oxidizing behaviour of H_2SO_4 ?



6. Which of the following statements are correct concerning redox properties?

- (i) A metal M for which E° for the half life reaction $\text{M}^{n+} + n\text{e}^- \rightleftharpoons \text{M}$ is very negative will be a good reducing agent.
 (ii) The oxidizing power of the halogens decreases from chlorine to iodine.
 (iii) The reducing power of hydrogen halides increases from hydrogen chloride to hydrogen iodide
 (a) (i), (ii) and (iii) (b) (i) and (ii)
 (c) (i) only (d) (ii) and (iii)

7. In the following balanced reaction,



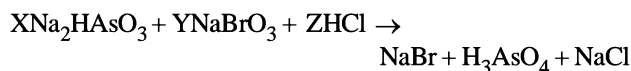
values of X, Y and Z respectively are

- (a) 2, 5, 16 (b) 8, 2, 5
 (c) 5, 2, 16 (d) 5, 8, 4

8. Arrange the following in the order of their decreasing electrode potentials : Mg, K, Ba and Ca

- (a) K, Ca, Ba, Mg (b) Ba, Ca, K, Mg
 (c) Ca, Mg, K, Ba (d) Mg, Ca, Ba, K

9. Given :



The values of X, Y and Z in the above redox reaction are respectively :

- (a) 2, 1, 2 (b) 2, 1, 3
 (c) 3, 1, 6 (d) 3, 1, 4

10. Oxidation state of sulphur in anions SO_3^{2-} , $\text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the orders :

- (a) $\text{S}_2\text{O}_6^{2-} < \text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-}$
 (b) $\text{SO}_3^{2-} < \text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-}$
 (c) $\text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-} < \text{S}_2\text{O}_6^{2-}$
 (d) $\text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-} < \text{SO}_3^{2-}$

11. Amongst the following, identify the species with an atom in +6 oxidation state:

- (a) $[\text{MnO}_4]^-$ (b) $[\text{Cr}(\text{CN})_6]^{3-}$
 (c) Cr_2O_3 (d) CrO_2Cl_2

12. Which one of the following cannot function as an oxidising agent ?

- (a) I^- (b) $\text{S}(\text{s})$
 (c) $\text{NO}_3^-(\text{aq})$ (d) $\text{Cr}_2\text{O}_7^{2-}$

13. A compound of Xe and F is found to have 53.5% of Xe. What is oxidation number of Xe in this compound ?

- (a) -4 (b) 0
 (c) +4 (d) +6

14. Copper becomes green when exposed to moist air for a long period. This is due to:

- (a) the formation of a layer of cupric oxide on the surface of copper.
 (b) the formation of a layer of basic carbonate of copper on the surface of copper.
 (c) the formation of a layer of cupric hydroxide on the surface of copper.
 (d) the formation of basic copper sulphate layer on the surface of the metal.

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. In the standardization of $\text{Na}_2\text{S}_2\text{O}_3$ using $\text{K}_2\text{Cr}_2\text{O}_7$ by iodometry, the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ is
 (a) (molecular weight)/2
 (b) (molecular weight)/6
 (c) (molecular weight)/3
 (d) same as molecular weight
16. Consider the reaction:

$$\text{H}_2\text{SO}_3(\text{aq}) + \text{Sn}^{4+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Sn}^{2+}(\text{aq}) + \text{HSO}_4^-(\text{aq}) + 3\text{H}^+(\text{aq})$$

 Which of the following statements is correct?
 (a) Sn^{4+} is the oxidizing agent because it undergoes oxidation
 (b) Sn^{4+} is the reducing agent because it undergoes oxidation
 (c) H_2SO_3 is the reducing agent because it undergoes oxidation
 (d) H_2SO_3 is the reducing agent because it undergoes reduction
17. The species that undergoes disproportionation in an alkaline medium are
 (a) Cl_2 (b) MnO_4^{2-}
 (c) NO_2 (d) All of these
18. How many electrons are involved in the following redox reaction?

$$\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{2+} + \text{C}_2\text{O}_4^{2-} \rightarrow \text{Cr}^{3+} + \text{Fe}^{3+} + \text{CO}_2 \text{ (Unbalanced)}$$

 (a) 3 (b) 4
 (c) 6 (d) 5
19. H_2S acts only as a reducing agent while SO_2 can act both as a reducing and oxidizing agent because
 (a) oxygen is more – ve in SO_2
 (b) hydrogen in H_2S is more + ve than oxygen
 (c) S in SO_2 has one oxidation state
 (d) S in H_2S has – 2 oxidation state
20. Which of the following cannot act as reducing agent?
 (a) NO_2 (b) SO_2
 (c) CO_2 (d) ClO_2
21. $\text{C}_2\text{H}_6(\text{g}) + n\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 In this equation, the ratio of the coefficients of CO_2 and H_2O is
 (a) 1 : 1 (b) 2 : 3
 (c) 3 : 2 (d) 1 : 3
22. Which substance serves as reducing agent in the following reaction ?

$$14\text{H}^+ + \text{Cr}_2\text{O}_7^{2-} + 3\text{Ni} \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{Ni}^{2+}$$

 (a) H_2O (b) Ni
 (c) H^+ (d) $\text{Cr}_2\text{O}_7^{2-}$
23. Which of the following reactions depict the oxidising behaviour of H_2SO_4 :
 (a) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \longrightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$
 (b) $2\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 (c) $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HCl}$
 (d) $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
24. Which one of the following reactions involves disproportionation?
 (a) $2\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow \text{CuSO}_4 + 2\text{H}_2\text{O} + \text{SO}_2$
 (b) $\text{As}_2\text{O}_3 + 3\text{H}_2\text{S} \rightarrow \text{As}_2\text{S}_3 + 3\text{H}_2\text{O}$
 (c) $2\text{KOH} + \text{Cl}_2 \rightarrow \text{KCl} + \text{KOC} + \text{H}_2\text{O}$
 (d) $\text{Ca}_3\text{P}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Ca}(\text{OH})_2 + 2\text{PH}_3$

RESPONSE
GRID

15. (a) (b) (c) (d) 16. (a) (b) (c) (d) 17. (a) (b) (c) (d) 18. (a) (b) (c) (d) 19. (a) (b) (c) (d)
 20. (a) (b) (c) (d) 21. (a) (b) (c) (d) 22. (a) (b) (c) (d) 23. (a) (b) (c) (d) 24. (a) (b) (c) (d)

Space for Rough Work

25. Point out the correct statement of the following about $\text{Na}_2\text{S}_4\text{O}_6$.
- Average oxidation number of S atom is +2
 - Oxidation number of two S atoms is zero each and that of other two is +5 each
 - Oxidation number of two S atoms is +1 each and that of other two is +4 each
 - None of these
26. The pair of compounds in which both the metals are in the highest possible oxidation state is
- $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{CN})_6]^{3-}$
 - CrO_2Cl_2 , MnO_4^-
 - TiO_3 , MnO_2
 - $[\text{Co}(\text{CN})_6]^{3-}$, MnO_3
27. Thiosulphate reacts differently with iodine and bromine in the reactions given below.
- $$2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$$
- $$\text{S}_2\text{O}_3^{2-} + 2\text{Br}_2 + 5\text{H}_2\text{O} \rightarrow 2\text{SO}_4^{2-} + 2\text{Br}^- + 10\text{H}^+$$
- Which of the following statements justifies the above dual behaviour of thiosulphate?
- Bromine is a stronger oxidant than iodine
 - Bromine is a weaker oxidant than iodine
 - Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reactions
 - Bromine undergoes oxidation and iodine undergoes reduction in these reactions
28. Standard reduction potentials of the half reactions are given below :
- $$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-(\text{aq}); E^\circ = +2.85 \text{ V}$$
- $$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq}); E^\circ = +1.36 \text{ V}$$
- $$\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq}); E^\circ = +1.06 \text{ V}$$
- $$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq}); E^\circ = +0.53 \text{ V}$$
- The strongest oxidising and reducing agents respectively are :
- F_2 and I^-
 - Br_2 and Cl^-
 - Cl_2 and Br^-
 - Cl_2 and I_2
29. A mixture of potassium chlorate, oxalic acid and sulphuric acid is heated. During the reaction which element undergoes maximum change in the oxidation number ?
- S
 - H
 - Cl
 - C
30. Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because
- Zn acts as an oxidising agent when it reacts with HNO_3
 - HNO_3 is weaker acid than H_2SO_4 and HCl
 - In electrochemical series, Zn is above hydrogen
 - NO_3^- is reduced in preference to hydronium ion

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 8 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	40	Qualifying Score	60
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC09

SYLLABUS : Hydrogen

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Which of the following will not displace hydrogen
 - Ba
 - Pb
 - Hg
 - Sn
- Which statement is correct for hydrogen ?
 - It has a very high ionisation potential
 - It is always collected at cathode
 - It can exist both in +1 as well as –1 oxidation state
 - It has same electronegativity as halogens
- The adsorption of hydrogen by metals is called
 - Dehydrogenation
 - Hydrogenation
 - Occlusion
 - Adsorption
- The low density of ice compared to water is due to
 - hydrogen bonding interactions
 - dipole – dipole interactions
 - dipole – induced dipole interactions
 - induced dipole – induced dipole interactions

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Which of the following terms is not correct for hydrogen ?
- Its molecule is diatomic
 - It exists both as H^+ and H^- in different chemical compounds
 - It is the only species which has no neutrons in the nucleus
 - Heavy water is unstable because hydrogen is substituted by its isotope deuterium
6. Hydrogen can be fused to form helium at
- High temperature and high pressure
 - High temperature and low pressure
 - Low temperature and high pressure
 - Low temperature and low pressure
7. The hydride ion, H^- , is a stronger base than the hydroxide ion, OH^- . Which one of the following reactions will occur if sodium hydride (NaH) is dissolved in water?
- $H^-(aq) + H_2O(l) \rightarrow H_3O^-(aq)$
 - $H^-(aq) + H_2O(l) \rightarrow OH^-(aq) + H_2(g)$
 - $H^-(aq) + H_2O(l) \rightarrow OH^-(aq) + 2H^+(aq) + 2e^-$
 - $H^-(aq) + H_2O(l) \rightarrow$ No reaction
8. Blackened oil painting can be restored into original form by the action of
- Chlorine
 - BaO_2
 - H_2O_2
 - MnO_2
9. Which of the following reactions is an example of use of water gas in the synthesis of other compounds?
- $CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270\text{ K}} CO(g) + H_2(g)$
 - $CO(g) + H_2O(g) \xrightarrow[\text{Catalyst}]{673\text{ K}} CO_2(g) + H_2(g)$
 - $C_nH_{2n+2} + nH_2O(g) \xrightarrow[Ni]{1270\text{ K}} nCO + (2n+1)H_2$
 - $CO(g) + 2H_2(g) \xrightarrow[\text{Catalyst}]{\text{Cobalt}} CH_3OH(l)$
10. Elements of which of the following group(s) of periodic table do not form hydrides?
- Groups 7, 8, 9
 - Group 13
 - Groups 15, 16, 17
 - Group 14
11. Which one of the following pairs of substances will not produce hydrogen when reacted together?
- Copper and conc. nitric acid
 - Ethanol and metallic sodium
 - Magnesium and steam
 - Phenol and metallic sodium
12. Which of the following is not a use of dihydrogen ?
- It used in fuel cells for generating electrical energy.
 - Atomic hydrogen and oxy-hydrogen torches are used for cutting and welding purposes.
 - It used in the synthesis of hydroquinone and tartaric acid.
 - Both (b) and (c)
13. Calculate the normality of 10 volume H_2O_2 ?
- 1.7 N
 - 12 N
 - 30.3 N
 - 0.0303 N
14. At its melting point ice is lighter than water because
- H_2O molecules are more closely packed in solid state
 - ice crystals have hollow hexagonal arrangement of H_2O molecules.
 - on melting of ice the H_2O molecule shrinks in size
 - ice forms mostly heavy water on first melting.

RESPONSE
GRID

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

11. (a)(b)(c)(d)

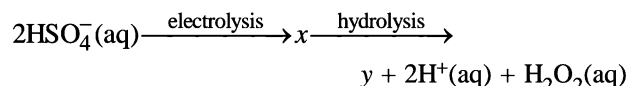
12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

Space for Rough Work

15. Identify x and y in following reaction:



- (a) $x = \text{H}_2\text{SO}_4(\text{aq})$, $y = 2\text{HSO}_4^-(\text{aq})$
 (b) $x = \text{HO}_3\text{SOOSO}_3\text{H}(\text{aq})$, $y = 2\text{HSO}_4^-(\text{aq})$
 (c) $x = \text{HO}_3\text{SOOSO}_3\text{H}(\text{aq})$, $y = \text{H}_2\text{SO}_4(\text{aq})$
 (d) $x = \text{H}_2\text{SO}_4(\text{aq})$, $y = \text{HO}_3\text{SOOSO}_3\text{H}(\text{aq})$
16. Which of the following species is diamagnetic in nature?
- (a) H_2^- (b) H_2^+
 (c) H_2 (d) He_2^+
17. The number of moles of H_2 in 0.224 litre of hydrogen gas at STP (273 K, 1 atm) is :
- (a) 0.1
 (b) 0.01
 (c) 0.001
 (d) 1
18. Hydrogen is not obtained when Zn reacts with
- (a) cold water
 (b) dil H_2SO_4
 (c) dil. HCl
 (d) 20% NaOH
19. Which of the following statements in relation to the hydrogen atom is correct ?
- (a) 3s, 3p and 3d orbitals all have the same energy
 (b) 3s and 3p orbitals are of lower energy than 3d orbital
 (c) 3p orbital is lower in energy than 3d orbital
 (d) 3s orbital is lower in energy than 3p orbital

20. Hydrogen molecules differs from chlorine molecule in the following respect

- (a) Hydrogen molecule is non-polar but chlorine molecule is polar
 (b) Hydrogen molecule is polar while chlorine molecule is non-polar
 (c) Hydrogen molecule can form intermolecular hydrogen bonds but chlorine molecule does not
 (d) Hydrogen molecule cannot participate in coordination bond formation but chlorine molecule can
21. Which of the following chemicals is not present in clear hard water?
- (a) MgCO_3 (b) MgSO_4
 (c) $\text{Mg}(\text{HCO}_3)_2$ (d) CaCl_2
22. True peroxide is
- (a) BaO_2 (b) MnO_2
 (c) PbO_2 (d) NO_2
23. Very pure hydrogen (99.9) can be made by which of the following processes ?
- (a) Reaction of methane with steam
 (b) Mixing natural hydrocarbons of high molecular weight
 (c) Electrolysis of water
 (d) Reaction of salts like hydrides with water
24. An inorganic compound gives off O_2 when heated, turns an acidic solution of KI violet and reduces acidified KMnO_4 . The compound is
- (a) SO_3 (b) KNO_3
 (c) H_2O_2 (d) All of these

RESPONSE
GRID

15. (a)(b)(c)(d)
20. (a)(b)(c)(d)

16. (a)(b)(c)(d)
21. (a)(b)(c)(d)

17. (a)(b)(c)(d)
22. (a)(b)(c)(d)

18. (a)(b)(c)(d)
23. (a)(b)(c)(d)

19. (a)(b)(c)(d)
24. (a)(b)(c)(d)

Space for Rough Work

C-36

DPP/ CC09

25. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium?
- (a) MnO_4^- (b) $\text{Cr}_2\text{O}_7^{2-}$
 (c) SO_3^{2-} (d) KI
26. Choose the correct statement :
 The reason for use of polyphosphates as water softening agents is, that
- (a) they form soluble complexes with anionic species
 (b) they precipitate out cationic species
 (c) they precipitate out anionic species
 (d) they form soluble complexes with cationic species
27. Which statement is wrong?
- (a) Ordinary hydrogen is an equilibrium mixture of ortho and para hydrogen
 (b) In ortho hydrogen spin of two nuclei is in same direction
 (c) Ortho and para forms do not resemble in their chemical properties
 (d) In para hydrogen spin of two nuclei is in opposite direction.
28. From the following statements regarding H_2O_2 , choose the incorrect statement :
- (a) It has to be stored in plastic or wax lined glass bottles in dark
 (b) It has to be kept away from dust
 (c) It can act only as an oxidizing agent
 (d) It decomposes on exposure to light
29. When zeolite (hydrated sodium aluminium silicate) is treated with hard water the sodium ions are exchanged with
- (a) H^+ ions (b) Ca^{2+} ions
 (c) SO_4^{2-} ions (d) OH^- ions
30. Identify x and y in following reaction. What is the mixture of x and y called?
- $$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \xrightarrow[\text{Ni}]{1270\text{K}} x + y$$
- (a) $x = \text{CO}_2$, $y = \text{H}_2\text{O}$, water gas
 (b) $x = \text{CO}$, $y = \text{H}_2\text{O}$, syn gas
 (c) $x = \text{CO}$, $y = \text{H}_2$, water gas
 (d) $x = \text{CO}_2$, $y = \text{H}_2$, syn gas

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 9 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	40	Qualifying Score	60
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC10

SYLLABUS : The s-Block Elements

Max. Marks : 120

Marking Scheme : (+ 4) for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Alkali metals are generally extracted by
 - reduction methods
 - double decomposition methods
 - displacement methods
 - electrolytic methods
- Which of the following represents a correct sequence of reducing power of the following elements?
 - $\text{Li} > \text{Cs} > \text{Rb}$
 - $\text{Rb} > \text{Cs} > \text{Li}$
 - $\text{Cs} > \text{Li} > \text{Rb}$
 - $\text{Li} > \text{Rb} > \text{Cs}$
- Strongest bond is in between
 - CsF
 - NaCl
 - Both (a) and (b)
 - None of above
- A metal salt solution forms a yellow precipitate with potassium chromate in acetic acid, a white precipitate with dil H_2SO_4 , but gives no precipitate with NaCl. The metal salt solution will consist of
 - PbCO_3
 - BaCO_3
 - MgCO_3
 - CaCO_3

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Which of the following has lowest thermal stability ?
 (a) Li_2CO_3 (b) Na_2CO_3
 (c) K_2CO_3 (d) Rb_2CO_3
6. The first (IE_1) and second (IE_2) ionisation energies (kJ/mol) of a few elements designated by Roman numerals are given below. Which of these would be an alkali metal?
- | | IE_1 | IE_2 |
|---------|---------------|---------------|
| (a) I | 2372 | 5251 |
| (b) II | 520 | 7300 |
| (c) III | 900 | 1760 |
| (d) IV | 16803 | 380 |
7. The solubilities of carbonates decrease down the magnesium group due to a decrease in
 (a) hydration energies of cations
 (b) inter-ionic attraction
 (c) entropy of solution formation
 (d) lattice energies of solids
8. KO_2 (potassium super oxide) is used in oxygen cylinders in space and submarines because it
 (a) absorbs CO_2 and increases O_2 content
 (b) eliminates moisture
 (c) absorbs CO_2
 (d) produces ozone.
9. Which one of the following salts does not impart colour to the flame ?
 (a) KI (b) LiCl
 (c) CaCl_2 (d) MgCl_2
10. Amongst LiCl, RbCl, BeCl_2 and MgCl_2 the compounds with the greatest and least ionic character respectively are :
 (a) LiCl and RbCl (b) MgCl_2 and BeCl_2
 (c) RbCl and BeCl_2 (d) RbCl and MgCl_2
11. Which of the following statements about Na_2O_2 is not correct?
 (a) It is diamagnetic in nature
 (b) It is derivative of H_2O_2
 (c) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium.
 (d) It is the super oxide of sodium
12. All of the following substances react with water. The pair that gives the same gaseous product is
 (a) K and KO_2
 (b) Na and Na_2O_2
 (c) Ca and CaH_2
 (d) Ba and BaO_2
13. Which is not correctly matched?
 (1) Basic strength $\text{Cs}_2\text{O} < \text{Rb}_2\text{O} < \text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Li}_2\text{O}$ of oxides
 (2) Stability of $\text{Na}_2\text{O}_2 < \text{K}_2\text{O}_2 < \text{Rb}_2\text{O}_2 < \text{Cs}_2\text{O}_2$ peroxides
 (3) Stability of $\text{LiHCO}_3 < \text{NaHCO}_3 < \text{KHCO}_3$ bicarbonates $< \text{RbHCO}_3 < \text{CsHCO}_3$
 (4) Melting point $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$
 (a) 1 and 4
 (b) 1 and 3
 (c) 1 and 2
 (d) 2 and 3
14. If NaOH is added to an aqueous solution of Zn^{2+} ions, a white precipitate appears and on adding excess NaOH, the precipitate dissolves. In this solution zinc exists in the :
 (a) both in cationic and anionic parts
 (b) there is no zinc left in the solution
 (c) cationic part
 (d) anionic part.

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. The compound A on heating gives a colourless gas and a residue that is dissolved in water to obtain B. Excess of CO_2 is bubbled through aqueous solution of B, C is formed which is recovered in the solid form. Solid C on gentle heating gives back A. The compound is
 (a) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (b) CaCO_3
 (c) Na_2CO_3 (d) K_2CO_3
16. In Castner-Kellner cell for production of sodium hydroxide:
 (a) Brine is electrolyzed with Pt electrodes
 (b) Brine is electrolyzed using graphite electrodes
 (c) Molten sodium chloride is electrolysed
 (d) Sodium amalgam is formed at mercury cathode
17. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour. Y is
 (a) $\text{Mg}(\text{NO}_3)_2$ (b) Mg_3N_2
 (c) NH_3 (d) MgO
18. The metals A and B form oxide but B also forms nitride when both burn in air. The A and B are
 (a) Cs, K (b) Mg, Ca
 (c) Li, Na (d) K, Mg
19. The melting point of lithium (181°C) is just double the melting point of sodium (98°C) because
 (a) down the group, the hydration energy decreases
 (b) down the group, the ionization energy decreases
 (c) down the group the cohesive energy decreases
 (d) None of these
20. Which of the following are arranged in increasing order of solubilities?
 (a) $\text{CaCO}_3 < \text{KHCO}_3 < \text{NaHCO}_3$
 (b) $\text{NaHCO}_3 < \text{KHCO}_3 < \text{CaCO}_3$
 (c) $\text{KHCO}_3 < \text{NaHCO}_3 < \text{CaCO}_3$
 (d) $\text{CaCO}_3 < \text{NaHCO}_3 < \text{KHCO}_3$
21. For a good quality cement what should be the ratio of following :
 I. Silica to alumina
 II. CaO to the total of oxides of SiO_2 , Al_2O_3 and Fe_2O_3
 (a) I = 2.5 to 4
 II = Greater than 2
 (b) I = Nearly 4
 II = Less than 2
 (c) I = 2.5
 II = Closer to 2
 (d) I = 2.5 to 4
 II = Closer to 2
22. Which one of the following does not react with water even under red hot condition?
 (a) Na (b) Be
 (c) Ca (d) K
23. Which of the following are found in biological fluids Na^+ , Mg^{2+} , Ca^{2+} , K^+ , Sr^{2+} , Li^+ and Ba^{2+}
 (a) Mg^{2+} , Ca^{2+} , and Sr^{2+}
 (b) Na^+ and K^+
 (c) Na^+ , K^+ , Mg^{2+} and Ca^{2+}
 (d) Sr^+ , Li and Ba^{2+}
24. Which of the following statements is not correct for alkali metals?
 (a) Alkali metals are the most electropositive metals.
 (b) Alkali metals exist in free state in nature.
 (c) These metals have the largest size in a particular period of the periodic table.
 (d) Both (b) and (c)

RESPONSE
GRID

15. (a) (b) (c) (d)

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

22. (a) (b) (c) (d)

23. (a) (b) (c) (d)

24. (a) (b) (c) (d)

Space for Rough Work

C-40

DPP/ CC10

25. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ?
 (a) BaSO_4 (b) SrSO_4
 (c) CaSO_4 (d) BeSO_4
26. The metallic sodium dissolves in liquid ammonia to form a deep blue coloured solution. The deep blue colour is due to formation of:
 (a) solvated electron, $e(\text{NH}_3)_x^-$
 (b) solvated atomic sodium, $\text{Na}(\text{NH}_3)_y$
 (c) $(\text{Na}^+ + \text{Na}^-)$
 (d) $\text{NaNH}_2 + \text{H}_2$
27. A firework gives out crimson coloured light. It contains a salt of
 (a) Ca (b) Na
 (c) Sr (d) Ba
28. Magnesium wire burns in the atmosphere of CO_2 because
 (a) Magnesium acts as an oxidising agent
 (b) Magnesium has 2 electrons in the outermost orbit.
 (c) Magnesium acts as a reducing agent and removes oxygen from CO_2
 (d) None of these
29. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be
 (a) -2.55 eV (b) -5.1 eV
 (c) -10.2 eV (d) $+2.55 \text{ eV}$
30. Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of :
 (a) $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$ (b) $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$
 (c) $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$ (d) $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 10 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	56
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC11

SYLLABUS : The p-Block Elements (Group 13 & 14)

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Boric acid is polymeric due to
 - its acidic nature
 - the presence of hydrogen bonds
 - its monobasic nature
 - its geometry
- Which of the following statements about anhydrous aluminium chloride is correct?
 - It exists as AlCl_3 molecules
 - It is not easily hydrolysed
 - It sublimes at 180°C
 - It is a strong Lewis base
- The approximate percentage of silica in cement is:
 - 5 – 10%
 - 15 – 20%
 - 20 – 25%
 - 25 – 30%
- An aqueous solution of potash alum gives :
 - two types of ions
 - only one type of ion
 - four types of ions
 - three types of ions

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

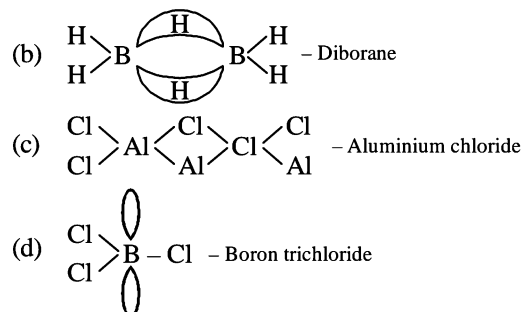
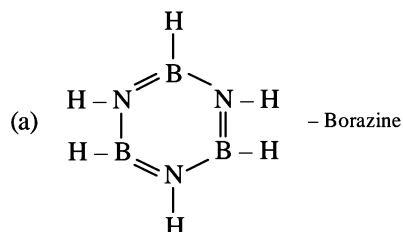
5. The IE_1 among the group 13 member follows as
 (a) $B > Al < Ga < Tl$ (b) $B > Al > Ga > Tl$
 (c) $B > Ga > Al > Tl$ (d) $B > Ga < Al < Tl$
6. Non-oxide ceramics can be
 (a) B_4C (b) SiC
 (c) Si_3N_4 (d) All of these
7. Be_2C and Al_4C_3 are called –
 (a) ethanides (b) methanides
 (c) carbonides (d) acetylides
8. Anhydrous $AlCl_3$ cannot be obtained from which of the following reactions ?
 (a) Heating $AlCl_3 \cdot 6H_2O$
 (b) By passing dry HCl over hot aluminium powder
 (c) By passing dry Cl_2 over hot aluminium powder
 (d) By passing dry Cl_2 over a hot mixture of alumina and coke
9. Aluminium is extracted from alumina (Al_2O_3) by electrolysis of a molten mixture of :
 (a) $Al_2O_3 + HF + NaAlF_4$
 (b) $Al_2O_3 + CaF_2 + NaAlF_4$
 (c) $Al_2O_3 + Na_3AlF_6 + CaF_2$
 (d) $Al_2O_3 + KF + Na_3AlF_6$
10. Alum helps in purifying water by
 (a) forming Si complex with clay particles
 (b) sulphate part which combines with the dirt and removes it
 (c) coagulating the mud particles
 (d) making mud water soluble.
11. Carbon and silicon belong to group 14. The maximum coordination number of carbon in commonly occurring compounds is 4, whereas that of silicon is 6. This is due to
 (a) large size of silicon
 (b) more electropositive nature of silicon
 (c) availability of d -orbitals in silicon
 (d) Both (a) and (b)
12. Soldiers of Napoleon army while at alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey powder. This transformation is related to
 (a) a change in the partial pressure of oxygen in the air
 (b) a change in the crystalline structure of tin
 (c) an interaction with nitrogen of the air at very low temperatures
 (d) an interaction with water vapours contained in the humid air
13. On adding ammonium hydroxide solution to $Al_2(SO_4)_3(aq)$:
 (a) A precipitate is formed which does not dissolve in excess of ammonium hydroxide
 (b) A precipitate is formed which dissolves in excess of ammonia solution
 (c) No precipitate is formed
 (d) None of these
14. Gas A is bubbled through slaked lime when a white precipitate is formed. On prolonged bubbling the precipitate is dissolved. On heating the resultant solution, the white precipitate appears with evolution of gas B. The gases A and B respectively are
 (a) CO and CO (b) CO_2 and CO
 (c) CO and CO_2 (d) CO_2 and CO_2

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. The straight chain polymer is formed by
- hydrolysis of CH_3SiCl_3 followed by condensation polymerisation
 - hydrolysis of $(\text{CH}_3)_4\text{Si}$ by addition polymerisation
 - hydrolysis of $(\text{CH}_3)_2\text{SiCl}_2$ followed by condensation polymerisation
 - hydrolysis of $(\text{CH}_3)_3\text{SiCl}$ followed by condensation polymerisation
16. It is because of inability of ns^2 electrons of the valence shell to participate in bonding that:-
- Sn^{2+} is oxidising while Pb^{4+} is reducing
 - Sn^{2+} and Pb^{2+} are both oxidising and reducing
 - Sn^{4+} is reducing while Pb^{4+} is oxidising
 - Sn^{2+} is reducing while Pb^{4+} is oxidising
17. A group 14 element is oxidised to form corresponding oxide which is gaseous in nature, when dissolved in water pH of the water decreases further addition of group 2 hydroxides leads to precipitation. This oxide can be
- GeO_2
 - CO
 - CO_2
 - SnO_2
18. In borax bead test which compound is formed?
- Ortho-borate
 - Meta-borate
 - Double oxide
 - Tetra-borate
19. Orthoboric acid when heated to red hot gives
- metaboric acid
 - pyroboric acid
 - boron and water
 - boric anhydride
20. Which of the following compounds is not matched correctly with its structure?



21. PbF_4 , PbCl_4 exist but PbBr_4 and PbI_4 do not exist because of
- large size of Br^- and I^-
 - strong oxidising character of Pb^{4+}
 - strong reducing character of Pb^{4+}
 - low electronegativity of Br^- and I^-
22. In silicon dioxide
- there are double bonds between silicon and oxygen atoms
 - silicon atom is bonded to two oxygen atoms
 - each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
 - each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms.
23. Which one of the following is the correct statement?
- Boric acid is a protonic acid
 - Beryllium exhibits coordination number of six
 - Chlorides of both beryllium and aluminium have bridged structures in solid phase
 - $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ is known as 'inorganic benzene'
24. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite
- is an allotropic form of diamond
 - has molecules of variable molecular masses like polymers
 - has carbon atoms arranged in large plates of rings of strongly bound carbon atoms with weak interplate bonds
 - is a non-crystalline substance

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

Space for Rough Work

C-44

DPP/ CC11

25. The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4 , on hydrolysis gives a white gelatinous precipitate. The precipitate is:
 (a) hydrofluosilicic acid (b) silica gel
 (c) silicic acid (d) calciumfluorosilicate
26. Glass is a
 (a) super-cooled liquid (b) gel
 (c) polymeric mixture (d) micro-crystalline solid
27. Identify the incorrect statement :
 (a) In $(\text{Si}_3\text{O}_9)^{6-}$, tetrahedral SiO_4 units share two oxygen atoms.
 (b) Trialkylchlorosilane on hydrolysis gives R_3SiOH .
 (c) SiCl_4 undergoes hydrolysis to give H_4SiO_4 .
 (d) $(\text{Si}_3\text{O}_9)^{6-}$ has cyclic structure.
28. The catenation tendency of C, Si and Ge is in the order $\text{Ge} < \text{Si} < \text{C}$. The bond energies (in kJ mol^{-1}) of C–C, Si–Si and Ge–Ge bonds are respectively;
 (a) 348, 297, 260 (b) 297, 348, 260
 (c) 348, 260, 297 (d) 260, 297, 348
29. Example of a three-dimensional silicate is:
 (a) Zeolites (b) Ultramarines
 (c) Feldspars (d) Beryls
30. CO_2 and N_2 are non-supporters of combustion. However for putting out fires CO_2 is preferred over N_2 because CO_2
 (a) does not burn
 (b) forms non-combustible products with burning substances
 (c) is denser than nitrogen
 (d) is a more reactive gas

RESPONSE
GRID

25. (a) (b) (c) (d) 26. (a) (b) (c) (d) 27. (a) (b) (c) (d) 28. (a) (b) (c) (d) 29. (a) (b) (c) (d)
 30. (a) (b) (c) (d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 11 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC12

SYLLABUS : Organic Chemistry : Some Basic Principles & Techniques

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

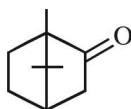
- In crystallisation process impurities which impart colour to the solution are removed by which of the following ?
 - Repeated crystallisation
 - Activated charcoal
 - Bleaching agent
 - Both (a) and (b)
- The IUPAC name of
$$\begin{array}{c} \text{CH}_3 \quad \text{O} \\ | \quad || \\ \text{CH}_3 - \text{CH} - \text{C} - \text{CH}_2 - \text{CH}_2\text{OH} \end{array}$$
 is
 - 1-hydroxy-4-methyl-3-pentanone
 - 2-methyl-5-hydroxy-3-pentanone
 - 4-methyl-3-oxo-1-pentanol
 - Hexanol-1-one-3
- The prussian blue colour obtained in the Lassaigne's test for nitrogen is due to the formation of
 - Iron (II) hexacyanoferrate (III)
 - Iron (III) hexacyanoferrate (II)
 - Iron (III) hexacyanoferrate (III)
 - Iron (II) hexacyanoferrate (II)
- Which one of the following orders is correct regarding the –I effect of the substituents ?
 - $-\text{NR}_2 < -\text{OR} > -\text{F}$
 - $-\text{NR}_2 > -\text{OR} > -\text{F}$
 - $-\text{NR}_2 < -\text{OR} < -\text{F}$
 - $-\text{NR}_2 > -\text{OR} < -\text{F}$

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

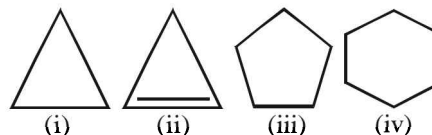
Space for Rough Work

5. Aniline is purified by
 (a) steam distillation (b) simple distillation
 (c) vacuum distillation (d) extraction with a solvent
6. Vinylcarbinol is
 (a) $\text{HO}-\text{CH}_2-\text{CH}=\text{CH}_2$ (b) $\text{CH}_3\text{C}(\text{OH})=\text{CH}_2$
 (c) $\text{CH}_3-\text{CH}=\text{CH}-\text{OH}$ (d) $\text{CH}_3-\text{C}(\text{CH}_2\text{OH})=\text{CH}_2$
7. Which of the following is true regarding hyperconjugation, also known as no-bond resonance?
 (i) Like inductive effect it involves donation of electrons through s bonds
 (ii) Hyperconjugation involves overlapping of filled orbitals with the empty p orbital of the carbocation
 (iii) Like resonance, it involves displacement of π or lone pair of electrons to the carbon bearing positive charge
 (iv) It involves delocalisation of s and π - electrons
 (a) (ii) and (iv) (b) (ii)
 (c) (iii) (d) none
8. The accepted IUPAC name of the camphor is



- (a) 1, 7, 7 - trimethyl bicyclo [2. 2. 1] heptan - 2 - one
 (b) 1, 7, 7, - trimethyl bicyclo [2. 1. 2] heptan - 2 - one
 (c) 1, 2, 2 - trimethyl bicyclo [2. 2. 1] heptan - 6 - one
 (d) None of these
9. Fractional distillation is used when
 (a) there is a large difference in the boiling point of liquids
 (b) there is a small difference in the boiling points of liquids
 (c) boiling points of liquids are same
 (d) liquids form a constant boiling mixture

10. The stability of the compounds



- (a) (iv) > (iii) > (i) > (ii) (b) (i) > (iii) > (ii) > (iv)
 (c) (ii) > (iii) > (i) > (iv) (d) (iv) > (i) > (iii) > (ii)
11. The reason for the loss of optical activity of lactic acid when OH group is changed by H is that :
 (a) Chiral centre of the molecule is destroyed
 (b) Molecule acquires asymmetry
 (c) Due to change in configuration
 (d) Structural change occurs
12. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :
 (a) 18.20 (b) 16.76
 (c) 15.76 (d) 17.36
13. Sodium nitroprusside, when added to an alkaline solution of sulphide ions, produces purple colour ion due to the formation of
 (a) $\text{Na}[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]$ (b) $\text{Na}_2[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]$
 (c) $\text{Na}_3[\text{Fe}(\text{CN})_5\text{NOS}]$ (d) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
14. The most satisfactory method to separate mixture of sugars is
 (a) fractional crystallisation
 (b) sublimation
 (c) chromatography
 (d) benedict reagent

RESPONSE
GRID

5. (a) (b) (c) (d)

6. (a) (b) (c) (d)

7. (a) (b) (c) (d)

8. (a) (b) (c) (d)

9. (a) (b) (c) (d)

10. (a) (b) (c) (d)

11. (a) (b) (c) (d)

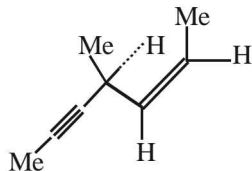
12. (a) (b) (c) (d)

13. (a) (b) (c) (d)

14. (a) (b) (c) (d)

Space for Rough Work

15.



Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives

- (a) optically active compound
 (b) an optically inactive compound
 (c) a racemic mixture
 (d) a diastereomeric mixture

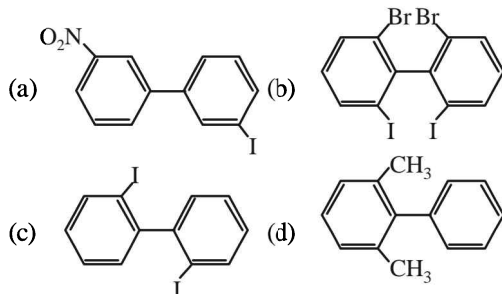
16. Base strength of :

- (A) $\text{H}_3\text{C}\bar{\text{C}}\text{H}_2$, (B) $\text{H}_2\text{C}=\bar{\text{C}}\text{H}$ and (C) $\text{H}-\text{C}\equiv\bar{\text{C}}$

is in the order of :

- (a) (B) > (A) > (C) (b) (C) > (B) > (A)
 (c) (A) > (C) > (B) (d) (A) > (B) > (C)

17. Which of the following biphenyls is optically active ?



18. Which of the following compounds does not show Lassaigne's test for nitrogen ?

- (a) Urea (b) Hydrazine
 (c) Phenylhydrazine (d) Azobenzene

19. The correct statement regarding the comparison of staggered and eclipsed conformation of ethane, is

- (a) The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain
 (b) The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain
 (c) The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain
 (d) The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.

20. The number of asymmetric C-atom created and number of possible stereoisomers when benzil (PhCOCOPh) is reduced with LiAlH_4 .

- (a) 2, 3 (b) 2, 2 (c) 2, 4 (d) 3, 2

21. The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is :

- (a) Chromatography
 (b) Crystallisation
 (c) Steam distillation
 (d) Sublimation

22. In Kjeldahl's method nitrogen present is estimated as

- (a) N_2 (b) NH_3
 (c) NO_2 (d) None of these

23. Which of the following sequence of T and F is correct for given statements. Here T stands for correct and F stands for false statement

- (i) The more the number of contributing structures, the more is the resonance energy.
 (ii) The resonance structures have different positions of nuclei but same number of unpaired electrons
 (iii) The energy of actual structure of the molecule (the resonance hybrid) is lower than that of any of the canonical structures.
 (a) TTT (b) TFT
 (c) FFF (d) TFF

RESPONSE
GRID

15. (a)(b)(c)(d)
20. (a)(b)(c)(d)

16. (a)(b)(c)(d)
21. (a)(b)(c)(d)

17. (a)(b)(c)(d)
22. (a)(b)(c)(d)

18. (a)(b)(c)(d)
23. (a)(b)(c)(d)

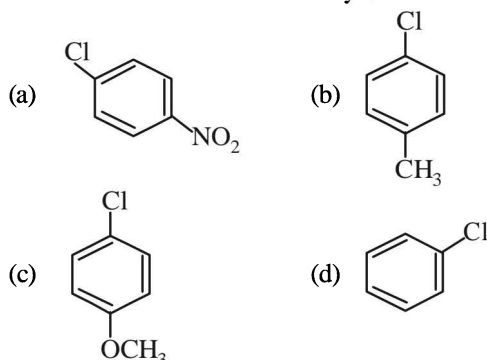
19. (a)(b)(c)(d)

Space for Rough Work

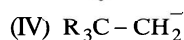
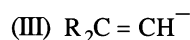
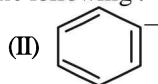
C-48

DPP/ CC12

24. Which of the following compounds undergoes nucleophilic substitution reaction most easily ?



25. The stability of carbanions in the following :



is in the order of :

- (a) (I) > (II) > (III) > (IV) (b) (II) > (III) > (IV) > (I)
 (c) (IV) > (II) > (III) > (I) (d) (I) > (III) > (II) > (IV)
26. The percentage of sulphur in an organic compound whose 0.32 g produces 0.233 g of BaSO_4 [At. wt. Ba = 137, S = 32] is
- (a) 1.0 (b) 10.0
 (c) 23.5 (d) 32.1

27. A similarity between optical and geometrical isomerism is that

- (a) each forms equal number of isomers for a given compound
 (b) if in a compound one is present then so is the other
 (c) both are included in stereoisomerism
 (d) they have no similarity.

28. Arrange the carbanions,



in order of their decreasing stability :

- (a) $(\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_3\text{C}^-$
 (b) $\text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_2\text{CH}^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_3\text{C}^-$
 (c) $(\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > \text{C}_6\text{H}_5\text{CH}_2^-$
 (d) $\text{C}_6\text{H}_5\text{CH}_2^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^-$
29. Chlorine in vinyl chloride is less reactive because
- (a) sp^2 - hybridised carbon has more acidic character than sp^3 - hybridised carbon
 (b) C - Cl bond develops partial double bond character
 (c) of resonance
 (d) All are correct
30. A compound contains 38.8% C, 16% H and 45.2% N. The formula of compound would be :
- (a) CH_3NH_2 (b) CH_3CN
 (c) $\text{C}_2\text{H}_5\text{CN}$ (d) $\text{CH}_2(\text{NH}_2)_2$

RESPONSE
GRID

24. (a) (b) (c) (d)
26. (a) (b) (c) (d)

25. (a) (b) (c) (d)
30. (a) (b) (c) (d)

26. (a) (b) (c) (d)

27. (a) (b) (c) (d)

28. (a) (b) (c) (d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 12 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	36	Qualifying Score	53
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC13

SYLLABUS : Hydrocarbons

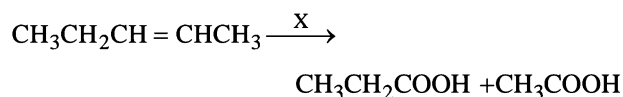
Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. In the given reaction



The X is

- (a) $\text{C}_2\text{H}_5\text{ONa}$
(b) Conc. $\text{HCl} + \text{Anhy. ZnCl}_2$
(c) Anh. AlCl_3
(d) $\text{KMnO}_4/\text{OH}^-$
2. Ethyl hydrogen sulphate is obtained by reaction of H_2SO_4 on
(a) Ethylene (b) Ethane
(c) Ethyl chloride (d) Ethanol

3. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is

- (a) propene
(b) 1-butene
(c) 2-butene
(d) ethene

4. Benzene on reaction with ozone forms _____.

- (a) 2 molecules of aldehyde and 1 molecule of ketone
(b) 2 molecules of ketone and 1 molecule of aldehyde
(c) triozone
(d) hexaozone

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

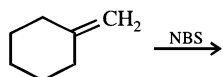
5. When neo-pentyl bromide is subjected to Wurtz reaction, the product formed is

- (a) 2,2,4,4-tetramethylhexane
(b) 2,2,4,4-tetramethylpentane
(c) 2,2,5,5-tetramethylhexane
(d) 2,2,3,3-tetramethylhexane

6. Which one of the following reactions is expected to readily give a hydrocarbon product in good yields ?

- (a) $\text{RCOOK} \xrightarrow[\text{oxidation}]{\text{Electrolytic}}$
(b) $\text{RCOO}^- \text{Ag}^+ \xrightarrow{\text{Br}_2}$
(c) $\text{CH}_3\text{CH}_3 \xrightarrow[\text{h}\nu]{\text{Cl}_2}$
(d) $(\text{CH}_3)_3\text{CCl} \xrightarrow{\text{C}_2\text{H}_5\text{OH}}$

7. What will be the product in the following reaction?



- (a) (b)
(c) (d)

8. A hydrocarbon contains 10.5 g carbon and 1 g hydrogen. Its 0.36 g has 1 L volume at 1 atm and 127°C, hydrocarbon is :

- (a) C_6H_7 (b) C_7H_8
(c) C_5H_6 (d) None of these

9. The conversion of 2, 3-dibromobutane to 2-butene with Zn and alcohol is

- (a) redox reaction (b) α -elimination
(c) β -elimination (d) Both (a) and (b)

10. 1, 2-Dibromocyclohexane on dehydrobromination gives :

- (a) (b)
(c) (d) None of these

11. $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow{\text{CrO}_2\text{Cl}_2} \text{Z}$

In the given sequence Z is:

- (a) benzaldehyde
(b) toluic acid
(c) phenyl acetic acid
(d) benzoic acid

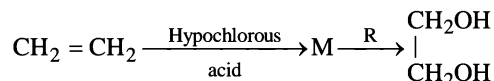
12. Acid catalyzed hydration of alkenes except ethene leads to the formation of

- (a) mixture of secondary and tertiary alcohols
(b) mixture of primary and secondary alcohols
(c) secondary or tertiary alcohol
(d) primary alcohol

13. Which of the following organic compounds has same hybridization as its combustion product (CO_2)?

- (a) Ethane
(b) Ethyne
(c) Ethene
(d) Ethanol

14. In reaction sequence



molecule 'M' and reagent 'R' respectively are

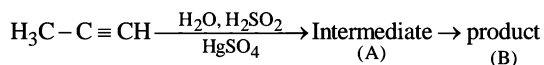
- (a) $\text{CH}_3\text{CH}_2\text{Cl}$ and NaOH
(b) $\text{CH}_3\text{CH}_2\text{OH}$ and H_2SO_4
(c) $\text{CH}_2\text{Cl} \cdot \text{CH}_2\text{OH}$ and aqueous NaHCO_3
(d) and heat

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

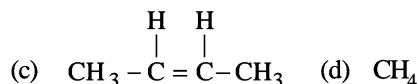
Space for Rough Work

15. Predict the correct intermediate and product in the following reaction :

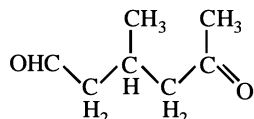


- (a) A : $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_2$ B : $\text{H}_3\text{C}-\text{C}(\text{SO}_4)=\text{CH}_2$
 (b) A : $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_3$ B : $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$
 (c) A : $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_2$ B : $\text{H}_3\text{C}-\text{C}(=\text{O})=\text{CH}_2$
 (d) A : $\text{H}_3\text{C}-\text{C}(\text{SO}_4)=\text{CH}_2$ B : $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$

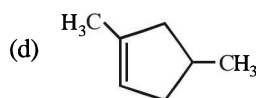
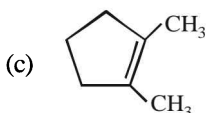
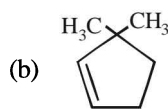
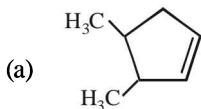
16. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C}-\text{H}$ produces



17. A single compound of the structure :



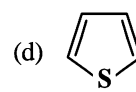
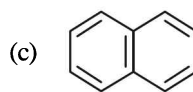
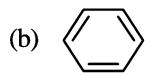
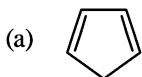
is obtainable from ozonolysis of which of the following cyclic compounds ?



18. Which of the following reagents convert propene to 1-propanol?

- (a) $\text{H}_2\text{O}, \text{H}_2\text{SO}_4$ (b) Aqueous KOH
 (c) $\text{MgSO}_4, \text{NaBH}_4/\text{H}_2\text{O}$ (d) $\text{B}_2\text{H}_6, \text{H}_2\text{O}_2, \text{OH}^-$

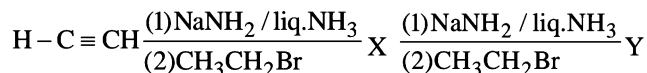
19. Which of the following chemical system is non aromatic?



20. A group which deactivates the benzene ring towards electrophilic substitution but which directs the incoming group principally to the o- and p-positions is

- (a) $-\text{NH}_2$ (b) $-\text{Cl}$
 (c) $-\text{NO}_2$ (d) $-\text{C}_2\text{H}_5$

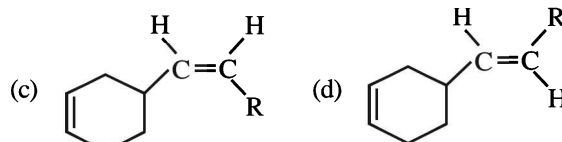
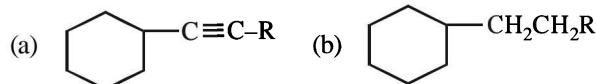
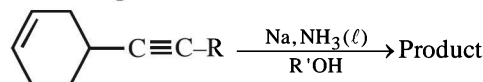
21. In the reaction



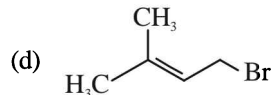
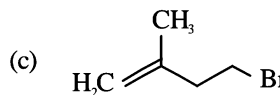
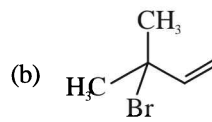
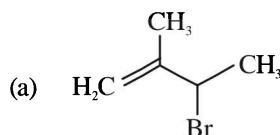
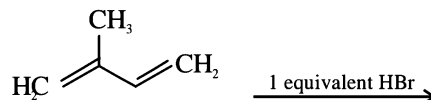
X and Y are

- (a) X = 1-Butyne ; Y = 3-Hexyne
 (b) X = 2-Butyne ; Y = 3-Hexyne
 (c) X = 2-Butyne ; Y = 2-Hexyne
 (d) X = 1-Butyne ; Y = 2-Hexyne

22. The main product of the reaction is :



23. In the following reaction, the major product is



24. Which of these will not react with acetylene?

- (a) NaOH (b) ammonical AgNO_3
 (c) Na (d) HCl

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

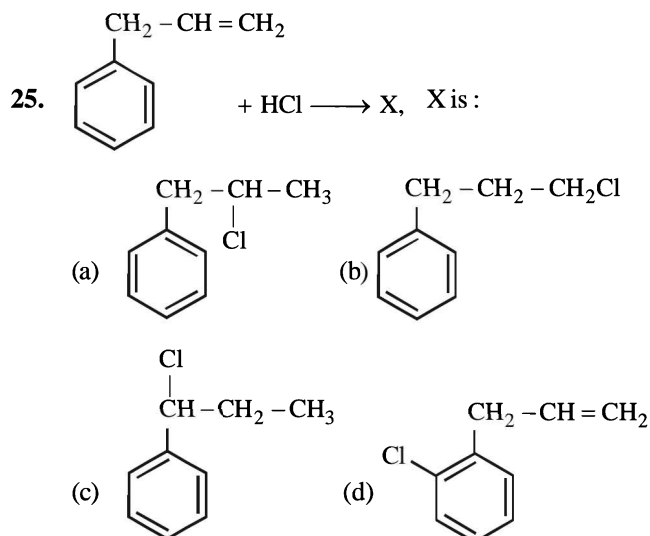
16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

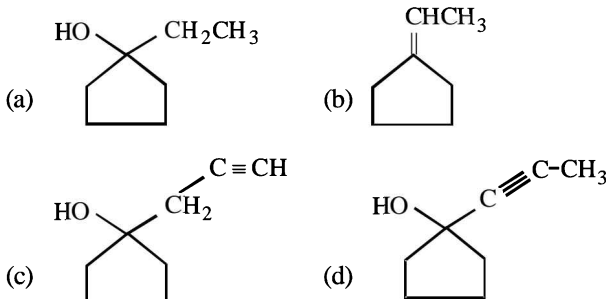
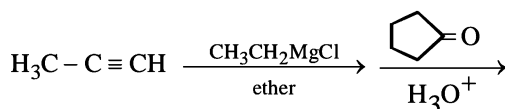
18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

Space for Rough Work



26. The major product of the following reaction

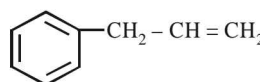


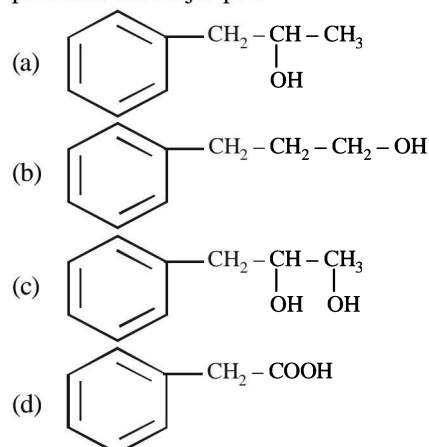
27. The gas liberated by the electrolysis of Dipotassium succinate solution is:

- (a) Ethane (b) Ethyne
(c) Ethene (d) Propene

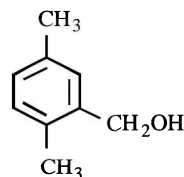
28. 1, 3-Butadiene when treated with Br_2 gives

- (a) 1, 4-dibromo-2-butene (b) 1, 3-dibromo-2-butene
(c) 3, 4-dibromo-1-butene (d) 2, 3-dibromo-2-butene

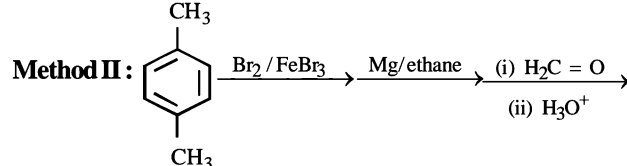
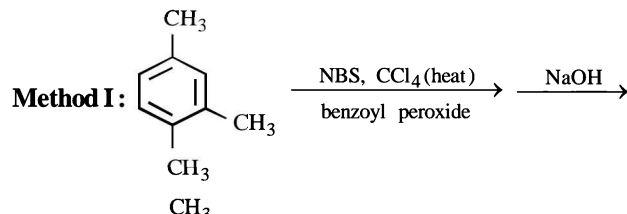
29.  on mercuration-demercuration produces the major product:



30. For the synthesis of the following compound :



Which method is best ?



- (a) Method I (b) Method II
(c) Both (I) and (II) are equally good (d) Neither (I) nor (II)

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 13 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	50
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC14

SYLLABUS : Environmental Chemistry

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- High concentration of which of the following in atmosphere leads to stiffness of flower buds which eventually fall off from plants?
(a) NO_2 (b) SO_2
(c) CFC (d) Smog
- Green house gases
(a) allow shorter wavelength to enter earth's atmosphere while doesn't allow longer wavelength to leave the earth's atmosphere.
(b) allow longer wavelength to enter earth atmosphere while doesn't allow shorter wavelength to leave the surface
(c) don't have wavelength specific character.
(d) show wavelength specific behaviour near the earth while far from earth these have wavelength independent behaviour.
- What is the concentration of dissolved oxygen in cold water ?
(a) 5 ppm (b) 10 ppm
(c) 200,000 ppm (d) 100 ppm
- Soil is polluted by
I. pesticides
II. synthetic fertilizers
III. green manure
Choose the correct option.
(a) I and III (b) I and II
(c) II and III (d) I, II and III

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Water is often treated with chlorine to
 (a) remove hardness
 (b) increase oxygen content
 (c) kill germs
 (d) remove suspended particles
6. The substance which is a primary pollutant?
 (a) H_2SO_4 (b) CO
 (c) PAN (d) Aldehydes
7. Which of the following is/are the hazardous pollutant(s) present in automobile exhaust gases?
 (i) N_2 (ii) CO
 (iii) CH_4 (iv) Oxides of nitrogen
 (a) (ii) and (iii) (b) (i) and (ii)
 (c) (ii) and (iv) (d) (i) and (iii)
8. "Reducing potentially hazardous waste through smarter production".
 This represents a great step forward for
 (a) green revolution (b) green chemistry
 (c) industrial revolution (d) green biotechnology
9. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value
 (a) slightly lower than that of rain water without thunderstorm
 (b) slightly higher than that when the thunderstorm is not there
 (c) uninfluenced by occurrence of thunderstorm
 (d) which depends upon the amount of dust in air
10. The irritant red haze in the traffic and congested places is due to presence of which of the following ?
 (i) Oxides of sulphur
 (ii) Oxides of nitrogen
 (iii) Carbon dioxide
 (iv) Mists, smoke and dust
 (v) Smog
 (a) (i), (iv) and (v) (b) (iii) only
 (c) (ii) only (d) (ii) and (v)
11. Formation of photochemical smog takes place in
 (a) winter during day time
 (b) summer during morning time
 (c) summer during day time
 (d) winter during morning time
12. B.O.D. test or biochemical oxygen demand test is made for measuring
 (a) air pollution (b) water pollution
 (c) noise pollution (d) soil pollution
13. The smog is essentially caused by the presence of
 (a) Oxides of sulphur and nitrogen
 (b) O_2 and N_2
 (c) O_2 and O_3
 (d) O_3 and N_2
14. Which of the following statements about the depletion of ozone layer is correct?
 (a) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $\text{ClO}^\bullet$ radicals.
 (b) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone of $\text{Cl}^\bullet$ and $\text{ClO}^\bullet$ radicals.
 (c) Freons, chlorofluorocarbons, are inert. Chemically, they do not react with ozone in stratosphere.
 (d) Oxides of nitrogen also do not react with ozone in stratosphere.

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. Ozone layer of stratosphere requires protection from indiscriminate use of
 (a) balloons
 (b) pesticides
 (c) atomic explosions
 (d) aerosols and high flying jets
16. Identify the incorrect statement from the following :
 (a) Ozone absorbs the intense ultraviolet radiation of the sun.
 (b) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.
 (c) Ozone absorbs infrared radiation.
 (d) Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer.
17. Which of the following statements about polar stratosphere clouds (PSCs) is not correct?
 (a) PSCs do not react with chlorine nitrate and HCl
 (b) Type I clouds are formed at about -77°C and contain solid $\text{HNO}_3 \cdot 3\text{H}_2\text{O}$
 (c) Type II clouds are formed at about -85°C and contain some ice
 (d) A tight whirlpool of wind called Polar Vortex is formed which surrounds Antarctica
18. Which of the following practices will not come under green chemistry?
 (a) If possible, making use of soap made of vegetable oils instead of using synthetic detergents.
 (b) Using H_2O_2 for bleaching purpose instead of using chlorine based bleaching agents
 (c) Using bicycles for travelling small distances instead of using petrol/ diesel based vehicles
 (d) Using plastic cans for neatly storing substances
19. Phosphate fertilizers when added to water leads to
 (a) increased growth of decomposers
 (b) reduced algal growth
 (c) increased algal growth
 (d) nutrient enrichment (eutrophication)
20. Thermal pollution affects mainly –
 (a) vegetation (b) aquatic creature
 (c) rocks (d) air
21. Roasting of sulphides give the gas X as a by product. This is colorless gas with choking smell of burnt sulphur and caused great damage to respiratory organs as a result of acid rain. Its aqueous solution is acidic, acts as a reducing agent and its acid has never been isolated. The gas X is :
 (a) SO_2 (b) CO_2
 (c) SO_3 (d) H_2S
22. Sewage water is purified by
 (a) aquatic plants (b) micro-organisms
 (c) light (d) fishes
23. Carbon monoxide (CO) is harmful to man because
 (a) it forms carbolic acid
 (b) it generates excess CO_2
 (c) it is carcinogenic
 (d) it competes with O_2 for haemoglobin
24. Which one of the following statements is not true?
 (a) Dissolved oxygen (DO) in cold water can reach a concentration upto 10 ppm.
 (b) Clean water would have a BOD value of 5 ppm.
 (c) Fluoride deficiency in drinking water is harmful. Soluble fluoride is often used to bring its concentration upto 1 ppm.
 (d) When the pH of rain water is higher than 6.5, it is called acid rain.

**RESPONSE
GRID**

 15. (a) (b) (c) (d)
 20. (a) (b) (c) (d)

 16. (a) (b) (c) (d)
 21. (a) (b) (c) (d)

 17. (a) (b) (c) (d)
 22. (a) (b) (c) (d)

 18. (a) (b) (c) (d)
 23. (a) (b) (c) (d)

 19. (a) (b) (c) (d)
 24. (a) (b) (c) (d)

Space for Rough Work

C-56

DPP/ CC14

25. Frequent occurrence of water blooms in a lake indicates
 (a) nutrient deficiency
 (b) oxygen deficiency
 (c) excessive nutrient availability
 (d) absence of herbivores in the lake
26. Which of the following is not a common component of photochemical smog?
 (a) Ozone
 (b) Acrolein
 (c) Peroxyacetyl nitrate
 (d) Chlorofluorocarbons
27. Which of the following is a sink for CO ?
 (a) Microorganism present in the soil
 (b) Oceans
 (c) Plants
 (d) Haemoglobin
28. Which one of the following statement is **not** true ?
 (a) pH of drinking water should be between 5.5 – 9.5.
 (b) Concentration of DO below 6 ppm is good for the growth of fish.
 (c) Clean water would have a BOD value of less than 5 ppm.
 (d) Oxides of sulphur, nitrogen and carbon are the most widespread air pollutant.
29. A dental disease characterised by mottling of teeth is due to presence of a certain chemical element in drinking water. Which is that element?
 (a) Boron (b) Chlorine
 (c) Fluorine (d) Mercury
30. Minamata disease is due to pollution of
 (a) arsenic into the atmosphere
 (b) organic waste into drinking water
 (c) oil spill in water
 (d) industrial waste mercury into fishing water

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 14 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	40	Qualifying Score	60
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC15

SYLLABUS : The Solid State

Max. Marks : 120 **Marking Scheme :** (+ 4) for correct & (–1) for incorrect answer

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. If Germanium crystallises in the same way as diamond, then which of the following statement is not correct?
 - (a) Every atom in the structure is tetrahedrally bonded to 4 atoms.
 - (b) Unit cell consists of 8 Ge atoms and co-ordination number is 4.
 - (c) All the octahedral voids are occupied.
 - (d) All the octahedral voids and 50% tetrahedral voids remain unoccupied.
2. The cubic unit cell of a metal (molar mass = 63.55 g mol^{-1}) has an edge length of 362 pm. Its density is 8.92 g cm^{-3} . The type of unit cell is
 - (a) primitive
 - (b) face centered
 - (c) body centered
 - (d) end centered
3. Doping of AgCl crystals with CdCl_2 results in
 - (a) Frenkel defect
 - (b) Schottky defect
 - (c) Substitutional cation vacancy
 - (d) Formation of F - centres
4. A metal crystallizes in 2 cubic phases fcc and bcc whose unit cell lengths are $3.5 \text{ \AA}$ and $3.0 \text{ \AA}$ respectively. The ratio of their densities is
 - (a) 0.72
 - (b) 2.04
 - (c) 1.46
 - (d) 3.12

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Coordination numbers of Zn^{2+} and S^{2-} in the crystal structure of wurtzite are
 (a) 4, 4 (b) 6, 6
 (c) 8, 4 (d) 8, 8
6. The coordination number X (.....) of each ion in KBr is changed to Y (.....) by.....
 (a) X = 6 to Y = 8 applying higher temperature
 (b) X = 8 to Y = 6 applying high pressure
 (c) X = 6 to Y = 8 applying high pressure
 (d) None of these
7. The radius of Li^+ ion is 60pm and that of F^- is 136 pm. Structure of LiF and coordination number is
 (a) Like NaCl, C.No. = 6 (b) Like CsCl, C.No. = 8
 (c) Anti fluoride, C.No. = 8 (d) None of these
8. The number of carbon atoms per unit cell of diamond unit cell is :
 (a) 8 (b) 6
 (c) 1 (d) 4
9. The pycnometric density of sodium chloride crystal is $2.165 \times 10^3 \text{ kg m}^{-3}$ while its X-ray density is $2.178 \times 10^3 \text{ kg m}^{-3}$. The fraction of unoccupied sites in sodium chloride crystal is
 (a) 5.96×10^{-3} (b) 5.96×10^4
 (c) 5.96×10^{-2} (d) 5.96×10^{-1}
10. Which of the following metal oxides is anti-ferromagnetic in nature?
 (a) MnO_2 (b) TiO_2
 (c) VO_2 (d) CrO_2
11. Which of the following solids is not an electrical conductor?
 (a) Mg (s) (b) TiO (s)
 (c) I_2 (s) (d) H_2O (s)
12. The r_+/r_- ratio of ZnS is 0.402. Pick out the false statements of the following
 (a) ZnS is 4 : 4 coordination compound.
 (b) ZnS does not crystallize in rock salt type lattice because r_+/r_- is too small to avoid overlapping of S^{2-} ions.
 (c) Zn^{2+} ion is too small to fit precisely into the octahedral voids of S^{2-} ions.
 (d) Zn^{2+} ion is too large to fit into the octahedral voids of S^{2-} ions.
13. The unit cell dimensions of a cubic lattice (edges a, b, c and the angles between them, α , β and γ) are
 (a) $a = b = c, \alpha = \beta = \gamma = 90^\circ$
 (b) $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$
 (c) $a = b = c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
 (d) $a \neq b \neq c, \alpha = \beta = 90^\circ, \gamma \neq 90^\circ$
14. The pure crystalline substance on being heated gradually first forms a turbid liquid at constant temperature and still at higher temperature turbidity completely disappears. The behaviour is a characteristic of substance forming.
 (a) Allotropic crystals (b) Liquid crystals
 (c) Isomeric crystals (d) Isomorphous crystals.

RESPONSE
GRID

5. (a) (b) (c) (d) 6. (a) (b) (c) (d) 7. (a) (b) (c) (d) 8. (a) (b) (c) (d) 9. (a) (b) (c) (d)
 10. (a) (b) (c) (d) 11. (a) (b) (c) (d) 12. (a) (b) (c) (d) 13. (a) (b) (c) (d) 14. (a) (b) (c) (d)

Space for Rough Work

15. To get a n -type semiconductor from silicon, it should be doped with a substance with valency
 (a) 2 (b) 1
 (c) 3 (d) 5
16. The second order Bragg diffraction of X-rays with $\lambda = 1.0 \text{ \AA}$ from a set of parallel planes in a metal occurs at an angle of 60° . The distance between the scattering planes in the crystal is
 (a) $0.575 \text{ \AA}$ (b) $1.00 \text{ \AA}$
 (c) $2.00 \text{ \AA}$ (d) $1.15 \text{ \AA}$
17. Which of the following oxides shows electrical properties like metals?
 (a) SiO_2 (b) MgO
 (c) SO_2 (s) (d) CrO_2
18. Al (at. wt 27) crystallizes in the cubic system with a cell edge of $4.05 \text{ \AA}$. Its density is 2.7 g per cm^3 . Determine the unit cell type calculate the radius of the Al atom
 (a) fcc, $2.432 \text{ \AA}$ (b) bcc, $2.432 \text{ \AA}$
 (c) bcc, $1.432 \text{ \AA}$ (d) fcc, $1.432 \text{ \AA}$
19. The ratio of inner planar distances of three types of planes ($d_{100}, d_{110}, d_{111}$) for simple cubic lattice are
 (a) $1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}} : \frac{1}{\sqrt{2}} : 1$
 (c) $\sqrt{3} : \sqrt{2} : 1$ (d) $1 : \sqrt{2} : \sqrt{3}$
20. For a cubic geometry the limiting $\frac{r^+}{r^-}$ is :
 (a) 0.414 (b) 0.155
 (c) >0.731 (d) 0.731
21. The incorrect statement for the sulphur atom of ZnS is
 (a) As S^{2-} is larger than Zn^{2+} only 4 rather than 6 or 8 S^{2-} can be packed around Zn^{2+}
 (b) Its structure is similar to diamond except that alternate atoms are Zn and S
 (c) As S^{2-} is larger than Zn^{2+} only 6 rather than 8 or 4 sulphide ions can be placed around Zn^{2+} ions
 (d) ZnS is a covalent compound
22. AB; crystallizes in a body centred cubic lattice with edge length ' a ' equal to 387 pm . The distance between two oppositely charged ions in the lattice is :
 (a) 335 pm (b) 250 pm (c) 200 pm (d) 300 pm
23. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00 g ?
 [Atomic masses : Na = 23, Cl = 35.5]
 (a) 5.14×10^{21} unit cells (b) 1.28×10^{21} unit cells
 (c) 1.71×10^{21} unit cells (d) 2.57×10^{21} unit cells
24. The interionic distance for cesium chloride crystal will be
 (a) a (b) $\frac{a}{2}$
 (c) $\frac{\sqrt{3}a}{2}$ (d) $\frac{2a}{\sqrt{3}}$

RESPONSE
GRID

15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
 20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

C-60

DPP/ CC15

25. For *fcc* if AB is just like the rock salt like structure then, A^+ and B^- are located at –
 (a) A^+ - Tetrahedral voids ; B^- - Corner
 (b) A^+ - Corner and faces ; B^- - Octahedral voids
 (c) A^+ - Octahedral voids ; B^- - Corner and faces
 (d) A^+ - Corner and faces ; B^- - Octahedral and tetrahedral voids
26. Which of the following exists as covalent crystals in the solid state ?
 (a) Iodine (b) Silicon
 (c) Sulphur (d) Phosphorus
27. The packing fraction for a body-centred cubic is
 (a) 0.42 (b) 0.53
 (c) 0.68 (d) 0.82
28. Consider the radii 0.095 nm (Na^+), 0.181 nm (Cl^-), 0.074 nm (Zn^{2+}), 0.184 nm (S^{2-}), 0.068 nm (Ti^{4+}), 0.140 nm (O^{2-}), 0.169 nm (Cs^+). Choose the correct option from among the following. (Use radius ratio rules)
- (a) Na^+ ions are packed in octahedral holes between the planes of close-packed Cl^- ions.
 (b) Zn^{2+} ions are packed in tetrahedral holes
 (c) Cs^+ ions are packed in a simple cubic array of Cl^- ions
 (d) All of these
29. What type of crystal defect is indicated in the diagram below?
- $Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-$
 $Cl^- \square Cl^- Na^+ \square Na^+$
 $Na^+ Cl^- \square Cl^- Na^+ Cl^-$
 $Cl^- Na^+ Cl^- Na^+ \square Na^+$
- (a) Interstitial defect
 (b) Schottky defect
 (c) Frenkel defect
 (d) Frenkel & Schottky defects
30. What is the coordination number of sodium in Na_2O ?
 (a) 6 (b) 4
 (c) 8 (d) 2

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 15 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	50
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC16

SYLLABUS : Solutions

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. The solubility of a solid in a liquid is significantly affected by temperature changes.

Solute + Solvent $\rightleftharpoons$ Solution.

The system being in a dynamic equilibrium must follow Le-chatelier's principle. Considering the Le-chatelier's principle which of the following is correct?

- (a) $\Delta H_{\text{sol}} > 0$; solubility $\uparrow$; temperature $\downarrow$
- (b) $\Delta H_{\text{sol}} < 0$; solubility $\downarrow$; temperature $\uparrow$
- (c) $\Delta H_{\text{sol}} > 0$; solubility $\downarrow$; temperature $\uparrow$

- (d) $\Delta H_{\text{sol}} < 0$; solubility $\uparrow$; temperature $\uparrow$

2. The vapour pressure of a solution of the liquids A ($p^\circ = 80$ mm Hg and $x_A = 0.4$) and B ($p^\circ = 120$ mm Hg and $x_B = 0.6$) is found to be 100 mm Hg. It shows that the solution exhibits

- (a) positive deviation from ideal behaviour
- (b) negative deviation from ideal behaviour
- (c) ideal behaviour
- (d) positive deviation for lower conc. and negative for higher conc.

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d)

Space for Rough Work

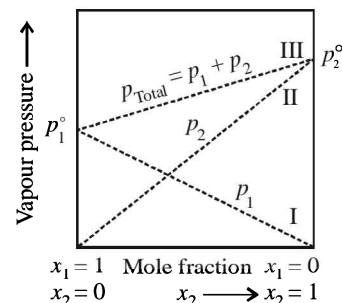
3. Plot of $\frac{1}{x_A}$ Vs $\frac{1}{y_A}$ (x_A mole fraction of A in liquid state and y_A in vapour state) is linear whose slope and intercept respectively are given
- (a) p_B° / p_A° , $\frac{p_B^\circ - p_A^\circ}{p_B^\circ}$
 (b) $p_A^\circ - p_B^\circ$, $\frac{p_A^\circ - p_B^\circ}{p_B^\circ}$
 (c) $p_B^\circ - p_A^\circ$, $\frac{p_B^\circ - p_A^\circ}{p_B^\circ}$
 (d) p_B° / p_A° , $\frac{p_A^\circ - p_B^\circ}{p_B^\circ}$
4. Coolant used in car radiator is aqueous solution of ethylene glycol. In order to prevent the solution from freezing at -0.3°C . How much ethylene glycol must be added to 5 kg of water? ($K_f = 1.86 \text{ K kg mol}^{-1}$)
- (a) 50 kg (b) 50 g
 (c) 45 g (d) 40 g
5. A solution contains non-volatile solute of molecular mass M_2 . Which of the following can be used to calculate the molecular mass of solute in terms of osmotic pressure?
- (a) $M_2 = \left(\frac{m_2}{\pi}\right) VRT$ (b) $M_2 = \left(\frac{m_2}{V}\right) \frac{RT}{\pi}$
 (c) $M_2 = \left(\frac{m_2}{V}\right) \pi RT$ (d) $M_2 = \left(\frac{m_2}{V}\right) \frac{\pi}{RT}$
6. Henry's law constant of oxygen is $1.4 \times 10^{-3} \text{ mol. lit}^{-1} \cdot \text{atm}^{-1}$ at 298 K. How much of oxygen is dissolved in 100 ml at 298 K when the partial pressure of oxygen is 0.5 atm?
- (a) 1.4 g (b) 3.2 g
 (c) 22.4 mg (d) 2.24 mg
7. What is the degree of dissociation of sodium chloride, if the molar mass determined by a cryoscopic method was found to be 31.80 g mol^{-1} [Atomic mass Na = 23 g mol $^{-1}$ Cl = 35.5 g mol $^{-1}$]
- (a) 0.58 (b) 0.73
 (c) 0.83 (d) 0.92
8. A solution containing components A and B follows Raoult's law when
- (a) A – B attraction force is greater than A – A and B – B
 (b) A – B attraction force is less than A – A and B – B
 (c) A – B attraction force remains same as A – A and B – B
 (d) Volume of solution is different from sum of volume of solute and solvent
9. Two 1-litre flask A and B are connected to each other by a valve which is closed. Flask A has benzene in equilibrium with its vapours at 30°C . The flask B, is evacuated, and the valve is opened. Which of the following is true. If temperature is kept constant.
- (a) Some of the benzene molecules would move to flask B from flask A.
 (b) Vapour pressure will be half the initial value.
 (c) The vapour pressure remains unchanged
 (d) Some more of the liquid benzene in flask A would evaporate.
10. For a solution of two liquids A and B it was proved that $P_S = x_A (p_A^\circ - p_B^\circ) + p_B^\circ$. The resulting solution will be
- (a) Non-ideal (b) ideal
 (c) semi-ideal (d) None of these
11. A 0.0020 m aqueous solution of an ionic compound $\text{Co}(\text{NH}_3)_5(\text{NO}_2)\text{Cl}$ freezes at -0.00732°C . Number of moles of ions which 1 mol of ionic compound produces on being dissolved in water will be ($K_f = -1.86^\circ\text{C/m}$)
- (a) 3 (b) 4
 (c) 1 (d) 2
12. A solution of urea (mol. mass 56 g mol^{-1}) boils at 100.18°C at the atmospheric pressure. If K_f and K_b for water are 1.86 and $0.512 \text{ K kg mol}^{-1}$ respectively, the above solution will freeze at
- (a) 0.654°C (b) -0.654°C
 (c) 6.54°C (d) -6.54°C

RESPONSE
GRID

3. (a)(b)(c)(d) 4. (a)(b)(c)(d) 5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d)
 8. (a)(b)(c)(d) 9. (a)(b)(c)(d) 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d)

Space for Rough Work

13. In mixture A and B components show -ve deviation as
 (a) $\Delta V_{\text{mix}} > 0$
 (b) $\Delta H_{\text{mix}} < 0$
 (c) A – B interaction is weaker than A – A and B – B interaction
 (d) A – B interaction is stronger than A – A and B – B interaction.
14. All form ideal solution except
 (a) C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$
 (b) C_2H_6 and $\text{C}_2\text{H}_5\text{I}$
 (c) $\text{C}_6\text{H}_5\text{Cl}$ and $\text{C}_6\text{H}_5\text{Br}$
 (d) $\text{C}_2\text{H}_5\text{I}$ and $\text{C}_2\text{H}_5\text{OH}$.
15. A binary liquid solution is prepared by mixing *n*-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution?
 (a) The solution is non-ideal, showing – ve deviation from Raoult's Law.
 (b) The solution is non-ideal, showing + ve deviation from Raoult's Law.
 (c) *n*-heptane shows + ve deviation while ethanol shows – ve deviation from Raoult's Law.
 (d) The solution formed is an ideal solution.
16. We have three aqueous solutions of NaCl labelled as 'A', 'B' and 'C' with concentrations 0.1M, 0.01M and 0.001M, respectively. The value of van't Hoff factor for these solutions will be in the order _____.
 (a) $i_A < i_B < i_C$ (b) $i_A > i_B > i_C$
 (c) $i_A = i_B = i_C$ (d) $i_A < i_B > i_C$
17. During depression of freezing point in a solution the following are in equilibrium
 (a) liquid solvent, solid solvent
 (b) liquid solvent, solid solute
 (c) liquid solute, solid solute
 (d) liquid solute, solid solvent
18. A plot of p_1 or p_2 vs the mole fractions x_1 and x_2 is given as.



- In this figure, lines I and II pass through the point for which.
 (a) $x_1 \neq 1; x_2 = 1$ (b) $x_1 = x_2 \neq 1$
 (c) $x_1 = 1; x_2 \neq 1$ (d) $x_1 = x_2 = 1$
19. Which of the following modes of expressing concentration is independent of temperature?
 (a) Molarity (b) Molality
 (c) Formality (d) Normality
20. How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0M HNO_3 ? The concentrated acid is 70% HNO_3
 (a) 90.0 g conc. HNO_3 (b) 70.0 g conc. HNO_3
 (c) 54.0 g conc. HNO_3 (d) 45.0 g conc. HNO_3
21. Which among the following will show maximum osmotic pressure?
 (a) 1 M NaCl (b) 1 M MgCl_2
 (c) 1 M $(\text{NH}_4)_3\text{PO}_4$ (d) 1 M Na_2SO_4
22. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case?
 (a) Molecular mass of X is greater than the molecular mass of Y.
 (b) Molecular mass of X is less than the molecular mass of Y
 (c) Y is undergoing dissociation in water while X undergoes no change.
 (d) X is undergoing dissociation in water.

RESPONSE
GRID

13. (a) (b) (c) (d)
18. (a) (b) (c) (d)

14. (a) (b) (c) (d)
19. (a) (b) (c) (d)

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

Space for Rough Work

C-64

DPP/ CC16

23. Which of the following 0.10 m aqueous solutions will have the lowest freezing point ?
 (a) $\text{Al}_2(\text{SO}_4)_3$ (b) $\text{C}_6\text{H}_{12}\text{O}_6$
 (c) KCl (d) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
24. If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is ($K_f = 1.86 \text{ K kg mol}^{-1}$)
 (a) 0.372 K (b) 0.0558 K
 (c) 0.0744 K (d) 0.0186 K
25. Which one of the following salts will have the same value of van't Hoff factor (i) as that of $\text{K}_4[\text{Fe}(\text{CN})_6]$.
 (a) $\text{Al}_2(\text{SO}_4)_3$ (b) NaCl
 (c) $\text{Al}(\text{NO}_3)_3$ (d) Na_2SO_4
26. Mole fraction of the solute in a 1.00 molal aqueous solution is :
 (a) 0.1770 (b) 0.0177
 (c) 0.0344 (d) 1.7700
27. 25.3 g of sodium carbonate, Na_2CO_3 is dissolved in enough water to make 250 mL of solution. If sodium carbonate dissociates completely, molar concentration of sodium ions, Na^+ and carbonate ions, CO_3^{2-} are respectively (Molar mass of $\text{Na}_2\text{CO}_3 = 106 \text{ g mol}^{-1}$)
 (a) 0.955 M and 1.910 M
 (b) 1.910 M and 0.955 M
 (c) 1.90 M and 1.910 M
 (d) 0.477 M and 0.477 M
28. Azeotropic mixture of HCl and H_2O has
 (a) 48% HCl (b) 22.2% HCl
 (c) 36% HCl (d) 20.2% HCl
29. Freezing point of an aqueous solution is -0.186°C . If the values of K_b and K_f of water are respectively $0.52 \text{ K kg mol}^{-1}$ and $1.86 \text{ K kg mol}^{-1}$, then the elevation of boiling point of the solution in K is
 (a) 0.52 (b) 1.04
 (c) 1.34 (d) 0.052
30. Which of the following statements, regarding the mole fraction (x) of a component in solution, is incorrect?
 (a) $0 \leq x \leq 1$
 (b) $x \leq 1$
 (c) x is always non-negative
 (d) $-2 \leq x \leq 2$

RESPONSE
GRID

23. (a)(b)(c)(d) 24. (a)(b)(c)(d) 25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d)
 28. (a)(b)(c)(d) 29. (a)(b)(c)(d) 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 16 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	50
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC17

SYLLABUS : Electrochemistry

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. Which of the following expressions correctly represents the equivalent conductance at infinite dilution of $\text{Al}_2(\text{SO}_4)_3$,
Given that $\Lambda_{\text{Al}^{3+}}^\circ$ and $\Lambda_{\text{SO}_4^{2-}}^\circ$ are the equivalent conductances at infinite dilution of the respective ions?
- (a) $\frac{1}{3}\Lambda_{\text{Al}^{3+}}^\circ + \frac{1}{2}\Lambda_{\text{SO}_4^{2-}}^\circ$ (b) $2\Lambda_{\text{Al}^{3+}}^\circ + 3\Lambda_{\text{SO}_4^{2-}}^\circ$
(c) $\Lambda_{\text{Al}^{3+}}^\circ + \Lambda_{\text{SO}_4^{2-}}^\circ$ (d) $(\Lambda_{\text{Al}^{3+}}^\circ + \Lambda_{\text{SO}_4^{2-}}^\circ) \times 6$
2. The equivalent conductance of $\frac{M}{32}$ solution of a weak monobasic acid is 8.0 mho cm^2 and at infinite dilution is 400 mho cm^2 . The dissociation constant of this acid is:
- (a) 1.25×10^{-6} (b) 6.25×10^{-4}
(c) 1.25×10^{-4} (d) 1.25×10^{-5}
3. Aqueous solution of which of the following compounds is the best conductor of electric current ?
(a) Acetic acid, $\text{C}_2\text{H}_4\text{O}_2$
(b) Hydrochloric acid, HCl
(c) Ammonia, NH_3
(d) Fructose, $\text{C}_6\text{H}_{12}\text{O}_6$
4. The standard EMF of Daniell cell is 1.10 volt. The maximum electrical work obtained from the Daniell cell is
(a) 212.3 kJ
(b) 175.4 kJ
(c) 106.15 kJ
(d) 53.07 kJ

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Which of the following reaction occurs at the cathode during the charging of lead storage battery?
- (a) $\text{Pb}^{2+} + 2\text{e}^- \longrightarrow \text{Pb}$
 (b) $\text{Pb}^{2+} + \text{SO}_4^{2-} \longrightarrow \text{PbSO}_4$
 (c) $\text{Pb} \longrightarrow \text{Pb}^{2+} + 2\text{e}^-$
 (d) $\text{PbSO}_4 + 2\text{H}_2\text{O} \longrightarrow 2\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^-$
6. Molar ionic conductivities of a two-bivalent electrolytes x^{2+} and y^{2-} are 57 and 73 respectively. The molar conductivity of the solution formed by them will be
- (a) $130 \text{ S cm}^2 \text{ mol}^{-1}$
 (b) $65 \text{ S cm}^2 \text{ mol}^{-1}$
 (c) $260 \text{ S cm}^2 \text{ mol}^{-1}$
 (d) $187 \text{ S cm}^2 \text{ mol}^{-1}$
7. Kohlrausch's law states that at :
- (a) finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte.
 (b) infinite dilution each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of the electrolyte.
 (c) infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.
 (d) infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte.
8. Standard free energies of formation (in kJ/mol) at 298 K are -237.2 , -394.4 and -8.2 for $\text{H}_2\text{O}(l)$, $\text{CO}_2(g)$ and pentane (g), respectively. The value E°_{cell} for the pentane-oxygen fuel cell is :
- (a) 1.968 V
 (b) 2.0968 V
 (c) 1.0968 V
 (d) 0.0968 V
9. If the E°_{cell} for a given reaction has a negative value, then which of the following gives the correct relationships for the values of ΔG° and K_{eq} ?
- (a) $\Delta G^\circ > 0$; $K_{\text{eq}} > 1$ (b) $\Delta G^\circ < 0$; $K_{\text{eq}} > 1$
 (c) $\Delta G^\circ < 0$; $K_{\text{eq}} < 1$ (d) $\Delta G^\circ > 0$; $K_{\text{eq}} < 1$
10. Standard electrode potentials are : Fe^{+2}/Fe [$E^\circ = -0.44$]; $\text{Fe}^{+3}/\text{Fe}^{+2}$ $E^\circ = +0.77$; If Fe^{+2} , Fe^{+3} and Fe blocks are kept together, then
- (a) Fe^{+3} increases
 (b) Fe^{+3} decreases
 (c) $\frac{\text{Fe}^{+2}}{\text{Fe}^{+3}}$ remains unchanged
 (d) Fe^{+2} decreases
11. An electrolytic cell contains a solution of Ag_2SO_4 and has platinum electrodes. A current is passed until 1.6 gm of O_2 has been liberated at anode. The amount of silver deposited at cathode would be
- (a) 107.88 gm (b) 1.6 gm
 (c) 0.8 gm (d) 21.60 gm
12. If ϕ denotes reduction potential, then which is true?
- (a) $E^\circ_{\text{cell}} = \phi_{\text{right}} - \phi_{\text{left}}$ (b) $E^\circ_{\text{cell}} = \phi_{\text{left}} + \phi_{\text{right}}$
 (c) $E^\circ_{\text{cell}} = \phi_{\text{left}} - \phi_{\text{right}}$ (d) $E^\circ_{\text{cell}} = -(\phi_{\text{left}} + \phi_{\text{right}})$
13. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is
- $$\text{CH}_3\text{OH}(l) + 3/2\text{O}_2(g) \longrightarrow \text{CO}_2(g) + 2\text{H}_2\text{O}(l)$$
- At 298 K standard Gibb's energies of formation for $\text{CH}_3\text{OH}(l)$, $\text{H}_2\text{O}(l)$ and $\text{CO}_2(g)$ are -166.2 , -237.2 and $-394.4 \text{ kJ mol}^{-1}$ respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1} , efficiency of the fuel cell will be:
- (a) 87% (b) 90%
 (c) 97% (d) 80%
14. For the cell reaction,
 $\text{Cu}^{2+}(\text{C}_1, \text{aq}) + \text{Zn}(s) = \text{Zn}^{2+}(\text{C}_2, \text{aq}) + \text{Cu}(s)$ of an electrochemical cell, the change in free energy, ΔG , at a given temperature is a function of
- (a) $\ln(\text{C}_1)$ (b) $\ln(\text{C}_2/\text{C}_1)$
 (c) $\ln(\text{C}_2)$ (d) $\ln(\text{C}_1 + \text{C}_2)$

RESPONSE
GRID

5. (a)(b)(c)(d)
10. (a)(b)(c)(d)

6. (a)(b)(c)(d)
11. (a)(b)(c)(d)

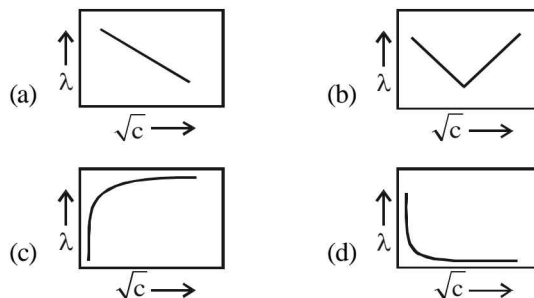
7. (a)(b)(c)(d)
12. (a)(b)(c)(d)

8. (a)(b)(c)(d)
13. (a)(b)(c)(d)

9. (a)(b)(c)(d)
14. (a)(b)(c)(d)

Space for Rough Work

15. When electric current is passed through acidified water, 112 ml of hydrogen gas at STP collected at the cathode in 965 seconds. The current passed in amperes is
(a) 1.0 (b) 0.5 (c) 0.1 (d) 2.0
16. The electrode potential $E_{(Zn^{2+}/Zn)}$ of a zinc electrode at 25°C with an aqueous solution of 0.1 M $ZnSO_4$ is
[$E_{(Zn^{2+}/Zn)}^\circ = -0.76$ V. Assume $\frac{2.303RT}{F} = 0.06$ at 298 K].
(a) +0.73 (b) -0.79
(c) -0.82 (d) -0.70
17. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and MZ^- at 25°C. If the reduction potential of $Z > Y > X$, then,
(a) Y will oxidize X and not Z
(b) Y will oxidize Z and not X
(c) Y will oxidize both X and Z
(d) Y will reduce both X and Z
18. For the electrochemical cell, $M | M^+ || X^- | X$,
 $E^\circ M^+ / M = 0.44$ V and $E^\circ (X/X^-) = 0.33$ V.
From this data one can deduce that
(a) $M + X \rightarrow M^+ + X^-$ is the spontaneous reaction
(b) $M^+ + X^- \rightarrow M + X$ is the spontaneous reaction
(c) $E_{cell} = 0.77$ V
(d) $E_{cell} = -0.77$ V
19. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half cell reactions and their standard potentials are given below:
 $MnO_4^-(aq.) + 8H^+(aq.) + 5e^- \rightarrow Mn^{2+}(aq.) + 4H_2O(l)$
 $E^\circ = 1.51$ V
 $Cr_2O_7^{2-}(aq.) + 14H^+(aq.) + 6e^- \rightarrow 2Cr^{3+}(aq.) + 7H_2O(l)$
 $E^\circ = 1.38$ V
 $Fe^{3+}(aq.) + e^- \rightarrow Fe^{2+}(aq.)$ $E^\circ = 0.77$ V
 $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq.)$ $E^\circ = 1.40$ V
Identify the only incorrect statement regarding the quantitative estimation of aqueous $Fe(NO_3)_2$
(a) MnO_4^- can be used in aqueous HCl
(b) $Cr_2O_7^{2-}$ can be used in aqueous HCl
(c) MnO_4^- can be used in aqueous H_2SO_4
(d) $Cr_2O_7^{2-}$ can be used in aqueous H_2SO_4
20. A hypothetical electrochemical cell is shown below
 $A | A^+(xM) || B^+(yM) | B$
The emf measured is +0.20 V. The cell reaction is
(a) $A^+ + e^- \rightarrow A$; $B^+ + e^- \rightarrow B$
(b) The cell reaction cannot be predicted
(c) $A + B^+ \rightarrow A^+ + B$
(d) $A^+ + B \rightarrow A + B^+$
21. Conductance of 0.1 M KCl (conductivity = $X \text{ Ohm}^{-1} \text{ cm}^{-1}$) filled in a conductivity cell is $Y \text{ Ohm}^{-1}$. If the conductance of 0.1 M NaOH filled in the same cell is $Z \text{ Ohm}^{-1}$, the molar conductance of NaOH will be
(a) $10^3 \frac{XZ}{Y}$ (b) $10^4 \frac{XZ}{Y}$ (c) $10 \frac{XZ}{Y}$ (d) $0.1 \frac{XZ}{Y}$
22. On the basis of the following E° values, the strongest oxidizing agent is:
 $Fe(CN)_6^{4-} \rightarrow [Fe(CN)_6]^{3-} + e^-$; $E^\circ = -0.35$ V
 $Fe^{2+} \rightarrow Fe^{3+} + e^-$; $E^\circ = -0.77$ V
(a) $[Fe(CN)_6]^{4-}$ (b) Fe^{2+}
(c) Fe^{3+} (d) $[Fe(CN)_6]^{3-}$
23. The mathematical expression for law of independent migration of ions and Ostwald's dilution law are given by
(a) $\Lambda = \Lambda_m^\circ - BC^{1/2}$ (b) $\Lambda^\circ = F(U_+ + U_-)$
(c) $\Lambda_m^\circ = v_+ \lambda_+ + v_- \lambda_-$ (d) $\frac{\Lambda^\circ}{\Lambda_m} = \frac{1}{\Lambda_m^\circ} + \frac{\Lambda_m c}{K_a (\Lambda_m^\circ)^2}$
24. The variation of equivalent conductance of strong electrolyte with $(\text{concentration})^{1/2}$ is represented by



RESPONSE
GRID

15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

C-68

DPP/ CC17

25. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as :
 (a) Electrolytic cell (b) Dynamo
 (c) Ni-Cd cell (d) Fuel Cell
26. In acidic medium MnO_2 is an oxidant as

$$\text{MnO}_2(s) + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$$
 If the pH of solution is decreased by one unit, the electrode potential of the half cell Pt : $\text{MnO}_2, \text{Mn}^{2+}$ will change by
 (a) 0.236 V (b) -0.236 V
 (c) -0.118 V (d) 0.118 V
27. Consider the following cell reaction:

$$2\text{Fe}(s) + \text{O}_2(g) + 4\text{H}^+(aq) \rightarrow 2\text{Fe}^{2+}(aq) + 2\text{H}_2\text{O}(l); E^\circ = 1.67\text{V}$$
 At $[\text{Fe}^{2+}] = 10^{-3}\text{M}$, $p(\text{O}_2) = 0.1\text{ atm}$ and $\text{pH} = 3$, the cell potential at 25°C is
 (a) 1.47 V (b) 1.77 V
 (c) 1.87 V (d) 1.57 V
28. In a hydrogen-oxygen fuel cell, combustion of hydrogen occurs to
 (a) produce high purity water
 (b) create potential difference between two electrodes
 (c) generate heat
 (d) remove adsorbed oxygen from electron surfaces
29. Consider the following relations for emf of an electrochemical cell:
 (i) $\text{emf of cell} = (\text{Oxidation potential of anode}) - (\text{Reduction potential of cathode})$
 (ii) $\text{emf of cell} = (\text{Oxidation potential of anode}) + (\text{Reduction potential of cathode})$
 (iii) $\text{emf of cell} = (\text{Reduction potential of anode}) + (\text{Reduction potential of cathode})$
 (iv) $\text{emf of cell} = (\text{Oxidation potential of anode}) - (\text{Oxidation potential of cathode})$
 Which of the above relations are correct?
 (a) (ii) and (iv)
 (b) (iii) and (i)
 (c) (i) and (ii)
 (d) (iii) and (iv)
30. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of $\text{pH} = 10$ and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be ?
 (a) 0.59 V (b) 0.118 V
 (c) 1.18 V (d) 0.059 V

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 17 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	53
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC18

SYLLABUS : Chemical Kinetics

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- The temperature dependence of rate constant (k) of a chemical reaction is written in terms of Arrhenius equation, $k = A.e^{-E_a/RT}$. Activation energy (E_a) of the reaction can be calculated by plotting
 - k vs. $\frac{1}{\log T}$
 - $\log k$ vs. $\frac{1}{T}$
 - $\log k$ vs. $\frac{1}{\log T}$
 - k vs. T
- The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. If the concentration of A is 5 M then concentration of B after 20 minutes is :
 - 1.08 M
 - 3.60 M
 - 0.36 M
 - 0.72 M
- The half life period for catalytic decomposition of AB_3 at 50 mm Hg is 4 hrs and at 100 mm Hg it is 2 hrs. The order of reaction is
 - 1
 - 2
 - 3
 - 0
- For the first order reaction
 $C_2H_4O(g) \rightarrow CH_4(g) + CO(g)$, the initial pressure of $C_2H_4O(g)$ is 80 torr and total pressure at the end of 20 minutes is 120 torr. The time needed for 75% decomposition of C_2H_4O would be :
 - 20 minutes
 - 40 minutes
 - 80 minutes
 - 120 minutes

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. When a biochemical reaction is carried out in laboratory in the absence of enzyme then rate of reaction obtained is 10^{-6} times, then activation energy of reaction in the Presence of enzyme is
- $\frac{6}{RT}$
 - Different from E_a obtained in laboratory
 - P is required
 - Can't say anything
6. In a reaction $A \rightarrow \text{Products}$, when start is made from 8.0×10^{-2} M of A, half-life is found to be 120 minute. For the initial concentration 4.0×10^{-2} M, the half-life of the reaction becomes 240 minute. The order of the reaction is :
- zero
 - one
 - two
 - 0.5
7. In a first-order reaction $A \rightarrow B$, if k is rate constant and initial concentration of the reactant A is 0.5 M, then the half-life is
- $\frac{\log 2}{k}$
 - $\frac{\log 2}{k\sqrt{0.5}}$
 - $\frac{\ln 2}{k}$
 - $\frac{0.693}{0.5k}$
8. The rate expression for the reaction $A(g) + B(g) \rightarrow C(g)$ is $\text{rate} = kC_A^2 C_B^{1/2}$. What changes in the initial concentrations of A and B will cause the rate of reaction to increase by a factor of eight?
- $C_A \times 2; C_B \times 2$
 - $C_A \times 2; C_B \times 4$
 - $C_A \times 1; C_B \times 4$
 - $C_A \times 4; C_B \times 1$
9. The reaction $A(g) + 2B(g) \rightarrow C(g) + D(g)$ is an elementary process. In an experiment involving this reaction, the initial partial pressures of A and B are $p_A = 0.60$ atm and $p_B = 0.80$ atm respectively. When $p_C = 0.20$ atm, the rate of the reaction relative to the initial rate is
- 1/6
 - 1/12
 - 1/36
 - 1/18
10. Select the rate law that corresponds to data shown for the following reaction
- $$A + B \longrightarrow \text{products.}$$
- | Exp. | [A] | [B] | Initial rate |
|------|-------|-------|--------------|
| 1 | 0.012 | 0.035 | 0.1 |
| 2 | 0.024 | 0.070 | 0.8 |
| 3 | 0.024 | 0.035 | 0.1 |
| 4 | 0.012 | 0.070 | 0.8 |
- $\text{rate} = k[B]^3$
 - $\text{rate} = k[B]^4$
 - $\text{rate} = k[A][B]^3$
 - $\text{rate} = k[A]^2[B]^2$
11. A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial velocity is ten times the permissible value, after how many days will it be safe to enter the room?
- 100 days
 - 1000 days
 - 300 days
 - 10 days
12. The energies of activation for forward and reverse reactions for $A_2 + B_2 \rightleftharpoons 2AB$ are 180 kJ mol^{-1} and 200 kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol^{-1} . The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of a catalyst will be (in kJ mol^{-1})
- 20
 - 300
 - 120
 - 280
13. Consider the reaction :
- $$\text{Cl}_2(aq) + \text{H}_2\text{S}(aq) \rightarrow \text{S}(s) + 2\text{H}^+(aq) + 2\text{Cl}^-(aq)$$
- The rate equation for this reaction is
- $$\text{rate} = k[\text{Cl}_2][\text{H}_2\text{S}]$$
- Which of these mechanisms is/are consistent with this rate equation?
- A. $\text{Cl}_2 + \text{H}_2\text{S} \rightarrow \text{H}^+ + \text{Cl}^- + \text{Cl}^+ + \text{HS}^-$ (slow)
- $$\text{Cl}^+ + \text{HS}^- \rightarrow \text{H}^+ + \text{Cl}^- + \text{S}$$
- (fast)

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d)

Space for Rough Work



- (a) B only (b) Both A and B
(c) Neither A nor B (d) A only

14. The initial rates of reaction

$3\text{A} + 2\text{B} + \text{C} \longrightarrow \text{Products}$, at different initial concentrations are given below:

Initial rate, Ms^{-1}	$[\text{A}]_0, \text{M}$	$[\text{B}]_0, \text{M}$	$[\text{C}]_0, \text{M}$
5.0×10^{-3}	0.010	0.005	0.010
5.0×10^{-3}	0.010	0.005	0.015
1.0×10^{-2}	0.010	0.010	0.010
1.25×10^{-3}	0.005	0.005	0.010

The order with respect to the reactants, A, B and C are respectively

- (a) 3, 2, 0 (b) 3, 2, 1 (c) 2, 2, 0 (d) 2, 1, 0

15. In a reversible reaction the energy of activation of the forward reaction is 50 kcal. The energy of activation for the reverse reaction will be

- (a) < 50 kcal
(b) either greater than or less than 50 kcal
(c) 50 kcal
(d) > 50 kcal

16. Half life period of a first-order reaction is 1386 seconds. The specific rate constant of the reaction is:

- (a) $0.5 \times 10^{-2} \text{ s}^{-1}$ (b) $0.5 \times 10^{-3} \text{ s}^{-1}$
(c) $5.0 \times 10^{-2} \text{ s}^{-1}$ (d) $5.0 \times 10^{-3} \text{ s}^{-1}$

17. $t_{\frac{1}{4}}$ can be taken as the time taken for the concentration of a reactant to drop to $\frac{3}{4}$ of its initial value. If the rate constant for a first order reaction is k , the $t_{\frac{1}{4}}$ can be written as

- (a) $0.75/k$ (b) $0.69/k$
(c) $0.29/k$ (d) $0.10/k$

18. Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperatures (T_1 and T_2) are related by :

(a) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ (b) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$

(c) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} + \frac{1}{T_1} \right)$ (d) $\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

19. Consider a reaction $a\text{G} + b\text{H} \rightarrow \text{Products}$. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is

- (a) 0 (b) 1
(c) 2 (d) 3

20. The rate law for a reaction between the substances A and B is given by

$\text{Rate} = k [\text{A}]^n [\text{B}]^m$

On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as

- (a) $(m+n)$ (b) $(n-m)$
(c) $2^{(n-m)}$ (d) $\frac{1}{2^{(m+n)}}$

21. The rate constants k_1 and k_2 for two different reactions are $10^{16} \cdot e^{-2000/T}$ and $10^{15} \cdot e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is :

- (a) 1000 K (b) $\frac{2000}{2.303}$ K (c) 2000 K (d) $\frac{1000}{2.303}$ K

RESPONSE
GRID

14. (a)(b)(c)(d)
19. (a)(b)(c)(d)

15. (a)(b)(c)(d)
20. (a)(b)(c)(d)

16. (a)(b)(c)(d)
21. (a)(b)(c)(d)

17. (a)(b)(c)(d)

18. (a)(b)(c)(d)

Space for Rough Work

22. Match the columns

Column - I

(A) $k = \frac{[R_0] - [R]}{t}$

(B) $k = \frac{2.303}{t} \log \frac{[R_0]}{[R]}$

(C) Value of k for first order reaction when $t = t_{1/2}$ and $[R] = \frac{[R]_0}{2}$ (D) Value of k for zero order reaction when $t = t_{1/2}$ and $[R] = \frac{[R]_0}{2}$

(a) A - (s), B - (q), C - (p), D - (r)

(b) A - (q), B - (s), C - (p), D - (r)

(c) A - (q), B - (p), C - (s), D - (r)

(d) A - (q), B - (s), C - (p), D - (t)

23. The slope in Arrhenius plot, is equal to:

(a) $-\frac{E_a}{2.303R}$

(b) $\frac{E_a}{R}$

(c) $-\frac{R}{2.303 E_a}$

(d) None of these

24. For the reaction $A + B \longrightarrow$ products, it is observed that:

(1) On doubling the initial concentration of A only, the rate of reaction is also doubled and

(2) On doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.

The rate of this reaction is given by:

(a) $\text{rate} = k[A][B]^2$

(b) $\text{rate} = k[A]^2[B]^2$

(c) $\text{rate} = k[A][B]$

(d) $\text{rate} = k[A]^2[B]$

25. For the reaction $2N_2O_5 \rightarrow 4NO_2 + O_2$, rate and rate constant are $1.02 \times 10^{-4} \text{ mol lit}^{-1} \text{ sec}^{-1}$ and $3.4 \times 10^{-5} \text{ sec}^{-1}$ respectively then concentration of N_2O_5 at that time will be

(a) 1.732M

(b) 3M

(c) $3.4 \times 10^5 \text{M}$

(d) $1.02 \times 10^{-4} \text{M}$

26. The given reaction



is an example of

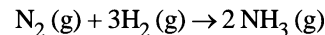
(a) first order reaction

(b) second order reaction

(c) third order reaction

(d) None of these

27. Consider the reaction

The equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is

(a) $+\frac{d[NH_3]}{dt} = -\frac{2}{3} \frac{d[H_2]}{dt}$

(b) $+\frac{d[NH_3]}{dt} = -\frac{3}{2} \frac{d[H_2]}{dt}$

(c) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

(d) $\frac{d[NH_3]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt}$

28. Which of the following statements is incorrect?

(a) Activation energy for the forward reaction equals to activation energy for the reverse reaction

(b) For a reversible reaction, an increase in temperature increases the reaction rate for both the forward and the backward reaction

(c) The larger the initial reactant concentration for a second order reaction, the shorter is its half-life.

(d) When Δt is infinitesimally small, the average rate equals the instantaneous rate29. $3A \rightarrow B + C$, it would be a zero order reaction when

(a) the rate of reaction is proportional to square of concentration of A

(b) the rate of reaction remains same at any concentration of A

(c) the rate remains unchanged at any concentration of B and C

(d) the rate of reaction doubles if concentration of B is increased to double

30. The decomposition of a substance follows first order kinetics.

Its concentration is reduced to 1/8th of its initial value in 24 minutes. The rate constant of the decomposition process is

(a) $1/24 \text{ min}^{-1}$

(b) $\frac{0.692}{24} \text{ min}^{-1}$

(c) $\frac{2.303}{24} \log \left(\frac{1}{8} \right) \text{ min}^{-1}$

(d) $\frac{2.303}{24} \log \left(\frac{8}{1} \right) \text{ min}^{-1}$

RESPONSE
GRID

22. (a)(b)(c)(d)

23. (a)(b)(c)(d)

24. (a)(b)(c)(d)

25. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 18 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	56
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC19

SYLLABUS : Surface Chemistry

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

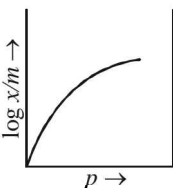
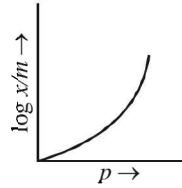
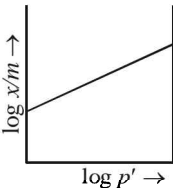
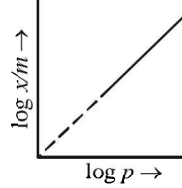
INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Which of the following electrolytes is least effective in coagulating ferric hydroxide solution?
(a) KBr (b) K_2SO_4
(c) K_2CrO_4 (d) $K_4[Fe(CN)_6]$
- Cod Liver oil is
(a) fat dispersed in water (b) water dispersed in fat
(c) water dispersed in oil (d) fat dispersed in fat.
- One desires to prepare a positively charged sol of silver iodide. This can be achieved by
(a) adding small amount of $AgNO_3$ solution to KI solution in slight excess
(b) adding small amount of KI solution to $AgNO_3$ solution in slight excess
(c) mixing equal volumes of equimolar solutions of $AgNO_3$ and KI
(d) None of these
- Which of the following statements is *not* true about the oil-in-water type emulsion?
(a) On addition of small amount of water, no separate layer of water appears
(b) On addition of oil, separate layer of oil is formed
(c) Addition of an electrolyte causes the conductivity of the emulsion to increase
(d) Addition of small amount of oil soluble dye renders the entire emulsion coloured.

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Freundlich equation for adsorption of gases (in amount of x g) on a solid (in amount of m g) at constant temperature can be expressed as
- (a) $\log \frac{x}{m} = \log p + \frac{1}{n} \log k$ (b) $\log \frac{x}{m} = \log k + \frac{1}{n} \log p$
- (c) $\frac{x}{m} \propto p^n$ (d) $\frac{x}{m} = \log p + \frac{1}{n} \log k$
6. The size of colloidal particles is of the order
- (a) $> 0.1 \mu$ (b) $1 \text{ m}\mu$ to 0.2μ
- (c) $< 1 \text{ m}\mu$ (d) $> 10 \text{ m}\mu$
7. Which of the following constitutes irreversible colloidal system in water as dispersion medium?
- (a) Clay (b) Platinum
- (c) $\text{Fe}(\text{OH})_3$ (d) All of these
8. Which of the following forms cationic micelles above certain concentration?
- (a) Sodium dodecyl sulphate
- (b) Sodium acetate
- (c) Urea
- (d) Cetyl trimethyl ammonium bromide
9. Position of non-polar and polar part in micelle is
- (a) polar at outer surface and non-polar at inner surface
- (b) polar at inner surface and non-polar at outer surface
- (c) distributed all over the surface
- (d) present in the surface only
10. The cause of Brownian movement is
- (a) Heat changes in liquid state
- (b) Convectional currents
- (c) The impact of molecules of the dispersion medium on the colloidal particles.
- (d) Attractive forces between the colloidal particles and molecules of dispersion medium.
11. Which of the following curves is in accordance with Freundlich adsorption isotherm ?
- (a) 
- (b) 
- (c) 
- (d) 
12. A colloidal solution is subjected to an electrical field. The particles move towards anode. The coagulation of same sol is studied using NaCl , BaCl_2 and AlCl_3 solutions. Their coagulating power should be
- (a) $\text{NaCl} > \text{BaCl}_2 > \text{AlCl}_3$
- (b) $\text{BaCl}_2 > \text{AlCl}_3 > \text{NaCl}$
- (c) $\text{AlCl}_3 > \text{BaCl}_2 > \text{NaCl}$
- (d) $\text{BaCl}_2 > \text{NaCl} > \text{AlCl}_3$
13. Which of the following is most effective in causing the coagulation of ferric hydroxide sol?
- (a) KCl (b) KNO_3
- (c) K_2SO_4 (d) $\text{K}_3[\text{Fe}(\text{CN})_6]$
14. Alum helps in purifying water by
- (a) forming Si complex with clay particles
- (b) sulphate part which combines with the dirt and removes it
- (c) aluminium which coagulates the mud particles
- (d) making mud water soluble

RESPONSE
GRID

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

11. (a)(b)(c)(d)

12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

Space for Rough Work

15. Which is not correct regarding the adsorption of a gas on surface of solid?
- On increasing temperature, adsorption increases continuously
 - Enthalpy and entropy changes are -ve
 - Adsorption is more for some specific substances
 - This phenomenon is reversible
16. The heats of adsorption in physisorption lie in the range (in kJ/mol)
- 40 - 400
 - 40 - 100
 - 10 - 40
 - 1 - 10
17. Example of intrinsic colloid is
- glue
 - sulphur
 - Fe
 - As₂S₃
18. For adsorption of a gas on a solid, the plot of $\log x/m$ vs $\log P$ is linear with slope equal to (n being whole number)
- k
 - $\log k$
 - n
 - $\frac{1}{n}$
19. Which of the following statements is incorrect?
- Colloidal gold is used for intramuscular injection.
 - Colloidal solution of latex is used in preparation of rubber.
 - Photographic films are prepared by coating an emulsion of AgBr in gelatin over glass plate.
 - Tannin used in leather industry contains positively charged colloidal particles.
20. The disperse phase in colloidal iron (III) hydroxide and colloidal gold is positively and negatively charged, respectively. Which of the following statements is NOT correct?
- Coagulation in both sols can be brought about by electrophoresis
 - Mixing the sols has no effect
 - Sodium sulphate solution causes coagulation in both sols
 - Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxide sol.
21. Which of the following will show Tyndall effect?
- Aqueous solution of soap below critical micelle concentration.
 - Aqueous solution of soap above critical micelle concentration.
 - Aqueous solution of sodium chloride.
 - Aqueous solution of sugar.
22. If x is amount of adsorbate and m is amount of adsorbent, which of the following relations is not related to adsorption process ?
- $x/m = f(p)$ at constant T .
 - $x/m = f(T)$ at constant p .
 - $p = f(T)$ at constant (x/m) .
 - $\frac{x}{m} = p^n T$
23. In Langmuir's model of adsorption of a gas on a solid surface
- the mass of gas striking a given area of surface is proportional to the pressure of the gas
 - the mass of gas striking a given area of surface is independent of the pressure of the gas
 - the rate of dissociation of adsorbed molecules from the surface does not depend on the surface covered
 - the adsorption at a single site on the surface may involve multiple molecules at the same time
24. Which of the following gas molecules have maximum value of enthalpy of physisorption?
- C₂H₆
 - Ne
 - H₂O
 - H₂

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

Space for Rough Work

C-76

DPP/ CC19

25. Which one of the following statements is not correct?
 (a) The value of equilibrium constant is changed in the presence of a catalyst in the reaction at equilibrium
 (b) Enzymes catalyse mainly bio-chemical reactions
 (c) Coenzymes increase the catalytic activity of enzyme
 (d) Catalyst does not initiate any reaction
26. Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 sol is
 (a) Na_2SO_4 (b) CaCl_2
 (c) $\text{Al}_2(\text{SO}_4)_3$ (d) NH_4Cl
27. Which of the following is **not** a favourable condition for physical adsorption ?
 (a) High pressure
 (b) Negative ΔH
 (c) Higher critical temperature of adsorbate
 (d) High temperature
28. At high pressure, the entire surface gets covered by a monomolecular layer of the gas follows
 (a) three-halved order (b) second-order
 (c) first-order (d) zero-order
29. At the Critical Micelle Concentration (CMC) the surfactant molecules
 (a) decompose
 (b) dissociate
 (c) associate
 (d) become completely soluble
30. Peptization involves
 (a) precipitation of colloidal particles
 (b) disintegration of colloidal aggregates
 (c) evaporation of dispersion medium
 (d) impact of molecules of the dispersion medium on the colloidal particles

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 19 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	40	Qualifying Score	60
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC20

SYLLABUS : General Principles and Processes of Isolation of Elements

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. Identity x , y , z for the following metallurgical process.

Metal sulphide $\xrightarrow{x}$ Metal oxide $\xrightarrow{y}$ Impure metal

$\xrightarrow{z}$ Pure metal.

x , y and z are respectively

- (a) roasting, smelting, electrolysis
(b) roasting, calcination, smelting
(c) calcination, auto-reduction, bassemmerisation
(d) none of the above is correct
2. Bronze is a mixture of

- (a) $\text{Pb} + \text{Sn}$ (b) $\text{Cu} + \text{Sn}$
(c) $\text{Cu} + \text{Zn}$ (d) $\text{Pb} + \text{Zn}$

3. Which reagent is used in Bayer's process?

- (a) Na_2CO_3 (b) Carbon
(c) NaOH (d) Silica

4. Thomas slag is

- (a) $\text{Ca}_3(\text{PO}_4)_2$
(b) CaSiO_3
(c) Mixture of (a) and (b)
(d) FeSiO_3

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Which of the following condition favours the reduction of a metal oxide to metal?
- $\Delta H = +ve$, $T\Delta S = +ve$ at low temperature
 - $\Delta H = +ve$, $T\Delta S = -ve$ at any temperature
 - $\Delta H = -ve$, $T\Delta S = -ve$ at high temperature
 - $\Delta H = -ve$, $T\Delta S = +ve$ at any temperature
6. Which of the following fluxes is used to remove acidic impurities in metallurgical process?
- Silica
 - Lime stone
 - Sodium chloride
 - Sodium carbonate
7. A basic lining is given to a furnace by using
- Calcined dolomite
 - Lime stone
 - haematite
 - silica
8. Electromagnetic separation is used in the concentration of
- copper pyrites
 - bauxite
 - cassiterite
 - cinnabar
9. Extraction of gold and silver involves leaching with CN^- ion. Silver is later recovered by
- distillation
 - zone refining
 - displacement with Zn
 - liquation
10. Which of the following reactions is an example for calcination process ?
- $2Ag + 2HCl + (O) \rightarrow 2AgCl + H_2O$
 - $2Zn + O_2 \rightarrow 2ZnO$
 - $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$
 - $MgCO_3 \rightarrow MgO + CO_2$
11. ΔG° vs T plot in the Ellingham's diagram slopes downward for the reaction
- $Mg + \frac{1}{2}O_2 \rightarrow MgO$
 - $2Ag + \frac{1}{2}O_2 \rightarrow Ag_2O$
 - $C + \frac{1}{2}O_2 \rightarrow CO$
 - $CO + \frac{1}{2}O_2 \rightarrow CO_2$
12. Which of the following elements is present as the impurity to the maximum extent in the pig iron ?
- Manganese
 - Carbon
 - Silicon
 - Phosphorus
13. Among the following statements the incorrect one is
- Calamine and siderite are carbonates
 - Argentite and cuprite are oxides
 - Zinc blende and iron pyrites are sulphides
 - Malachite and azurite are ores of copper.
14. Match list I with list II and select the correct answer using the codes given below the lists:
- | <i>List I</i> | <i>List II</i> |
|-----------------------------|---------------------|
| I. Cyanide process | A. Ultrapure Ge |
| II. Floatation process | B. Pine oil |
| III. Electrolytic reduction | C. Extraction of Al |
| IV. Zone refining | D. Extraction of Au |
- Codes:
- I-C, II-A, III-D, IV-B
 - I-D, II-B, III-C, IV-A
 - I-C, II-B, III-D, IV-A
 - I-D, II-A, III-C, IV-B
15. A coupled reaction takes place as follow—
 $A + B \longrightarrow C + D$, $\Delta G^\circ = +x \text{ kJ}$
 $D + E \longrightarrow F$, $\Delta G^\circ = -y \text{ kJ}$
 for the spontaneity of reaction $A + B + E \longrightarrow C + F$, which of the following is correct?
- $2x = y$
 - $x < y$
 - $x > y$
 - $x = (y) \times T\Delta S$
16. Aluminium is extracted from alumina (Al_2O_3) by electrolysis

**RESPONSE
GRID**

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

11. (a)(b)(c)(d)

12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

15. (a)(b)(c)(d)

Space for Rough Work

- of a molten mixture of
- $\text{Al}_2\text{O}_3 + \text{HF} + \text{NaAlF}_4$
 - $\text{Al}_2\text{O}_3 + \text{CaF}_2 + \text{NaAlF}_4$
 - $\text{Al}_2\text{O}_3 + \text{Na}_3\text{AlF}_6 + \text{CaF}_2$
 - $\text{Al}_2\text{O}_3 + \text{KF} + \text{Na}_3\text{AlF}_6$
17. Which of the following statements, about the advantage of roasting of sulphide ore before reduction is not true?
- The ΔG_f° of the sulphide is greater than those for CS_2 and H_2S .
 - The ΔG_f° is negative for roasting of sulphide ore to oxide.
 - Roasting of the sulphide to the oxide is thermodynamically feasible.
 - Carbon and hydrogen are suitable reducing agents for reduction of metal sulphides.
18. In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with :
- Copper (I) sulphide (Cu_2S)
 - Sulphur dioxide (SO_2)
 - Iron sulphide (FeS)
 - Carbon monoxide (CO)
19. The method of zone refining of metals is based on the principle of
- Greater solubility of the impurity in the molten state than in the solid
 - Greater mobility of the pure metal than that of the impurity
 - Higher melting point of the impurity than that of the pure metal
 - Greater noble character of the solid metal than that of the impurity
20. In the cyanide extraction process of silver from argentite ore, the oxidising and reducing agents used are
- O_2 and CO respectively
 - O_2 and Zn dust respectively
 - HNO_3 and Zn dust respectively
 - HNO_3 and CO respectively
21. The metal oxide which cannot be reduced to metal by carbon is
- Fe_2O_3
 - Al_2O_3
 - PbO
 - ZnO
22. Which of the following reaction takes place in blast furnace during extraction of copper ?
- $2\text{Cu}_2\text{S} + 3\text{O}_2 \longrightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$
 - $2\text{FeS} + 3\text{O}_2 \longrightarrow 2\text{FeO} + 2\text{SO}_2$
 - $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \longrightarrow 6\text{Cu} + \text{SO}_2$
 - All of these
23. Which process represents the change,
- $$\text{Ti} + 2\text{I}_2 \rightarrow \text{TiI}_4 \rightarrow \text{Ti} + 2\text{I}_2$$
- Cupellation
 - Van Arkel
 - Polling
 - Zone Refining
24. Among the following groups of oxides, the group containing oxides that cannot be reduced by carbon to give the respective metals is
- $\text{Cu}_2\text{O}, \text{SnO}_2$
 - $\text{Fe}_2\text{O}_3, \text{ZnO}$
 - $\text{CaO}, \text{K}_2\text{O}$
 - $\text{PbO}, \text{Fe}_3\text{O}_4$
25. Which of the following pairs of metals is purified by van Arkel method ?
- Ga and In
 - Zr and Ti
 - Ag and Au
 - Ni and Fe

RESPONSE
GRID

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

22. (a) (b) (c) (d)

23. (a) (b) (c) (d)

24. (a) (b) (c) (d)

25. (a) (b) (c) (d)

26. Which of the following reaction(s) occur in temperature range 500 – 800 K in blast furnace.
- (i) $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow 2\text{FeO} + \text{CO}_2$
 (ii) $\text{Fe}_3\text{O}_4 + 4\text{CO} \longrightarrow 3\text{Fe} + 4\text{CO}_2$
 (iii) $\text{FeO} + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
 (iv) $\text{C} + \text{CO}_2 \longrightarrow 2\text{CO}$
 (a) (i) and (ii) (b) (i), (ii) and (iii)
 (c) (iii) and (iv) (d) (iv) only
27. Which of the following pair is incorrectly matched ?
 (a) Magnetite – Fe_3O_4 (b) Copper glance – Cu_2S
 (c) Calamine – ZnCO_3 (d) Zincite – ZnS
28. Before introducing FeO in blast furnace, it is converted to Fe_2O_3 by roasting so that
 (a) it may not be removed as slag with silica
 (b) it may not evaporate in the furnace
 (c) presence of it may increase the m.pt. of charge
 (d) None of these.
29. Which of the following statements is correct ?
 (a) Gangues are carefully chosen to combine with the slag present in the ore to produce easily fusible flux to carry away the impurities
 (b) Slags are carefully chosen to combine with the flux present in the ore to produce easily fusible gangue to carry away the impurities
 (c) Gangues are carefully chosen to combine with the flux present in the ore to produce easily fusible slag to carry away the impurities
 (d) Fluxes are carefully chosen to combine with the gangue present in the ore to produce easily fusible slag to carry away the impurities
30. The value of $\Delta_f G^\circ$ for formation of Cr_2O_3 is -540 kJ mol^{-1} and that of Al_2O_3 is -827 kJ mol^{-1} . What is the value of $\Delta_r G^\circ$ for the reaction?
- $$\frac{4}{3} \text{Al(s)} + \frac{2}{3} \text{Cr}_2\text{O}_3(\text{s}) \rightarrow \frac{2}{3} \text{Al}_2\text{O}_3(\text{s}) + \frac{4}{3} \text{Cr(s)}.$$
- (a) -574 kJ mol^{-1} (b) -287 kJ mol^{-1}
 (c) $+574 \text{ kJ mol}^{-1}$ (d) $+287 \text{ kJ mol}^{-1}$

RESPONSE GRID

26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d) 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 20 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	56
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC21

SYLLABUS : The p-Block Elements (Group 15, 16, 17 and 18)

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- The oxyacid of phosphorous in which phosphorous has the lowest oxidation state is
 - hypophosphorous acid
 - orthophosphoric acid
 - pyrophosphoric acid
 - metaphosphoric acid
- The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down group-15 of the periodic table is due to
 - decreasing lp-bp repulsion
 - decreasing electronegativity
 - increasing bp-bp repulsion
 - increasing p-orbital character in sp^3
- The deep blue colour produced on adding excess of ammonia to copper sulphate is due to presence of
 - Cu^{2+}
 - $\text{Cu}(\text{NH}_3)_4^{2+}$
 - $\text{Cu}(\text{NH}_3)_6^{2+}$
 - $\text{Cu}(\text{NH}_3)_2^{2+}$
- In a cyclotrimetaphosphoric acid molecule, how many single and double bonds are present?
 - 3 double bonds; 9 single bonds
 - 6 double bonds; 6 single bonds
 - 3 double bonds; 12 single bonds
 - Zero double bonds; 12 single bonds

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. In nitroprusside ion, the iron and NO exist as Fe^{++} and NO^+ rather than Fe^{+++} and NO. These forms can be differentiated by
- estimating the concentration of iron
 - measuring the concentration of CN^-
 - measuring the solid state magnetic moment
 - thermally decomposing the compound**
6. The brown ring test for NO_2^- and NO_3^- is due to the formation of complex ion with a formula
- $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 - $[\text{Fe}(\text{NO})(\text{CN})_5]^{2+}$
 - $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
 - $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2+}$
7. Which compound is used in photography?
- Na_2SO_5
 - $\text{Na}_2\text{S}_2\text{O}_8$
 - $\text{Na}_2\text{S}_2\text{O}_6$
 - $\text{Na}_2\text{S}_2\text{O}_3$
8. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that
- oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas
 - strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke.
 - due to strong affinity for water, concentrated hydrochloric acid pulls moisture of air towards itself. This moisture forms droplets of water and hence the cloud.
 - concentrated hydrochloric acid emits strongly smelling HCl gas all the time.
9. The hybridization in ICl_7 is
- sp^3d^3
 - d^2sp^3
 - sp^3d
 - sp^3
10. The compound of sulphur that can be used as refrigerant is
- SO_2
 - SO_3
 - S_2Cl_2
 - H_2SO_4
11. Which of the following is a saline oxide ?
- Na_2O_2
 - BaO_2
 - Na_2O
 - Fe_2O_3
12. Which one of the following substances has the highest proton affinity ?
- H_2S
 - NH_3
 - PH_3
 - H_2O
13. Which one of the following statement regarding helium is **incorrect** ?
- It is used to produce and sustain powerful superconducting magnets
 - It is used as a cryogenic agent for carrying out experiments at low temperatures
 - It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable
 - It is used in gas-cooled nuclear reactors
14. Caro's acid is
- H_2SO_3
 - $\text{H}_3\text{S}_2\text{O}_5$
 - H_2SO_5
 - $\text{H}_2\text{S}_2\text{O}_8$

RESPONSE
GRID

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

11. (a)(b)(c)(d)

12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

Space for Rough Work

15. The acid which forms two series of salts is
 (a) H_3PO_4 (b) H_3PO_3
 (c) H_3BO_3 (d) H_3PO_2^-
16. Which of the following species is not a pseudo halide
 (a) CNO^- (b) RCOO^-
 (c) OCN^- (d) NNN^-
17. Gaseous HCl is a poor conductor of electricity while its aqueous solution is a good conductor this is because
 (a) H_2O is a good conductor of electricity
 (b) a gas cannot conduct electricity but a liquid can
 (c) HCl gas does not obey Ohm's law, whereas the solution does
 (d) HCl ionises in aqueous solution
18. The number of hydrogen atom(s) attached to phosphorus atom in hypophosphorous acid is
 (a) three (b) one
 (c) two (d) zero
19. What products are expected from the disproportionation reaction of hypochlorous acid?
 (a) HCl and Cl_2O
 (b) HCl and HClO_3
 (c) HClO_3 and Cl_2O
 (d) HClO_2 and HClO_4
20. Shape of XeOF_4 is
 (a) octahedral
 (b) square pyramidal
 (c) pyramidal
 (d) T-shaped
21. Yellow ammonium sulphide is
 (a) $(\text{NH}_4)_2\text{S}_8$
 (b) $(\text{NH}_4)_2\text{S}$
 (c) $(\text{NH}_4)_2\text{S}_x$
 (d) $(\text{NH}_4)_2\text{S}_4$
22. The correct order of increasing bond angles in the following species are :
 (a) $\text{Cl}_2\text{O} < \text{ClO}_2 < \text{ClO}_2^-$
 (b) $\text{ClO}_2 < \text{Cl}_2\text{O} < \text{ClO}_2^-$
 (c) $\text{Cl}_2\text{O} < \text{ClO}_2^- < \text{ClO}_2$
 (d) $\text{ClO}_2^- < \text{Cl}_2\text{O} < \text{ClO}_2$
23. Which one of the following reactions of xenon compounds is not feasible?
 (a) $3\text{XeF}_4 + 6\text{H}_2\text{O} \longrightarrow 2\text{Xe} + \text{XeO}_3 + 12\text{HF} + 1.5\text{O}_2$
 (b) $2\text{XeF}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
 (c) $\text{XeF}_6 + \text{RbF} \longrightarrow \text{Rb}[\text{XeF}_7]$
 (d) $\text{XeO}_3 + 6\text{HF} \longrightarrow \text{XeF}_6 + 3\text{H}_2\text{O}$
24. Trigonal bipyramidal geometry is shown by :
 (a) XeO_3F_2
 (b) XeO_3F_2
 (c) FXeOSO_2F
 (d) $[\text{XeF}_8]^{2-}$

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

Space for Rough Work

25. Which of the following is used to produce and sustain powerful superconducting magnets to form an essential part of NMR spectrometer ?
 (a) Ar (b) Ne
 (c) Rn (d) He
26. Oxidation of thiosulphate by iodine gives
 (a) tetrathionate ion (b) sulphide ion
 (c) sulphate ion (d) sulphite ion
27. The crystals of ferrous sulphate on heating give :
 (a) $\text{FeO} + \text{SO}_2 + \text{H}_2\text{O}$
 (b) $\text{FeO} + \text{SO}_3 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
 (c) $\text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
 (d) $\text{Fe}_2\text{O}_3 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
28. Which one of the following orders correctly represents the increasing acid strengths of the given acids?
 (a) $\text{HOClO} < \text{HOCl} < \text{HOClO}_3 < \text{HOClO}_2$
 (b) $\text{HOClO}_2 < \text{HOClO}_3 < \text{HOClO} < \text{HOCl}$
 (c) $\text{HOClO}_3 < \text{HOClO}_2 < \text{HOClO} < \text{HOCl}$
 (d) $\text{HOCl} < \text{HOClO} < \text{HOClO}_2 < \text{HOClO}_3$
29. Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated?
 (a) IF_7 : pentagonal bipyramid
 (b) BrF_5 : trigonal bipyramid
 (c) BrF_3 : planar T-shaped
 (d) ICl_3 : planar dimeric
30. The nitrogen oxides that contain(s) N–N bond(s) is /are
 (i) N_2O (ii) N_2O_3
 (iii) N_2O_4 (iv) N_2O_5
 (a) (i), (ii) (b) (ii), (iii), (iv)
 (c) (iii), (iv) (d) (i), (ii) and (iii)

**RESPONSE
GRID**

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 21 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	51
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC22

SYLLABUS : The *d*- and *f*- Block Elements

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them?
(a) $(n-1)d^3 ns^2$
(b) $(n-1)d^5 ns^1$
(c) $(n-1)d^8 ns^2$
(d) $(n-1)d^5 ns^2$
- In which of the following cases, the stability of two oxidation states is correctly represented
(a) $Ti^{3+} > Ti^{4+}$
(b) $Mn^{2+} > Mn^{3+}$
(c) $Fe^{2+} > Fe^{3+}$
(d) $Cu^+ > Cu^{2+}$
- $(n-1)d^{10} ns^2$ is the general electronic configuration of
(a) Fe, Co, Ni
(b) Cu, Ag, Au
(c) Zn, Cd, Hg
(d) Se, Y, La
- Super conductors are derived from compounds of
(a) p-Block elements
(b) lanthanides
(c) actinides
(d) transition elements

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?
 (a) $3d^5 4s^1$ (b) $3d^5 4s^2$
 (c) $3d^2 4s^2$ (d) $3d^3 4s^2$
6. The catalytic activity of transition metals and their compounds is mainly due to :
 (a) their magnetic behaviour
 (b) their unfilled d -orbitals
 (c) their ability to adopt variable oxidation state
 (d) their chemical reactivity
7. Arrange Ce^{+3} , La^{+3} , Pm^{+3} and Yb^{+3} in increasing order of their ionic radii.
 (a) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$
 (b) $\text{Ce}^{+3} < \text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3}$
 (c) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3}$
 (d) $\text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3} < \text{Yb}^{+3}$
8. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid?
 (a) $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 (b) CrO_4^{2-} is reduced to +3 state of Cr
 (c) CrO_4^{2-} is oxidized to +7 state of Cr
 (d) Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
9. On the basis of data given below,
 $E_{\text{Sc}^{3+}/\text{Sc}^{2+}}^\ominus = -0.37$, $E_{\text{Mn}^{3+}/\text{Mn}^{2+}}^\ominus = +1.57$
 $E_{\text{Cr}^{2+}/\text{Cr}}^\ominus = -0.90$, $E_{\text{Cu}^{2+}/\text{Cu}}^\ominus = 0.34$
 Which of the following statements is incorrect?
 (a) Sc^{3+} has good stability due of $[\text{Ar}]3d^0 4s^0$ configuration.
 (b) Mn^{3+} is more stable than Mn^{2+} .
 (c) Cr^{2+} is reducing in nature.
 (d) Copper does not give H_2 on reaction with dil. H_2SO_4 .
10. When KMnO_4 acts as an oxidising agent and ultimately forms $[\text{MnO}_4]^{-1}$, MnO_2 , Mn_2O_3 , Mn^{+2} then the number of electrons transferred in each case respectively is
 (a) 4, 3, 1, 5 (b) 1, 5, 3, 7 (c) 1, 3, 4, 5 (d) 3, 5, 7, 1.
11. For which of the following species $d-d$ transition does not account for its colour ?
 (a) $\text{Cr}_2\text{O}_7^{2-}$ (b) CrO_4^{2-}
 (c) CrO_2Cl_2 (d) All
12. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. Ni = 28)
 (a) 6 (b) 1.73
 (c) 2.82 (d) 4.90
13. Heating mixture of Cu_2O and Cu_2S will give
 (a) Cu_2SO_3 (b) $\text{CuO} + \text{CuS}$
 (c) $\text{Cu} + \text{SO}_3$ (d) $\text{Cu} + \text{SO}_2$
14. In which of the following lanthanides oxidation state +2 is most stable?
 (a) Ce (b) Eu
 (c) Tb (d) Dy
15. Which one of the following nitrates will leave behind a metal on strong heating ?
 (a) Copper nitrate (b) Manganese nitrate
 (c) Silver nitrate (d) Ferric nitrate
16. The black compound formed during the reaction between sodium thiosulphate and silver nitrate is
 (a) silver thiosulphate ($\text{Ag}_2\text{S}_2\text{O}_3$)
 (b) silver sulphide (Ag_2S)
 (c) silver sulphate (Ag_2SO_4)
 (d) silver sulphite (Ag_2SO_3)

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)
 15. (a)(b)(c)(d) 16. (a)(b)(c)(d)

Space for Rough Work

17. Which of the following element is responsible for oxidation of water to O_2 in biological process?
 (a) Fe (b) Mn
 (c) Cu (d) Mo
18. Which of the following arrangements does not represent the correct order of the property stated against it ?
 (a) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour
 (b) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
 (c) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
 (d) $Sc < Ti < Cr < Mn$: number of oxidation states
19. Which of the following factors may be regarded as the main cause of lanthanoid contraction?
 (a) Greater shielding of $5d$ electrons by $4f$ electrons
 (b) Poorer shielding of $5d$ electrons by $4f$ electrons
 (c) Effective shielding of one of $4f$ electrons by another in the subshell
 (d) Poor shielding of one of $4f$ electron by another in the subshell
20. Which of the following elements shows maximum number of different oxidation states in its compounds?
 (a) Eu (b) Ld
 (c) Gd (d) Am
21. An explosion take place when conc. H_2SO_4 is added to $KMnO_4$. Which of the following is formed?
 (a) Mn_2O_7 (b) MnO_2
 (c) $MnSO_4$ (d) M_2O_3
22. $AgCl$ is soluble in NH_4OH solution. The solubility is due to the formation of
 (a) $AgOH$ (b) Ag_2O
 (c) $[Ag(NH_3)_2]^+$ (d) NH_4Cl
23. Consider the complexes given below and choose the correct order of paramagnetic moments (spin only)
 I. $[FeF_6]^{3-}$; II. $[CrF_6]^{3-}$;
 III. $[V(H_2O)_6]^{3+}$; IV. $[Ti(H_2O)_6]^{3+}$
 (a) $I > II > III > IV$ (b) $I < II < III < IV$
 (c) $I > III > IV > II$ (d) $III > I > IV > II$
24. The value of the 'spin only' magnetic moment for one of the following configurations is 2.82 BM. The correct one is
 (a) d^5 (in strong ligand field)
 (b) d^3 (in weak as well as in strong fields)
 (c) d^4 (in weak ligand fields)
 (d) d^4 (in strong ligand fields)
25. In neutral or faintly alkaline medium, thiosulphate is quantitatively oxidized by $KMnO_4$ to
 (a) SO_3^{2-} (b) SO_4^{2-}
 (c) SO_2 (d) SO_5^{2-}
26. Parke's process is used
 (a) to extract Ag using NaCN
 (b) to extract Cu using $CuFeS_2$
 (c) to extract Ag from argentiferous lead
 (d) to extract Ag by forming amalgam
27. A metal which is not affected by conc. H_2SO_4 , HNO_3 or alkalis forms a compound X. This compound X can be used to give a complex which finds its application for toning

RESPONSE
GRID

17. (a) (b) (c) (d)
 22. (a) (b) (c) (d)
 27. (a) (b) (c) (d)

18. (a) (b) (c) (d)
 23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
 24. (a) (b) (c) (d)

20. (a) (b) (c) (d)
 25. (a) (b) (c) (d)

21. (a) (b) (c) (d)
 26. (a) (b) (c) (d)

Space for Rough Work

C-88

DPP/ CC22

in photography. The metal is

- (a) Au (b) Ag
(c) Hg (d) Cu

28. Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solutions. What is the reason for it ?

- (a) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
(b) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
(c) Copper hydroxide is an amphoteric substance

(d) In acidic solutions hydration protects copper ions

29. A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of a metal appear on the cooler parts of the test tube. The red solid is

- (a) HgI_2 (b) HgO
(c) Pb_3O_4 (d) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

30. Four successive members of the first series of the transition metals are listed below. For which one of them the standard

potential $(E_{\text{M}^{2+}/\text{M}}^\circ)$ value has a positive sign?

- (a) Co (Z = 27) (b) Ni (Z = 28)
(c) Cu (Z = 29) (d) Fe (Z = 26)

RESPONSE
GRID

28. (a)(b)(c)(d) 29. (a)(b)(c)(d) 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 22 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC23

SYLLABUS : Coordination Compounds

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Which of the following is incorrect regarding spectrochemical series?
(a) $\text{NH}_3 > \text{H}_2\text{O}$ (b) $\text{F}^- > \text{C}_2\text{O}_4^{2-}$
(c) $\text{NCS}^- > \text{SCN}^-$ (d) $\text{en} > \text{edta}^{4-}$
- $\text{C}_{63}\text{H}_{88}\text{CoN}_{14}\text{O}_{14}\text{P}$ is the formulae of the Cyanocobalamine, (vitamin B_{12}) it contain CN^- and CN^- is very poisonous, than why this compound does not prove to be fatal for us? (it inhibit the electron transport chain ?
(a) CN^- forms covalent bond
(b) CN^- is coordinating to the cobalt as the ligand
(c) CN^- hydrolysis immediately
(d) All of these
- An example of double salt is
(a) Bleaching powder
(b) $\text{K}_4[\text{Fe}(\text{CN})_6]$
(c) Hypo
(d) Potash alum
- Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively :
(a) one, square planar
(b) two, square planar
(c) one, tetrahedral
(d) two, tetrahedral

RESPONSE GRID

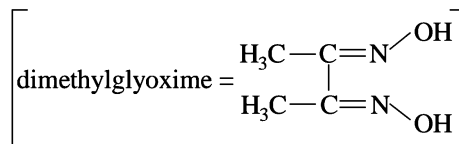
1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

|C-90|

DPP/ CC23

5. The crystal field splitting energy for octahedral (Δ_0) and tetrahedral (Δ_t) complexes is related as
- (a) $\Delta_t = -\frac{1}{2}\Delta_0$ (b) $\Delta_t = -\frac{4}{9}\Delta_0$
 (c) $\Delta_t = -\frac{3}{5}\Delta_0$ (d) $\Delta_t = -\frac{2}{5}\Delta_0$
6. A square planar complex is formed by hybridisation of which atomic orbitals?
- (a) s, p_x, p_y, d_{yz} (b) $s, p_x, p_y, d_{x^2-y^2}$
 (c) s, p_x, p_y, d_{z^2} (d) s, p_y, p_z, d_{xy}
7. Consider the following complex $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{ClO}_4$. The coordination number, oxidation number, number of d -electrons and number of unpaired d -electrons on the metal are respectively
- (a) 6, 3, 6, 0 (b) 7, 2, 7, 1
 (c) 7, 1, 6, 4 (d) 6, 2, 7, 3
8. How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion?
- (a) One (b) Two
 (c) Six (d) Three
9. Which of the following are inner orbital complex (i.e., involving d^2sp^3 hybridisation) and is paramagnetic in nature?
- (a) $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (b) $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$, $[\text{CoF}_6]^{3-}$
 (c) $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$
 (d) $[\text{MnCl}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
10. Which of the following is organo-metallic compound?
- (a) $\text{Ti}(\text{C}_2\text{H}_4)_4$ (b) $\text{Ti}(\text{OC}_2\text{H}_5)_4$
 (c) $\text{Ti}(\text{OCOCH}_3)_4$ (d) $\text{Ti}(\text{OC}_6\text{H}_5)_4$
11. Which is not π -bonded complex?
- (a) Zeise's salt (b) Ferrocene
 (c) Dibenzene chromium (d) Tetraethyl lead
12. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is (At. mass of Ag = 108 u)
- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$
 (c) $[\text{CoCl}_3(\text{NH}_3)_3]$ (d) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
13. The type of isomerism present in Pentamminenitrochromium (III) chloride is
- (a) optical (b) linkage
 (c) ionisation (d) polymerisation.
14. Red precipitate is obtained when ethanol solution of dimethylglyoxime is added to ammoniacal Ni(II). Which of the following statements is not true?
- (a) Red complex has a square planar geometry.
 (b) Complex has symmetrical H-bonding
 (c) Red complex has a tetrahedral geometry.
 (d) Dimethylglyoxime functions as bidentate ligand.



RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. An excess of AgNO_3 is added to 100 mL of a 0.01 M solution of dichlorotetraaquachromium (iii) chloride. The number of moles of AgCl precipitated would be :

(a) 0.002 (b) 0.003
(c) 0.01 (d) 0.001

16. Magnetic moment of Gd^{3+} ion ($Z = 64$) is

(a) 3.62 BM (b) 9.72 BM
(c) 7.9 BM (d) 10.60 BM

17. Which of the following species represent the example of dsp^2 - hybridisation ?

(a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$
(c) $[\text{Ag}(\text{CN})_2]^-$ (d) $[\text{Co}(\text{CN})_6]^{3-}$

18. Which one of the following has largest number of isomers ?

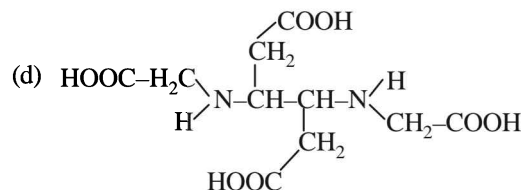
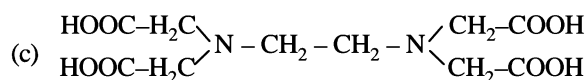
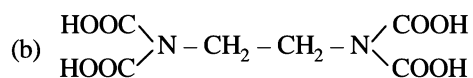
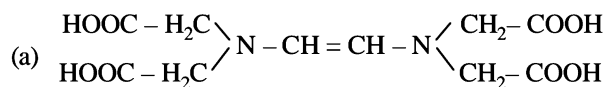
(a) $[\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})]^{2+}$ (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$

(c) $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$ (d) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$

19. In $\text{Fe}(\text{CO})_5$, the Fe – C bond possesses

(a) ionic character (b) σ -character only
(c) π -character (d) both σ and π characters

20. The correct structure of ethylenediaminetetraacetic acid (EDTA) is



21. The structure of which of the following chloro species can be explained on the basis of dsp^2 hybridization ?

(a) PdCl_4^{2-} (b) FeCl_4^{2-}

(c) CoCl_4^{2-} (d) NiCl_4^{2-}

22. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:

(a) $L_4 < L_3 < L_2 < L_1$ (b) $L_1 < L_3 < L_2 < L_4$

(c) $L_3 < L_2 < L_4 < L_1$ (d) $L_1 < L_2 < L_4 < L_3$

23. Which one of the following coordination compounds is used to inhibit the growth of tumours?

(a) Trans-platin (b) EDTA complex of calcium
(c) $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ (d) Cis-platin

24. $[\text{NiCl}_2 \{ \text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)_2 \}_2]$ exhibits temperature dependent magnetic behaviour (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively

(a) tetrahedral and tetrahedral
(b) square planar and square planar
(c) tetrahedral and square planar
(d) square planar and tetrahedral

RESPONSE
GRID

15. (a) (b) (c) (d)

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

22. (a) (b) (c) (d)

23. (a) (b) (c) (d)

24. (a) (b) (c) (d)

Space for Rough Work

C-92

DPP/ CC23

25. Which will give a white precipitate with AgNO_3 in aqueous solution

- (a) $[\text{Co}(\text{NH}_3)_5\text{Cl}](\text{NO}_2)_2$ (b) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$
 (c) $[\text{Pt}(\text{en})\text{Cl}_2]$ (d) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

26. Which of the following name formula combinations is not correct?

Formula		
(a)	$\text{K}_2[\text{Pt}(\text{CN})_4]$	Potassium tetracyanoplatinate (II)
(b)	$[\text{Mn}(\text{CN})_5]^{2-}$	Pentacyanomagnate (II) ion
(c)	$\text{K}[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]$	Potassium diammine tetrachlorochromate (III)
(d)	$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})]\text{SO}_4$	Magnetron value.

27. Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[\text{NiL}_4]^{2-}$. The

hybridisation involved and the number of unpaired electrons present in the complex are respectively:

- (a) sp^3 , two (b) dsp^2 , zero
 (c) dsp^2 , one (d) sp^3 , zero

28. Which of the following species is not expected to be a ligand?

- (a) NO (b) NH_4^+
 (c) $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (d) Both (a) and (b)

29. Which of the following organometallic compound is σ and π -bonded?

- (a) $[\text{Fe}(\eta^5-\text{C}_5\text{H}_5)_2]$ (b) $[\text{PtCl}_3(\eta^2-\text{C}_2\text{H}_4)]$
 (c) $[\text{Co}(\text{CO})_5\text{NH}_3]^{2+}$ (d) $\text{Al}(\text{CH}_3)_3$

30. The most stable complex among the following is

- (a) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$ (b) $[\text{Pt}(\text{en})_2]\text{Cl}$
 (c) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ (d) $\text{K}_2[\text{Ni}(\text{EDTA})]$

RESPONSE
GRID

25. (a)(b)(c)(d)
30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 23 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	36	Qualifying Score	54
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC24

SYLLABUS : Haloalkanes and Haloarenes

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- In the preparation of chlorobenzene from aniline, the most suitable reagent is
 - Chlorine in the presence of ultraviolet light
 - Chlorine in the presence of AlCl_3
 - Nitrous acid followed by heating with Cu_2Cl_2
 - HCl and Cu_2Cl_2
- Which reagent cannot be used to prepare an alkyl halide from an alcohol ?
 - $\text{HCl} + \text{ZnCl}_2$
 - NaCl
 - PCl_5
 - SOCl_2
- Vapour density of an organic compound is 23. It contains 52.17% of carbon and 13% of hydrogen. The compound gives iodoform test. The compound is :
 - ethanol
 - dimethyl ether
 - acetone
 - methanol
- Reaction of *trans* 2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces
 - 1-phenylcyclopentene
 - 3-phenylcyclopentene
 - 4-phenylcyclopentene
 - 2-phenylcyclopentene

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

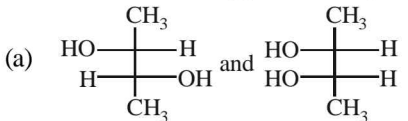
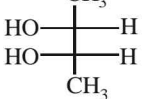
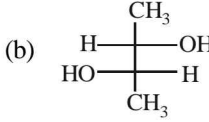
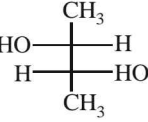
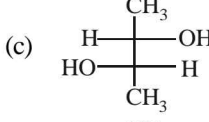
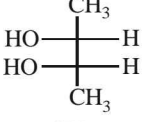
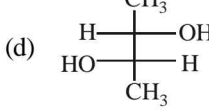
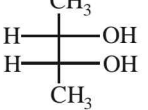
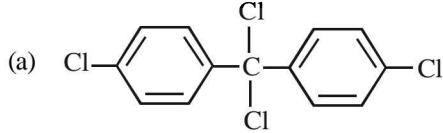
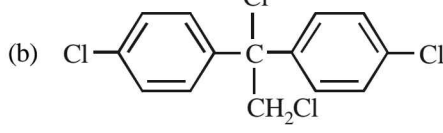
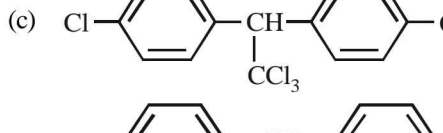
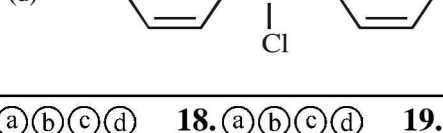
Space for Rough Work

5. Which of the following is an example of S_N2 reaction?
- $\text{CH}_3\text{Br} + \text{OH}^- \longrightarrow \text{CH}_3\text{OH} + \text{Br}^-$
 - $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_3 + \text{OH}^- \longrightarrow \text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{-\text{H}_2\text{O}} \text{CH}_2 = \text{CH}_2$
 - $(\text{CH}_3)_3\text{C}-\text{Br} + \text{OH}^- \longrightarrow (\text{CH}_3)_3\text{COH} + \text{Br}^-$
6. The reaction of $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ with CuCl gives
- $\text{C}_6\text{H}_5\text{Cl}$
 - C_6H_6
 - $\text{C}_6\text{H}_5-\text{C}_6\text{H}_5$
 - $\text{C}_6\text{H}_4\text{Cl}_2$
7. On sulphonation of $\text{C}_6\text{H}_5\text{Cl}$
- m*-Chlorobenzenesulphonic acid is formed
 - Benzenesulphonic acid is formed
 - o*-Chlorobenzenesulphonic acid is formed
 - o*- and *p*-Chlorobenzenesulphonic acid is formed
8. The structure of the major product formed in the following reaction
-
- -
 -
 -
9. Which of the following will have a mesoisomer also?
- 2, 3- Dichloropentane
 - 2, 3-Dichlorobutane
 - 2-Chlorobutane
 - 2-Hydroxypropanoic acid
10. The pesticide DDT slowly changes to
- $\text{CCl}_3\text{-CHO}$ and chlorobenzene
 - p, p'*-Dichlorodiphenylethene
 - p, p'*-Dichlorodiphenyldichloroethane
 - p, p'*-Dichlorodiphenyldichloroethene
11. *o*-Methoxybromobenzene is treated with sodamide and then with ammonia. The product formed is
- o*-Methoxyaniline
 - Aniline
 - Methoxybenzene
 - m*-Methoxyaniline
12. The starting substance for the preparation of iodoform is any one of the following, except
- $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{OH}$
 - HCH_2OH
 - CH_3COCH_3
13. Which of the following statements is wrong?
- Ethyl chloride on reduction with Zn-Cu couple and alcohol gives ethane.
 - The reaction of methyl magnesium bromide with acetone gives butanol-2.
 - Alkyl halides follow the following reactivity sequence on reaction with alkenes.
 $\text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-I}$
 - $\text{C}_2\text{H}_4\text{Cl}_2$ may exist in two isomeric forms
14. An organic compound A ($\text{C}_4\text{H}_9\text{Cl}$) on reaction with Na /diethyl ether gives a hydrocarbon which on monochlorination gives only one chloro derivative, then A is
- tert-butyl chloride
 - sec-butyl chloride
 - isobutyl chloride
 - n-butyl chloride

RESPONSE
GRID

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. If chloroform is left open in air in the presence of sunlight, it gives
 (a) carbon tetrachloride (b) carbonyl chloride
 (c) mustard gas (d) lewisite
16. The reaction of toluene with Cl_2 in presence of FeCl_3 gives 'X' and reaction in presence of light gives 'Y'. Thus, 'X' and 'Y' are :
 (a) X = Benzal chloride, Y = *o*-Chlorotoluene
 (b) X = *m*-Chlorotoluene, Y = *p*-Chlorotoluene
 (c) X = *o*- and *p*-Chlorotoluene, Y = Trichloromethylbenzene
 (d) X = Benzyl chloride, Y = *m*-Chlorotoluene
17. $\text{CH}_3\text{Br} + \text{Nu}^- \longrightarrow \text{CH}_3 - \text{Nu} + \text{Br}^-$ The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is
 [Nu^- = (A) PhO^- , (B) AcO^- , (C) HO^- , (D) CH_3O^-]
 (a) $\text{A} > \text{B} > \text{C} > \text{D}$ (b) $\text{B} > \text{D} > \text{C} > \text{A}$
 (c) $\text{D} > \text{C} > \text{A} > \text{B}$ (d) $\text{D} > \text{C} > \text{B} > \text{A}$
18. The Wurtz-Fittig reaction involves condensation of :
 (a) two molecules of aryl halides
 (b) one molecule of each of aryl-halide and alkyl-halide.
 (c) one molecule of each of aryl-halide and phenol.
 (d) two molecules of aralkyl-halides.
19. The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is:
 (a) Acetylene (b) Ethene
 (c) 2-Butyne (d) 2-Butene
20. Which of the following pairs of compounds are enantiomers?
 (a)  and 
- (b)  and 
- (c)  and 
- (d)  and 
21. Iodoform can be prepared from all except :
 (a) Ethyl methyl ketone
 (b) Isopropyl alcohol
 (c) 3-Methyl 2-butanone
 (d) Isobutyl alcohol
22. The reagent(s) for the following conversion,
 $\text{Br}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{Br} \xrightarrow{?} \text{H}-\text{C}\equiv\text{C}-\text{H}$
 is/are
 (a) alcoholic KOH
 (b) alcoholic KOH followed by NaNH_2
 (c) aqueous KOH followed by NaNH_2
 (d) $\text{Zn}/\text{CH}_3\text{OH}$
23. In a nucleophilic substitution reaction:
 $\text{R}-\text{Br} + \text{Cl}^- \xrightarrow{\text{DMF}} \text{R}-\text{Cl} + \text{Br}^-$,
 which one of the following undergoes complete inversion of configuration?
 (a) $\text{C}_6\text{H}_5\text{CHClC}_6\text{H}_5$ (b) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$
 (c) $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br}$ (d) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br}$
24. Chlorobenzene reacts with trichloro acetaldehyde in the presence of H_2SO_4 .
 $2 \text{C}_6\text{H}_5\text{Cl} + \text{H}-\text{C}(=\text{O})-\text{CCl}_3 \xrightarrow{\text{H}_2\text{SO}_4}$
 The major product formed is:
 (a) 
 (b) 
 (c) 
 (d) 

RESPONSE
GRID

15. (a) (b) (c) (d)

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

22. (a) (b) (c) (d)

23. (a) (b) (c) (d)

24. (a) (b) (c) (d)

Space for Rough Work

C-96

DPP/ CC24

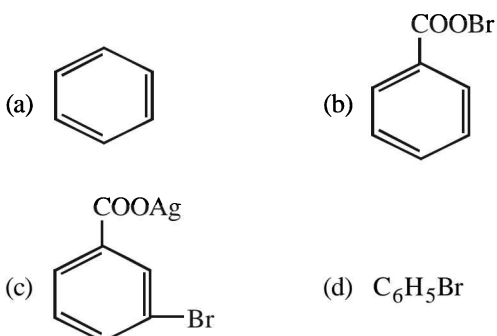
25. Benzene reacts with n-propyl chloride in the presence of anhydrous AlCl_3 to give

- (a) 3-Propyl-1-chlorobenzene
(b) n-Propylbenzene
(c) No reaction
(d) Isopropylbenzene

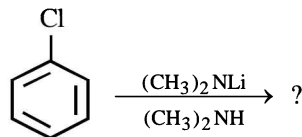
26. Replacement of Cl of chlorobenzene to give phenol requires drastic conditions but chlorine of 2,4-dinitrochlorobenzene is readily replaced. This is because

- (a) NO_2 makes the ring electron rich at *ortho* and *para* positions
(b) NO_2 withdraws e^- from *meta*-position
(c) NO_2 donates e^- at *m*-position
(d) NO_2 withdraws e^- from *ortho/para*-positions

27. Silver benzoate reacts with bromine to form



28. What is the product of the following reaction?

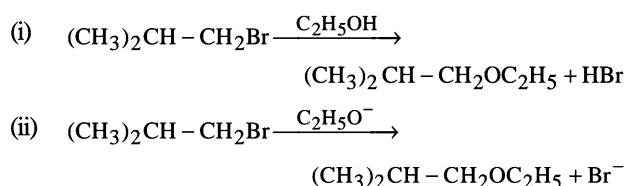


- (a) N,N-dimethyl aniline
(b) phenyl lithium ($\text{C}_6\text{H}_5\text{Li}$)
(c) para chloro-N,N-dimethyl aniline
(d) meta chloro-N,N-dimethyl aniline

29. The major product formed when 1,1,1-trichloropropane is treated with aqueous potassium hydroxide is:

- (a) Propyne (b) 1-Propanol
(c) 2-Propanol (d) Propionic acid

30. Consider the reactions :



The mechanisms of reactions (i) and (ii) are respectively :

- (a) $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ (b) $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}1$
(c) $\text{S}_{\text{N}}2$ and $\text{S}_{\text{N}}2$ (d) $\text{S}_{\text{N}}2$ and $\text{S}_{\text{N}}1$

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 24 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	54
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC25

SYLLABUS : Alcohols, Phenols and Ethers

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- Lucas reagent is
 - Conc. HCl and anhydrous ZnCl_2
 - Conc. HNO_3 and hydrous ZnCl_2
 - Conc. HCl and hydrous ZnCl_2
 - Conc. HNO_3 and anhydrous ZnCl_2
- Diethyl ether reacts, inspite of its usual inert nature, with :
 - Dilute sulphuric acid
 - Dilute sodium hydroxide
 - Boron trifluoride
 - Metallic sodium
- Vinyl carbinol is
 - $\text{HO}-\text{CH}_2-\text{CH}=\text{CH}_2$
 - $\text{CH}_3\text{C}(\text{OH})=\text{CH}_2$
 - $\text{CH}_3-\text{CH}=\text{CH}-\text{OH}$
 - $\text{CH}_3-\text{C}(\text{CH}_2\text{OH})=\text{CH}_2$
- Among the following compounds which can be dehydrated very easily is
 - $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_2\text{CH}_3 \\ | \\ \text{OH} \end{array}$$
 - $$\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_3 \end{array}$$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 - $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{CH}_2\text{CHCH}_2\text{CH}_2\text{OH} \end{array}$$

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. $\text{CH}_3\text{CH}_2\text{OH}$ can be converted into CH_3CHO by _____
- catalytic hydrogenation
 - treatment with LiAlH_4
 - treatment with pyridinium chlorochromate
 - treatment with KMnO_4
6. Epichlorohydrin is
- 3-Chloropropane
 - 3-Chloropropan-1-ol
 - 3-Chloro-1, 2-epoxypropane
 - None of these
7. 2-Phenylethanol may be prepared by the reaction of phenylmagnesium bromide with
- HCHO
 - CH_3CHO
 - CH_3COCH_3
 - 
8. HBr reacts with $\text{CH}_2 = \text{CH} - \text{OCH}_3$ under anhydrous conditions at room temperature to give
- $\text{BrCH}_2 - \text{CH}_2 - \text{OCH}_3$
 - $\text{H}_3\text{C} - \text{CHBr} - \text{OCH}_3$
 - CH_3CHO and CH_3Br
 - BrCH_2CHO and CH_3OH
9. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous ZnCl_2 , is
- 2-Butanol
 - 2-Methylpropan-2-ol
 - 2-Methylpropanol
 - 1-Butanol
10. Which of the following cannot be made by using Williamson's synthesis?
- Methoxybenzene
 - Benzyl p-nitrophenyl ether
 - Methyl tertiary butyl ether
 - Di-tert-butyl ether
11. Rectified spirit is a mixture of
- 95% ethyl alcohol + 5% water
 - 94% ethyl alcohol + 4.53% water
 - 94.4% ethyl alcohol + 5.43% water
 - 95.87% ethyl alcohol + 4.13% water
12. Absolute alcohol (100% alcohol) is prepared by distilling rectified spirit over
- Na
 - CaCl_2
 - Mg
 - $\text{Mg}(\text{OC}_2\text{H}_5)_2$
13. Reagent used to convert allyl alcohol to acrolein is
- MnO_2
 - H_2O_2
 - OsO_4
 - KMnO_4
14. In Williamson synthesis of mixed ether having a primary and a tertiary alkyl group if tertiary halide is used, then :
- Rate of reaction will be slow due to slow cleavage of carbon-halogen bond.
 - Alkene will be the main product.
 - Simple ether will form instead of mixed ether.
 - Expected mixed ether will be formed.

**RESPONSE
GRID**

 5. (a)(b)(c)(d)
10. (a)(b)(c)(d)

 6. (a)(b)(c)(d)
11. (a)(b)(c)(d)

 7. (a)(b)(c)(d)
12. (a)(b)(c)(d)

 8. (a)(b)(c)(d)
13. (a)(b)(c)(d)

 9. (a)(b)(c)(d)
14. (a)(b)(c)(d)

Space for Rough Work

15. Arrange the following in increasing order of their acidity?
o-cresol(a), salicylic acid(b), phenol(c)
 (a) $c < a < b$ (b) $b < c < a$
 (c) $a < b < c$ (d) $a < c < b$
16. Zerevitinov's determination of active hydrogen in a compound is based upon its reaction with
 (a) Na (b) CH_3MgI
 (c) Zn (d) Al
17. When wine is put in air, it becomes sour due to
 (a) bacteria
 (b) oxidation of $\text{C}_2\text{H}_5\text{OH}$ to CH_3COOH
 (c) virus
 (d) formic acid formation
18. Osmium tetroxide is a reagent used for
 (a) hydroxylation of acetylenes
 (b) hydroxylation of olefins to give *cis*-diols
 (c) hydroxylation of olefins to form *trans*-diols
 (d) hydroxylation of carbonyl compounds
19. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acetylation produces C
-
- The major product C would be
- (a) (b)
 (c) (d)
20. Williamson's synthesis is used to prepare
 (a) acetone (b) diethyl ether
 (c) P.V.C. (d) bakelite
21. Aspirin is an acetylation product of
 (a) *p*-Dihydroxybenzene
 (b) *o*-Hydroxybenzoic acid
 (c) *o*-Dihydroxybenzene
 (d) *m*-Hydroxybenzoic acid
22. *o*-Xylene $\xrightarrow{\text{HNO}_3}$ X $\xrightarrow[\text{H}_2\text{SO}_4]{\text{Phenol}}$ Y. The product Y is
 (a) Phthalic acid
 (b) Isophthalic acid
 (c) Phenolphthalein
 (d) *o*-Hydroxysulphonic acid
23. The following reaction
-
- is known as:
 (a) Perkin reaction
 (b) Gatterman-Koch Formylation
 (c) Kolbe's reaction
 (d) Gattermann reaction
24. Denaturation of alcohol is the
 (a) mixing of CuSO_4 (a foul smelling solid) and pyridine (to give the colour) to make the commercial alcohol unfit for drinking
 (b) mixing of CuSO_4 (to give the colour) and pyridine (a foul smelling solid) to make the commercial alcohol unfit for drinking
 (c) mixing of $\text{Cu}(\text{OAc})_2$ and ammonia to make the commercial alcohol unfit for drinking
 (d) mixing of $\text{Cu}(\text{OAc})_2$ and pyridine to make the commercial alcohol unfit for drinking

RESPONSE
GRID

15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
 20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

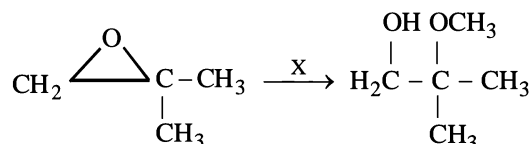
C-100

DPP/ CC25

25. Formation of which compound given below from 1 - butanol needs an oxidising agent?

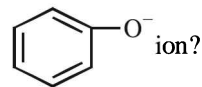
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
 (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{O}$
 (c) $(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{O}$
 (d) $\text{CH}_3-\text{CH}_2\text{CH}=\text{CH}_2$

26. What is X in the following reaction ?



- (a) $\text{CH}_3\text{OH}, \text{H}_2\text{SO}_4$
 (b) $\text{CH}_3\text{OH}, \text{CH}_3\text{O}^- \text{Na}^+$
 (c) $\text{H}_2\text{O} / \text{H}_2\text{SO}_4$ followed by CH_3OH
 (d) CH_3MgBr / ether followed by H_3O^+

27. Which one of the following substituents at *para*-position is most effective in stabilizing the phenoxide



- (a) $-\text{CH}_3$ (b) $-\text{OCH}_3$
 (c) $-\text{COCH}_3$ (d) $-\text{CH}_2\text{OH}$

28. Williamson synthesis of ether is an example of:

- (a) Nucleophilic addition
 (b) Electrophilic addition
 (c) Electrophilic substitution
 (d) Nucleophilic substitution

29. Widespread deaths due to liquor poisoning occurs due to

- (a) presence of carbonic acid in liquor
 (b) presence of ethyl alcohol in liquor
 (c) presence of methyl alcohol in liquor
 (d) presence of lead compounds in liquor

30. Which of the following diols would cleave into two fragments with HIO_4

- (a) 1,3-hexanediol (b) 2,4-hexanediol
 (c) 1,6-hexanediol (d) 3,4-hexanediol

RESPONSE
GRID

25. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

26. (a)(b)(c)(d)

27. (a)(b)(c)(d)

28. (a)(b)(c)(d)

29. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 25 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	36	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC26

SYLLABUS : Aldehydes, Ketones and Carboxylic acids

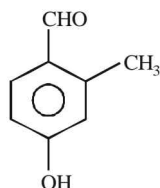
Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. IUPAC name of following will be



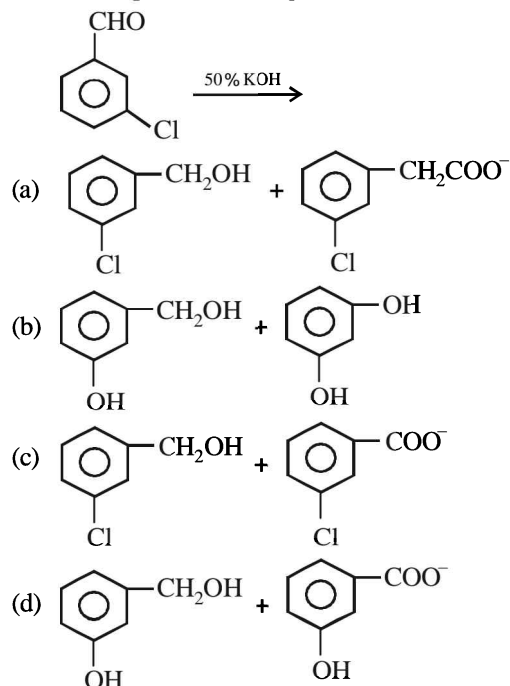
- (a) 4-formyl 3-methyl 1-hydroxy benzene
(b) 4-formyl 3-methyl phenol
(c) 4-hydroxy 2-methyl benzaldehyde
(d) 4-hydroxy 2-methyl carbaldehyde
2. In which of the following, the number of carbon atoms does not remain same when carboxylic acid is obtained by oxidation
- (a) CH_3COCH_3 (b) $\text{CCl}_3\text{CH}_2\text{CHO}$
(c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (d) $\text{CH}_3\text{CH}_2\text{CHO}$
3. Acetone oxime is obtained by reacting acetone with
(a) NH_3 (b) NH_2OH
(c) NH_2Na (d) $\text{NH}_2\cdot\text{NH}_2$
4. Which alkene on ozonolysis gives $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{C}(=\text{O})\text{CH}_3$
- (a) $\text{CH}_3\text{CH}_2\text{CH}=\text{C}(\text{CH}_3)_2$ (b) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_3$
(c) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$ (d) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CHCH}_3$

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

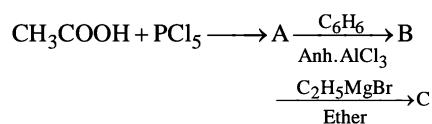
Space for Rough Work

5. Predict the product in the given reaction.



6. The carboxyl functional group ($-\text{COOH}$) is present in
 (a) picric acid (b) barbituric acid
 (c) ascorbic acid (d) aspirin
7. Which one of the following is reduced with zinc and hydrochloric acid to give the corresponding hydrocarbon?
 (a) Acetamide (b) Acetic acid
 (c) Ethyl acetate (d) Butan-2-one
8. Carboxylic acid group does not give the usual addition and elimination reactions of aldehydes and ketones because
 (a) O-H bond is more polar than $\text{C}=\text{O}$ group
 (b) carboxylate ion gets ionised
 (c) carboxylate ion gets stabilised by resonance
 (d) it exists as $-\text{COOH}$ and there is no carbonyl group
9. Phenylmethyl ketone can be converted into ethylbenzene in one step by which of the following reagents?
 (a) LiAlH_4 (b) Zn-Hg/HCl
 (c) NaBH_4 (d) CH_3MgI

10. In a set of the given reactions, acetic acid yielded a product C.



Product C would be -

- (a) $\text{CH}_3-\text{C}(\text{OH})(\text{C}_2\text{H}_5)\text{C}_6\text{H}_5$ (b) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$
 (c) $\text{CH}_3\text{COC}_6\text{H}_5$ (d) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$
11. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C', 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is:
 (a) CH_3COOH (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (c) $\text{CH}_3-\text{CH}(\text{CH}_3)\text{COOH}$ (d) $\text{CH}_3\text{CH}_2\text{COOH}$

12. Match the compounds given in List-I with List-II and select the suitable option using the code given below :

List-I

- (A) Benzaldehyde
 (B) Phthalic anhydride
 (C) Phenyl benzoate
 (D) Methyl salicylate

List-II

- (i) Phenolphthalein
 (ii) Benzoin condensation
 (iii) Oil of wintergreen
 (iv) Fries rearrangement

Code :

- | | | | |
|----------|-------|-------|-------|
| (A) | (B) | (C) | (D) |
| (a) (iv) | (i) | (iii) | (ii) |
| (b) (iv) | (ii) | (iii) | (i) |
| (c) (ii) | (iii) | (iv) | (i) |
| (d) (ii) | (i) | (iv) | (iii) |

13. $\text{C}_6\text{H}_5\text{CH}=\text{CHCHO} \xrightarrow{\text{X}} \text{C}_6\text{H}_5\text{CH}=\text{CHCH}_2\text{OH}$
 In the above sequence X can be :

- (a) H_2/Ni (b) NaBH_4
 (c) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ (d) Both (a) and (b)

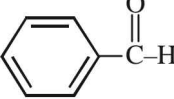
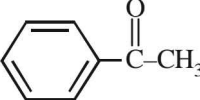
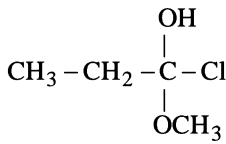
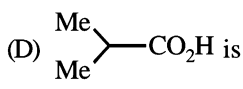
14. Acetaldehyde reacts with semicarbazide and forms semicarbazone. Its structure is

- (a) $\text{CH}_3\text{CH}=\text{NNHCON}=\text{CHCH}_3$
 (b) $\text{CH}_3\text{CH}=\text{NNHCONH}_2$
 (c) $\text{CH}_3\text{CH}=\text{N}-\text{N}(\text{OH})-\text{CONH}_2$
 (d) $\text{CH}_3\text{CH}=\text{N}-\text{CONHNH}_2$

**RESPONSE
GRID**

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

Space for Rough Work

15. Which one of the following can be oxidised to the corresponding carbonyl compound?
 (a) 2-hydroxy-propane
 (b) Ortho-nitro-phenol
 (c) Phenol
 (d) 2-methyl-2 hydroxy-propane
16. Benzoic acid may be converted to ethyl benzoate by reaction with :
 (a) Sodium ethoxide
 (b) Ethyl chloride
 (c) Dry $\text{HCl}-\text{C}_2\text{H}_5\text{OH}$
 (d) Ethanol
17. Heating mixture of sodium benzoate and soda-lime gives
 (a) benzene (b) methane
 (c) sodium phenoxide (d) calcium benzoate
18. Which of the following compounds is most reactive towards nucleophilic addition reactions?
 (a) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
 (b) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$
 (c) 
 (d) 
19. The increasing order of the rate of HCN addition to compound A – D is
 (A) HCHO (B) CH_3COCH_3
 (C) PhCOCH_3 (D) PhCOPh
 (a) $\text{D} < \text{C} < \text{B} < \text{A}$ (b) $\text{C} < \text{D} < \text{B} < \text{A}$
 (c) $\text{A} < \text{B} < \text{C} < \text{D}$ (d) $\text{D} < \text{B} < \text{C} < \text{A}$
20. Sodium formate on heating yields
 (a) Oxalic acid and H_2
 (b) Sodium oxalate and H_2
 (c) CO_2 and NaOH
 (d) Sodium oxalate
21. 5 - methyl -2 -hexanone can be synthesised from acetoacetic ester and RX. Which of the following RX is used ?
 (a) $(\text{CH}_3)_2\text{CHBr}$
 (b) $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$
 (c) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3$
 (d) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{Br}$
22. The following tetrahedral intermediate breaks down to

 (a) propanal and HCl
 (b) methyl propanoate and HCl
 (c) propanoic acid and CH_3Cl
 (d) propyl chloride and CH_3OH
23. The correct order of increasing acid strength of the compounds
 (A) $\text{CH}_3\text{CO}_2\text{H}$ (B) $\text{MeOCH}_2\text{CO}_2\text{H}$
 (C) $\text{CF}_3\text{CO}_2\text{H}$ (D)  is
 (a) $\text{D} < \text{A} < \text{B} < \text{C}$ (b) $\text{A} < \text{D} < \text{B} < \text{C}$
 (c) $\text{B} < \text{D} < \text{A} < \text{C}$ (d) $\text{D} < \text{A} < \text{C} < \text{B}$
24. The compound that neither forms semicarbazone nor oxime is
 (a) HCHO
 (b) $\text{CH}_3\text{COCH}_2\text{Cl}$
 (c) CH_3CHO
 (d) $\text{CH}_3\text{CONHCH}_3$

RESPONSE
GRID

15. (a)(b)(c)(d) 16. (a)(b)(c)(d) 17. (a)(b)(c)(d) 18. (a)(b)(c)(d) 19. (a)(b)(c)(d)
 20. (a)(b)(c)(d) 21. (a)(b)(c)(d) 22. (a)(b)(c)(d) 23. (a)(b)(c)(d) 24. (a)(b)(c)(d)

Space for Rough Work

C-104

DPP/ CC26

25. Which of the following cannot reduce Fehling solution?
 (a) Formic acid (b) Acetic acid
 (c) Formaldehyde (d) Acetaldehyde
26. Pinacolone is
 (a) 2, 3-Dimethyl-2, 3-butanediol
 (b) 3,3-Dimethyl-2-butanone
 (c) 1-Phenyl-2-propanone
 (d) 1, 1-Diphenyl-1, 2-ethandiol
27. Conversion of acetaldehyde into ethyl acetate in presence of aluminium ethoxide is called
 (a) Aldol condensation (b) Cope reaction
 (c) Tischenko reaction (d) Benzoin condensation
28. Self condensation of two moles of ethyl acetate in presence of sodium ethoxide yields
 (a) acetoacetic ester
 (b) methyl acetoacetate
 (c) ethyl propionate
 (d) ethyl butyrate
29. The reagent which can be used to distinguish acetophenone from benzophenone is
 (a) 2,4- dinitrophenylhydrazine
 (b) aqueous solution of NaHSO_3
 (c) benedict reagent
 (d) I_2 and Na_2CO_3
30. The compound formed when malonic ester is heated with urea is
 (a) Cinnamic acid (b) Butyric acid
 (c) Barbituric acid (d) Crotonic acid.

**RESPONSE
GRID**

 25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 26 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	35	Qualifying Score	51
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC27

SYLLABUS : Amines

Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

- An organic amino compound reacts with aqueous nitrous acid at low temperature to produce an oily nitrosoamine. The compound is
(a) CH_3NH_2 (b) $\text{CH}_3\text{CH}_2\text{NH}_2$
(c) $\text{CH}_3\text{CH}_2\text{NHCH}_2\text{CH}_3$ (d) $(\text{CH}_3\text{CH}_2)_3\text{N}$
- Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 in the mixture, nitric acid acts as a/an:
(a) acid (b) base
(c) catalyst (d) reducing agent
- Which of the following statements are correct ?
(i) In Sandmeyer reaction nucleophiles like Cl^- , Br^- and CN^- are introduced in benzene ring in the presence of Cu^+ ion
(ii) In Gattermann reaction nucleophiles are introduced in benzene ring in the presence of copper powder and HCl .
(iii) The yield in Gattermann reaction is found to be better than Sandmeyer reaction.
(a) (i) and (ii) (b) (i), (ii) and (iii)
(c) (ii) and (iii) (d) (i) and (iii)
- Consider the following sequence of reactions :
$$\text{Compound [A]} \xrightarrow{\text{Reduction}} [\text{B}] \xrightarrow{\text{HNO}_2} \text{CH}_3\text{CH}_2\text{OH}$$

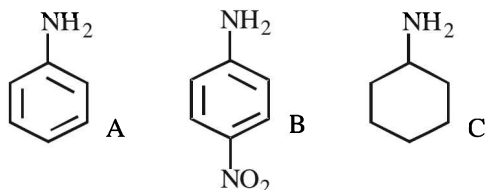
The compound [A] is
(a) $\text{CH}_3\text{CH}_2\text{CN}$ (b) CH_3NO_2
(c) CH_3NC (d) CH_3CN

RESPONSE GRID

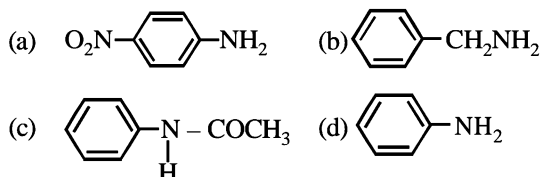
1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

5. The correct order of basicity of the following compounds



- (a) $B > A > C$ (b) $A > B > C$
 (c) $C > A > B$ (d) $C > B > A$
6. The constituent of the powerful explosive RDX is formed during the nitration of
 (a) toluene (b) phenol
 (c) glycerol (d) urotropine
7. $C_6H_5C \equiv N$ and $C_6H_5N \equiv C$ exhibit which type of isomerism?
 (a) Position (b) Functional
 (c) Dextroisomerism (d) Position isomerism
8. Which of the following reagents will convert p-methylbenzenediazonium chloride into p-cresol?
 (a) Cu powder (b) H_2O
 (c) H_3PO_2 (d) C_6H_5OH
9. Replacement of $-N_2^+Cl^-$ from benzene diazonium chloride by iodine can be done by using
 (a) HI (b) NaOI
 (c) PI_3 (d) KI
10. Which of the following compounds is most basic?



11. $CH_3CH_2Cl \xrightarrow{NaCN} X \xrightarrow{Ni/H_2} Y \xrightarrow{\text{acetic anhydride}} Z$
 Z in the above reacting sequence is

- (a) $CH_3CH_2CH_2NHCOCH_3$
 (b) $CH_3CH_2CH_2NH_2$
 (c) $CH_3CH_2CH_2CONHCH_3$
 (d) $CH_3CH_2CH_2CONHCOCH_3$
12. Match the compounds given in List - I with their characteristic reactions given in List - II. Select the correct option.

List - I (Compounds) List - II (Reactions)

- (1) $CH_3CH_2CH_2CH_2NH_2$ (i) alkaline hydrolysis
 (2) $CH_3C \equiv CH$ (ii) with KOH (alcohol) and $CHCl_3$ produces bad smell
 (3) $CH_3CH_2COOCH_3$ (iii) gives white ppt. with ammonical $AgNO_3$
 (4) $CH_3CH(OH)CH_3$ (iv) with Lucas reagent cloudiness appears after 5 minutes

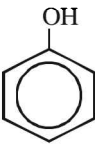
Options :

- (1) (2) (3) (4)
 (a) (iv) (ii) (iii) (i)
 (b) (ii) (i) (iv) (iii)
 (c) (iii) (ii) (i) (iv)
 (d) (ii) (iii) (i) (iv)
13. Nitrosoamines ($R_2N - N = O$) are insoluble in water. On heating with conc. H_2SO_4 , they give secondary amines. The reaction is called
 (a) Liebermann nitroso reaction
 (b) Etard reaction
 (c) Fries reaction
 (d) Perkin reaction
14. All three amines 1° , 2° , 3° react with
 1. H_2O 2. $R-X$
 3. HCl 4. $(CH_3CO)_2O$
 (a) 1, 2 (b) 4 only
 (c) 1, 2, 4 (d) 1, 2, 3

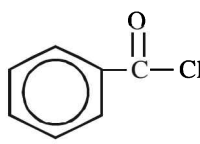
**RESPONSE
GRID**

5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d) 8. (a)(b)(c)(d) 9. (a)(b)(c)(d)
 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d) 13. (a)(b)(c)(d) 14. (a)(b)(c)(d)

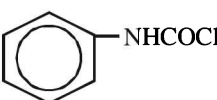
Space for Rough Work

15. Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?
 (a) Methyl isocyanide
 (b) Acetamide
 (c) Methyl cyanide
 (d) Nitroethane
16. Aniline reacts with phosgene and KOH to form
- 

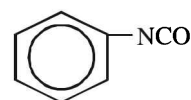
(a)

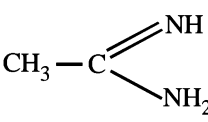


(b)



(c)

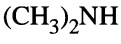


(d)
17. When aniline reacts with oil of bitter almonds (C_6H_5CHO) condensation takes place and benzal derivative is formed. This is known as
 (a) Million's base (b) Schiff's reagent
 (c) Schiff's base (d) Benedict's reagent
18. Which reaction sequence would be best to prepare 3-chloroaniline from benzene ?
 (a) Chlorination, nitration, reduction
 (b) Nitration, chlorination, reduction
 (c) Nitration, reduction, chlorination
 (d) Nitration, reduction, acetylation, chlorination, hydrolysis
19. A nitro alkane reacts with HONO to give insoluble product in alkali which turns blue on treatment with an alkali. The nitro alkane is
 (a) $CH_3CH_2NO_2$ (b) $CH_3-\underset{\substack{| \\ CH_3}}{CH}-CH_2NO_2$
 (c) $(CH_3)_2CHNO_2$ (d) $(CH_3)_3CNO_2$
20. The reduction of nitro compounds is most preferred in the presence of
 (a) Pd/H_2 in ethanol (b) $Sn + HCl$
 (c) finely divided Ni (d) iron scrap and HCl
21. Ethyl isocyanide on hydrolysis in acidic medium generates
 (a) propanoic acid and ammonium salt
 (b) ethanoic acid and ammonium salt
 (c) methylamine salt and ethanoic acid
 (d) ethylamine salt and methanoic acid
22. Acetamide is treated with the following reagents separately. Which one of these would yield methylamine?
 (a) $NaOH - Br_2$ (b) Sodalime
 (c) Hot conc. H_2SO_4 (d) PCl_5
23. Match the compounds in List I with their nature from List II, as seen in aqueous medium
- | List I | List II |
|----------------------------|----------------------------|
| I. Acetamide | A. Acidic |
| II. Benzonitrile | B. Basic |
| III. Triethylamine | C. Neutral |
| IV. Phenol | |
| (a) I-C; II-C; III-B; IV-A | (b) I-B; II-C; III-C; IV-A |
| (c) I-C; II-B; III-B; IV-C | (d) I-A; II-A; III-C; IV-B |
24. The correct order of basicities of the following compounds is
- 

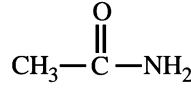
1



2



3



4
- (a) $2 > 1 > 3 > 4$ (b) $1 > 3 > 2 > 4$
 (c) $3 > 1 > 2 > 4$ (d) $1 > 2 > 3 > 4$

RESPONSE
GRID

15. (a) (b) (c) (d)
20. (a) (b) (c) (d)

16. (a) (b) (c) (d)
21. (a) (b) (c) (d)

17. (a) (b) (c) (d)
22. (a) (b) (c) (d)

18. (a) (b) (c) (d)
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)
24. (a) (b) (c) (d)

C-108

DPP/ CC27

25. Which of the following reactions will not give a primary amine?

- (a) $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2 / \text{KOH}}$
 (b) $\text{CH}_3\text{CN} \xrightarrow{\text{LiAlH}_4}$
 (c) $\text{CH}_3\text{NC} \xrightarrow{\text{LiAlH}_4}$
 (d) $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{LiAlH}_4}$

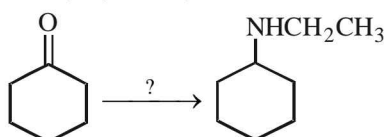
26. Indicate which nitrogen compound amongst the following would undergo Hoffmann reaction (i.e., reaction with Br_2 and strong KOH) to furnish the primary amine ($\text{R}-\text{NH}_2$)

- (a) RCONHCH_3 (b) RCOONH_4
 (c) RCONH_2 (d) $\text{R}-\text{CO}-\text{NHOH}$

27. In the reaction $\text{C}_6\text{H}_5\text{NH}_2 + \text{CS}_2 \xrightarrow[\Delta]{\text{HgCl}_2}$ the product obtained is

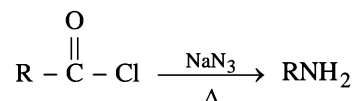
- (a) phenyl isocyanide (b) phenyl cyanide
 (c) p-amino benzene sulphonic acid (d) phenyl isothiocyanate

28. Reagents capable of converting cyclohexanone to N-ethyl cyclohexylamine is



- (a) $\text{CH}_3\text{CH}_2\text{Br}$ and NH_3
 (b) $\text{CH}_3\text{CH}_2\text{NH}_2$ and H_2 / Pt
 (c) $\text{CH}_3\text{CH}=\text{O}$ and NH_3
 (d) LiAlH_4 followed by H_2O and then $\text{CH}_3\text{CH}_2\text{Br}$

29. The intermediate obtained in the following reaction



- (a) $\text{R}-\text{C}(=\text{O})-\text{N}^+\equiv\text{N}^-$ (b) $\text{R}-\text{N}=\text{C}=\text{O}$
 (c) Both (a) and (b) (d) None of these

30. Which of the following gives propylamine upon hydrolysis?

- (a) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{N}$
 (b) $(\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{NH}$

- (c) $\text{CH}_3-\text{C}(=\text{O})-\text{NH}\cdot\text{CH}_2\text{CH}_2\text{CH}_3$
 (d) $\text{CH}_3\text{CH}_2-\text{CH}=\text{NH}$

RESPONSE
GRID

25. (a)(b)(c)(d) 26. (a)(b)(c)(d) 27. (a)(b)(c)(d) 28. (a)(b)(c)(d) 29. (a)(b)(c)(d)
 30. (a)(b)(c)(d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 27 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	36	Qualifying Score	52
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC28

SYLLABUS : Biomolecules

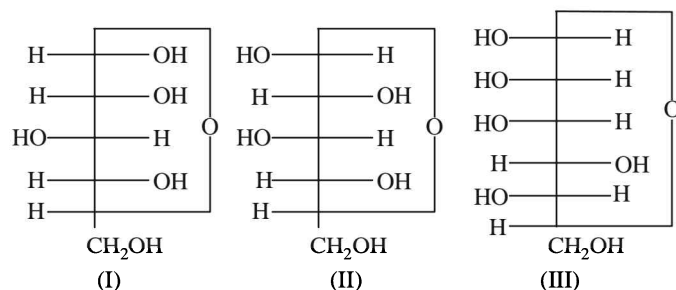
Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. Three cyclic structures of monosaccharides are given below which of these are anomers



- (a) I and II
(b) II and III
(c) I and III
(d) III is anomer of I and II

2. Which of the following statements about vitamin B₁₂ is incorrect ?

- (a) It has a cobalt atom
(b) It also occurs in plants
(c) It is also present in rain water
(d) It is needed for human body in very small amounts

3. Proteins when heated with conc. HNO₃ give a yellow colour. This is

- (a) Oxidizing test
(b) Xanthoproteic test
(c) Hoppe's test
(d) Acid base test

4. The change in the optical rotation (with time) of freshly prepared solution of sugar is known as

- (a) Specific rotation
(b) Inversion
(c) Rotatory motion
(d) Mutarotation

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

C-110

DPP/ CC28

5. Table sugar is a
 - (a) disaccharide of d-glucose and d-fructose
 - (b) a monosaccharide
 - (c) a disaccharide containing two glucose units
 - (d) D-glucose
6. The pair of compounds in which both the compounds give positive test with Tollen's reagent is
 - (a) Glucose and Sucrose
 - (b) Fructose and Sucrose
 - (c) Acetophenone and Hexanal
 - (d) Glucose and Fructose
7. Hydrolysis of sucrose is called
 - (a) hydration
 - (b) saponification
 - (c) esterification
 - (d) inversion
8. Which is not a true statement?
 - (a) α - carbon of α -amino acid is asymmetric
 - (b) all proteins are found in L-form
 - (c) human body can synthesize all proteins they need
 - (d) at pH=7 both amino acids and carboxylic groups exist in the ionised form
9. The best solvent for removing butter stain from cloth is
 - (a) CHCl_3
 - (b) $\text{C}_2\text{H}_5\text{OH}$
 - (c) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
 - (d) H_2O
10. The number of amino acids found in proteins that a human body can synthesize is
 - (a) 20
 - (b) 10
 - (c) 5
 - (d) 14
11. Chemically amylose is a _____ with 200–1000 α -D-(+)-glucose units held by _____ glycosidic linkage
 - (a) long unbranched chain, C1–C6.
 - (b) branched chain, C1 – C4.
 - (c) long unbranched chain, C1– C4.
 - (d) branched chain, C1–C6
12. Which is wrongly matched?
 - (a) Insulin - steroid hormone
 - (b) Estrone - control the uterine cycle in women
 - (c) Oxytocin - contraction of uterus
 - (d) Pot. metabisulphite - food preservative
13. If one strand of DNA has the sequence ATGCTTGA, the sequence in the complimentary strand would be
 - (a) TACGAACT
 - (b) TCCGAACT
 - (c) TACGTACT
 - (d) TACGTAGT
14. In nucleic acids, the sequence is
 - (a) phosphate - base - sugar
 - (b) sugar- base-phosphate
 - (c) base- sugar - phosphate
 - (d) base- phosphate - sugar

**RESPONSE
GRID**

5. (a)(b)(c)(d)

6. (a)(b)(c)(d)

7. (a)(b)(c)(d)

8. (a)(b)(c)(d)

9. (a)(b)(c)(d)

10. (a)(b)(c)(d)

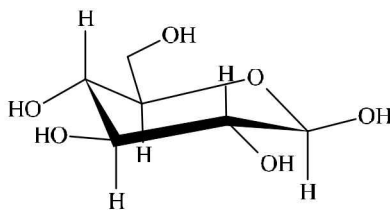
11. (a)(b)(c)(d)

12. (a)(b)(c)(d)

13. (a)(b)(c)(d)

14. (a)(b)(c)(d)

 Space for Rough Work

15. Which statement is incorrect about peptide bond?
- C–N bond length in proteins is longer than usual C–N bond length
 - Spectroscopic analysis shows planar structure of $\begin{array}{c} \text{—C—NH—} \\ \parallel \\ \text{O} \end{array}$ bond
 - C–N bond length in proteins is smaller than usual C–N bond length
 - None of these
16. An acidic amino acid among the following is
- glycine
 - valine
 - proline
 - leucine
17. In an amino acid, the carboxyl group ionises at $\text{pK}_{a1} = 2.34$ and ammonium ion at $\text{pK}_{a2} = 9.60$. The isoelectric point of the amino acid is at pH
- 5.97
 - 2.34
 - 9.60
 - 6.97
18. For osazone formation, the effective structural unit necessary is
- $\begin{array}{c} \text{CH}_2\text{OCH}_3 \\ | \\ \text{CO} \\ | \end{array}$
 - $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CO} \\ | \end{array}$
 - $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CHOCH}_3 \\ | \end{array}$
 - $\begin{array}{c} \text{CHO} \\ | \\ \text{CHOCH}_3 \\ | \end{array}$
19. The following carbohydrate is
- 
20. A reagent suitable for the determination of N-terminal residue of a peptide is
- p-Toluenesulphonyl chloride
 - 2, 4-Dinitrophenylhydrazine
 - Carboxypeptidase
 - 2, 4-Dinitrofluorobenzene
21. Amino acids generally exist in the form of Zwitter ions. This means they contain
- basic— NH_2 group and acidic — COOH group
 - the basic— NH_3^+ group and acidic — COO^- group
 - basic— NH_2 and acidic — H^+ group
 - basic— COO^- group and acidic — NH_3^+ group
22. The two functional groups present in a typical carbohydrate are:
- CHO and —COOH
 - $>\text{C}=\text{O}$ and —OH
 - OH and —CHO
 - OH and —COOH
23. Enzymes are essential as biocatalysts. They function in
- aqueous medium, temp = $30\text{--}35^\circ\text{C}$; pH=7
 - organic medium
 - aqueous medium under extreme pH conditions
 - None of these
24. α - Helix is found in
- DNA
 - RNA
 - lipid
 - Carbohydrates

RESPONSE
GRID

15. (a) (b) (c) (d)

20. (a) (b) (c) (d)

16. (a) (b) (c) (d)

21. (a) (b) (c) (d)

17. (a) (b) (c) (d)

22. (a) (b) (c) (d)

18. (a) (b) (c) (d)

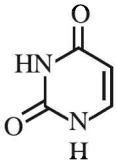
23. (a) (b) (c) (d)

19. (a) (b) (c) (d)

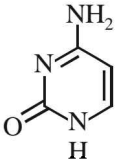
24. (a) (b) (c) (d)

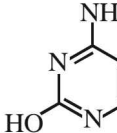
C-112

DPP/ CC28

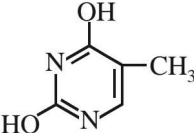
25. Lack of essential amino acids in the diet leads to the disease called
 (a) Night blindness (b) Pernicious anaemia
 (c) Kwashiorkor (d) Sickle cell anaemia
26. Which of the following structures represents thymine ?
- 

(a)



(b)
- 

(c)



(d)
27. The segment of DNA which acts as the instrumental manual for the synthesis of the protein is:
 (a) ribose (b) gene
 (c) nucleoside (d) nucleotide
28. Identify the correct statement regarding enzymes
 (a) enzymes are specific biological catalysts that cannot be poisoned
 (b) enzymes are normally heterogeneous catalysts that are very specific in their action
 (c) enzymes are specific biological catalysts that can normally function at very high temperatures ($t \sim 1000\text{K}$)
 (d) enzymes are specific biological catalysts that possess well-defined active sites
29. In both DNA and RNA, heterocyclic base and phosphate ester linkages are at –
 (a) C_5' and C_1' respectively of the sugar molecule
 (b) C_1' and C_5' respectively of the sugar molecule
 (c) C_2 and C_5' respectively of the sugar molecule
 (d) C_5 and C_2 respectively of the sugar molecule
30. Glucose molecule reacts with X number of molecules of phenylhydrazine to yield osazone. The value of X is
 (a) three (b) two
 (c) one (d) four

**RESPONSE
GRID**

 25. (a) (b) (c) (d)
 30. (a) (b) (c) (d)

26. (a) (b) (c) (d)

27. (a) (b) (c) (d)

28. (a) (b) (c) (d)

29. (a) (b) (c) (d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 28 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	56
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct $\times$ 4) – (Incorrect $\times$ 1)			

Space for Rough Work

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC29

SYLLABUS : Polymers

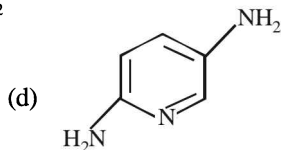
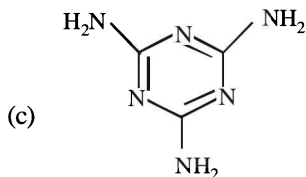
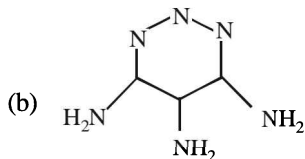
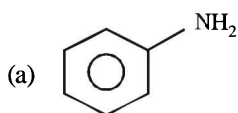
Max. Marks : 120

Marking Scheme : + 4 for correct & (–1) for incorrect

Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. Which of the following compound is used for preparation of melamine formaldehyde polymer ?



2. Which one of the following is not an example of chain growth polymer?

(a) Neoprene (b) Buna-S
(c) PMMA (d) Glyptal

3. Urethane is

(a) $\text{H}_2\text{N}-\text{C}\equiv\text{N}$ (b) $\text{H}_2\text{N}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{OH}$
(c) $\text{HO}-\text{C}\equiv\text{N}$ (d) $\text{H}_2\text{N}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{OC}_2\text{H}_5$

4. The turbidity of a polymer solution measures

(a) the light scattered by solution
(b) the light absorbed by a solution
(c) the light transmitted by a solution
(d) None of these

RESPONSE GRID

1. (a)(b)(c)(d) 2. (a)(b)(c)(d) 3. (a)(b)(c)(d) 4. (a)(b)(c)(d)

Space for Rough Work

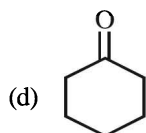
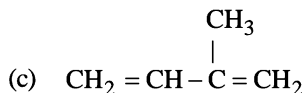
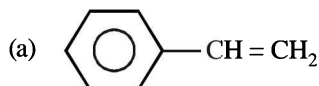
C-114

DPP/ CC29

5. Which of the following polymers do not involve cross linkages?

- (a) Melmac (b) Bakelite
(c) Polythene (d) Vulcanised rubber

6. Which compound/set of compounds is used in the manufacture of nylon 6?



7. Ebonite is

- (a) natural rubber (b) synthetic rubber
(c) highly vulcanized rubber (d) polypropene

8. Monomer of $\left[\begin{array}{c} \text{CH}_3 \\ | \\ -\text{C}-\text{CH}_2- \\ | \\ \text{CH}_3 \end{array} \right]_n$ is

- (a) 2-Methylpropene (b) Styrene
(c) Propylene (d) Ethene

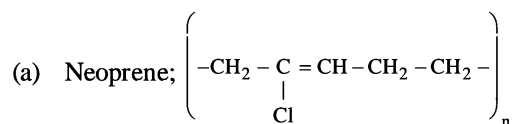
9. Caprolactam polymerises to give

- (a) terylene (b) teflon
(c) glyptal (d) Nylon-6

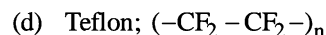
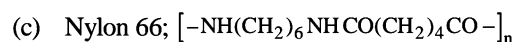
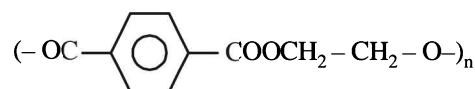
10. Polymer used in bullet proof glass is

- (a) PMMA (b) Lexan
(c) Nomex (d) Kevlar

11. Structures of some common polymers are given. Which one is **not** correctly presented?

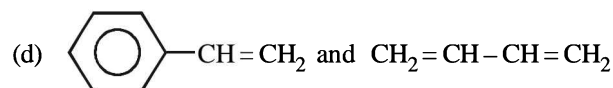


(b) Terylene;



12. Which one of the following sets forms the biodegradable polymer?

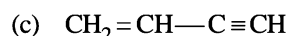
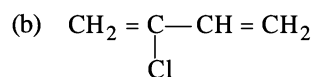
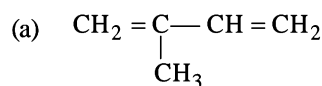
- (a) $\text{CH}_2 = \text{CH} - \text{CN}$ and $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$
(b) $\text{H}_2\text{N} - \text{CH}_2 - \text{COOH}$ and $\text{H}_2\text{N} - (\text{CH}_2)_5 - \text{COOH}$
(c) $\text{HO} - \text{CH}_2 - \text{CH}_2 - \text{OH}$ and



13. Which of the following catalyst is used in preparation of high density polythene ?

- (a) Peroxide catalyst
(b) Ziegler - Natta catalyst
(c) Wilkinson's catalyst
(d) Pd - catalyst

14. Which is the monomer of Neoprene in the following ?



RESPONSE
GRID

5. (a) (b) (c) (d)

6. (a) (b) (c) (d)

7. (a) (b) (c) (d)

8. (a) (b) (c) (d)

9. (a) (b) (c) (d)

10. (a) (b) (c) (d)

11. (a) (b) (c) (d)

12. (a) (b) (c) (d)

13. (a) (b) (c) (d)

14. (a) (b) (c) (d)

Space for Rough Work

15. Buna-N synthetic rubber is a copolymer of :

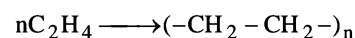
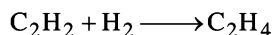
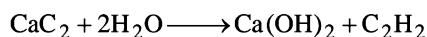
- (a) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$ and $\text{H}_5\text{C}_6-\text{CH}=\text{CH}_2$
 (b) $\text{H}_2\text{C}=\text{CH}-\text{CN}$ and $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
 (c) $\text{H}_2\text{C}=\text{CH}-\text{CN}$ and $\text{H}_2\text{C}=\text{CH}-\underset{\text{CH}_3}{\text{C}}=\text{CH}_2$

- (d) $\text{H}_2\text{C}=\text{CH}-\underset{\text{Cl}}{\text{C}}=\text{CH}_2$ and $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$

16. The polymer containing strong intermolecular forces e.g. hydrogen bonding, is

- (a) teflon (b) nylon 6, 6
 (c) polystyrene (d) natural rubber

17. Formation of polyethylene from calcium carbide takes place as follows



The amount of polyethylene obtained from 64.1 kg of CaC_2 is

- (a) 7 kg (b) 14 kg
 (c) 21 kg (d) 28 kg

18. Ziegler-Natta catalyst is

- (a) $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)]$ (b) $(\text{Ph}_3\text{P})_3\text{RhCl}$
 (c) $\text{Al}_2(\text{C}_2\text{H}_5)_6 + \text{TiCl}_4$ (d) $\text{Fe}(\text{C}_5\text{H}_5)_2$

19. Nylon 6,6 is a polyamide obtained by the reaction of

- (a) $\text{COOH}(\text{CH}_2)_4\text{COOH} + \text{NH}_2\text{C}_6\text{H}_4\text{NH}_2$
 (b) $\text{COOH}(\text{CH}_2)_4\text{COOH} + \text{NH}_2(\text{CH}_2)_6\text{NH}_2$
 (c) $\text{COOH}(\text{CH}_2)_6\text{COOH} + \text{NH}_2(\text{CH}_2)_4\text{NH}_2$
 (d) $\text{COOHC}_6\text{H}_4\text{COOH}-(p) + \text{NH}_2(\text{CH}_2)_6\text{NH}_2$

20. Which one of the following monomers gives the polymer neoprene on polymerization?

- (a) $\text{CCl}_2=\text{CCl}_2$ (b) $\text{CH}_2=\underset{\text{Cl}}{\text{C}}-\text{CH}=\text{CH}_2$
 (c) $\text{CF}_2=\text{CF}_2$ (d) $\text{CH}_2=\text{CHCl}$

21. Which compound/set of compounds is used in the manufacture of nylon-6,6?

- (a) $\text{HOOC}(\text{CH}_2)_4\text{COOH} + \text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$
 (b) $\text{CH}_2=\text{CH}-\text{C}(\text{CH}_3)=\text{CH}_2$
 (c) $\text{CH}_2=\text{CH}_2$

- (d) $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH} + \text{HOCH}_2-\text{CH}_2\text{OH}$

22. Which of the following is not correctly matched?

- (a) Neoprene; $\left[-\text{CH}_2-\underset{\text{Cl}}{\text{C}}=\text{CH}-\text{CH}_2- \right]_n$

- (b) Nylon-66: $\left[-\text{NH}-(\text{CH}_2)_6-\text{NH}-\text{CO}-(\text{CH}_2)_4-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}- \right]_n$

- (c) Terylene; $\left[-\text{OCH}_2-\text{CH}_2-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_4-\overset{\text{O}}{\parallel}{\text{C}}- \right]_n$

- (d) PMMA: $\left[-\text{CH}_2-\underset{\text{COOCH}_3}{\overset{\text{CH}_3}{\text{C}}}- \right]_n$

23. Plexiglas (PMMA) is a polymer of

- (a) acrylic acid
 (b) methyl acrylate
 (c) methyl methacrylate
 (d) None of these

24. Orlon is a polymer of

- (a) styrene
 (b) tetrafluoroethylene
 (c) vinyl chloride
 (d) acrylonitrile

RESPONSE
GRID

15. (a) (b) (c) (d)

16. (a) (b) (c) (d)

17. (a) (b) (c) (d)

18. (a) (b) (c) (d)

19. (a) (b) (c) (d)

20. (a) (b) (c) (d)

21. (a) (b) (c) (d)

22. (a) (b) (c) (d)

23. (a) (b) (c) (d)

24. (a) (b) (c) (d)

Space for Rough Work

- 25.** Synthetic human hair wigs are made from a copolymer of vinyl chloride and acrylonitrile and is called
 (a) PVC (b) Polyacrylonitrile
 (c) Cellulose (d) Dynel
- 26.** Melamine plastic crockery is a condensation polymer of
 (a) HCHO and melamine (b) HCHO and ethylene
 (c) melamine and ethylene (d) None of these
- 27.** Match List-I (Monomer) with List II (Polymer) and select the correct answer using the codes given below the lists:
- | List I | List II |
|-------------------------|----------------|
| I. Hexamethylenediamine | A. Bakelite |
| II. Phenol | B. Dacron |
| III. Phthalic acid | C. Glyptal |
| IV. Terephthalic acid | D. Melamine |
| | E. Nylon |
- Codes:**
 (a) I-E, II-A, III-B, IV-C (b) I-E, II-A, III-C, IV-B
 (c) I-D, II-C, III-A, IV-B (d) I-D, II-C, III-A, IV-B
- 28.** Among cellulose, poly(vinyl chloride), nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is
 (a) nylon (b) poly(vinyl chloride)
 (c) cellulose (d) natural Rubber
- 29.** Which of the following structures represents neoprene polymer?
- (a) $\left\{ \text{CH}_2 - \underset{\begin{array}{c} | \\ \text{Cl} \end{array}}{\text{C}} = \text{CH} - \text{CH}_2 \right\}_n$

(b) $\left\{ \text{CH}_2 - \underset{\begin{array}{c} | \\ \text{CN} \end{array}}{\text{CH}} \right\}_n$

(c) $\left\{ \text{CH}_2 - \underset{\begin{array}{c} | \\ \text{Cl} \end{array}}{\text{CH}} \right\}_n$

(d) $\left\{ \text{CH} - \text{CH}_2 \right\}_n$
 $\quad \quad \quad |$
 $\quad \quad \quad \text{C}_6\text{H}_5$
- 30.** Which one of the following is used to make ‘non-stick’ cookware?
 (a) PVC
 (b) Polystyrene
 (c) Poly(ethylene terephthalate)
 (d) Polytetrafluoroethylene

RESPONSE GRID	25. (a) (b) (c) (d)	26. (a) (b) (c) (d)	27. (a) (b) (c) (d)	28. (a) (b) (c) (d)	29. (a) (b) (c) (d)
	30. (a) (b) (c) (d)				

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 29 -CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	38	Qualifying Score	57
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work -

DPP - Daily Practice Problems

Chapter-wise Sheets

Date :

Start Time :

End Time :

CHEMISTRY

CC30

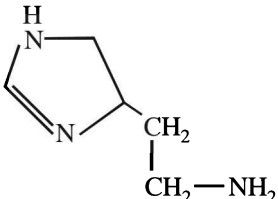
SYLLABUS : Chemistry in Everyday Life

Max. Marks : 120

Marking Scheme : + 4 for correct & (−1) for incorrect

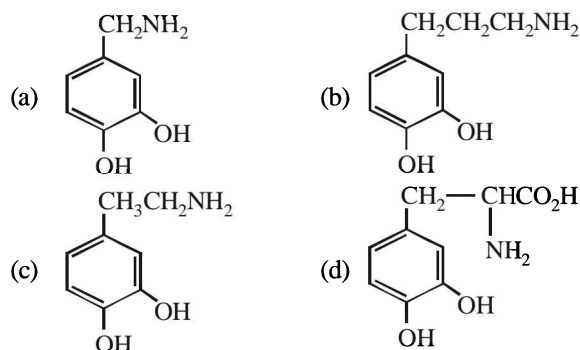
Time : 60 min.

INSTRUCTIONS : This Daily Practice Problem Sheet contains 30 MCQ's. For each question only one option is correct. Darken the correct circle/ bubble in the Response Grid provided on each page.

1. The drug  is used as

- (a) Antacid (b) Analgesic
(c) Antimicrobial (d) Antiseptic

2. Parkinson's disease is linked to abnormalities in the levels of dopamine in the body. The structure of dopamine is



RESPONSE GRID

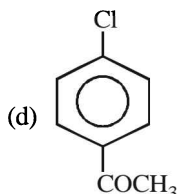
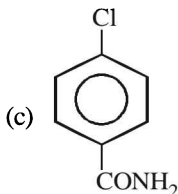
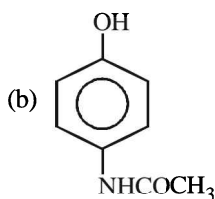
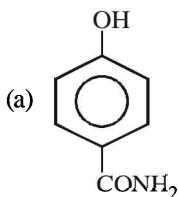
1. (a)(b)(c)(d) 2. (a)(b)(c)(d)

Space for Rough Work

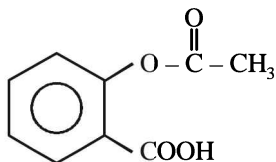
C-118

DPP/ CC30

3. The correct structure of the drug paracetamol is



4. Arsenic drugs are mainly used in the treatment of
 (a) Jaundice (b) Typhoid
 (c) Syphilis (d) Cholera
5. Antiseptic chloroxylenol is
 (a) 4-chloro-3,5-dimethylphenol
 (b) 3-chloro-4,5-dimethylphenol
 (c) 4-chloro-2,5-dimethylphenol
 (d) 5-chloro-3,4-dimethylphenol
6. The following compound is used as



- (a) an anti-inflammatory compound
 (b) analgesic
 (c) hypnotic
 (d) antiseptic
7. Which of the following ceramics possesses good strength, translucency and very low porosity
 (a) whiteware (b) earthenware
 (c) stoneware (d) bricks and tiles

8. Which one of the following is employed as a tranquilizer drug?

(a) Promethazine (b) Valium
 (c) Naproxen (d) Mifepristone

9. Which of the following substances cannot be used as a fixative in perfumes?

(a) Benzoin
 (b) Sandalwood
 (c) Glycerol dinitrate
 (d) Glycerol diacetate

10. Detergents are prepared by the action of H_2SO_4 followed by neutralization by starting with

(a) chloesterol (b) lauryl alcohol
 (c) cyclohexanol (d) p-nitrophenol

11. Which of the following is not correctly matched?

(i) Proteins that are crucial to body's communication process. – Receptors
 (ii) Drugs that mimic the natural messenger by switching on the receptor. – Antagonists
 (iii) Drugs that binds to the receptor site and inhibit its natural function. – Agonists

(a) (ii) only (b) (iii) only
 (c) (i) and (iii) (d) (ii) and (iii)

12. Commonly used antiseptic 'Dettol' is a mixture of

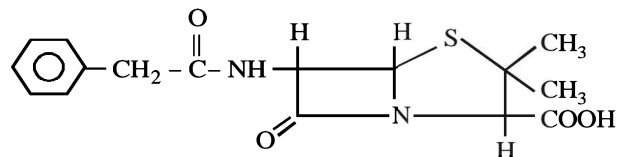
(a) o-chlorophenoxylenol + terpeneol
 (b) o-cresol + terpeneol
 (c) phenol + terpeneol
 (d) chloroxylenol + terpeneol

RESPONSE
GRID

3. (a)(b)(c)(d) 4. (a)(b)(c)(d) 5. (a)(b)(c)(d) 6. (a)(b)(c)(d) 7. (a)(b)(c)(d)
 8. (a)(b)(c)(d) 9. (a)(b)(c)(d) 10. (a)(b)(c)(d) 11. (a)(b)(c)(d) 12. (a)(b)(c)(d)

Space for Rough Work

13. Which one of the following can possibly be used as analgesic without causing addiction and mood modification ?
 (a) Diazepam (b) Morphine
 (c) N-Acetyl-para-aminophenol
 (d) Tetrahydrocannabinol
14. The indicator used in the titration of a strong acid and a strong base is
 (a) phenolphthalein (b) methyl orange
 (c) alizarin yellow (d) red litmus
15. Benzalkonium chloride is a
 (a) cationic surfactant and antiseptic
 (b) anionic surfactant and soluble in most of organic solvents
 (c) cationic surfactant and insoluble in most of organic solvents
 (d) cationic surfactant and antimalarial
16. During the deepening of the colour of a dye, the absorption shifts towards red. This shifting is called
 (a) bathochromic shift (b) hypsochromic shift
 (c) hyperchromic shift (d) auxochromic shift
17. Amoxillin is semi-synthetic modification of
 (a) penicillin (b) streptomycin
 (c) tetracycline (d) chloramphenicol
18. Select the incorrect statement.
 (a) Equanil is used to control depression and hypertension.
 (b) Mifepristone is a synthetic steroid used as "morning after pill".
 (c) 0.2 percent solution of phenol is an antiseptic while its 1.0 percent solution is a disinfectant.
 (d) A drug which kills the organism in the body is called bacteriostatic.
19. Antibiotics that are effective mainly against Gram-positive or Gram-negative bacteria X. Antibiotics that are effective against a single organism or disease are Y
 What is X and Y ?
 (a) X = Broad spectrum antibiotics.
 Y = Narrow spectrum antibiotics.
 (b) X = Broad spectrum antibiotics.
 Y = Limited spectrum antibiotics.
 (c) X = Narrow spectrum antibiotics.
 Y = Limited spectrum antibiotics.
 (d) X = Narrow spectrum antibiotics.
 Y = Broad spectrum antibiotics.
20. Structurally biodegradable detergent should contain
 (a) normal alkyl chain (b) branched alkyl chain
 (c) phenyl side chain (d) cyclohexyl side chain
21. H_1 – Receptor antagonists is a term associated with :
 (a) Antiseptics (b) Antihistamins
 (c) Antacids (d) Analgesics
22. Which of the following is not true ?
 (a) Some disinfectants can be used as antiseptics
 (b) Sulphadiazine is a synthetic antibacterial
 (c) Aspirin is analgesic as well as antipyretic
 (d) Polystyrene is used to make non-stick cookware
23. Which of the followings forms the base of talcum powder?
 (a) Zine stearate
 (b) Sodium aluminium silicate
 (c) Magnesium hydrosilicate
 (d) Chalk
24. The structure given below is known as



- (a) Penicillin F (b) Penicillin G
 (c) Penicillin K (d) Ampicillin

RESPONSE
GRID

13. (a) (b) (c) (d) 14. (a) (b) (c) (d) 15. (a) (b) (c) (d) 16. (a) (b) (c) (d) 17. (a) (b) (c) (d)
 18. (a) (b) (c) (d) 19. (a) (b) (c) (d) 20. (a) (b) (c) (d) 21. (a) (b) (c) (d) 22. (a) (b) (c) (d)
 23. (a) (b) (c) (d) 24. (a) (b) (c) (d)

Space for Rough Work

C-120

DPP/ CC30

25. Most of the deodorants contain aluminium salts because they
 (a) act as antiperspirants
 (b) act as anti-bacterial agents
 (c) mask body odour
 (d) All the these
26. Chemical substances used by insects for communication are called
 (a) pheromones (b) hormones
 (c) enzymes (d) nucleoproteins
27. Antiseptics and disinfectants either kill or prevent growth of microorganisms. Identify which of the following statements **is not true**:
 (a) Chlorine and iodine are used as strong disinfectants.
 (b) Dilute solutions of Boric acid and Hydrogen Peroxide are strong antiseptics.
 (c) Disinfectants harm the living tissues.
 (d) A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant.
28. The substance which is not an artificial sweetener
 (a) sucralose (b) alitame
 (c) saccharin (d) sucrose
29. An antibiotic with a broad spectrum
 (a) kills the antibodies
 (b) acts on a specific antigen
 (c) acts on different antigens
 (d) acts on both the antigens and antibodies
30. SLV -3 uses _____ propellants
 (a) solid (b) liquid
 (c) solid-liquid (d) biliquid

**RESPONSE
GRID**

 25. (a) (b) (c) (d) 26. (a) (b) (c) (d) 27. (a) (b) (c) (d) 28. (a) (b) (c) (d) 29. (a) (b) (c) (d)
 30. (a) (b) (c) (d)

DAILY PRACTICE PROBLEM DPP CHAPTERWISE 30 - CHEMISTRY

Total Questions	30	Total Marks	120
Attempted		Correct	
Incorrect		Net Score	
Cut-off Score	37	Qualifying Score	55
Success Gap = Net Score – Qualifying Score			
Net Score = (Correct × 4) – (Incorrect × 1)			

Space for Rough Work

1. (b)

2. (a) No. of moles = $\frac{\text{Wt. in g}}{\text{Mol. wt}}$

$$\text{No. of moles in 200 mg} = \frac{200}{1000 \times 44}$$

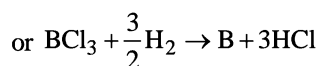
$$= 4.5 \times 10^{-3} \text{ moles}$$

No. of moles in 10^{21} molecules

$$= \frac{10^{21}}{6.02 \times 10^{23}} = 1.67 \times 10^{-3} \text{ moles}$$

$$\text{No. of moles left} = (4.5 - 1.67) \times 10^{-3} = 2.88 \times 10^{-3}$$

3. (a) $2\text{BCl}_3 + 3\text{H}_2 \rightarrow 2\text{B} + 6\text{HCl}$



Now, since 10.8 gm boron requires hydrogen

$$= \frac{3}{2} \times 22.4 \text{ L at N.T.P}$$

hence 21.6 gm boron requires hydrogen

$$\frac{3}{2} \times \frac{22.4}{10.8} \times 21.6 = 67.2 \text{ L at N.T.P.}$$

4. (c) M. Wt of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ is 322 g which contains 224 g oxygen. $\therefore$ 32.2 g will contain 22.4 g oxygen.

5. (b) Moles of urea present in 100 ml of sol. = $\frac{6.02 \times 10^{-20}}{6.02 \times 10^{-23}}$

$$\therefore M = \frac{6.02 \times 10^{-20} \times 1000}{6.02 \times 10^{-23} \times 100} = 0.01 \text{ M}$$

[$\therefore$ M = Moles of solute present in 1L of solution]

6. (d) Relative atomic mass =

$$\frac{\text{Mass of one atom of the element}}{1/12^{\text{th}} \text{ part of the mass of one atom of Carbon - 12}}$$

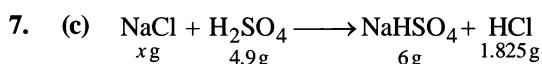
$$\text{or } \frac{\text{Mass of one atom of the element}}{\text{mass of one atom of the C - 12}} \times 12$$

Now if we use $1/6$ in place of $1/12$ the formula becomes

Relative atomic mass =

$$\frac{\text{Mass of one atom of element}}{\text{Mass of one atom of carbon}} \times 6$$

$\therefore$ Relative atomic mass decrease twice



According to law of conservation of mass "mass is neither created nor destroyed during a chemical change"

$\therefore$ Mass of the reactants = Mass of products

$$x + 4.9 = 6 + 1.825$$

$$\text{or } x = 2.925 \text{ g}$$

8. (c) (a) Weight of H_2 = mole $\times$ molecular wt.
 $= 0.2 \times 2 = 0.4 \text{ g}$

(b) $6.023 \times 10^{23} = 1 \text{ mole}$

Thus $6.023 \times 10^{22} = 0.1 \text{ mole}$

Weight of $\text{N}_2 = 0.1 \times 28 = 2.8 \text{ g}$

(c) Weight of silver = 0.1 g

(d) Weight of oxygen = $32 \times 0.1 = 3.2 \text{ g}$

9. (d) 1 Mole of $\text{Mg}_3(\text{PO}_4)_2$ contains 8 mole of oxygen atoms
 $\therefore$ 8 mole of oxygen atoms $\equiv$ 1 mole of $\text{Mg}_3(\text{PO}_4)_2$ mole of $\text{Mg}_3(\text{PO}_4)_2$

$$0.25 \text{ mole of oxygen atom} \equiv \frac{1}{8} \times 0.25 \text{ mole of } \text{Mg}_3(\text{PO}_4)_2$$

$$= 3.125 \times 10^{-2} \text{ mole of } \text{Mg}_3(\text{PO}_4)_2$$

10. (d) Since molarity of solution is 3.60 M. It means 3.6 moles of H_2SO_4 is present in its 1 litre solution.

Mass of 3.6 moles of H_2SO_4

= Moles $\times$ Molecular mass

$$= 3.6 \times 98 \text{ g} = 352.8 \text{ g}$$

$\therefore$ 1000 ml solution has 352.8 g of H_2SO_4

Given that 29 g of H_2SO_4 is present in = 100 g of solution

$\therefore$ 352.8 g of H_2SO_4 is present in

$$= \frac{100}{29} \times 352.8 \text{ g of solution}$$

$$= 1216 \text{ g of solution}$$

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{1216}{1000} = 1.216 \text{ g/ml} = 1.22 \text{ g/ml}$$

$$11. \quad (b) \quad \text{From } \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{V_1 \times 640}{(273 + 47)} = \frac{620 \times 300}{(273 + 27)}$$

$$V_1 = \frac{620 \times 300 \times 320}{640 \times 300} = 310 \text{ cc}$$

$$12. \quad (d) \quad \text{Weight of Iron in 67200} = \frac{0.33}{100} \times 67200 = 221.76$$

$$\text{Number of atoms of Iron} = \frac{221.76}{56} = 3.96 \approx 4$$

13. (a) 20 volume H_2O_2 means that 1 mL of this H_2O_2 solutions produces 20 mL of O_2 at N.T.P. on decomposition by heat.

$\therefore$ For 20 mL of O_2 , the volume of 20 volume H_2O_2 required = 1 mL

For 1 mL of O_2 , the volume of 20 volume

$$\text{H}_2\text{O}_2 \text{ required} = \frac{1}{20} \text{ mL}$$

For 5000 mL or 5L of O_2 , the volume of 20

$$\text{volume H}_2\text{O}_2 \text{ required} = \frac{1}{20} \times 5000 \text{ mL} = 250 \text{ mL}$$

14. (a) $2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{Al}^{3+}(\text{aq}) + 6\text{Cl}^{-}(\text{aq}) + 3\text{H}_2(\text{g})$
 $\therefore$ 6 moles of HCl produces = 3 moles of H_2
 $= 3 \times 22.4 \text{ L of H}_2 \text{ at S.T.P}$

$$\therefore 1 \text{ mole of HCl produces} = \frac{3 \times 22.4}{6} \text{ L}$$

$$\text{of H}_2 \text{ at S.T.P}$$

$$= 11.2 \text{ L of H}_2 \text{ at STP}$$

15. (a) 95% H₂SO₄ by weight means 100g H₂SO₄ solution contains 95g H₂SO₄ by mass.

Molar mass of H₂SO₄ = 98g mol⁻¹

$$\text{Moles in 95g} = \frac{95}{98} = 0.969 \text{ mole}$$

Volume of 100g H₂SO₄

$$= \frac{\text{mass}}{\text{density}} = \frac{100\text{g}}{1.834\text{g cm}^{-3}}$$

$$= 54.52 \text{ cm}^3 = 54.52 \times 10^{-3} \text{ L}$$

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solute in L}}$$

$$= \frac{0.969}{54.52 \times 10^{-3}} = 17.8 \text{ M}$$

16. (c) 50 ml of 16.9% solution of AgNO₃

$$\left(\frac{16.9}{100} \times 50 \right) = 8.45 \text{ g of Ag NO}_3$$

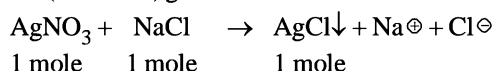
$$n_{\text{mole}} = \frac{8.45\text{g}}{(107.8 + 14 + 16 \times 3) \text{ g/mol}} = \left(\frac{8.45 \text{ g}}{169.8 \text{ g/mol}} \right)$$

$$= 0.0497 \text{ moles}$$

50 ml of 5.8% solution of NaCl contain

$$\text{NaCl} = \left(\frac{5.8}{100} \times 50 \right) = 2.9 \text{ g}$$

$$n_{\text{NaCl}} = \frac{2.9\text{g}}{(23 + 35.5) \text{ g/mol}} = 0.0495 \text{ moles}$$



$\therefore$ 0.049 mole 0.049 mole 0.049 mole of AgCl

$$n = \frac{w}{M} \rightarrow w = (n_{\text{AgCl}}) \times \text{Molecular Mass}$$

$$= (0.049) \times (107.8 + 35.5)$$

$$= 7.02 \text{ g}$$

17. (b) Number of valence electrons in a N₃⁻ ion = 1

Now, 1 mol or 42 g of N₃⁻ has = 6.023 × 10²³ ions

So, 42 g of N₃⁻ has 6.023 × 4 × 10²³ valence e⁻

$$1 \text{ g of N}_3^- \text{ has } \frac{6.023 \times 1 \times 10^{23}}{42} \text{ valence e}^-$$

$$4.2 \text{ g of N}_3^- \text{ has } \frac{4.2 \times 6.023 \times 1 \times 10^{23}}{42} \text{ valence e}^- \text{ i.e.,}$$

0.1 N_A valence e⁻.

18. (c) 74.75% of chlorine means 74.75g chlorine is present in 100g of metal chloride.

$$\text{Weight of metal} = 100\text{g} - 74.75\text{g}$$

$$= 25.25\text{g}$$

Equivalent weight

$$= \frac{\text{weight of metal}}{\text{weight of chlorine}} \times 35.5$$

$$= \frac{25.25}{74.75} \times 35.5 = 12$$

$$\text{Valency of metal} = \frac{2 \times \text{V.D.}}{\text{Equivalent wt. of metal} + 35.5}$$

$$= \frac{2 \times 94.8}{12 + 35.5} = 4$$

$\therefore$ Formula of compound = MCl₄

19. (d) $\therefore$ 18 gm, H₂O contains = 2 gm H

$\therefore$ 0.72 gm H₂O contains

$$= \frac{2}{18} \times 0.72 \text{ gm} = 0.08 \text{ gm H}$$

$\therefore$ 44 gm CO₂ contains = 12 gm C

$\therefore$ 3.08 gm CO₂ contains

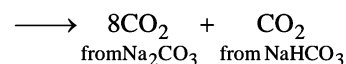
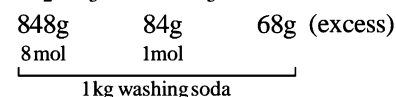
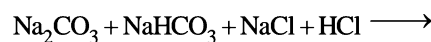
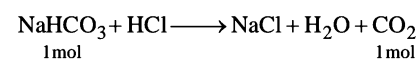
$$= \frac{12}{44} \times 3.08 = 0.84 \text{ gm C}$$

$$\therefore \text{C : H} = \frac{0.84}{12} : \frac{0.08}{1}$$

$$= 0.07 : 0.08 = 7 : 8$$

$\therefore$ Empirical formula = C₇H₈

20. (a) $\text{Na}_2\text{CO}_3 + 2\text{HCl} \longrightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$



Thus, on complete reaction with HCl, 1kg of washing soda will evolve 9 mol of CO₂.

21. (a) 2.6 has two significant figures.

0.260 has three significant figures.

0.002600 has four significant figures.

2.6000 has five significant figures.

22. (b) Given

mass of solute (w) = 120 g

mass of solvent (w) = 1000 g

Mol. mass of solute = 60 g

density of solution = 1.12 g/ml

From the given data,

Mass of solution = 1000 + 120 = 1120 g

$$\therefore d = \frac{\text{Mol. mass}}{V} \text{ or } V = \frac{\text{Mol. mass}}{d}$$

$$\text{Volume of solution } V = \frac{1120}{1.12} = 1000 \text{ ml or } = 1 \text{ litre}$$

$$\text{Now molarity (M)} = \frac{W}{\text{Mol. mass} \times V(\text{lit})} = \frac{120}{60 \times 1} = 2\text{M}$$

23. (d) In an unknown compounds containing N and H given % of H = 12.5%

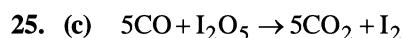
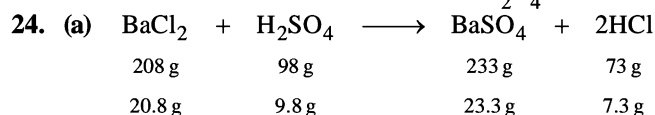
$$\therefore \% \text{ of N} = 100 - 12.5 = 87.5\%$$

$$2 \times \text{vapour density} = \text{Mol. wt} = \text{mol wt.} = 16 \times 2 = 32.$$

Molecular formula = $n \times$ empirical formula mass

$$n = \frac{32}{16} = 2$$

$\therefore$ Molecular formula of the compound will be $= (\text{NH}_2)_2$
 $= \text{N}_2\text{H}_4$



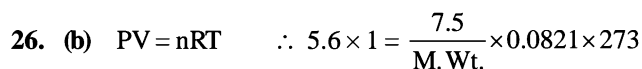
$$\text{Moles of I}_2\text{O}_5 = \frac{25.4}{254}$$

$$= 0.01 \equiv 0.05 \text{ moles of CO}$$

Weight of CO = $0.05 \times 28 = 1.4$ g;

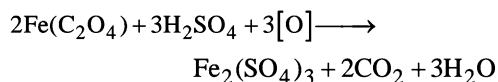
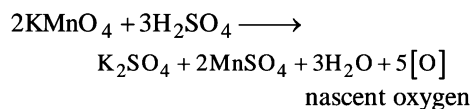
Weight of $\text{CO}_2 = 2 - 1.4 = 0.6 \text{ g}$

$$\text{Hence \% of CO}_2 = \frac{0.6}{2} \times 100 = 30\%$$



M. Wt = 30.12 Hence gas is NO.

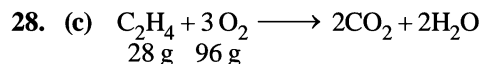
27. (b) The required equation is



[O] required for 1 mol. of $\text{Fe}(\text{C}_2\text{O}_4)$ is 1.5, 5 [O] are obtained from 2 moles of KMnO_4

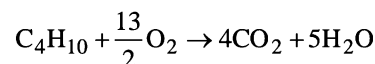
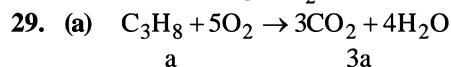
$\therefore$ 1.5 [O] will be obtained from

$$= \frac{2}{5} \times 1.5 = 0.6 \text{ moles of KMnO}_4.$$



$\therefore$ 28 g of C_2H_4 undergo complete combustion by
= 96 g of O_2

$\therefore 2.8 \text{ kg of } \text{C}_2\text{H}_4 \text{ undergo complete combustion by}$
 $= 9.6 \text{ kg of } \text{O}_2.$



$$(3-a) \qquad 4(3-a)$$

But, $3a + 4(3 - a) = 10$

$\therefore a = 2$ (Propane) and $3 - 2 = 1$ (Butane)

Element	%	Relative no. of atoms	Simplest ratio of atoms
C	49.3	$49.3/12 = 4.1$ $1.5 \times 2 = 3$	$4.1/2.74 = 1.5$
H	6.84	$6.84/1 = 6.84$ $= 2.5 \times 2$ $= 5$	$6.84/2.74 = 2.5$
O	43.86	$43.86/16 = 2.74$ $1 \times 2 = 2$	$2.74/2.74 = 1$

$$\therefore \text{Empirical formula} = \text{C}_3\text{H}_5\text{O}_2$$

Empirical formula mass

$$= (3 \times 12) + (5 \times 1) + (2 \times 16) = 36 + 5 + 32 = 73$$

Molecular mass = $2 \times$ Vapour density

$$= 2 \times 73 = 146$$

$$n = \frac{\text{molecular mass}}{\text{empirical formula mass}} = 146/73 = 2$$

$$\text{Molecular formula} = \text{Empirical formula} \times 2$$

$$= (\overset{1}{\text{C}_3\text{H}_5\text{O}_2}) \times 2 = \text{C}_6\text{H}_{10}\text{O}_4$$

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC02

1. (c) The species CO, NO⁺, CN⁻ and C₂²⁻ contain 14 electrons each.

2. (d) NaCl: No. of e⁻ in Na⁺ = At. No. of Na - 1
= 11 - 1 = 10
No. of e⁻ in Cl⁻ = At. No. of Cl + 1
= 17 + 1 = 18

CsF : No. of e⁻ in Cs⁺ = 55 - 1 = 54
No. of e⁻ in F⁻ = 9 + 1 = 10

NaI : No. of e⁻ in Na⁺ = 11 - 1 = 10
No. of e⁻ in I⁻ = 53 + 1 = 54

K₂S : No. of e⁻ in K⁺ = 19 - 1 = 18
No. of e⁻ in S²⁻ = 16 + 2 = 18

3. (c) For electron in the ground state,

$$mvr = \frac{h}{2\pi} \Rightarrow mv = \frac{h}{2\pi r}$$

$$\text{Now, } mv = \frac{h}{\lambda}$$

$$\text{So, } \frac{h}{\lambda} = \frac{h}{2\pi r} \Rightarrow \lambda = 2\pi r$$

$$\lambda = 2 \times 3.14 \times 0.53 \text{ \AA} = 3.328 \text{ \AA}$$

$$= 3.328 \times 10^{-10} \text{ m}$$

$$= 0.3328 \times 10^{-9} \text{ m} = 0.3328 \text{ nm}$$

4. (a) For He⁺,

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$= R_H (2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R_H \left(\frac{1}{(1)^2} - \frac{1}{(2)^2} \right)$$

For H,

$$\bar{\nu} = \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For same frequency,

$$R_H \left(\frac{1}{(1)^2} - \frac{1}{(2)^2} \right) = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\therefore \frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{1^2} - \frac{1}{2^2}$$

$$\therefore n_1 = 1 \text{ \& } n_2 = 2$$

5. (d) For Balmer $n_1 = 2$ and $n_2 = 3$;

$$\nu = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36} \text{ cm}^{-1}$$

6. (c) Series limit is the last line of the series, i.e. $n_2 = \infty$.

$$\therefore \bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = R \left[\frac{1}{n_1^2} - \frac{1}{\infty^2} \right] = \frac{R}{n_1^2}$$

$$\therefore \bar{\nu} = 12186.3 = \frac{109677.76}{n_1^2}$$

$$\Rightarrow n_1^2 = \frac{109677.76}{12186.3} = 9 \Rightarrow n_1 = 3$$

$\therefore$ The line belongs to Paschen series.

7. (d) de Broglie wavelength, $\lambda = \frac{h}{mv}$

$$\frac{\lambda_1}{\lambda_2} = \frac{m_2 v_2}{m_1 v_1}; \frac{1}{4} = \frac{1}{9} \times \frac{v_2}{v_1}$$

$$\frac{v_2}{v_1} = \frac{9}{4}$$

$$\frac{v_1}{v_2} = \frac{4}{9}$$

$$KE = \frac{1}{2} mv^2$$

$$\frac{KE_1}{KE_2} = \frac{m_1}{m_2} \times \frac{v_1^2}{v_2^2} = \frac{9}{1} \times \left(\frac{4}{9} \right)^2 = \frac{16}{9}$$

8. (c) Fe (III) = [Ar] 3d⁵ unpaired electrons = 5;

$$\text{Magnetic moment} = \sqrt{5(5+2)};$$

$$\text{Ratio} = \sqrt{7} : \sqrt{3}$$

$$\text{Co(II)} = [\text{Ar}] 3d^7 \text{ unpaired electrons} = 3;$$

$$\text{Magnetic moment} = \sqrt{3(3+2)}$$

$$\text{Ratio} = \sqrt{7} : \sqrt{3}$$

9. (b) $E = h\nu = \frac{ch}{\lambda}$; and $\nu = \frac{c}{\lambda}$

$$8 \times 10^{15} = \frac{3.0 \times 10^8}{\lambda}$$

$$\therefore \lambda = \frac{3.0 \times 10^8}{8 \times 10^{15}} = 0.37 \times 10^{-7} = 37.5 \times 10^{-9} \text{ m} = 4 \times 10^1$$

10. (d) $\lambda_p = \frac{h}{\sqrt{2eVm_p}}; \lambda_{Li} = \frac{h}{\sqrt{2 \times 3eVm_{Li}}}$

$$= \frac{h}{\sqrt{2 \times 3eV \times 9m_p}}$$

$$\text{Hence, } \frac{\lambda_{Li}^{3+}}{\lambda_p} = \sqrt{\frac{2eVm_p}{2 \times 3eV \times 9m_p}} = \frac{1}{3\sqrt{3}}$$

11. (b) From the given data, we have

$$(E_C - E_B) + (E_B - E_A) = (E_C - E_A)$$

$$\text{or } \left(\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \right) = \frac{hc}{\lambda_3} \quad \left[\text{or } \frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3} \right]$$

$$\text{or } \boxed{\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}} \quad \left[\because \frac{\lambda_1 + \lambda_2}{\lambda_1 \lambda_2} = \frac{1}{\lambda_3} \right]$$

12. (a) We know $\Delta p \cdot \Delta x \geq \frac{h}{4\pi}$
since $\Delta p = \Delta x$ (given)

$$\therefore \Delta p \cdot \Delta p = \frac{h}{4\pi}$$

$$\text{or } m\Delta v \cdot m\Delta v = \frac{h}{4\pi} \quad [\because \Delta p = m\Delta v]$$

$$\text{or } (\Delta v)^2 = \frac{h}{4\pi m^2}$$

$$\text{or } \Delta v = \sqrt{\frac{h}{4\pi m^2}} = \frac{1}{2m} \sqrt{\frac{h}{\pi}}$$

13. (a) $(n + l)$ rule the higher the value of $(n + l)$, the higher is the energy. When $(n + l)$ value is the same see value of n .

	I	II	III	IV
$(n + l)$	$(4 + 1)$	$(4 + 0)$	$(3 + 2)$	$(3 + 1)$
	5	4	5	4

$$\therefore \text{IV} < \text{II} < \text{III} < \text{I}$$

14. (b) I. $E = \frac{Z^2}{n^2} \times 13.6 \text{ eV}$... (i)

$$\text{or } \frac{I_1}{I_2} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2} \quad \dots \text{(ii)}$$

$$\text{Given } I_1 = -19.6 \times 10^{-18}, Z_1 = 2,$$

$$n_1 = 1, Z_2 = 3 \text{ and } n_2 = 1$$

Substituting these values in equation (ii).

$$-\frac{19.6 \times 10^{-18}}{I_2} = \frac{4}{1} \times \frac{1}{9}$$

$$\text{or } I_2 = -19.6 \times 10^{-18} \times \frac{9}{4}$$

$$= -4.41 \times 10^{-17} \text{ J/atom}$$

15. (c) As per Bohr's postulate,

$$mvr = \frac{nh}{2\pi}$$

$$\text{So, } v = \frac{nh}{2\pi mr}$$

$$\text{KE} = \frac{1}{2}mv^2$$

$$\text{So, KE} = \frac{1}{2}m \left(\frac{nh}{2\pi mr} \right)^2$$

$$\text{Since, } r = \frac{a_0 \times n^2}{z}$$

So, for 2nd Bohr orbit

$$r = \frac{a_0 \times 2^2}{1} = 4a_0$$

$$\text{KE} = \frac{1}{2}m \left(\frac{2^2 h^2}{4\pi^2 m^2 \times (4a_0)^2} \right)$$

$$\text{KE} = \frac{h^2}{32\pi^2 m a_0^2}$$

$$16. (a) \Delta E = 2.178 \times 10^{-18} \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{hc}{\lambda}$$

$$\Rightarrow 2.178 \times 10^{-18} \times \frac{3}{4} = \frac{hc}{\lambda}$$

$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\lambda = \frac{6.62 \times 10^{-34} \times 3 \times 10^8 \times 4}{2.178 \times 10^{-18} \times 3}$$

$$= 1.214 \times 10^{-7} \text{ m}$$

17. (c) Not more than two electrons can be present in same atomic orbital. This is Pauli's exclusion principle.

18. (a) 2nd excited state will be the 3rd energy level.

$$E_n = \frac{13.6}{n^2} \text{ eV} \quad \text{or} \quad E = \frac{13.6}{9} \text{ eV} = 1.51 \text{ eV}.$$

19. (b) Calculating number of electrons

$$\left. \begin{array}{l} \text{BO}_3^{3-} \longrightarrow 5 + 8 \times 3 + 3 = 32 \\ \text{CO}_3^{2-} \longrightarrow 6 + 8 \times 3 + 2 = 32 \\ \text{NO}_3^- \longrightarrow 7 + 8 \times 3 + 1 = 32 \end{array} \right\} \text{iso-electronic species}$$

$$\left. \begin{array}{l} \text{SO}_3^{2-} \longrightarrow 16 + 8 \times 3 + 2 = 42 \\ \text{CO}_3^{2-} \longrightarrow 32 \\ \text{NO}_3^- \longrightarrow 32 \end{array} \right\} \text{not iso-electronic species}$$

$$\left. \begin{array}{l} \text{CN}^- \longrightarrow 6 + 7 + 1 = 14 \\ \text{N}_2 \longrightarrow 7 \times 2 = 14 \\ \text{C}_2^- \longrightarrow 6 \times 2 + 2 = 14 \end{array} \right\} \text{iso-electronic species}$$

$$\left. \begin{array}{l} \text{PO}_4^{3-} \longrightarrow 15 + 8 \times 4 + 3 = 50 \\ \text{SO}_4^{2-} \longrightarrow 16 + 8 + 2 = 50 \\ \text{ClO}_4^- \longrightarrow 17 + 8 \times 4 + 1 = 50 \end{array} \right\} \text{iso-electronic species}$$

Hence the species in option (b) are not isoelectronic.

20. (d) (ΔE), The energy required to excite an electron in an atom of hydrogen from $n = 1$ to $n = 2$ is ΔE (difference in energy E_2 and E_1)

Values of E_2 and E_1 are,

$$E_2 = \frac{-1.312 \times 10^6 \times (1)^2}{(2)^2} = -3.28 \times 10^5 \text{ J mol}^{-1}$$

ΔE is given by the relation,

$$E_1 = -1.312 \times 10^6 \text{ J mol}^{-1}$$

$$\therefore \Delta E = E_2 - E_1 = [-3.28 \times 10^5] - [-1.312 \times 10^6] \text{ J mol}^{-1}$$

$$= (-3.28 \times 10^5 + 1.312 \times 10^6) \text{ J mol}^{-1}$$

$$= 9.84 \times 10^5 \text{ J mol}^{-1}$$

Thus the correct answer is (d)

$$21. (a) \quad \bar{\nu} = \frac{1}{\lambda} = R_H Z = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

In Balmer series $n_1 = 2$ & $n_2 = 3, 4, 5, \dots$. Last line of the spectrum is called series limit.

Limiting line is the line of shortest wavelength and high energy when $n_2 = \infty$

$$\therefore \bar{\nu} = \frac{1}{\lambda} = \frac{R_H}{n_1^2} = \frac{3.29 \times 10^{15}}{2^2} = \frac{3.29 \times 10^{15}}{4}$$

$$= 8.22 \times 10^{14} \text{ s}^{-1}$$

22. (d) Energy required to break one mole of Cl – Cl bonds in Cl_2

$$= \frac{242 \times 10^3}{6.023 \times 10^{23}} = \frac{hc}{\lambda}$$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{242 \times 10^3}$$

$$= 0.4947 \times 10^{-6} \text{ m} = 494.7 \text{ nm}$$

$$23. (b) \quad \lambda = \frac{h}{mv}$$

$$h = 6.6 \times 10^{-34} \text{ J/s}$$

$$m = 1000 \text{ kg}$$

$$v = 36 \text{ km/hr} = \frac{36 \times 10^3}{60 \times 60} \text{ m/sec} = 10 \text{ m/sec}$$

$$\therefore \lambda = \frac{6.6 \times 10^{-34}}{10^3 \times 10} = 6.6 \times 10^{-38} \text{ m}$$

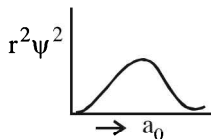
$$24. (c) \quad r_n = a_0 n^2$$

$$r = a_0 \times (3)^2 = 9a_0$$

$$mvr = \frac{nh}{2\pi}; \quad mv = \frac{nh}{2\pi r} = \frac{3h}{2\pi \times 9a_0} = \frac{h}{6\pi a_0}$$

$$\lambda = \frac{h}{mv} = \frac{n}{h} \times 6\pi a_0 = 6\pi a_0$$

25. (c) $l = 2$ represent d orbital for which



26. (b) de – Broglie wavelength is given by :

$$\lambda = \frac{h}{mv} \quad \dots (i)$$

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$v^2 = \frac{2KE}{m}$$

$$v = \sqrt{\frac{2KE}{m}}$$

Substituting this in equation (i)

$$\lambda = \frac{h}{m} \sqrt{\frac{m}{2KE}}$$

$$\lambda = h \sqrt{\frac{1}{2m(KE)}} \quad \dots (i)$$

$$\text{i.e. } \lambda \propto \frac{1}{\sqrt{KE}}$$

$\therefore$ when KE become 4 times wavelength become $1/2$.

27. (a) The electronic configuration of Rubidium ($\text{Rb} = 37$) is

$$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$$

Since last electron enters in $5s$ orbital

$$\text{Hence } n = 5, l = 0, m = 0, s = \pm \frac{1}{2}$$

28. (c) The kinetic energy of the ejected electron is given by the equation

$$h\nu = h\nu_o + \frac{1}{2}mv^2 \quad \therefore v = \frac{c}{\lambda}$$

$$\text{or } \frac{hc}{\lambda} = \frac{hc}{\lambda_o} + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = \frac{hc}{\lambda} - \frac{hc}{\lambda_o}$$

$$= hc \left(\frac{\lambda_o - \lambda}{\lambda \lambda_o} \right)$$

$$\therefore v^2 = \frac{2hc}{m} \left(\frac{\lambda_o - \lambda}{\lambda \lambda_o} \right)$$

$$\text{or } v = \sqrt{\frac{2hc}{m} \left(\frac{\lambda_o - \lambda}{\lambda \lambda_o} \right)}$$

29. (d) Total energy of a revolving electron is the sum of its kinetic and potential energy.

Total energy = K.E. + P.E.

$$= \frac{e^2}{2r} + \left(-\frac{e^2}{r} \right)$$

$$= -\frac{e^2}{2r}$$

30. (b) Energy of 1 mole of photons,

$$E = N_0 \times h\nu$$

$$= \frac{N_0 \times h \times c}{\lambda}$$

$$= \frac{6.023 \times 10^{23} \times 6.63 \times 10^{-34} \times 3 \times 10^8}{253.7 \times 10^{-9}}$$

$$= 472.2 \text{ kJ}$$

Energy converted into KE = (472.2 – 430.53) kJ

$$\% \text{ of energy converted into KE} = \frac{(472.2 - 430.53)}{472.2}$$

$$= 8.76 \%$$

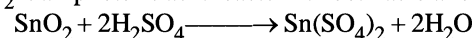
1. (d) All the given species contains $10 e^-$ each i.e. isoelectronic.
For isoelectronic species anion having high negative charge is largest in size and the cation having high positive charge is smallest.
2. (c) Covalent radius is radius of an atom in its bound state i.e., in fluorine it is half of distance between two covalently bonded fluorine atoms; van der Waal radii is one-half of the distance between the nuclei of two identical non-bonded isolated atoms. These atoms are attracted toward each other through weak van der Waal's force hence van der Waal radii are very large.
3. (c) All the given species are isoelectronic. In case of isoelectronic species ionic radii increases with increase in negative charge on anions.
4. (b) On moving along the period, ionization enthalpy increases.
In second period, the order of ionization enthalpy should be as follows :
 $F > O > N$.
But N has half-filled structure, therefore, it is more stable than O. That is why its ionization enthalpy is higher than O. Thus, the correct order of IE is
 $F > N > O$.
5. (c) In a period the value of ionisation potential increases from left to right with breaks where the atoms have some what stable configuration. In this case N has half filled stable orbitals. Hence has highest ionisation energy. Thus the correct order is
 $B < C < O < N$
and not as given in option (c)
6. (a) Atomic number (Z) = 120
IUPAC name = Unbinilium
Symbol = Ubn
7. (b) The acidic character of non metal oxides increases across a period from left to right and decreases down a group. So, acidic character will follow the order:
oxide of nitrogen > oxides of sulfur > oxides of carbon.
Among oxides of carbon acidic character increases with the oxidation number of carbon. So, $^{+4}CO_2$ is more acidic than CO. Hence the sequence of acidic character is $N_2O_5 > SO_2 > CO_2 > CO$
8. (c) The ionization energy increases with decrease in size. Further the element having stable configuration has higher ionisation energy than expected. Hence the ionization energy of nitrogen (Z = 7) is more than oxygen (Z = 8) and carbon (Z = 6) because it has half-filled p-orbitals.
 ${}^6C = 1s^2 2s^2 2p^2$; ${}^7N = 1s^2 2s^2 2p^3$; ${}^8O = 1s^2 2s^2 2p^4$
Hence the correct order should be $C < N > O$
9. (b) Along the period, I.P. generally increases but not regularly. Be and B are exceptions. First I.P. increases in moving from left to right in a period, but I.P. of B is lower than Be.
10. (d)
11. (d) The electronic configuration of the given ions are as follows.
 ${}_{12}Mg^{2+} = 1s^2, 2s^2 2p^6$ (No unpaired electron)
 ${}_{22}Ti^{3+} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^1$ (One unpaired electron)
 ${}_{23}V^{3+} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^2$ (Two unpaired electrons)
 ${}_{26}Fe^{2+} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6$ (Four unpaired electrons)
12. (b) IE_2 of Mg is lower than that of Na because in case of Mg^+ , 3s-electron has to be removed whereas in case of Na^+ , an electron is removed from the stable inert gas configuration which is difficult.
13. (b) According to modern periodic law, the properties of the elements are repeated after certain regular intervals when these elements are arranged in order of their increasing atomic numbers.
14. (a) For isoelectronic species, size of anion increases as negative charge increases whereas size of cation decreases with increase in positive charge. Further ionic radii of anions is more than that of cations. Thus the correct order is $Ca^{++} < K^+ < Ar < Cl^- < S^{--}$
15. (b) The alkali metals are highly reactive because their first ionisation potential is very low and hence they have great tendency to loses electron to form unipositive ion.
On moving down group- I from Li to Cs ionisation enthalpy decreases hence the reactivity increases. The halogens are most reactive elements due to their low bond dissociation energy, high electron affinity and high enthalpy of hydration of halide ion. However their reactivity decreases with increase in atomic number
16. (b) In hydrides of 15th group elements, basic character decreases on descending the group i.e.
 $NH_3 > PH_3 > AsH_3 > SbH_3$.
17. (b) The right sequence of IE_1 of $Li < B < Be < C$.
18. (b) As we move down in a group electron gain enthalpy becomes less negative because the size of the atom increases and the distance of added electron from the nucleus increases. Negative electron gain enthalpy of F is less than Cl. This is due to the fact that when an electron is added to F, the added electron goes to the smaller $n = 2$ energy level and experiences significant repulsion from the other electrons present in this level. In Cl, the electron goes to the larger $n = 3$ energy level and consequently occupies a larger region of space leading to much less electron-electron repulsion. So the correct order is $Cl > F > Br > I$.

19. (a) On moving along a period ionic radii decreases due to increase in effective nuclear charge.
20. (a) As the nuclear charge increases, the force of attraction between the nucleus and the incoming electron increases and hence the electron gain enthalpy becomes more negative, hence the correct order is
 $\text{Ca} < \text{Al} < \text{C} < \text{O} < \text{F}$
21. (a) First period has H and He only out of which He is inert, hence, H behaves as a highly electropositive as well as electronegative.
22. (a) O^{2-} and F^{-} are isoelectronic. Hence have same number of shells, therefore greater the nuclear charge smaller will be the size i.e.
 $\text{O}^{2-} > \text{F}^{-}$
further Li^{+} and B^{3+} are isoelectronic. therefore
 $\text{Li}^{+} > \text{B}^{3+}$
Hence the correct order of atomic size is.
 $\text{O}^{2-} > \text{F}^{-} > \text{Li}^{+} > \text{B}^{3+}$
23. (d) As the size increases the basic nature of oxides changes to acidic nature i.e., acidic nature increases.
 $\text{SO}_2 > \text{P}_2\text{O}_3 > \text{SiO}_2 > \text{Al}_2\text{O}_3$
Acidic Weak acidic Amphoteric
- SO_2 and P_2O_3 are acidic as their corresponding acids H_2SO_3 and H_3PO_3 are strong acids.
24. (a) For isoelectronic species ionic radii decreases as the charge on ion decreases. Further on moving down in a group ionic radii increases. Hence the correct order is
 $\text{O}^{2-} < \text{N}^{3-} < \text{S}^{2-} < \text{P}^{3-}$
25. (a) CaO is basic as it forms strong base $\text{Ca}(\text{OH})_2$ on reaction with water.
 $\text{CaO} + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2$
 CO_2 is acidic as it dissolves in water forming unstable carbonic acid.



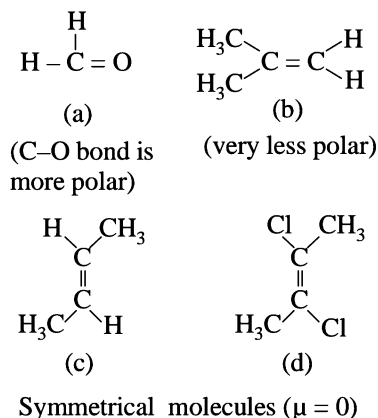
Silica (SiO_2) is insoluble in water and acts as a very weak acid.

SnO_2 is amphoteric as it reacts with both acid and base.

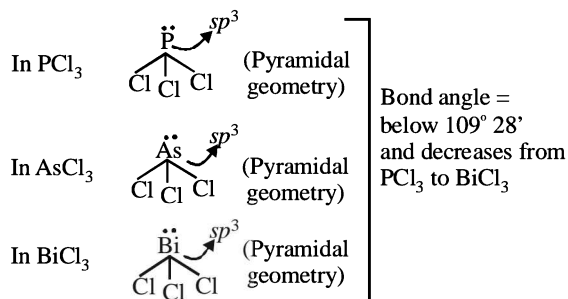


26. (a) Incoming electrons occupies the smaller $n = 2$ shell, also negative charge on oxygen (O^-) is another factor due to which incoming electron feel repulsion. Hence electron repulsion outweigh the stability gained by achieving noble gas configuration.
27. (b) In the isoelectronic species, all isoelectronic anions belong to the same period and cations to the next period.
28. (c) I represents Li, II represents K
III represents Br, IV represents I
V represents He
So, amongst these, II represents most reactive metal and V represents least reactive non-metal.
29. (d)
30. (d) Helium (He) $1s^2$ → Highest ionisation energy due to noble gas in nature.
- Fluorine (F) $1s^2, 2s^2 2p^3$ → High electronegativity in nature due to small size and -1 oxidation state.
- Rubidium (Rb) → Most electronegative element due to large atomic size.
- Lithium (Li) → Strongest reducing agent due to small size and positive oxidation state ($+1$)

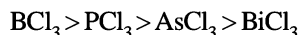
1. (c) CO_3^{2-} Its structure is $\left[\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \text{:}\ddot{\text{O}}\text{:} \end{array} \text{C} - \ddot{\text{O}}\text{:} \right]^{2-}$
2. (a) Greater the difference in electronegativity between the two atoms, larger will be polarity and hence dipole moment. Thus (a) has maximum dipole moment.



3. (c) NH_3 undergoes H-bonding and hence has the highest b.p. Among the remaining hydrides i.e. PH_3 , AsH_3 and SbH_3 as we move from PH_3 to BiH_3 , the molecular mass increases. As a result the van der waal's forces of attraction increases and the boiling point increases regularly from PH_3 to BiH_3 .
4. (b) BiCl_3 : $\text{Cl} - \text{Bi} \begin{array}{l} \nearrow \text{Cl} \\ \searrow \text{Cl} \end{array}$; sp^2 - Hybridisation
(Trigonal planar geometry); Bond angle = 120°

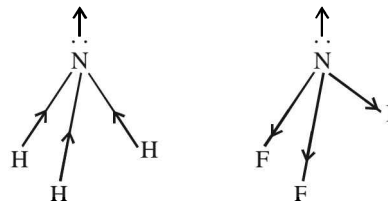


In these, order of bond angle :



5. (a) In NH_3 the atomic dipole (orbital dipole due to lone pair) and bond dipole are in the same direction whereas in NF_3 these are in opposite direction so in the former case they are added up whereas in the latter case net

result is reduction of dipole moment. It has been shown in the following figure :



6. (c) Since F form H-bond $[\text{HF}_2]^-$ exists. Therefore KHF_2 gives $\text{K}^+ + \text{HF}_2^-$
7. (b) On changing N_2 to N_2^+ , B.O. decreases from 3 to 2.5 whereas on changing O_2 to O_2^+ , B.O. increases from 2 to 2.5. In former case, the bond dissociation energy decreases and in the latter case, it increases.
8. (a) Molecular orbital configuration of

$$\text{N}_2^{2-} = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 -$$

$$\left\{ \begin{array}{l} \pi 2p_x^2 \\ \pi 2p_y^2 \end{array} \right\} \sigma 2p_z^2 \left\{ \begin{array}{l} \pi^* 2p_x^1 \\ \pi^* 2p_y^1 \end{array} \right\}$$

$$\text{Bond order} = \frac{10 - 6}{2} = 2$$

$$\text{N}_2^- = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \left\{ \begin{array}{l} \pi 2p_x^2 \\ \pi 2p_y^2 \end{array} \right\} \sigma 2p_z^2 \left\{ \begin{array}{l} \pi^* 2p_x^1 \\ \pi^* 2p_y^0 \end{array} \right\}$$

$$\text{Bond order} = \frac{10 - 5}{2} = 2.5$$

$$\text{N}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \left\{ \begin{array}{l} \pi 2p_x^2 \\ \pi 2p_y^2 \end{array} \right\} \sigma 2p_z^2$$

$$\text{Bond order} = \frac{10 - 4}{2} = 3$$

$$\therefore \text{The correct order is } = \text{N}_2^{2-} < \text{N}_2^- < \text{N}_2$$

9. (a) Hybridisation = $\frac{1}{2}$ [No. of valence electrons of central atom + No. of monovalent atoms attached to it + Negative charge if any - Positive charge if any]

$$\text{NO}_2^- \quad \text{H} = \frac{1}{2}[5 + 0 + 1 - 0] = 3 = sp^2$$

$$\text{NO}_3^- \quad \text{H} = \frac{1}{2}[5 + 0 + 1 - 0] = 3 = sp^2$$

$$\text{NH}_2^- \quad \text{H} = \frac{1}{2}[5 + 2 + 1 + 0] = 4 = sp^3$$

$$\text{NH}_4^+, H = \frac{1}{2}[5 + 4 + 0 - 1] = 4 = sp^3$$

$$\text{SCN}^- = sp$$

i.e., NO_2^- and NO_3^- have same hybridisation.

10. (b) Electronegativity difference is $4.0 - 1.20 = 2.8$ percentage ionic character is 72.24% when the electronegativity difference is 1.7, the % ionic character is approx 51%.

11. (c) In PO_4^{3-} ion, formal charge on each O-atom of P – O

$$\text{bond} = \frac{\text{Total charge}}{\text{Number of O atom}} = -\frac{3}{4} = -0.75$$

12. (c) The electronic configuration of the given molecules are:

$$\text{N}_2^+ = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi p_y^2, \sigma 2p_z^1;$$

1 unpaired e^-

$$\text{O}_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 \approx \pi 2p_y^2,$$

$$\pi^* 2p_x^1 \approx \pi^* 2p_y^1; 2 \text{ unpaired } e^-s$$

$$\text{O}_2^{2-} = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 \approx \pi 2p_y^2,$$

$$\pi^* 2p_x^2 \approx \pi^* 2p_y^2; \text{ no unpaired } e^-s$$

$$\text{B}_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^1 \approx \pi 2p_y^1;$$

2 unpaired e^-s

13. (c) ClO_3^- and SO_3^{2-} both have same number of electrons (42) and central atom in each being sp^3 hybridised. Both are having one lone pair on central atom hence they are pyramidal.

14. (a) The number of lone pairs of electrons on central atom in various given species are

Species Number of lone pairs on central atom

IF_7 0

IF_5 1

ClF_3 2

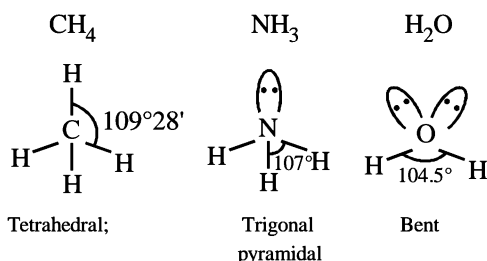
XeF_2 3

Thus the correct increasing order is

$$\text{IF}_7 < \text{IF}_5 < \text{ClF}_3 < \text{XeF}_2$$

0 1 2 3

15. (b)



Note: The geometry of H_2O should have been tetrahedral if there are all bond pairs. But due to presence of two lone pairs the shape is distorted tetrahedral. Hence bond angle reduced to 104.5° from 109.5° .

16. (a) According to VSEPR theory order of repulsion in between lp – lp, lp – bp and bp – bp is as under
lp – lp > lp – bp > bp – bp

17. (a) In NH_3 and BF_4^- the hybridisation is sp^3 and the bond angle is almost $109^\circ 28'$.

18. (d) In XeF_2 Total number of valence electrons of Xe = 8, two electrons shared with 2F atoms, 6 electrons left hence 3 lone pairs, in XeF_4 4 shared with 4 F atoms 4 left hence 2 lone pairs; in XeF_6 6 shared with 6 F atoms 2 left hence 1 lone pair.

$$19. (a) \text{Hybridisation} = \frac{1}{2} \left[\left(\begin{array}{l} \text{No. of electrons} \\ \text{in valence} \\ \text{shell of atom} \end{array} \right) + \right.$$

$$\left(\begin{array}{l} \text{No. of monovalent} \\ \text{atoms around it} \end{array} \right) - \left(\begin{array}{l} \text{charge on} \\ \text{cation} \end{array} \right) +$$

$$\left(\begin{array}{l} \text{charge on} \\ \text{anion} \end{array} \right)$$

- (a) For AlH_3 ,

$$\text{Hybridisation of Al atom} = \frac{1}{2}[3 + 3 - 0 + 0]$$

$$= 3 = sp^2$$

For AlH_4^- ,

$$\text{Hybridisation of Al atom} = \frac{1}{2}[3 + 4 - 0 + 1]$$

$$= 4 = sp^3$$

- (b) For H_2O ,

Hybridisation of O atom

$$= \frac{1}{2}[6 + 2 - 0 + 0] = 4 = sp^3$$

For H_3O^+ , Hybridisation of O atom

$$= \frac{1}{2}[6 + 3 - 1 + 0] = 4 = sp^3$$

- (c) For NH_3

Hybridisation of N atom

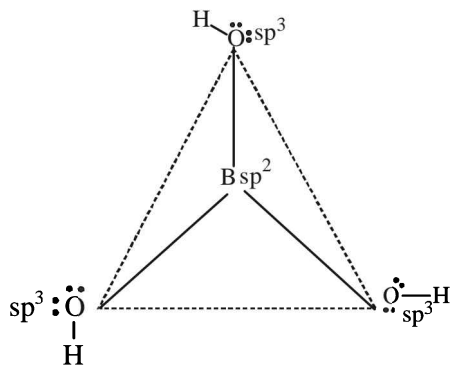
$$= \frac{1}{2}[5 + 3 - 0 + 0] = 4 = sp^3$$

For NH_4^+ , Hybridisation of N atom

$$= \frac{1}{2}[5 + 4 - 1 + 0] = 4 = sp^3$$

Thus hybridisation changes only in option (a).

20. (b)



21. (b) Hybridisation in $\text{PO}_4^{3-} = \frac{1}{2} [5 + 0 + 3 \times -0] = 4 \text{ sp}^3$. In π bonding only d orbital of P, p orbital of O can be involved. Since hybrid atomic orbitals do not form π bond.

22. (b) The bond angle decreases on moving down the group due to decrease in bond pair-bond pair repulsion.

NH_3	PH_3	AsH_3	SbH_3	BiH_3
107°	94°	92°	91°	90°

This can also be explained by the fact that as the size of central atom increases sp^3 hybrid orbital becomes more distinct with increasing size of central atom i.e. pure p-orbitals are utilized in M-H bonding

23. (a) For any species to have same bond order we can expect them to have same number of electrons. Calculating the number of electrons in various species.

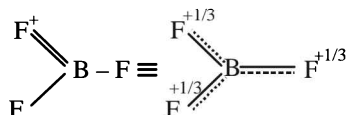
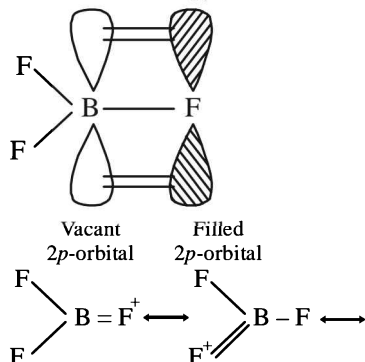
$$\text{O}_2^- (8+8+1=17); \text{CN}^- (6+7+1=14)$$

$$\text{NO}^+ (7+8-1=14); \text{CN}^+ (6+7-1=12)$$

We find CN^- and NO^+ both have 14 electrons so they have same bond order. Correct answer is (a).

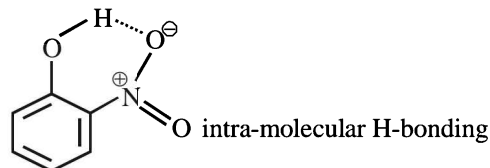
24. (d) $\sigma_b^2 \sigma_a^{*2} \sigma_b^2 \sigma_a^{*2} (\pi_b^2 = \pi_b^{*2}) \sigma_b^1$ ($\text{N}_2^+ = 13$ electrons) it contains one unpaired electron hence paramagnetic

25. (b) The delocalised $p\pi - p\pi$ bonding between filled p-orbital of F and vacant p-orbital of B leads to shortening of B-F bond length which results in higher bond dissociation energy of the B-F bond.



26. (b) Compounds involved in chelation become non-polar.

Consequently such compounds are soluble in non-polar solvents like ether, benzene etc. and are only sparingly soluble in water whereas meta and para isomers are more soluble in water & less soluble in non-polar solvents.



27. (b) According to Fajan's rules smaller, highly charged cation has greatest covalent character while large cation with smaller charge has greatest ionic character.

28. None of the given option is correct. The molecular orbital configuration of the given molecules is

$$\text{H}_2 = \sigma 1s^2 \text{ (no electron anti-bonding)}$$

$$\text{Li}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \text{ (two anti-bonding electrons)}$$

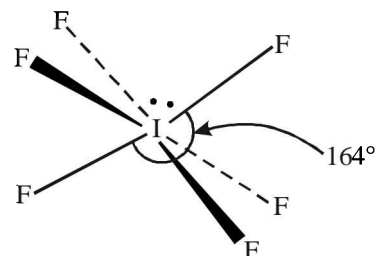
$$\text{B}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \{ \pi 2p_y^1 = \pi 2p_z^1 \}$$

(4 anti-bonding electrons)

Though the bond order of all the species are same (B.O = 1) but stability is different. This is due to difference in the presence of no. of anti-bonding electron.

Higher the no. of anti-bonding electron lower is the stability hence the correct order is $\text{H}_2 > \text{Li}_2 > \text{B}_2$

29. (a) The structure of IF_6^- is distorted octahedral. This is due to presence of a "weak" lone pair.



30. (b) The molecular orbital configuration of the molecules given is

$$\text{Total no. of electrons in NO} = 7(\text{N}) + 8(\text{O}) = 15$$

$$\text{Hence E.C. of NO} = KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma 2p_z]^2$$

$$[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^1$$

Due to presence of one unpaired electron NO is paramagnetic.

Except NO all are diamagnetic due to absence of unpaired electrons.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC05

1. (c) $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ at const. pressure

$$\Rightarrow \frac{22.4}{273} = \frac{V_2}{373}, V_2 = 30.6 \text{ litre}$$

2. (a) From the graph we can see the correct order of pressures $p_1 > p_3 > p_2$

3. (b) Gases become cooler during Joule Thomson's expansion only if they are below a certain temperature known as inversion temperature (T_i). The inversion temperature is characteristic of each gas and is given by

$$T_i = \frac{2a}{bR}, \text{ where } R \text{ is gas constant}$$

$$\text{Given } a = 0.244 \text{ atm L}^2 \text{ mol}^{-2}$$

$$b = 0.027 \text{ L mol}^{-1}$$

$$R = 0.0821 \text{ L atm deg}^{-1} \text{ mol}^{-1}$$

$$\therefore T_i = \frac{2 \times 0.244}{0.027 \times 0.0821} = 220 \text{ K}$$

4. (d) $u \propto \sqrt{T}$ or $u_1/u_2 = \sqrt{T_1/T_2}$

$$= \sqrt{\frac{27+273}{927+273}} = \sqrt{\frac{300}{1200}} = \frac{1}{2}$$

$$u_2 = 2u_1$$

5. (d) Let the mass of methane and oxygen = m gm.

Mole fraction of O_2

$$= \frac{\text{Moles of } O_2}{\text{Moles of } O_2 + \text{Moles of } CH_4}$$

$$= \frac{m/32}{m/32 + m/16} = \frac{m/32}{3m/32} = \frac{1}{3}$$

Partial pressure of O_2 = Total pressure $\times$ mole fraction of

$$O_2, P_{O_2} = P \times \frac{1}{3} = \frac{1}{3}P$$

6. (c) The expression of root mean square speed is

$$u_{rms} = \sqrt{\frac{3RT}{M}}$$

Hence,

$$\frac{u_{rms}(H_2)}{u_{rms}(O_2)} = \left[\frac{3R(50K)/(2 \text{ g mol}^{-1})}{3R(800K)/(32 \text{ g mol}^{-1})} \right]^{1/2} = 1$$

7. (a) Given $T = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$

$$V = 10.0 \text{ L}$$

$$\text{Mass of He} = 0.4 \text{ g}$$

$$\text{Mass of oxygen} = 1.6 \text{ g}$$

$$\text{Mass of nitrogen} = 1.4 \text{ g}$$

$$n_{\text{He}} = 0.4/4 = 0.1$$

$$n_{O_2} = 1.6/32 = 0.05$$

$$n_{N_2} = 1.4/28 = 0.05$$

$$n_{\text{total}} = n_{\text{He}} + n_{O_2} + n_{N_2} = 0.1 + 0.05 + 0.05 = 0.2$$

$$P = \frac{nRT}{V} = \frac{0.2 \times 0.082 \times 300}{10} = 0.492 \text{ atm}$$

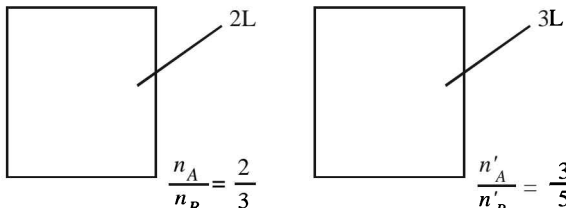
8. (b) Rate of diffusion $\propto \sqrt{\frac{1}{\text{Molecular mass}}}$

$\therefore$ Molecular mass of $HCl >$ Molecular mass of NH_3

$\therefore$ HCl diffuses at slower rate and white ammonium chloride is first formed near HCl bottle.

9. (a) $\frac{U_{CO_2}}{U_{N_2O}} = \sqrt{\frac{T_{CO_2} \times M_{N_2O}}{M_{CO_2} \times T_{N_2O}}}$

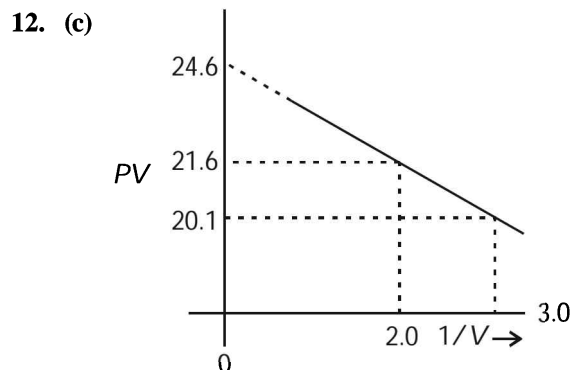
$$\Rightarrow \frac{x}{4x} = \sqrt{\frac{T_{CO_2} \times 44}{44 \times T_{N_2O}}} \Rightarrow T_{N_2O} = 16T_{CO_2}$$

10. (b) 

$$\text{Mean molar mass} = \frac{(n_A + n'_A)M_A + (n_B + n'_B)M_B}{n_A + n'_A + n_B + n'_B}$$

$$= \frac{77M_A + 123M_B}{200}$$

11. (d) $\frac{D_A}{D_B} = \sqrt{\frac{\rho_B}{\rho_A}} = \left[\frac{\rho_B}{\rho_A} \right]^{1/2}; \therefore D_A = D_B \left(\frac{\rho_B}{\rho_A} \right)^{1/2}$



$$\left(P + \frac{a}{V^2} \right) (V) = RT$$

$$PV + a/V = RT$$

$$PV = RT - \frac{a}{V}$$

$$y = RT - a(x)$$

$$\text{So, slope} = -a = \frac{21.6 - 20.1}{2 - 3} = -a = -1.5$$

$$a = 1.5$$

13. (a) $\frac{\text{K.E of neon at } 40^\circ\text{C}}{\text{K.E of neon at } 20^\circ\text{C}} = \frac{\frac{3}{2}K \times 313}{\frac{3}{2}K \times 293} = \frac{313}{293}$

14. (c) Higher the critical temperature more easily will be the gas liquify. Now since most easily liquifiable gas show larger deviation, NH_3 will show maximum deviation from ideal behaviour.

15. (c) According to Avogadro's law "At same temperature and pressure

Volume $\propto$ no. of moles"

$$n_{\text{H}_2} = \frac{w}{2}; \quad n_{\text{O}_2} = \frac{w}{32}; \quad n_{\text{CH}_4} = \frac{w}{16}$$

$$\therefore V_{\text{H}_2} : V_{\text{O}_2} : V_{\text{CH}_4} = n_{\text{H}_2} : n_{\text{O}_2} : n_{\text{CH}_4}$$

$$= \frac{w}{2} : \frac{w}{32} : \frac{w}{16} = 16 : 1 : 2$$

16. (a) Given, $n_{\text{H}_2} = n_{\text{O}_2}$ and $t_{\text{H}_2} = t_{\text{O}_2}$
According to Graham's law of diffusion for two different gases.

$$\frac{r_{\text{H}_2}}{r_{\text{O}_2}} = \frac{v_1/t_1}{v_2/t_2} \Rightarrow \sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}} = \sqrt{\frac{32}{2}}$$

$$\frac{1/2}{1/x} = \sqrt{16} = 4$$

$$\frac{x}{2} = 4$$

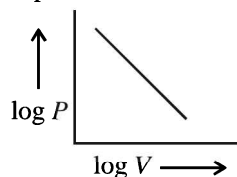
$$\therefore x = 8$$

$$\therefore \text{Fraction of } \text{O}_2 = 1/8$$

17. (c) As temperature rises the most probable speed increases and the fraction of molecules possessing most probable speed decreases.

18. (b) According to kinetic theory the gas molecules are in a state of constant rapid motion in all possible directions colliding in a random manner with one another and with the walls of the container and between two successive collisions molecules travel in a straight line path but show haphazard motion due to collisions.

19. (b) According to Boyle's law, $PV = \text{constant}$
 $\therefore \log P + \log V = \text{constant}$
 $\log P = -\log V + \text{constant}$
Hence, the plot of $\log P$ vs $\log V$ is straight line with negative slope.



20. (a) $\text{CO}_2 + \text{C} \longrightarrow 2\text{CO}$

Stoichiometry ratio is 1 : 2

AT STP, $P = 1 \text{ atm}$, $T = 273 \text{ K}$, $R = 0.0821$

$$\text{Initial moles of } \text{CO}_2; n(\text{CO}_2)_{\text{initial}} = \frac{PV}{RT}$$

$$= \frac{1 \times 0.5}{0.0821 \times 273} = 0.022 \text{ mole}$$

In final mixture no. of moles; $n(\text{CO}_2/\text{CO mixture})$

$$= \frac{1 \times 0.7}{0.0821 \times 273} = 0.031$$

Increase in volume is by $= 0.031 - 0.022$

$= 0.009$ mole of gas

Final no. of moles of CO i.e. $n_{(\text{CO final})}$

$$n_{(\text{CO final})} = 2n_{(\text{CO}_2 \text{ initial})} - n_{(\text{CO}_2 \text{ final})}$$

$$= 2(0.022 - n_{(\text{CO}_2 \text{ final})}) \quad \dots(i)$$

$$n_{(\text{CO final})} = 0.044 - 2n_{(\text{CO}_2 \text{ final})} \quad \dots(ii)$$

$$\therefore \text{Now, } n_{(\text{CO final})} + n_{(\text{CO final})} = 0.031$$

$$n_{(\text{CO}_2 \text{ final})} = 0.031 - n_{(\text{CO final})} \quad \dots(ii)$$

Substituting (ii) in eq. (i)

$$n_{(\text{CO final})} = 0.044 - 2[0.031 - n_{(\text{CO final})}]$$

$$n_{(\text{CO final})} = 0.044 - 0.062 + 2n_{(\text{CO final})}$$

$$n_{(\text{CO final})} = 0.018 \text{ mol.}$$

$$\text{Volume of CO} = V = \frac{nRT}{P} = \frac{0.018 \times 0.0821 \times 273}{1}$$

$$= 0.40 \text{ Litre}$$

and volume of $\text{CO}_2 = 0.7 \text{ litre} - 0.4 \text{ litre}$

$$= 0.3 \text{ litre}$$

$$\therefore \text{CO}_2 = 300 \text{ mL, CO} = 400 \text{ mL}$$

21. (c) Most probable speed (C^*) = $\sqrt{\frac{2RT}{M}}$

$$\text{Average Speed } (\bar{C}) = \sqrt{\frac{8RT}{\pi M}}$$

$$\text{Root mean square velocity } (C) = \sqrt{\frac{3RT}{M}}$$

$$C^* : \bar{C} : C = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}}$$

$$= 1 : \sqrt{\frac{4}{\pi}} : \sqrt{\frac{3}{2}} = 1 : 1.128 : 1.225$$

22. (b) According to Graham's Law Diffusion:

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}}$$

$$\text{Since rate of diffusion} = \frac{\text{Vol. of gas diffused } (V)}{\text{Time taken for diffusion } (t)}$$

$$\therefore \frac{r_1}{r_2} = \frac{V_1/t_1}{V_2/t_2}$$

$$\text{or } \frac{r_1}{r_2} = \frac{V_1/t_1}{V_2/t_2} = \sqrt{\frac{d_2}{d_1}}$$

$$= \frac{20/60}{V_2/30} = \sqrt{\frac{16/2}{32/2}} = \sqrt{\frac{1}{2}}$$

$$\therefore \text{Mol. wt} = 2 \times \text{V.D}$$

$$\therefore \text{V.D} = \frac{\text{Mol. wt}}{2}$$

On calculating,

$$V_2 = 14.1$$

23. (d) r.m.s. velocity $V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

$$\text{i.e., } \frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{5 \times 10^4}{10 \times 10^4} = \frac{1}{2} = \sqrt{\frac{T_1}{T_2}}$$

$$\therefore T_2 = 4T_1$$

24. (d) $V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

$$V_{\text{rms}(\text{O}_2)} = V_{\text{rms}(\text{He})}$$

$$\sqrt{\frac{3RT_{\text{O}_2}}{M_{\text{O}_2}}} = \sqrt{\frac{3RT_{\text{He}}}{M_{\text{He}}}}$$

or $\frac{T_{\text{O}_2}}{M_{\text{O}_2}} = \frac{T_{\text{He}}}{M_{\text{He}}}$

$$\therefore T_{\text{O}_2} = \frac{300 \times 32}{4} = 2400\text{K}$$

25. (b) Compressibility factor $(Z) = \frac{PV}{RT}$

(For one mole of real gas)

van der Waals equation

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

At low pressure, volume is very large and hence correction term b can be neglected in comparison to very large volume of V .

i.e. $V - b \approx V$

$$\left(P + \frac{a}{V^2}\right)V = RT$$

$$PV + \frac{a}{V} = RT$$

$$PV = RT - \frac{a}{V}$$

$$\frac{PV}{RT} = 1 - \frac{a}{VRT}$$

Hence, $Z = 1 - \frac{a}{VRT}$

26. (a) According to Boyle's law

$$\frac{V_1}{V_2} = \frac{P_2}{P_1}; \frac{750}{V_2} = \frac{360}{840}$$

$$V_2 = 1750 \text{ ml} = 1.750 \text{ L}$$

27. (d) On applying Dalton's law,

Partial pressure of a component

= Mole fraction $\times$ Total pressure

Given, mass of $\text{N}_2 = 56 \text{ g}$, mass of $\text{O}_2 = 96 \text{ g}$

Total pressure = 10 atm

$${}^n\text{N}_2 = \frac{56}{28} = 2, {}^n\text{O}_2 = \frac{96}{32} = 3$$

$${}^x\text{N}_2 = \frac{{}^n\text{N}_2}{{}^n\text{N}_2 + {}^n\text{O}_2} = \frac{2}{2+3} = 0.4,$$

$${}^x\text{O}_2 = \frac{{}^n\text{O}_2}{{}^n\text{N}_2 + {}^n\text{O}_2} = \frac{3}{2+3} = 0.6$$

$$\therefore P_{\text{N}_2} = 0.4 \times 10 = 4 \text{ atm}, P_{\text{O}_2} = 0.6 \times 10 = 6 \text{ atm}$$

28. (c) Moles of A, $(n_A) = \frac{P_A V_A}{RT} = \frac{8 \times 12}{RT} = \frac{96}{RT}$

$$\text{Moles of B, } (n_B) = \frac{P_B V_B}{RT} = \frac{8 \times 5}{RT} = \frac{40}{RT}$$

Total pressure $\times$ total volume = $(n_A + n_B) \times RT$

$$p \times (12 + 8) = \frac{1}{RT} (96 + 40) RT$$

$$p = 6.8$$

Partial pressure of A = $p \times$ mole fraction of A

$$= 6.8 \left(\frac{96}{RT} / \frac{96+40}{RT} \right)$$

$$= 4.8 \text{ atm}$$

Partial pressure of B = $6.8 - 4.8 = 2 \text{ atm}$.

29. (d) Number of moles of $\text{O}_2 = \frac{70.6 \text{ g}}{32 \text{ g mol}^{-1}} = 2.21 \text{ mol}$

$$\text{Number of moles of Ne} = \frac{167.5 \text{ g}}{20 \text{ g mol}^{-1}} = 8.375 \text{ mol}$$

$$\text{Mole fraction of } \text{O}_2 = \frac{2.21}{2.21 + 8.375} = 0.21$$

$$\text{Mole fraction of Ne} = 1 - 0.21 = 0.79$$

Partial pressure of a gas = Mole fraction $\times$ total pressure

$$\text{Partial pressure of } \text{O}_2 = 0.21 \times 25 = 5.25 \text{ bar}$$

$$\text{Partial pressure of Ne} = 0.79 \times 25 = 19.75 \text{ bar}$$

30. (b) $V_0 = \sqrt{\frac{2RT}{M}} = \sqrt{R(273+t)}$

$$U_{\text{rms}} = \sqrt{\frac{3(2t + 273 + 273)R}{1}}$$

$$= \sqrt{6(t + 273)R} = \sqrt{6} V_0$$

1. (d) Given $\Delta H = 41 \text{ kJ mol}^{-1} = 41000 \text{ J mol}^{-1}$
 $T = 100^\circ\text{C} = 273 + 100 = 373 \text{ K}$
 $n = 1$
 $\Delta U = \Delta H - \Delta nRT = 41000 - (1 \times 8.314 \times 373)$
 $= 37898.88 \text{ J mol}^{-1} \approx 37.9 \text{ kJ mol}^{-1}$

2. (a) Internal energy is dependent upon temperature and according to first law of thermodynamics total energy of an isolated system remains same, i.e., in a system of constant mass, energy can neither be created nor destroyed by any physical or chemical change but can be transformed from one form to another

$$\Delta E = q + W$$

For closed insulated container, $q = 0$, so, $\Delta E = +W$, as work is done by the system

3. (b) Maximum work done by a system at constant pressure and constant temperature is $-\Delta G$. It can be derived as follows :

From first law of thermodynamics,

$$q = \Delta E + w_{\text{exp.}} + w_{\text{non exp.}}$$

At constant pressure, $w_{\text{exp}} = P\Delta V$

$$q = \Delta E + P\Delta V + w_{\text{non-exp.}}$$

$$q = \Delta H + w_{\text{non exp.}} \quad \dots\dots (1)$$

For the reversible change at const temp.,

$$\Delta S = \frac{q_{\text{rev}}}{T} \text{ or } q_{\text{rev}} = T\Delta S$$

so from equ. (1), $T\Delta S = \Delta H + w_{\text{non exp.}}$

$$\Rightarrow \Delta H - T\Delta S = -w_{\text{non exp.}} \quad \dots\dots (2)$$

For a change taking place under constant pressure and constant temperature,

$$\Delta H - T\Delta S = \Delta G$$

so from equ (2)

$$\Delta G = -w_{\text{non exp.}}$$

(it means $w_{\text{non exp.}}$ or w_{useful})

$$\text{or } w_{\text{max}} = -\Delta G$$

4. (a) $\Delta S = \frac{\text{Latent heat of fusion}}{\text{Melting point}} = \frac{\Delta H}{T}$
 $= \frac{2930}{300} \text{ J K}^{-1} \text{ mol}^{-1} = 9.77 \text{ J K}^{-1} \text{ mol}^{-1}$

5. (b) Desired equation is $\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(g)}$, $\Delta H = ?$
 (Equation II - Equation I)
 $\Delta H = -245.5 \text{ kJ} - (-286 \text{ kJ}) = 40.5 \text{ kJ}$

6. (b) $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ $\Delta H = 2 \times -46.0 \text{ kJ mol}^{-1}$

Let x be the bond enthalpy of N - H bond then

[Note : Enthalpy of formation or bond formation enthalpy is given which is negative but the given reaction involves bond breaking hence values should be taken as positive.]

$$\Delta H = \Sigma \text{Bond energies of reactants} - \Sigma \text{Bond energies of products}$$

$$\begin{aligned} 2 \times -46 &= 712 + 3 \times (436) - 6x \\ -92 &= 2020 - 6x \\ 6x &= 2020 + 92 \\ 6x &= 2112 \\ x &= +352 \text{ kJ/mol} \end{aligned}$$

7. (b) $\Delta H_{\text{S-S}} + 2\Delta H_{\text{H-S}} = 239$ $2\Delta H_{\text{H-S}} = 175$

$$\text{Hence, } \Delta H_{\text{S-S}} = 239 - 175 = 64 \text{ kcal mol}^{-1}$$

Then, ΔH for $8\text{S}_{(\text{g})} \rightarrow \text{S}_{8(\text{g})}$ is $8 \times (-64) = -512 \text{ kcal}$

8. (b) Enthalpy of reaction

$$= B.E_{(\text{Reactant})} - B.E_{(\text{Product})}$$

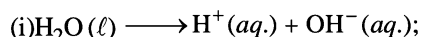
$$= [B.E_{(\text{C}=\text{C})} + 4B.E_{(\text{C-H})} + B.E_{(\text{H-H})}]$$

$$- [B.E_{(\text{C-C})} + 6B.E_{(\text{C-H})}]$$

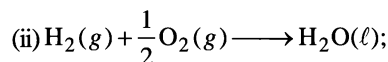
$$= [606.1 + (4 \times 410.5) + 431.37] - [336.49 + (6 \times 410.5)]$$

$$= -120.0 \text{ kJ mol}^{-1}$$

9. (a) Given, for reaction



$$\Delta H_r = 57.32 \text{ kJ}$$



$$\Delta H_r = -286.20 \text{ kJ}$$

For reaction (i)

$$\Delta H_r = \Delta H_f^\circ(\text{H}^+, \text{aq.}) + \Delta H_f^\circ(\text{OH}^-, \text{aq.}) -$$

$$\Delta H_f^\circ(\text{H}_2\text{O}, \ell)$$

$$57.32 = 0 + \Delta H_f^\circ(\text{OH}^-, \text{aq.}) - \Delta H_f^\circ(\text{H}_2\text{O}, \ell) \quad \dots(iii)$$

For reaction (ii)

$$\Delta H_r = \Delta H_f^\circ(\text{H}_2\text{O}, \ell) -$$

$$\Delta H_f^\circ(\text{H}_2, \text{g}) - \frac{1}{2}\Delta H_f^\circ(\text{O}_2, \text{g})$$

$$-286.20 = \Delta H_f^\circ(\text{H}_2\text{O}, \ell)$$

On replacing this value in eq. (iii) we have

$$57.32 = \Delta H_f^\circ(\text{OH}^-, \text{aq.}) - (-286.20)$$

$$\Delta H_f^\circ(\text{OH}^-, \text{aq.}) = -286.20 + 57.32 = -228.88 \text{ kJ}$$

10. (b) At equilibrium $\Delta G = 0$

$$\text{Hence, } \Delta G = \Delta H - T_e \Delta S = 0$$

$$\therefore \Delta H = T_e \Delta S \text{ or } T_e = \frac{\Delta H}{\Delta S}$$

For a spontaneous reaction

ΔG must be negative which is possible only if $\Delta H - T\Delta S < 0$

$$\therefore \Delta H < T\Delta S$$

$$\text{or } T > \frac{\Delta H}{\Delta S}; T_e < T$$

11. (b) $2\text{C}_8\text{H}_{18}(\text{g}) + 25\text{O}_2(\text{g}) \longrightarrow 16\text{CO}_2(\text{g}) + 18\text{H}_2\text{O}(\text{g})$
The given reaction is a combustion reaction. Hence this will be an exothermic reaction.

i.e. $\Delta H = -ve$

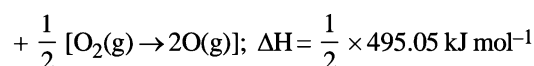
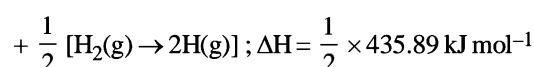
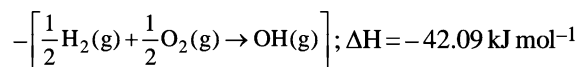
Further $\Delta n = +ve$ i.e. $\Delta S = +ve$

$\Delta G = \Delta H - T\Delta S = -\Delta H - T\Delta S = -ve$

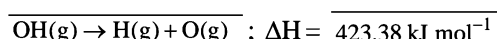
12. (a) $\Delta G = \Delta H - T\Delta S$, $\Delta H + ve$, ΔS is $+ve$; $T\Delta S > \Delta H$ for spontaneous process. It will make ΔG , $-ve$

13. (b) We have to calculate the enthalpy of the reaction
 $\text{OH}(\text{g}) \rightarrow \text{O}(\text{g}) + \text{H}(\text{g})$

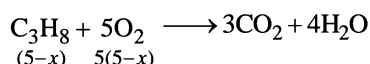
From the given reactions, this can be obtained as follows.



Add



14. (c) $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 $x \quad 2x$



$(5-x) \quad 5(5-x)$

$$2x + 5(5-x) = 16$$

$$\Rightarrow x = 3\text{L}$$

$\therefore$ Heat released

$$= \frac{3}{22.4} \times 890 + \frac{2}{22.4} \times 2220 = 317$$

15. (a) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + 393.5 \text{ kJ/mol}$
12g 44g
44g is formed from 12g of carbon

$$35.2\text{g is formed from } \frac{12 \times 35.2}{44} \text{ g of C}$$

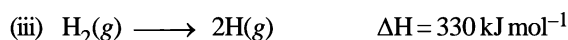
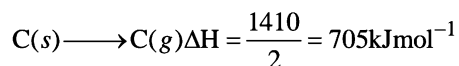
$$= 9.6 \text{ g of C} = 9.6/12 = 0.8 \text{ mole}$$

$$1 \text{ mole release heat } 393.5 \text{ kJ}$$

$$0.8 \text{ mole release heat} = 393.5 \times 0.8$$

$$= 314.8 \text{ kJ} \approx 315 \text{ kJ}$$

16. (d) (i) $2\text{C}(\text{s}) + \text{H}_2(\text{g}) \longrightarrow \text{H}-\text{C} \equiv \text{C}-\text{H}(\text{g})$
 $\Delta H = 225 \text{ kJ mol}^{-1}$



From equation (i) :

$$225 = \left[2 \times \Delta H_{\text{C}(\text{s}) \longrightarrow \text{C}(\text{g})} + 1 \times \text{BE}_{\text{H}-\text{H}} \right]$$

$$- [2 \times \text{BE}_{\text{C}-\text{H}} + 1 \times \text{BE}_{\text{C} \equiv \text{C}}]$$

$$225 = [1410 + 1 \times 330] - [2 \times 350 + 1 \times \text{BE}_{\text{C} \equiv \text{C}}]$$

$$225 = [1410 + 330] - [700 + \text{BE}_{\text{C} \equiv \text{C}}]$$

$$225 = 1740 - 700 - \text{BE}_{\text{C} \equiv \text{C}}$$

$$\text{BE}_{\text{C} \equiv \text{C}} = 815 \text{ kJ mol}^{-1}$$

17. (a) Given $\Delta H = 35.5 \text{ kJ mol}^{-1}$

$$\Delta S = 83.6 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\therefore \Delta G = \Delta H - T\Delta S$$

For a reaction to be spontaneous, $\Delta G = -ve$

i.e., $\Delta H < T\Delta S$

$$\therefore T > \frac{\Delta H}{\Delta S} = \frac{35.5 \times 10^3 \text{ J mol}^{-1}}{83.6 \text{ JK}^{-1}}$$

So, the given reaction will be spontaneous at $T > 425 \text{ K}$

18. (b) $\Delta G = \Delta H - T\Delta S$

Since $\Delta G = \Delta H - T\Delta S$ for an endothermic reaction,

$\Delta H = +ve$ and at low temperature $\Delta S = +ve$

Hence $\Delta G = (+) \Delta H - T(+)\Delta S$

and if $T\Delta S < \Delta H$ (at low temp)

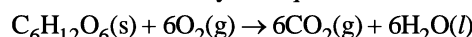
$\Delta G = +ve$ (non spontaneous)

But at high temperature, reaction becomes

spontaneous i.e. $\Delta G = -ve$.

because at higher temperature $T\Delta S > \Delta H$.

19. (c) The standard enthalpy of the combustion of glucose can be calculated by the eqn.



$$\Delta H_{\text{C}} = 6 \times \Delta H_{\text{f}}(\text{CO}_2) + 6 \times \Delta H_{\text{f}}(\text{H}_2\text{O}) - \Delta H_{\text{f}}[\text{C}_6\text{H}_{12}\text{O}_6]$$

$$\Delta H^\circ = 6(-400) + 6(-300) - (-1300)$$

$$\Delta H^\circ = -2900 \text{ kJ/mol}$$

For one gram of glucose, enthalpy of combustion

$$\Delta H^\circ = -\frac{2900}{180} = -16.11 \text{ kJ/gm}$$

20. (c) $\text{CH}_2 = \text{CH}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CH}_3 - \text{CH}_3$
Enthalpy change = Bond energy of reactants - Bond energy of products.

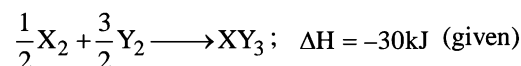
$$\Delta H = 1(\text{C}=\text{C}) + 4(\text{C}-\text{H}) + 1(\text{H}-\text{H}) - 1(\text{C}-\text{C}) - 6(\text{C}-\text{H})$$

$$= 1(\text{C}=\text{C}) + 1(\text{H}-\text{H}) - 1(\text{C}-\text{C}) - 2(\text{C}-\text{H})$$

$$= 615 + 435 - 347 - 2 \times 414 = 1050 - 1175 = -125 \text{ kJ}$$

21. (c) For a reaction to be at equilibrium $\Delta G = 0$. Since $\Delta G = \Delta H - T\Delta S$ so at equilibrium $\Delta H - T\Delta S = 0$ or $\Delta H = T\Delta S$

For the reaction



Calculating ΔS for the above reaction, we get

$$\Delta S = 50 - \left[\frac{1}{2} \times 60 + \frac{3}{2} \times 40 \right] \text{ JK}^{-1}$$

$$= 50 - (30 + 60) \text{ JK}^{-1} = -40 \text{ JK}^{-1}$$

At equilibrium, $T\Delta S = \Delta H$ [$\therefore \Delta G = 0$]

$$\therefore T \times (-40) = -30 \times 1000 [\therefore 1 \text{ kJ} = 1000 \text{ J}]$$

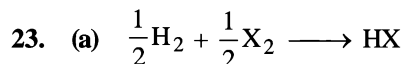
$$\text{or } T = \frac{-30 \times 1000}{-40} \text{ or } 750 \text{ K}$$

22. (a) $\text{I}_2(\text{s}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{ICl}(\text{g})$

$$\Delta_f H = [\Delta H(I_2(s) \rightarrow I_2(g)) + \Delta H_{I-I} + \Delta H_{Cl-Cl}] - [\Delta H_{I-Cl}]$$

$$= 151.0 + 242.3 + 62.76 - 2 \times 211.3 = 33.46$$

$$D_f H^\circ (ICl) = \frac{33.46}{2} = 16.73 \text{ kJ/mol}$$

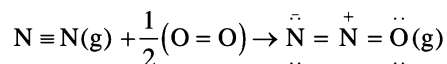
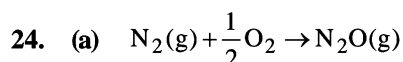


Let the bond enthalpy of X – X bond be x.

$$\Delta H_f (HX) = -50 = \frac{1}{2} \Delta H_{H-H} + \frac{1}{2} \Delta H_{X-X} - \Delta H_{H-X}$$

$$= \frac{1}{2} 2x + \frac{1}{2} x - 2x = \frac{-x}{2}$$

$$\therefore x = 50 \times 2 = 100 \text{ kJ mol}^{-1}$$



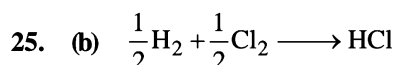
$$\Delta H_f^\circ = [\text{Energy required for breaking of bonds}]$$

$$- [\text{Energy released for forming of bonds}]$$

$$= (\Delta H_{N \equiv N} + \frac{1}{2} \Delta H_{O=O} - (\Delta H_{N=N} + \Delta H_{N=O}))$$

$$= (946 + \frac{1}{2} \times 498) - (418 + 607) = 170 \text{ kJ mol}^{-1}$$

$$\text{Resonance energy} = 170 - 82 = 88 \text{ kJ mol}^{-1}$$



$$\Delta H_{HCl} = \sum \text{B.E. of reactant}$$

$$- \sum \text{B.E. of products}$$

$$-90 = \frac{1}{2} \times 430 + \frac{1}{2} \times 240 - \text{B.E. of HCl}$$

$$\therefore \text{B.E. of HCl} = 215 + 120 + 90$$

$$= 425 \text{ kJ mol}^{-1}$$

26. (a) $-W_{\text{irreversible}} = P_{\text{ext}} (V_2 - V_1)$

$$= 10 \text{ atm} (2L - 1L)$$

$$= 10 \text{ atm-L}$$

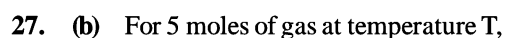
$$-W_{\text{reversible}} = \int_{V_1}^{V_2} P_{\text{ex}} dv$$

$$= 2.303 nRT \log \frac{V_2}{V_1}$$

$$= 1 \times 2.303 \times 0.0821 \text{ atm-L/K/mol} \times \log \frac{2}{1}$$

$$= 16.96 \text{ atm-L}$$

$$\frac{W_{\text{reversible}}}{W_{\text{irreversible}}} = \frac{16.96}{10.00} = 1.69 \approx 1.7$$



$$PV_1 = 5RT$$

For 5 moles of gas at temperature T – 2,

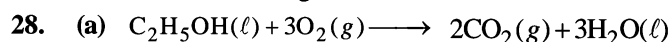
$$PV_2 = 5R(T - 2)$$

$$\therefore P(V_2 - V_1) = 5R(T - 2 - T);$$

$$P\Delta V = -10R,$$

$$-P\Delta V = 10R$$

When ΔV is negative, W is + ve.



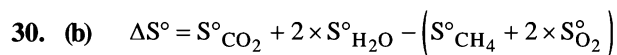
Bomb calorimeter gives ΔU of the reaction

$$\text{Given, } \Delta U = -1364.47 \text{ kJ mol}^{-1}$$

$$\Delta n_g = -1$$

$$\Delta H = \Delta U + \Delta n_g RT = -1364.47 - \frac{1 \times 8.314 \times 298}{1000}$$

$$= -1366.93 \text{ kJ mol}^{-1}$$



$$= (213.6 + 2 \times 69.9) - (186.2 + 2 \times 205.2)$$

$$= -242.8 \text{ J K}^{-1} \text{ mol}^{-1}$$

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC07

1. (b) $K_p = K_c (RT)^{\Delta n}$ since $\Delta n = 0$, $K_p = K_c$.

2. (a) K_w at $25^\circ\text{C} = 1 \times 10^{-14}$

At 25°C

$$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$$

At 100°C (given)

$$K_w = [\text{H}^+][\text{OH}^-] = 55 \times 10^{-14}$$

$\therefore$ for a neutral solution

$$[\text{H}^+] = [\text{OH}^-]$$

$$\therefore [\text{H}^+]^2 = 55 \times 10^{-14}$$

$$\text{or } [\text{H}^+] = (55 \times 10^{-14})^{1/2}$$

$$\therefore \text{pH} = -\log [\text{H}^+]$$

On taking log on both side

$$-\log [\text{H}^+] = -\log (55 \times 10^{-14})^{1/2}$$

$$\text{pH} = \frac{1}{2} - \log 55 + 14 \log 10$$

$$\text{pH} = 6.13$$

3. (c) $\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

$$5 = 4 + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad [\because \text{pK}_a = -\log K_a]$$

$$\text{Given, } K_a = 1 \times 10^{-4}$$

$$\therefore \text{pK}_a = -\log (1 \times 10^{-4}) = 4$$

Now from Handerson equation

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

Putting the values

$$5 = 4 + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$\log \frac{[\text{Salt}]}{[\text{Acid}]} = 5 - 4 = 1$$

Taking antilog

$$[\text{Salt}]/[\text{Acid}] = 10 = 10 : 1$$

4. (a) The reaction given is an exothermic reaction thus according to Le chatelier's principle lowering of temperature, addition of F_2 and Cl_2 favour the forward direction and hence the production of ClF_3 .

5. (c) Ag_2CrO_4

$$K_{sp} = [\text{Ag}^+]^2 [\text{Cr}_2\text{O}_4^{2-}] = 1.1 \times 10^{-12}$$

$$[\text{Ag}^+] = \sqrt{\frac{1.1 \times 10^{-12}}{[\text{Cr}_2\text{O}_4^{2-}]}}$$

AgCl

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = 1.8 \times 10^{-10}$$

$$[\text{Ag}^+] = \frac{1.8 \times 10^{-10}}{[\text{Cl}^-]}$$

AgBr

$$K_{sp} = [\text{Ag}^+][\text{Br}^-] = 5.0 \times 10^{-13}$$

$$[\text{Ag}^+] = \frac{5.3 \times 10^{-13}}{[\text{Br}^-]}$$

AgI

$$K_{sp} = [\text{Ag}^+][\text{I}^-] = 8.3 \times 10^{-17}$$

$$[\text{Ag}^+] = \frac{8.3 \times 10^{-17}}{[\text{I}^-]}$$

$$\text{If we take } [\text{Cr}_2\text{O}_4^{2-}] = [\text{Cl}^-] = [\text{Br}^-] = [\text{I}^-] = 1$$

than maximum $[\text{Ag}^+]$ will be required in case of Ag_2CrO_4 .

6. (b) $\text{MY} \rightleftharpoons \text{M}^+ + \text{Y}^-$

$$K_{sp} = s^2 = 6.2 \times 10^{-13}$$

$$s = \sqrt{6.2 \times 10^{-13}}$$

$$s = 7.87 \times 10^{-7} \text{ mol L}^{-1}$$

$$\text{NY}_3 \rightleftharpoons \text{N}^{3+} + 3\text{Y}^-$$

$$K_{sp} = s \times (3s)^3 = 27s^4 = 6.2 \times 10^{-13}$$

$$s = \left(\frac{6.2 \times 10^{-13}}{27} \right)^{1/4}$$

$$s = 3.89 \times 10^{-4} \text{ mol L}^{-1}$$

$\therefore$ molar solubility of NY_3 is more than MY in water.

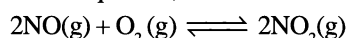
7. (b) $\Delta G^\circ_{\text{NO(g)}} = 86.6 \text{ kJ/mol} = 86600 \text{ J/mol}$

$$G^\circ_{\text{NO}_2(\text{g})} = x \text{ J/mol}$$

$$T = 298, K_p = 1.6 \times 10^{12}$$

$$\Delta G^\circ = -RT \ln K_p$$

Given equation,



$$\therefore 2\Delta G^\circ_{\text{NO}_2} - 2\Delta G^\circ_{\text{NO}} = -R(298) \ln (1.6 \times 10^{12})$$

$$2\Delta G^\circ_{\text{NO}_2} - 2 \times 86600 = -R(298) \ln (1.6 \times 10^{12})$$

$$2\Delta G^\circ_{\text{NO}_2} = 2 \times 86600 - R(298) \ln (1.6 \times 10^{12})$$

$$\Delta G^\circ_{\text{NO}_2} = \frac{1}{2} [2 \times 86600 - R(298) \ln (1.6 \times 10^{12})]$$

$$= 0.5 [2 \times 86600 - R(298) \ln (1.6 \times 10^{12})]$$

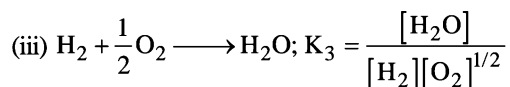
8. (d) Clausius – Clapeyron's equation

$$\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT^2}$$

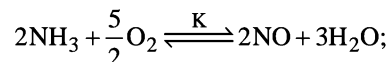
9. (a)

$$(i) \text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3; K_1 = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$$(ii) \text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}; K_2 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$



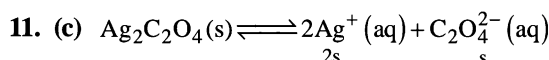
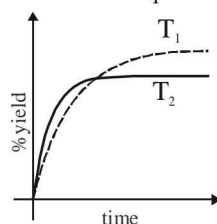
Applying (II + 3 × III - I) we will get



$$K = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} \times \frac{[\text{H}_2\text{O}]^3}{[\text{H}_2]^3 \times [\text{O}_2]^{3/2}} \bigg/ \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$$\therefore K = K_2 \times K_3^3 / K_1$$

10. (b) Initially on increasing temperature rate of reaction will increase, so % yield will also increase with time. But at equilibrium % yield at high temperature (T_2) would be less than at T_1 as reaction is exothermic so the graph is



$$K_{\text{SP}} = [\text{Ag}^+]^2 [\text{C}_2\text{O}_4^{2-}]$$

$$[\text{Ag}^+] = 2.2 \times 10^{-4} \text{ M}$$

Given that:

$\therefore$ Concentration of $\text{C}_2\text{O}_4^{2-}$ ions,

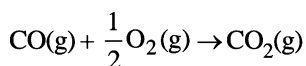
$$[\text{C}_2\text{O}_4^{2-}] = \frac{2.2 \times 10^{-4}}{2} \text{ M} \\ = 1.1 \times 10^{-4} \text{ M}$$

$$\therefore K_{\text{SP}} = (2.2 \times 10^{-4})^2 (1.1 \times 10^{-4}) \\ = 5.324 \times 10^{-12}$$

12. (b) For a gaseous phase reaction K_p and K_c are related as

$$K_p = K_c (RT)^{\Delta n_g}$$

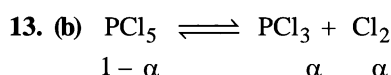
For the given reaction,



$$\Delta n_g = 1 - (1 + 0.5) = -0.5 \text{ or } -\frac{1}{2}$$

$$\therefore K_p = K_c (RT)^{-\frac{1}{2}}$$

$$\text{or } \frac{K_p}{K_c} = (RT)^{-\frac{1}{2}}$$



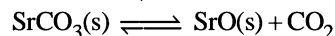
$$1 - \alpha \qquad \qquad \alpha \qquad \alpha$$

$$\therefore K_p = \frac{\frac{\alpha}{1+\alpha} P \times \frac{\alpha}{1+\alpha} P}{\frac{1-\alpha}{1+\alpha} P} = \frac{\alpha^2 P}{1-\alpha^2}$$

$$\text{or, } K_p = \alpha^2 P \quad \therefore \alpha = \sqrt{\frac{K_p}{P}} \text{ when } 1 - \alpha^2 = 1$$

14. (d) Max. pressure of CO_2 = Pressure of CO_2 at equilibrium

For reaction,



$$K_p = P_{\text{CO}_2} = 1.6 \text{ atm} = \text{maximum pressure of } \text{CO}_2$$

volume of container at this stage.

$$V = \frac{nRT}{P} \quad \dots(i)$$

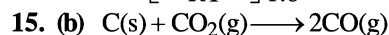
Since container is sealed and reaction was not earlier at equilibrium.

$\therefore n = \text{constant.}$

$$n = \frac{PV}{RT} = \frac{0.4 \times 20}{RT} \quad \dots(ii)$$

Put equation (ii) in equation (i)

$$V = \left[\frac{0.4 \times 20}{RT} \right] \frac{RT}{1.6} = 5 \text{ L}$$



Apply law of mass action,

$$K_p = \frac{(P_{\text{CO}})^2}{P_{\text{CO}_2}} \text{ or } 63 = \frac{(10P_{\text{CO}_2})^2}{P_{\text{CO}_2}}$$

$$(\text{Given } K_p = 63) \text{ and } P_{\text{CO}} = 10P_{\text{CO}_2}$$

$$\text{or } 63 = \frac{100(P_{\text{CO}_2})^2}{P_{\text{CO}_2}} \text{ or } 63 = 100 P_{\text{CO}_2}$$

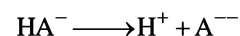
$$P_{\text{CO}_2} = \frac{63}{100} = 0.63 \text{ atm}$$

$$P_{\text{CO}} = 10P_{\text{CO}_2} = 10 \times 0.63 = 6.3 \text{ atm}$$

$$P_{\text{total}} = P_{\text{CO}_2} + P_{\text{CO}} = 0.63 + 6.3 = 6.93 \text{ atm.}$$



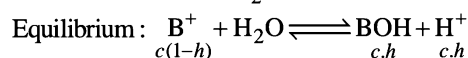
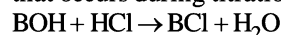
$$\therefore K_1 = 1.0 \times 10^{-5} = \frac{[\text{H}^+][\text{HA}^-]}{[\text{H}_2\text{A}]} \text{ (Given)}$$



$$\therefore K_2 = 5.0 \times 10^{-10} = \frac{[\text{H}^+][\text{A}^{2-}]}{[\text{HA}^-]} \text{ (Given)}$$

$$K = \frac{[\text{H}^+]^2 [\text{A}^{2-}]}{[\text{H}_2\text{A}]} = K_1 \times K_2 \\ = (1.0 \times 10^{-5}) \times (5 \times 10^{-10}) = 5 \times 10^{-15}$$

17. (d) Let the weak monoacidic base be BOH, then the reaction that occurs during titration is



Using the normality equation, $N_1 V_1 = N_2 V_2$
(acid) (base)

Substituting various given values, we get

$$\frac{2}{15} \times V_1 = 2.5 \times \frac{2}{5}$$

$$\text{or } V_1 = 2.5 \times \frac{2}{5} \times \frac{15}{2} = 2.5 \times 3 = 7.5 \text{ ml}$$

Then the concentration of BCl in resulting solution is given by

$$[\text{BCl}] = \frac{\frac{2}{5} \times 2.5}{10} = \frac{1}{10} \text{ or } 0.1 \text{ M}$$

[Total volume = 2.5 + 7.5 = 10 ml]

$$\text{Since } K_h = \frac{K_w}{K_b}$$

$$\therefore K_h = \frac{1 \times 10^{-14}}{1 \times 10^{-12}} = 10^{-2}$$

$$\text{Thus } K_h = \frac{0.1h^2}{(1-h)} \text{ or } 10^{-2} = \frac{0.1h^2}{(1-h)}$$

$$\text{or } 10^{-2} - 10^{-2}h = 0.1h^2$$

$$\text{or } 0.1h^2 + 10^{-2}h - 10^{-2} = 0$$

(Solving this quadratic equation for h , we get)

$$\text{Using } \left[x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$h = \frac{-10^{-2} \pm \sqrt{(10^{-2})^2 + 4 \times 10^{-1} \times 10^{-2}}}{2 \times 0.1}$$

$$= \frac{-10^{-2} \pm \sqrt{10^{-4} + 4 \times 10^{-3}}}{2 \times 0.1}$$

$$= \frac{-0.01 \pm \sqrt{0.0001 + 0.004}}{0.2}$$

$$= \frac{-0.01 \pm \sqrt{0.0041}}{0.2}$$

$$= \frac{-0.01 \pm 0.064}{0.2}$$

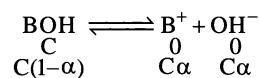
$$= \frac{0.54}{0.2} \quad [\text{Neglecting the negative term}]$$

$$= 0.27$$

$$\therefore [\text{H}^+] = c \cdot h = 0.1 \times 0.27 = 2.7 \times 10^{-2} \text{ M}$$

Thus the correct answer is [d].

18. (c) $M_1V_1 = M_2V_2$
 $1 \times 0.10 = M_2 \times 100$
 $M_2 = 0.001 = 10^{-3}$



$$K_b = \frac{C\alpha \times C\alpha}{C(1-\alpha)}$$

$$K_b = C\alpha^2 \quad (\because 1-\alpha \approx 1)$$

$$\alpha = \sqrt{K_b / C}$$

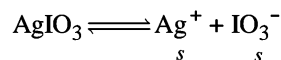
$$[\text{OH}^-] = C\alpha = \sqrt{\frac{K_b}{C}} \times C = \sqrt{K_b C}$$

$$= \sqrt{10^{-5} \times 10^{-3}} = 10^{-4}$$

$$\therefore \text{pH} + \text{pOH} = 14$$

$$\therefore \text{pH} = 14 - 4 = 10$$

19. (c) Let s = solubility



$$K_{sp} = [\text{Ag}^+][\text{IO}_3^-] = s \times s = s^2$$

$$\text{Given } K_{sp} = 1 \times 10^{-8}$$

$$\therefore s = \sqrt{K_{sp}} = \sqrt{1 \times 10^{-8}}$$

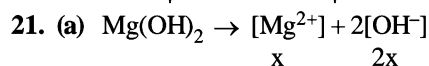
$$= 1.0 \times 10^{-4} \text{ mol/lit} = 1.0 \times 10^{-4} \times 283 \text{ g/lit}$$

$$(\because \text{Molecular mass of Ag IO}_3 = 283)$$

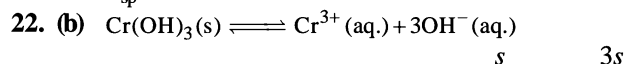
$$= \frac{1.0 \times 10^{-4} \times 283 \times 100}{1000} \text{ gm/100ml}$$

$$= 2.83 \times 10^{-3} \text{ gm/100 ml}$$

20. (a) $(\text{HSO}_4)^-$ can accept and donate a proton
 $(\text{HSO}_4)^- + \text{H}^+ \rightarrow \text{H}_2\text{SO}_4$ (acting as base)
 $(\text{HSO}_4)^- - \text{H}^+ \rightarrow \text{SO}_4^{2-}$ (acting as acid)



$$K_{sp} = [\text{Mg}][\text{OH}]^2 = [x][2x]^2 = x \cdot 4x^2 = 4x^3$$



$$(s) \quad (3s)^3 = K_{sp}$$

$$27s^4 = K_{sp}$$

$$s = \left(\frac{K_{sp}}{27} \right)^{1/4} = \left(\frac{1.6 \times 10^{-30}}{27} \right)^{1/4}$$

23. (d) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$
 involves lose and gain of electrons. H_2O is coordinated to Cu by donating electrons (LHS). It is then removed by withdrawing electrons.

24. (b) pH of an acidic solution should be less than 7. The reason is that from H_2O , $[\text{H}^+] = 10^{-7} \text{ M}$ which cannot be neglected in comparison to 10^{-8} M . The pH can be calculated as.

$$\text{from acid, } [\text{H}^+] = 10^{-8} \text{ M}$$

$$\text{from } \text{H}_2\text{O}, [\text{H}^+] = 10^{-7} \text{ M}$$

$$\therefore \text{Total } [\text{H}^+] = 10^{-8} + 10^{-7}$$

$$= 10^{-8}(1 + 10) = 11 \times 10^{-8}$$

$$\therefore \text{pH} = -\log [\text{H}^+] = -\log 11 \times 10^{-8}$$

$$= -[\log 11 + 8 \log 10]$$

$$= -[1.0414 - 8] = 6.9586$$

25. (d) $\therefore \text{pH} = 1$; $\text{H}^+ = 10^{-1} = 0.1 \text{ M}$

$$\text{pH} = 2$$
 ; $\text{H}^+ = 10^{-2} = 0.01 \text{ M}$

$$\therefore M_1 = 0.1 \quad V_1 = 1$$

$$M_2 = 0.01 \quad V_2 = ?$$

From

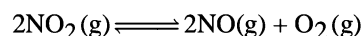
$$M_1V_1 = M_2V_2$$

$$0.1 \times 1 = 0.01 \times V_2$$

$$V_2 = 10 \text{ litres}$$

$$\therefore \text{Volume of water added} = 10 - 1 = 9 \text{ litres}$$

26. (d) For the reaction:-



Given $K_c = 1.8 \times 10^{-6}$ at 184°C

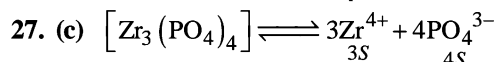
$R = 0.0831 \text{ kJ/mol.k}$

$$K_p = 1.8 \times 10^{-6} \times 0.0831 \times 457 = 6.836 \times 10^{-6}$$

$[\because 184^\circ\text{C} = (273 + 184) = 457 \text{ K}]$

$$\Delta n = (2 + 1, -1) = 1]$$

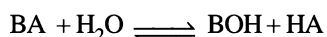
Hence it is clear that $K_p > K_c$



$$\begin{aligned} K_{sp} &= (3S)^3 (4S)^4 \\ &= 27S^3 \times 256S^4 \\ &= 6912S^7. \end{aligned}$$

$$\therefore S = \left(\frac{K_{sp}}{6912} \right)^{1/7}$$

28. (c) In aqueous solution BA(salt) hydrolyses to give



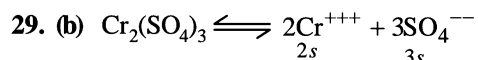
Base acid

Now pH is given by

$$\text{pH} = \frac{1}{2}\text{pK}_w + \frac{1}{2}\text{pK}_a - \frac{1}{2}\text{pK}_b$$

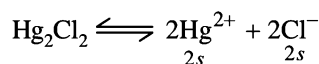
substituting given values, we get

$$\text{pH} = \frac{1}{2}(14 + 4.80 - 4.78) = 7.01$$



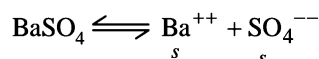
$$K_{sp} = (2s)^2 (3s)^3 = 4s^2 \times 27s^3 = 108s^5$$

$$s = \left(\frac{K_{sp}}{108} \right)^{1/5}$$



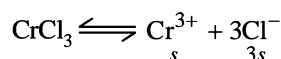
$$K_{sp} = (2s)^2 \times (2s)^2 = 16s^4$$

$$s = \left(\frac{K_{sp}}{16} \right)^{1/4}$$



$$K_{sp} = s^2$$

$$s = \sqrt{K_{sp}}$$



$$K_{sp} = s \times (3s)^3 = 27s^4$$

$$s = \left(\frac{K_{sp}}{27} \right)^{1/4}$$

Hence the correct order of solubilities of salts is

$$\sqrt{K_{sp}} > \left(\frac{K_{sp}}{16} \right)^{1/4} > \left(\frac{K_{sp}}{27} \right)^{1/4} > \left(\frac{K_{sp}}{108} \right)^{1/5}$$

30. (b) NaCN is a salt of strong base and weak acid ; pH

$$= 7 + \frac{1}{2}\text{pK}_a + \frac{1}{2}\log C$$

$$\text{pK}_a \text{ for HCN} = 14 - 4.70 = 9.30$$

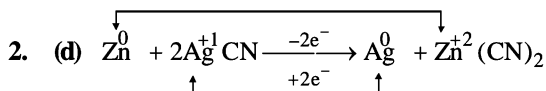
$$\therefore \text{pH} = 7 + \frac{1}{2} \times 9.30 + \frac{1}{2} \log 0.5; \text{pH} = 11.5$$

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC08

1. (d) O.N. of N in NO_2 and N_2O_4 is +4
 $\therefore$ difference is zero.
 O.N. of P in P_2O_5 and P_4O_{10} is +5
 $\therefore$ difference is zero
 O.N. of N in N_2O is +1 and in NO is +2. The difference is 1
 O.N. of S in SO_2 is +4 and in SO_3 is +6. The difference is +2.



The oxidation state shows a change only in (d)

3. (b) Magnesium provides cathodic protection and prevent rusting or corrosion.

4. (a) $2\overset{+7}{\text{KMnO}_4} + 3\text{H}_2\text{SO}_4 \rightarrow \overset{+2}{\text{K}_2\text{SO}_4} + 2\overset{+2}{\text{MnSO}_4} + 3\text{H}_2\text{O} + 5\text{O}$
 $2\overset{+2}{\text{FeSO}_4} + \text{H}_2\text{SO}_4 + \text{O} \rightarrow \overset{+3}{\text{Fe}_2(\text{SO}_4)_3} + \text{H}_2\text{O}$
 O.N. of Mn changes from +7 to +2 (Reduction)
 O.N. of Fe changes from +2 to +3 (Oxidation)

5. (c) $2\text{HI}^{-1} + \overset{+6}{\text{H}_2\text{SO}_4} \longrightarrow \overset{0}{\text{I}_2} + \overset{+4}{\text{SO}_2} + 2\text{H}_2\text{O}$. In this reaction oxidation number of S is decreasing from +6 to +4 hence undergoing reduction and for HI oxidation Number of I is increasing from -1 to 0 hence undergoing oxidation therefore H_2SO_4 is acting as oxidising agent.

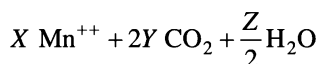
6. (a) (i) $\text{Mn}^{n+} + ne^- \rightleftharpoons \text{M}$, for this reaction, high negative value of E° indicates lower reduction potential, that means M will be a good reducing agent.

Stronger reducing agent $\Rightarrow$ Easy to oxidise				
$\Downarrow$				
Lower reduction potential $\Leftarrow$ higher oxidation potential				
(ii) Element	F	Cl	Br	I
Reduction potential (E° volt)	+2.87	+1.36	+1.06	+0.54

As reduction potential decreases from fluorine to iodine, oxidising nature also decreases from fluorine to iodine.

(iii) The size of halide ions increases from F^- to I^- . The bigger ion can loose electron easily. Hence the reducing nature increases from HF to HI.

7. (a) $X\text{MnO}_4^- + Y\text{C}_2\text{O}_4^{2-} + Z\text{H}^+ \rightleftharpoons$



First half reaction



On balancing



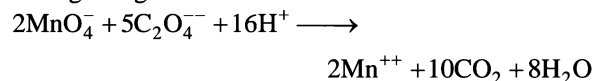
Second half reaction



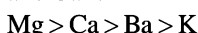
On balancing



On multiplying eqn. (ii) by 5 and (iv) by 2 and then adding we get



8. (d) Order of decreasing electrode potentials of Mg, K, Ba and Ca is



It can be explained by their standard reduction potentials.

$$E_{\text{K}^+|\text{K}}^0 = -2.925$$

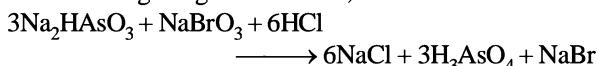
$$E_{\text{Ba}^{2+}|\text{Ba}}^0 = -2.90$$

$$E_{\text{Ca}^{2+}|\text{Ca}}^0 = -2.87$$

$$E_{\text{Mg}^{2+}|\text{Mg}}^0 = -2.37$$

Highly negative value of E_{red}^0 shows the least value of electrode potential.

9. (c) On balancing the given reaction, we find



10. (c) In SO_3^{--}

$$x + 3(-2) = -2; x = +4$$

$$\text{In } \text{S}_2\text{O}_4^{--}$$

$$2x + 4(-2) = -2$$

$$2x - 8 = -2$$

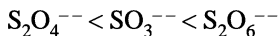
$$2x = 6; x = +3$$

$$\text{In } \text{S}_2\text{O}_6^{2-}$$

$$2x + 6(-2) = -2$$

$$2x = 10; x = +5$$

hence the correct order is



11. (a) CrO_2Cl_2

Let O. No. of Cr = x

$$\therefore x + 2(-2) + 2(-1) = 0$$

$$x - 4 - 2 = 0$$

$$\therefore x = +6$$

12. (a) If an electronegative element is in its lowest possible oxidation state in a compound or in free state. It can function as a powerful reducing agent.

e.g. I^-

13. (d) $\text{Xe} = 53.5\% \therefore \text{F} = 46.5\%$

Relative number of atoms Xe

$$= \frac{53.5}{131.2} = 0.4 \text{ and } \text{F} = \frac{46.5}{19} = 2.4$$

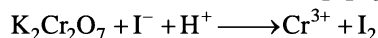
Simple ratio $\text{Xe} = 1$ and $\text{F} = 6$; Molecular formula is XeF_6

O.N. of Xe is +6

14. (d) Copper when exposed to moist air having CO_2 . It gets

superficially coated with a green layer of basic carbonate $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$.

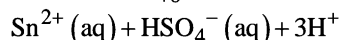
15. (b) In iodometry, $\text{K}_2\text{Cr}_2\text{O}_7$ liberates I_2 from iodides (NaI or KI). Which is titrated with $\text{Na}_2\text{S}_2\text{O}_3$ solution.



Here, one mole of $\text{K}_2\text{Cr}_2\text{O}_7$ accepts 6 mole of electrons.

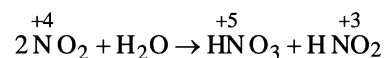
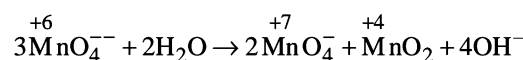
$$\therefore \text{Equivalent weight} = \frac{\text{molecular weight}}{6}$$

16. (c) $\text{H}_2\text{SO}_3(\text{aq}) + \text{Sn}^{4+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \longrightarrow$



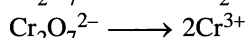
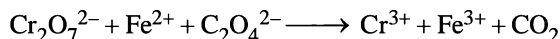
Hence H_2SO_3 is the reducing agent because it undergoes oxidation.

17. (d) $\text{Cl}_2^0 + 2\text{NaOH} \rightarrow \text{NaCl}^{-1} + \text{NaClO}^{+1} + \text{H}_2\text{O}$

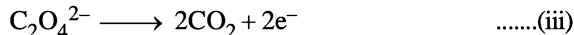
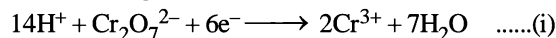


All undergo disproportionation

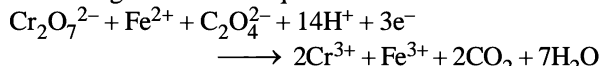
18. (a) The reaction given is



On balancing

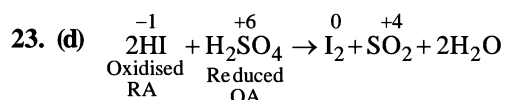


On adding all the three equations.



Hence the total no. of electrons involved in the reaction = 3

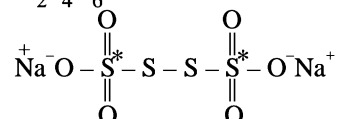
19. (d) H_2S , the oxidation state of S is -2. So it cannot accept more electrons because on accepting 2 electrons S acquires a noble gas configuration. So, it can act only as a reducing agent by losing electron. On the other hand, the oxidation state of S in SO_2 is +4 which is an intermediate oxidation state of sulphur so it can reduce as well oxidise.
20. (c) In all the given compounds oxidation number of non metal is +4. As C belongs to group IV and it is in its maximum oxidation state. So, reduction in oxidation number of nonmetal is not possible only in CO_2 . As we know that reduction is always accompanied by an increase in oxidation number of reducing agent. So, CO_2 cannot act as reducing agent among the given choices.
21. (b) The balanced equation is
 $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$
 Ratio of the coefficients of CO_2 and H_2O is 4 : 6 or 2 : 3.
22. (b) The compound undergoes oxidation itself and reduces others is known as reducing agent. In this reaction O. N. of Ni changes from 0 to +2 and hence Ni acts as a reducing agent.



24. (c) A reaction, in which a substance undergoes simultaneous oxidation and reduction, is called disproportionation reaction. In these reactions, the same substance simultaneously acts as an oxidising agent and as a reducing agent. Here Cl undergoes simultaneous oxidation and reduction.

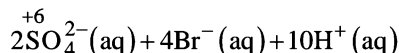
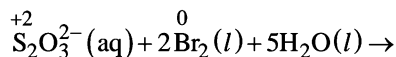
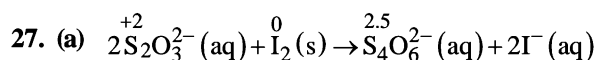


25. (b) $\text{Na}_2\text{S}_4\text{O}_6$ has the structure :



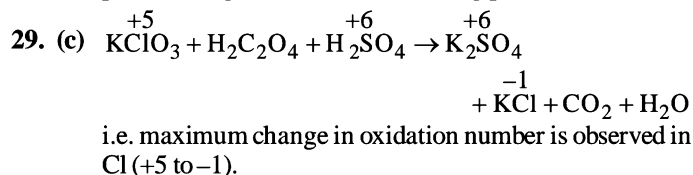
O.N. of two S^* atoms are +5 each and that of other two S atoms is zero each.

26. (b) CrO_2Cl_2 , MnO_4^- ; O.N. of Cr and Mn are +6 and +7 respectively.

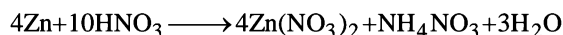
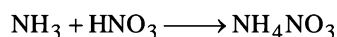
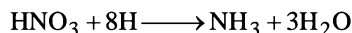
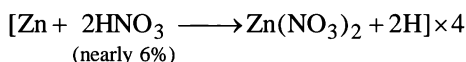


Hence, bromine is a stronger oxidising agent than I_2 , as it oxidises S of $\text{S}_2\text{O}_3^{2-}$ to SO_4^{2-} whereas I_2 oxidises it only into $\text{S}_4\text{O}_6^{2-}$ ion.

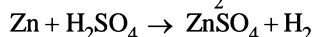
28. (a) Higher the value of reduction potential higher will be the oxidising power whereas the lower the value of reduction potential higher will be the reducing power.



30. (d) Zinc gives H_2 gas with dil $\text{H}_2\text{SO}_4/\text{HCl}$ but not with HNO_3 because in HNO_3 , NO_3^- ion is reduced and gives NH_4NO_3 , N_2O , NO and NO_2 (based upon the concentration of HNO_3)



Zn is on the top position of hydrogen in electrochemical series. So Zn displaces H_2 from dilute H_2SO_4 and HCl with liberation of H_2 .



DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC09

- (c) Hg will not displace hydrogen since it is present below hydrogen in ECS.
- (c) The correct option is (c). In H_2O , hydrogen has + 1 oxidation no. In CaH_2 , hydrogen has - 1 oxidation no.
- (c) Occlusion is the phenomenon of adsorption of hydrogen by metal.
- (a) It is due to hydrogen bonding when H_2O forms a cage like structure in solid ice and density is reduced.
- (d) Heavy water is stable.
- (a) A fusion reaction is difficult to occur because positively charged nuclei repel each other. At very high temperature of the order of 10^6 to 10^7 K, the nuclei may have sufficient energy to overcome the repulsive forces and fuse. It is for this reason, fusion reactions are also called thermonuclear reactions. Hence, hydrogen can be fused to form helium at high temperature and high pressure.
- (b)
$$\underset{\text{base 1}}{\text{H}^-}(\text{aq}) + \underset{\text{acid 1}}{\text{H}_2\text{O}}(\ell) \longrightarrow \underset{\text{base 2}}{\text{OH}^-}(\text{aq}) + \underset{\text{acid 2}}{\text{H}_2}(\text{g})$$

In this reaction H^- acts as bronsted base as it accepts one proton (H^+) from H_2O and form H_2 .
- (c) H_2O_2 . The key reactions are
$$\text{PbO} + \text{H}_2\text{S} \rightarrow \text{PbS} + \text{H}_2\text{O}$$

(black)

$$\text{PbS} + 4\text{H}_2\text{O}_2 \rightarrow \text{PbSO}_4 + 4\text{H}_2\text{O}$$

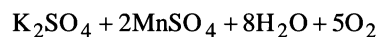
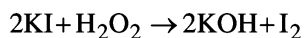
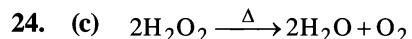
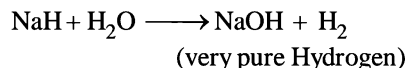
(white)

When blackened statues are treated with H_2O_2 , the PbS is oxidised to PbSO_4 , which is colourless (White).
- (d) The water gas is a combination of carbon and hydrogen. It is used in manufacturing of methanol.
$$\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \xrightarrow[\text{Catalyst}]{\text{Cobalt}} \text{CH}_3\text{OH}(\text{l})$$
- (a) Elements of group 7, 8 and 9 of *d*-block do not form hydrides at all. The inability of metals of group 7, 8 and 9 of periodic table to form hydrides is referred to as hydrides gap of *d*-block. In these compounds H atoms are supposed to occupy interstitial position in the metal lattices. They are also called non-stoichiometric hydrides.
- (a)
$$\text{Cu} + 4\text{HNO}_3(\text{conc.}) \longrightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$$
$$\text{C}_2\text{H}_5\text{OH} + \text{Na} \longrightarrow \text{C}_2\text{H}_5\text{O}^- \text{Na}^+ + 1/2\text{H}_2 \uparrow$$
$$\text{Mg} + 2\text{H}_2\text{O}(\text{steam}) \longrightarrow \text{Mg}(\text{OH})_2 + \text{H}_2 \uparrow$$
$$\text{C}_6\text{H}_5\text{OH} + \text{Na} \longrightarrow \text{C}_6\text{H}_5\text{O}^- \text{Na}^+ + 1/2\text{H}_2 \uparrow$$
$$\text{NaH} + \text{H}_2\text{O} \longrightarrow \text{NaOH} + \text{H}_2 \uparrow$$
- (c) Hydrogen is not used in the synthesis of hydroquinone and tartaric acid.
- (a) Normality of 10V of H_2O_2
$$\frac{68 \times 10}{22.4} = 17 \times N \quad \therefore N = 1.78$$
- (b) In the structure of ice each molecule of H_2O is surrounded by three H_2O molecules in hexagonal honey comb manner which results an open cage like structure. As a result there are a number of 'hole' or open spaces. In such a structure lesser number of molecules are packed per ml. When ice melts a large no. of hydrogen bonds are broken. The molecules therefore move into the holes or open spaces and come closer to each other than they were in solid state. This result sharp increase in the density. Therefore ice has lower density than water.
- (b)
$$2\text{HSO}_4^-(\text{aq}) \xrightarrow{\text{electrolysis}} \text{HO}_3\text{SOOSO}_3\text{H}(\text{aq})$$
$$\xrightarrow{\text{Hydrolysis}} 2\text{HSO}_4^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{H}_2\text{O}_2(\text{aq})$$
- (c) A diamagnetic substance contains no unpaired electron.
 H_2 is diamagnetic as it contains all paired electrons
$$\text{H}_2 = \sigma_b^2, \quad \text{H}_2^+ = \sigma_b^1, \quad \text{H}_2^- = \sigma_b^2,$$

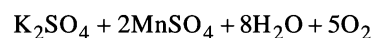
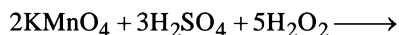
(diamagnetic) (paramagnetic) (paramagnetic)

$$\sigma_a^{*1}; \text{He}_2^+ = \sigma_b^2, \sigma_a^{*1}$$

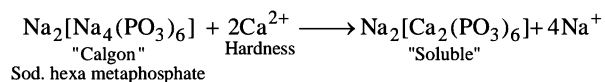
(paramagnetic) (paramagnetic)
- (b) $\therefore 22.4 \text{ L of } \text{H}_2 \text{ at STP} = 1 \text{ mole of } \text{H}_2$
$$\therefore 0.224 \text{ L of } \text{H}_2 \text{ at STP} = \frac{1}{22.4} \times 0.224 = 0.01 \text{ moles.}$$
- (a) Only elements having reduction potential less than - 0.41V liberate hydrogen with cold water.
- (a) In one electron species, such as H-atom, the energy of orbital depends only on the principal quantum number, *n*.
Hence answer (d)
i.e. is $< 2s = 2p < 3s = 3p = 3d < 4s$
 $= 4p \Rightarrow 4d = 4g$
- (d) Chlorine has lone pair which it can donate to form coordinate bond while hydrogen cannot.
- (a) MgCO_3 is insoluble in water.
- (a) True peroxide contains O-O linkage and O_2^{2-} ion. They give hydrogen peroxide with dil H_2SO_4 .
$$\text{BaO}_2 + \text{H}_2\text{SO}_4(\text{dil.}) \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$$
- (d) Very pure hydrogen can be prepared by the action of water on sodium hydride.



25. (a) H_2O_2 acts as a reducing agent only in presence of strong oxidising agents (i.e., MnO_4^-) in acidic as well as alkaline medium.



26. (d) Polyphosphates (e.g., sodium hexa metaphosphate) form soluble complexes with cations such as Ca^{2+} , Mg^{2+} and so they are used as water softening agents.



27. (c) Ortho and para forms of hydrogen resemble in their chemical properties.

28. (c) H_2O_2 has oxidizing and reducing properties both.

29. (b) $\text{Na}_2 \text{ zeolite} + \text{CaCl}_2 \rightarrow \text{Ca zeolite} + 2\text{NaCl}$

30. (c) $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \xrightarrow[\text{Ni}]{1270\text{K}} \text{CO} + 3\text{H}_2$

Mixture of CO and H_2 is called water gas.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC10

- (d) Alkali metals are highly electropositive and thus highly reducing. Therefore reduction, double decomposition and displacement methods for their extraction are not suitable. Only electrolytic methods are useful for their extraction.
- (a) A reducing agent is a substance which can lose electron and hence a reducing agent should have low ionisation energy. Now since ionisation energy decreases from Li to Cs, the reducing property should increase from Li to Cs. The only exception to this is lithium. This is because the net process of converting an atom to an ion takes place in 3 steps.
 - $M(s) \rightarrow M(g)$ $\Delta H = \text{Sublimation energy}$
 - $M(g) \rightarrow M^+(g) + e^-$ $\Delta H = \text{Ionisation energy}$
 - $M^+(g) + H_2O \rightarrow M^+(aq)$ $\Delta H = \text{Hydration energy}$
 The large amount of energy liberated in hydration of Li (because of its small size) makes the overall ΔH negative. This accounts for the higher oxidation potential of lithium i.e., its high reducing power.
- (a) According to Fajan's rules, ionic character increases with increase in size of the cation and decrease in size of the anion. Thus, CsF has higher ionic character than NaCl and hence bond in CsF is stronger than in NaCl.
- (b) $BaCO_3$ forms a yellow ppt of barium chromate. $BaCO_3$ forms a white precipitate of $BaSO_4$. $BaCl_2$ is soluble in water.
- (a) The weaker the base, the less stable is its carbonate. Since LiOH is the weakest base, hence Li_2CO_3 has the lowest thermal stability.
- (b) As outermost electronic configuration of alkali metals is ns^1 and also their size are largest in their respective periods so their 1st I.E. will be lowest among the given options. As second electron is to be removed from complete shell or noble gas core, so the 2nd I.E. must be highest among the given options. So, option (b) is correct choice.
- (a) As we move down the group, the lattice energies of carbonates remain approximately the same. However the hydration energies of the metal cation decreases from Be^{++} to Ba^{++} , hence the solubilities of carbonates of the alkaline earth metal decrease down the group mainly due to decreasing hydration energies of the cations from Be^{++} to Ba^{++} .
- (a) $4KO_2 + 2CO_2 \rightarrow 2K_2CO_3 + 3O_2$.
 KO_2 is used as an oxidising agent. It is used as air purifier in space capsules. Submarines and breathing masks as it produces oxygen and removes carbon dioxide.
- (d) Halides of group I and II impart characteristic colour to the flame due to low I.E. of the central atom. However, ionization energy of Be and Mg atoms is high due to their small size, hence they can't be excited to higher levels by Bunsen burner flame. Thus, their halides don't impart colour to flame.
- (c) According to Fajan's rule,
 Size of cation $\propto$ Ionic character.
 Recall that size of metal (ion) increases while going down the group, and decreases on crossing the periods from left to right. Thus Rb^+ (group I) is largest and Be^{2+} (group II) is smallest in size. Hence $RbCl$ has greatest, and $BeCl_2$ has lowest ionic character.
- (d) Na_2O_2 is peroxide of sodium not super oxide. The formula of sodium superoxide is NaO_2 .
- (c) Ca and CaH_2 both react with H_2O to form H_2 gas,

$$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$$

$$CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + 2H_2$$
 whereas K gives H_2 while KO_2 gives O_2 and H_2O_2

$$2K + 2H_2O \longrightarrow 2KOH + H_2$$

$$2KO_2 + 2H_2O \longrightarrow 2KOH + O_2 + H_2O_2$$
 Similarly, Na gives H_2 while Na_2O_2 gives H_2O_2

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

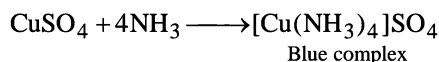
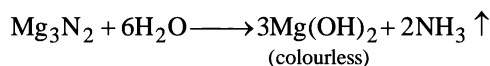
$$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$$
 Likewise Ba gives H_2 while BaO_2 gives H_2O_2

$$Ba + 2H_2O \longrightarrow Ba(OH)_2 + H_2$$

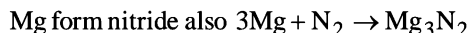
$$BaO_2 + 2H_2O \longrightarrow Ba(OH)_2 + H_2O_2$$
- (a) Basic strength of the oxides increases in the order $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$. The increase in basic strength is due to the decrease in I.E. and increase in electropositive character.
 The melting points of the halides decrease in the order $NaF > NaCl > NaBr > NaI$, as the size of the halide ion increases. The decrease in melting point is due to increase in the covalent character with increase in the size of anion according to Fajan's rule.
- (d) $Zn^{++} + 2NaOH \longrightarrow Zn(OH)_2 + 2Na^+$

$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

$$Na_2ZnO_2 \longrightarrow 2Na^+ + (ZnO_2)^{2-}$$
- (b) $CaCO_{3(s)} \xrightarrow{\Delta} CO_{2(g)} \uparrow + CaO_{(s)}$
 A colourless B
 $CaO_{(s)} + H_2O \longrightarrow Ca(OH)_{2(aq)}$
 B
 $Ca(OH)_2 + 2CO_2 \longrightarrow Ca(HCO_3)_{2(aq)}$
 C
 $Ca(HCO_3)_{2(s)} \xrightarrow{\Delta} CaCO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$
 A
 $\therefore$ Correct choice : (b)
- (d) In Castner Kellner cell, sodium amalgam is formed at mercury cathode.
- (b) $3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$
 X Y



18. (d) K and Mg, both form oxides



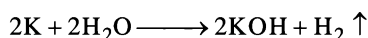
K does not form nitride.

19. (c) The atom becomes larger on descending the group, so the bonds become weaker (metallic bond), the cohesive force/energy decreases and accordingly melting point also decreases.

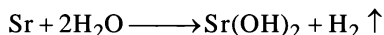
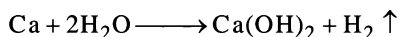
20. (d) Lesser the lattice energy, more will be the solubility in H_2O .

21. (d) For a good quality cement, the ratio of silica (SiO_2) to alumina (Al_2O_3) should be between 2.5 and 4 and the ratio of lime (CaO) to the total of the oxides of silicon (SiO_2), aluminium (Al_2O_3) and iron (Fe_2O_3) should be as close as possible to 2.

22. (b) $2\text{Na} + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2 \uparrow$



All alkali metals decompose water with the evolution of hydrogen.

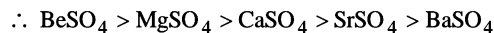


Ca, Sr, Ba and Ra decompose cold water readily with evolution of hydrogen. Mg decomposes boiling water but Be is not attacked by water even at high temperatures as its oxidation potential is lower than the other members.

23. (c) Monovalent sodium and potassium ions and divalent magnesium and calcium ions are found in large proportions in biological fluids.

24. (b) Alkali metals readily lose electron to give monovalent M^+ ion. Hence they are never found in free state in nature.

25. (d) In alkaline earth metals, ionic size increases down the group. The lattice energy remains constant because sulphate ion is so large, so that small change in cationic size does not make any difference. On moving down the group the degree of hydration of metal ions decreases very much leading to decrease in solubility.



26. (a) The alkali metals dissolve in liquid ammonia without evolution of hydrogen. The metal loses electrons and combine with ammonia molecule.



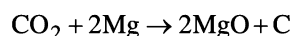
+ e^- (ammoniated)



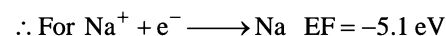
It is ammoniated electron which is responsible for colour.

27. (c)

28. (c) Mg is more reducing in nature than carbon



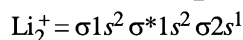
29. (b) $\therefore \text{For Na} \longrightarrow \text{Na}^+ + \text{e}^- \quad \text{IE}_1 = 5.1 \text{ eV}$



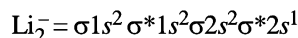
(because the reaction is reverse)

30. (b) $\text{Li}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2$

$$\therefore \text{Bond order} = \frac{1}{2}(4 - 2) = 1$$

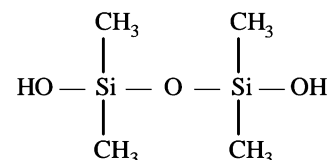


$$\text{B.O.} = \frac{1}{2}(3 - 2) = 0.5$$



$$\text{B.O.} = \frac{1}{2}(4 - 3) = 0.5$$

The bond order of Li_2^+ and Li_2^- is same but Li_2^+ is more stable than Li_2^- because Li_2^+ is smaller in size and has 2 electrons in antibonding orbitals whereas Li_2^- has 3 electrons in antibonding orbitals. Hence Li_2^+ is more stable than Li_2^- .



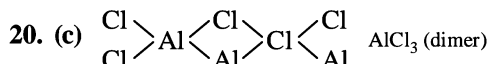
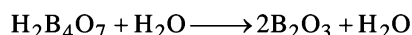
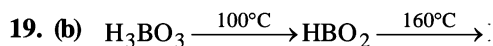
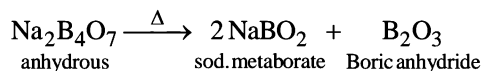
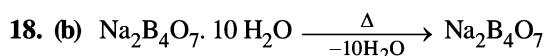
16. (d) Inertness of ns^2 electrons of the valence shell to participate in bonding on moving down the group in heavier p-block elements is called inert pair effect.

As a result, Pb(II) is more stable than Pb(IV)

Sn(IV) is more stable than Sn(II)

$\therefore$ Pb(IV) is easily reduced to Pb(II) and can act as an oxidising agent whereas Sn(II) is easily oxidised to Sn(IV) and can act as a reducing agent.

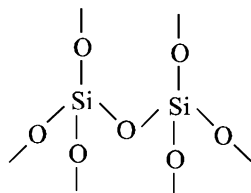
17. (c) CO_2 forms carbonic acid H_2CO_3 , when dissolved in water, CO is neutral, whereas other two GeO_2 and SnO_2 are solids.



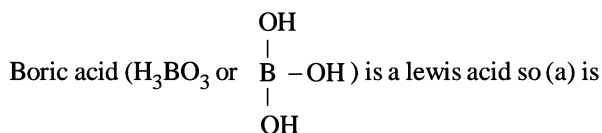
21. (b) F and Cl are more oxidising in nature and can achieve Pb in (IV) O.S. but Br_2 and I_2 can not achieve Pb in (IV)

O.S. secondly Pb^{4+} is strong in oxidising nature and in its presence, Br^- and I^- can not exist.

22. (d) In SiO_2 (quartz), each of O-atom is shared between two SiO_4^{4-} tetrahedra.



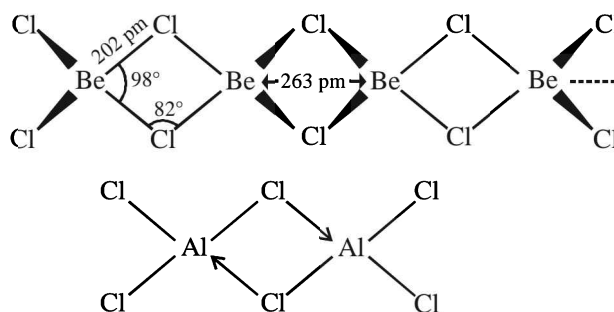
23. (c) The correct formula of inorganic benzene is $\text{B}_3\text{N}_3\text{H}_6$ so (d) is incorrect statement



incorrect statement.

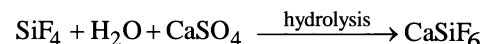
The coordination number exhibited by beryllium is 4 and not 6 so statement (b) is incorrect.

Both BeCl_2 and AlCl_3 exhibit bridged structures in solid state so (c) is correct statement.



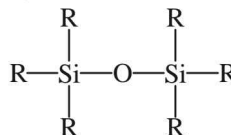
24. (c) In graphite, carbon is sp^2 hybridized. Each carbon is thus linked to three other carbon atoms forming hexagonal rings. Since only three electrons of each carbon are used in making hexagonal ring, fourth electron of each carbon is free to move. This makes graphite a good conductor of heat and electricity.

Further graphite has a two dimensional sheet like structure. These various sheets are held together by weak van der Waal's force of attraction. Due to these weak forces of attraction, one layer can slip over the other. This makes graphite soft and a good lubricating agent.



26. (a) Glass is a translucent or transparent amorphous supercooled solid solution or we can say super cooled liquid of silicates and borates having a general formula $\text{R}_2\text{O} \cdot \text{MO} \cdot 6\text{SiO}_2$ where $\text{R} = \text{Na}$ or K and $\text{M} = \text{Ca}, \text{Ba}, \text{Zn}$ or Pb .

27. (b) The hydrolysis of Trialkylchlorosilane R_3SiCl yields dimer:



28. (a) The linking of identical atoms with each other to form long chains is called catenation. However, this property decreases from carbon to lead. Decrease of this property is associated with M-M bond energy which decreases from carbon to lead.

29. (c) The feldspars are most abundant aluminosilicate minerals in the Earth's surface. The silicon atoms and aluminium atoms occupy the centres of interlinked tetrahedra of SiO_4^{4-} and

AlO_4^{5-} . These tetrahedra connect at each corner to other tetrahedra forming an intricate, three dimensional, negatively charged framework. The sodium cations sit within the voids in this structure.

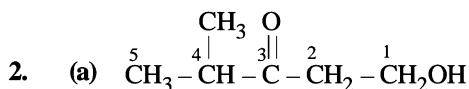
30. (c) CO_2 being more dense covers the igniting material more effectively than N_2 .

DAILY PRACTICE PROBLEMS

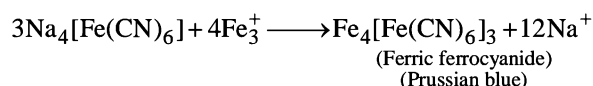
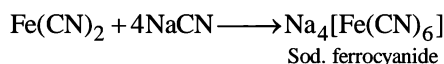
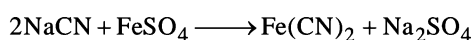
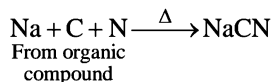
CHEMISTRY SOLUTIONS

DPP/CC12

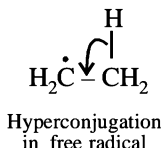
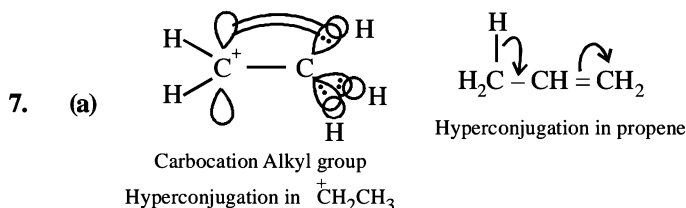
1. (b) Coloured impurities are removed by adsorbing over activated charcoal.



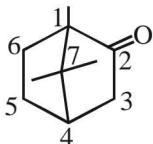
3. (b) The chemistry of the Lassaigne's test of nitrogen is



4. (c) The atom or group which has more power to attract electrons in comparison to hydrogen is said to have -I effect. Thus higher the electronegativity of atom stronger will be the -I effect. As electronegativity of N, O and F follow the order $\text{N} < \text{O} < \text{F}$ hence based upon electronegative character order of -I effect is $-\text{NR}_2 < -\text{OR} < -\text{F}$.
5. (a) This method is applied for the purification of substances which (i) are insoluble in water, (ii) are volatile in steam, (iii) are associated with non steam volatile impurities, (iv) have high molecular weights and (v) possess a fairly high vapour pressure at about the boiling point of water e.g. Aniline.
6. (a) Carbinol is methyl alcohol (CH_3OH), hence vinylcarbinol should be $\text{CH}_2 = \overset{3}{\text{CH}} - \overset{2}{\text{CH}} - \overset{1}{\text{CH}_2}\text{OH}$ (prop-2-enol)



8. (a) It is a bridge compound



1, 7, 7-trimethyl bicyclo [2. 2. 1] heptan-2-one

9. (b) If there is a small difference (10 or less) in the boiling points of liquids fractional distillation is used e.g. acetone b.p. 333 K and methanol b.p. 338 K.

10. (a) Cyclohexane (iv) is non-planar and has chair conformation. In this conformation, the bond angle is the normal tetrahedral angle ($109^\circ, 28'$) and thus has no angle strain and hence is most stable. The rest of the molecules are nearly planar and hence their stability depends upon the angle strain in accordance with *Baeyer's strain theory*. Since cyclopropane has higher

angle strain $\left(\frac{109^\circ 28' - 60^\circ}{2} = 24^\circ, 44' \right)$ than

cyclopentane $\left(\frac{109^\circ 28' - 108^\circ}{2} = 0^\circ, 44' \right)$. Therefore

cyclopentane (iii) is more stable than cyclopropane (i). Further, because of the presence of a double bond in a three membered ring, cyclopropene (ii) is the least stable. Thus the order of stability is (iv) > (iii) > (i) > (ii).

11. (a) $\text{CH}_3^* \text{CH(OH)COOH} \xrightarrow[\text{+H}]{\text{-OH}} \text{CH}_3\text{CH(H)COOH}$
Lactic acid
Chiral carbon is present
(optically active)
No chiral carbon
(optically inactive)

12. (b) Wt. of organic substance = 0.25 g

$$V_1 = 40 \text{ mL}, T_1 = 300 \text{ K}$$

$$P_1 = 725 - 25 = 700 \text{ mm of Hg}$$

$$P_2 = 760 \text{ mm of Hg (at STP)}$$

$$T_2 = 273 \text{ K}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 \text{ (Volume of nitrogen at STP)}$$

$$= \frac{273 \times 700 \times 40}{300 \times 760} = 33.52 \text{ mL}$$

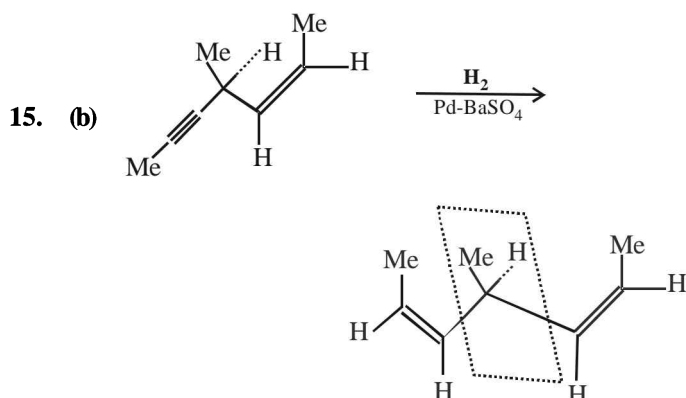
Percentage of nitrogen

$$= \frac{28 \times \text{volume of } \text{N}_2 \text{ at STP} \times 100}{22400 \times \text{wt. of organic substance}}$$

$$= \frac{28 \times 33.52 \times 100}{22400 \times 0.25} = 16.76\%$$

13. (d) $\text{Na}_2[\text{Fe(CN)}_5\text{NO}] + \text{S}^{2-} \rightarrow \text{Na}_4[\text{Fe(CN)}_5\text{NOS}]$
Sodium thio-nitroprusside
(violet)

14. (c) The mixture of sugars is a homogenous one. Homogeneous mixtures of a solvent and one or more solutes (dissolved substances) are often separated by chromatography. Chromatography works to separate a mixture because the components of a mixture distribute themselves differently when they are in contact with a "two phase system". One phase is stationary and the other is moving or mobile. The stationary phase may be a solid packed in a tube or a piece of paper. The mobile phase may be liquid or gaseous.

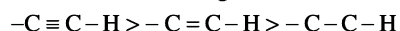


Due to cis-addition of H_2 to the triple bond, the reduced product has a plane of symmetry and hence is optically inactive.

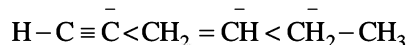
16. (d) The amount of s -character in various hybrid orbitals is as follows.

$$sp = 50\%, sp^2 = 33\% \text{ and } sp^3 = 25\%$$

Therefore s character of the $C-H$ bond in acetylene (sp) is greater than that of the $C-H$ bond in alkene (sp^2 hybridized) which in turn has greater s character of the $C-H$ bond than in alkanes. Thus owing to a high s character of the $C-H$ bond in alkynes, the electrons constituting this bond are more strongly held by the carbon nucleus with the result the hydrogen present on such a carbon atom can be easily removed as proton. The acidic nature of three types of $C-H$ bonds follows the following order



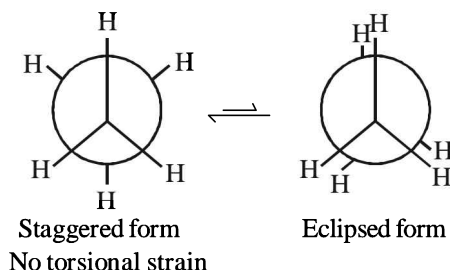
Further, as we know that conjugate base of a strong acid is a weak base, hence the correct order of basicity is



17. (b)
-
- is optically active
- due to absence of plane of symmetry and center of symmetry.

18. (b) Hydrazine (NH_2NH_2) does not contain carbon and hence on fusion with Na metal, it cannot form NaCN; consequently hydrazine does not show Lassaigne's test for nitrogen.

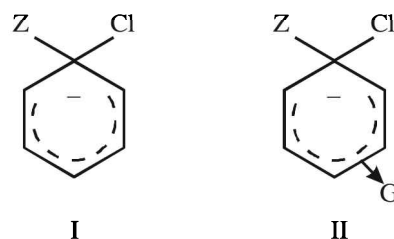
19. (d) In staggered conformation any two hydrogen atoms on adjacent carbon atoms are as far apart as possible there by minimising repulsion between the electron clouds of s -bonds of two non-bonded H-atomic (torsional strain)



20. (a) $PhCOCOPh \xrightarrow{LiAlH_4} Ph-\overset{\overset{OH}{|}}{\underset{\underset{H}{|}}{C^x}}-\overset{\overset{OH}{|}}{\underset{\underset{H}{|}}{C^x}}-Ph$ The

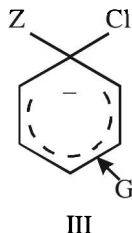
product contains two similar asymmetric carbon atoms and two optically active and one optically inactive meso form.

21. (c) Steam distillation is the most suitable method of separation of 1 : 1 mixture of ortho and para nitrophenols as there is intramolecular hydrogen bonding in o-nitrophenol.
22. (b) In Kjeldahl's method nitrogen is converted into $(NH_4)_2 SO_4$, then to NH_3
23. (b) The resonance structures have same positions of nuclei and same number of unpaired electrons.
24. (a) In S_NAr reactions, a carbanion is formed as an intermediate, so any substituent that increases the stability of carbanion and hence the transition state leading to its formation will enhance the S_NAr reactions. To compare the rates of substitution in chlorobenzene, chlorobenzene having electron-withdrawing group, and chlorobenzene having electron-releasing group, we compare the structures carbanion I (from chlorobenzene), II (from chlorobenzene containing electron-withdrawing group) and III (from chlorobenzene containing electron-releasing group).



Z withdraws electrons, neutralises (disperses) $-ve$ charge

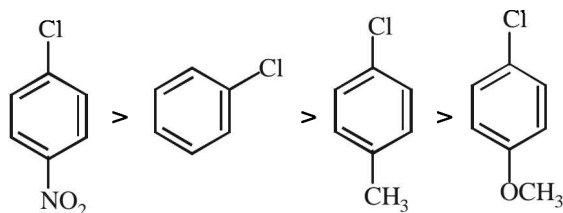
of the ring, stabilises carbanion, facilitates S_N reaction (activation effect)



G releases electrons, intensifies $-ve$ charge, destabilizes carbanion, retards S_N reaction (deactivation)

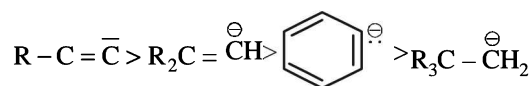
NO_2 is activating group and CH_3 and OCH_3 are deactivating group.

Hence, the correct order of nucleophilic substitution reactions



25. (d) The stability of carbanions is affected due to resonance, inductive effect and s -character of orbitals. Greater the number of groups having $+I$ group (alkyl group) lesser stable would be the carbanion.

Further stability of carbanion decreases with decrease in s -character. Benzene carbanions are stabilized due to resonance, hence the correct order is

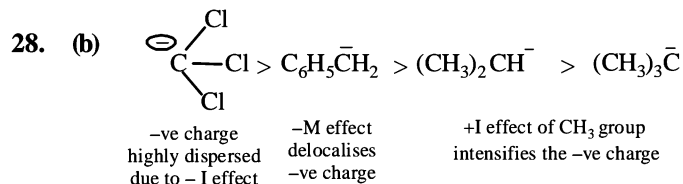


The correct order of stability of given carbanion is in the order $I > III > II > IV$.

26. (b) $\% \text{ of } S = \frac{32}{233} \times \frac{0.233}{0.32} \times 100 = 10\%$

27. (c) Stereoisomerism involve those isomers which contain same ligands in their co-ordination spheres but differ in the arrangement of these ligands in space. Stereoisomerism is of two type geometrical isomerism and optical isomerism. In geometrical isomerism ligands occupy different positions around the central metal atom or ion.

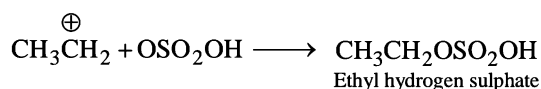
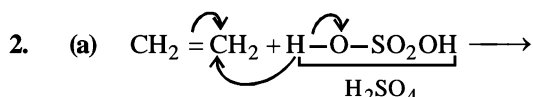
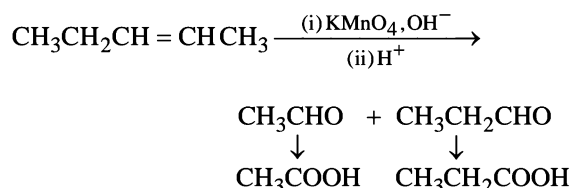
In optical isomerism isomers have same formula but differ in their ability to rotate directions of the plane of polarised light.



29. (c) $CH_2=CH-\ddot{Cl} \leftrightarrow \bar{C}H_2-CH=\overset{+}{Cl}$: a double bond is formed between C and Cl. Hence it is less reactive due to resonance

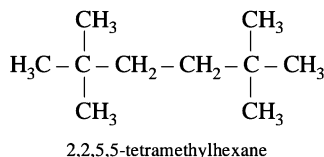
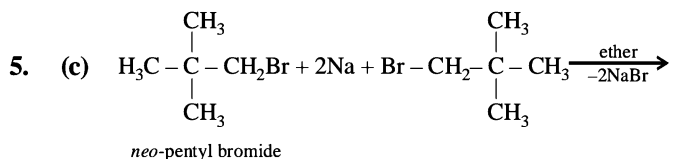
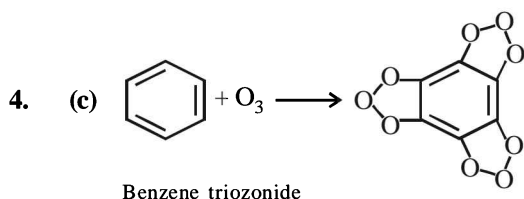
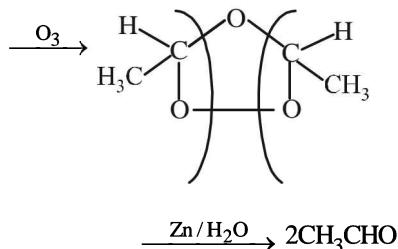
30. (a)	% of element	Relative no. of atoms	Simple ratio
C	38.8	$\frac{38.8}{12} = 3.2$	1
H	16.0	$\frac{16}{1} = 16.0$	5
N	45.28	$\frac{45.28}{14} = 3.2$	1

1. (d) A doubly bonded carbon atom having an alkyl group is oxidised to aldehyde which is further oxidised to carboxylic acid.

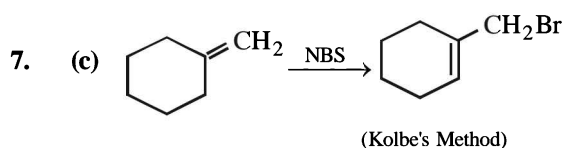
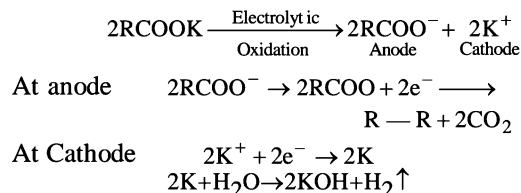


Addition of sulphuric acid takes place according to Markownikoff's rule. Alkanes do not absorb cold conc. H_2SO_4 .

3. (c) The given molecular formula suggests that the aldehyde formed will be acetaldehyde hence the alkene will be
- $$\text{CH}_3\text{CH}=\text{CHCH}_3$$
- 2-butene



6. (a) Electrolysis of a concentrated aqueous solution of either sodium or potassium salts of saturated monocarboxylic acids yields higher alkane at anode.



8. (b) Sum of masses of C and H = 10.5 + 1.0 = 11.5

$$\% \text{ of C} = \frac{10.5}{11.5} \times 100 = 91.3\%$$

$$\% \text{ of H} = \frac{1.0}{11.5} \times 100 = 8.7\%$$

Element	%	Ratio of atoms	Simplest ratio
C	91.3	91.3/12 = 7.61	7.61/7.61 = 1 × 7 = 7
H	8.7	8.7/1 = 8.7	8.7/7.61 = 1.14 × 7 = 8

∴ Empirical formula = C_7H_8

We know that $n = \frac{PV}{RT}$

$$\frac{\text{mass}}{\text{molar mass}} = \frac{PV}{RT}$$

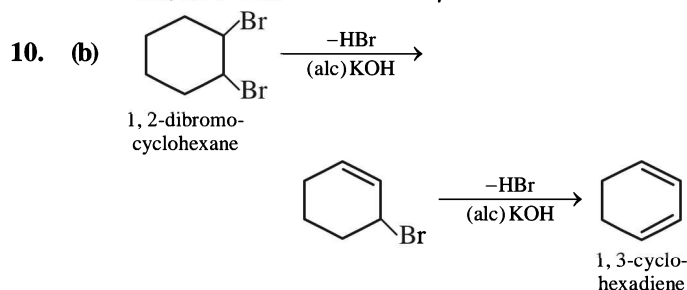
$$\frac{2.81}{M} = \frac{1 \times 1}{0.082 \times 400}; M = 92.168 \approx 92$$

$$\therefore n = \frac{84 + 8}{92} = 1$$

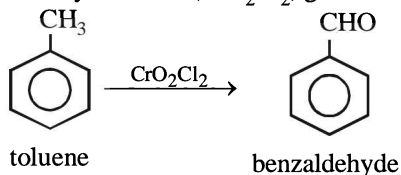
$$\therefore \text{Molecular formula} = (\text{C}_7\text{H}_8) \times 1 = \text{C}_7\text{H}_8$$

9. (c) $\text{CH}_3\text{CHBrCHBrCH}_3 \xrightarrow{\text{Zn / alcohol}} \text{CH}_3\text{CH}=\text{CHCH}_3$

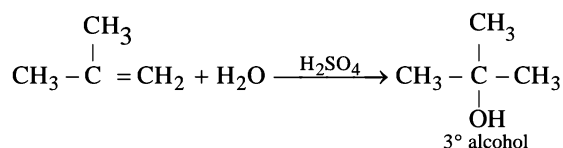
Since two bromine atoms are lost from different carbon atom, the reaction is known as β-elimination.



11. (a) Toluene on oxidation with mild oxidising agent like chromyl chloride (CrO_2Cl_2) gives benzaldehyde.

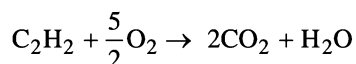


12. (c) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{H}_2\text{O} \xrightarrow{\text{H}_2\text{SO}_4} \text{CH}_3 - \underset{\text{OH}}{\underset{2^\circ \text{ alcohol}}{\text{CH}}} - \text{CH}_3$



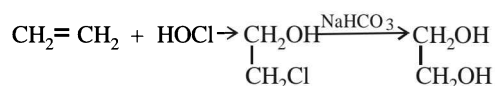
Addition follows Markownikoff's rule.

13. (b) The combustion reaction of ethylene is



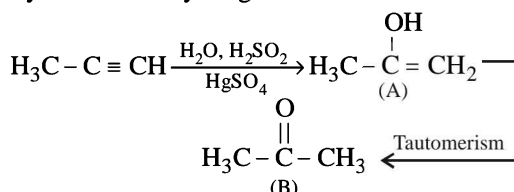
Both $\text{HC}\equiv\text{CH}$ and CO_2 have sp hybridization.

14. (c) We know that

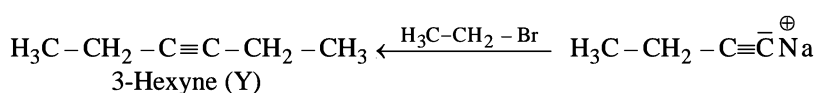


$\therefore \text{M} = \text{CH}_2\text{OH} - \text{CH}_2\text{Cl}$ and $\text{R} = \text{NaHCO}_3$

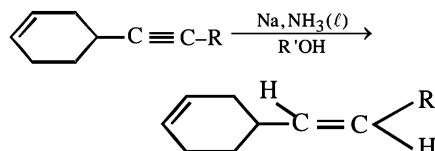
15. (c) Hydration of alkynes give ketones.



21. (a) $\text{HC}\equiv\text{CH} \xrightarrow[\text{liq. NH}_3]{\text{NaNH}_2} \text{HC}\equiv\text{C}^-\text{Na}^+ \xrightarrow{\text{H}_3\text{C}-\text{CH}_2-\text{Br}} \text{HC}\equiv\text{C}-\text{CH}_2-\text{CH}_3$ (X) 1-Butyne



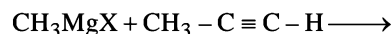
22. (d) It is a stereoselective reaction and in it a trans product is formed.



[It is a reduction. Metal dissolved reduction $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ reaction is stereo selective].

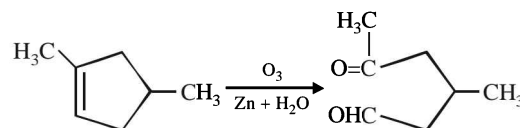
23. (d) $\text{H}_2\text{C}=\text{C}(\text{CH}_3)-\text{CH}=\text{CH}_2 \xrightarrow[\text{H}^+]{1 \text{ equiv. HBr}} \text{H}_3\text{C}-\text{C}^+-\text{CH}=\text{CH}_2 \rightarrow \text{H}_3\text{C}-\text{CH}=\text{CH}-\text{CH}_2^+ \xrightarrow{\text{Br}^-} \text{H}_3\text{C}-\text{CH}=\text{CH}-\text{CH}_2\text{Br}$ (Major)

16. (d) Writing the reaction we get

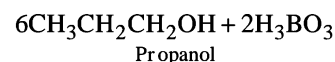
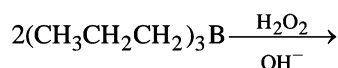


So we find that CH_4 is produced in this reaction.

17. (d)



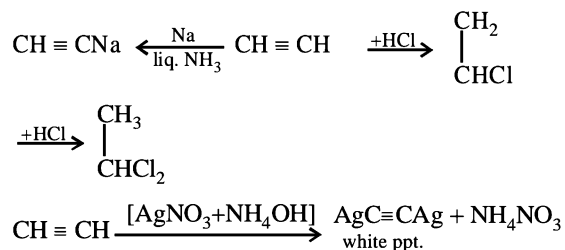
18. (d) $6\text{CH}_3 - \underset{1, \text{ Propene}}{\text{CH}} = \text{CH}_2 \xrightarrow[\text{ether, } 0^\circ\text{C}]{\text{B}_2\text{H}_6}$



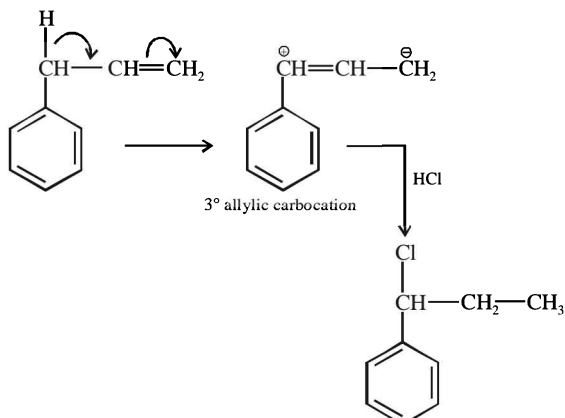
19. (a) Huckel rule is not obeyed. It has only four electrons. Further it does not have continuous conjugation.

20. (b) $-\text{Cl}$ group is o -, p -directing due to $+R$ effect ; however it is deactivating due to strong $-I$ effect of Cl (difference from other o -, p -directing groups which are activating). The net result is that chlorobenzene undergoes o -, p -substitution, but with difficulty

24. (a) Acetylene reacts with the other three as:

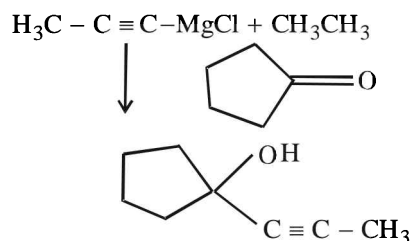


25. (c)

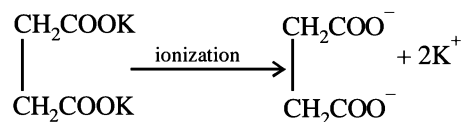


(Since tertiary carbocation is more stable)

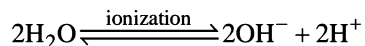
26. (d) $\text{H}_3\text{C} - \text{C} \equiv \text{C} - [\text{H} + \text{CH}_3\text{CH}_2] \text{MgCl} \longrightarrow$



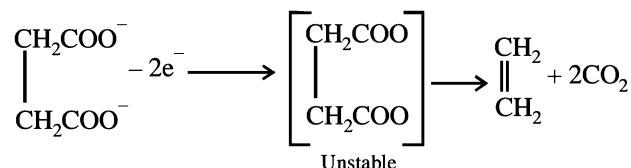
27. (c) Ethene is obtained by electrolysis of dipotassium succinate as follows



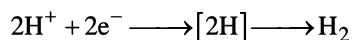
Pot. Succinate



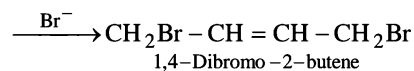
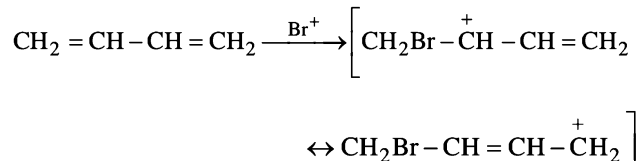
At anode :



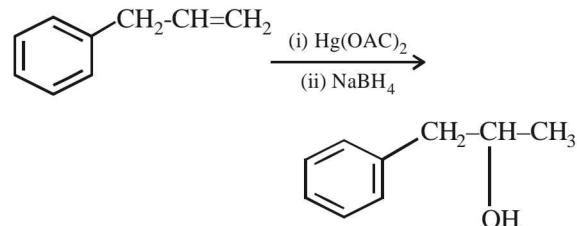
At cathode :



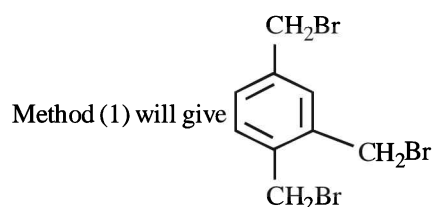
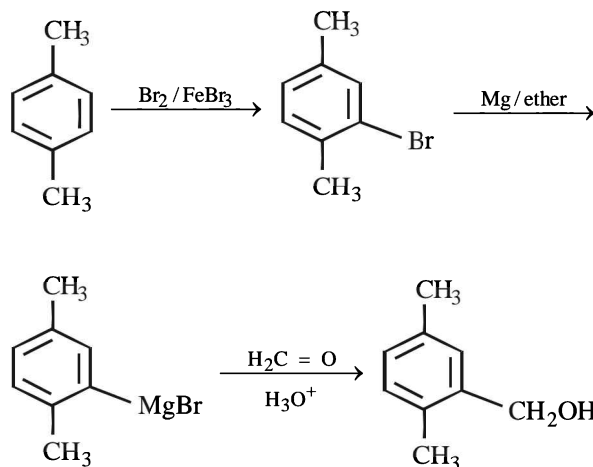
28. (a) The intermediate 2° carbocation shows resonance



29. (a)



30. (b)



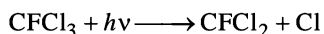
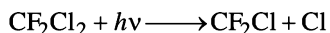
which is allylic bromination.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

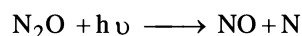
DPP/CC14

1. (b) High concentration of SO_2 leads to stiffness of flower buds.
2. (a) Radiation coming from sun or outerspace have high energy or short wavelength, which are allowed to enter by green house gases. However, radiation emitted by earth is in infrared region, having long wavelength, are reflected back by the envelope of green house gases.
3. (b) In cold water, dissolved oxygen can reach a concentration upto 10 ppm, whereas oxygen in air is about 200, 000 ppm.
4. (b) Pesticides and synthetic fertilizers pollute the soil.
5. (c) Water is often treated with Cl_2 to kill germs.
6. (b) CO is a primary pollutant.
7. (c) CO and oxides of Nitrogen are poisonous gases present in automobile exhaust gases.
8. (b) This represents a great step forward for green chemistry.
9. (a) Normal rain water has pH 5.6. Thunderstorm results in the formation of NO and HNO_3 which lowers the pH.
10. (c) The irritant red haze in the traffic and congested places is due to presence of oxides of nitrogen.
11. (c) Photochemical smog is formed in noon in summer.
12. (b) Strength of sewage or degree of water pollution is measured in terms of BOD (Biochemical oxygen demand) value.
13. (a) Photochemical smog is caused by oxides of sulphur and nitrogen.
14. (c) Freons and chlorofluorocarbon find their way to stratosphere through without being destroyed in the troposphere due to their low reactivity. In the stratosphere the Cl and Br atoms are liberated from the parent compounds by the action of ultraviolet light e.g.

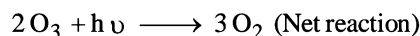
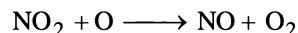
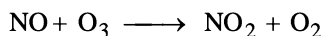


The Cl atoms can destroy ozone molecules through a variety of catalytic cycles.

15. (d) Aerosols use CFC and flying jets release NO. Both causes depletion of ozone layer.
16. (c) The ozone layer, existing between 20 to 35 km above the earth's surface, shield the earth from the harmful U. V. radiations from the sun. Depletion of ozone is caused by oxides of nitrogen



reactive nitric oxide



The presence of oxides of nitrogen increase the decomposition of O_3 .

17. (a) PSCs react with chlorine nitrate and HCl to give HOCl and Cl_2 .
18. (d) Using plastic cans for neatly storing substances will not come under green chemistry. The plastic materials are non-biodegradable.
19. (d) Addition of phosphate fertilizers to water leads to nutrient enrichment (eutrophication)
20. (b) Thermal pollution is caused by power plants. Power plant requires a larger quantity of water for cooling. The water after cooling is left in the water body. The temperature of left water is generally very high and affects aquatic life.
21. (a) Based on the features given gas must be SO_2 .
22. (b) Sewage water is purified by micro-organisms.
23. (d) CO is highly toxic and impairs respiration. CO combine with haemoglobin of blood and reduces its O_2 carry capacity.
24. (d) Acid rain is the rain water containing sulphuric acid and nitric acid which are formed from the oxides of sulphur and nitrogen present in the air as pollutants and rain water has a pH range of 4-5.
25. (b)
26. (d) The oxidised hydrocarbons and ozone in presence of humidity cause photochemical smog. Hydrocarbons + O_2 , NO_2 , NO, O, $\text{O}_3 \rightarrow$ Peroxides, formaldehyde, peroxyacetyl-nitrate (PAN), acrolein etc. Hence chlorofluorocarbons are not common component of photochemical smog.
27. (a) Microorganisms present in the soil is a sink for CO.
28. (b) The ideal value of D.O for growth of fishes is $8 \text{ mg}/\ell$. $7 \text{ mg}/\ell$ is desirable range, below this value fishes get susceptible to disease. A value of $2 \text{ mg}/\ell$ or below is lethal for fishes.
29. (c) The excess of fluorine in water causes fluorosis. The symptoms of fluorosis are mottling of teeth (yellowish streaks) and abnormal bones liable to fracture etc. It is an example of endemic disease.
30. (d) Minamata is caused by Hg poisoning.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC15

1. (c)
2. (b) $\rho = \frac{ZM}{N_A V}$

$$Z = \frac{\rho N_A V}{M} = \frac{8.92 \times 6.02 \times 10^{23} \times (362)^3 \times 10^{-30}}{63.55} = 4$$
 $\therefore$ It has *fcc* unit cell
3. (c)
4. (c) $\frac{d_1}{d_2} = \frac{(a_2)^3}{(a_1)^3} \times \frac{z_1}{z_2} = \left(\frac{3}{3.5}\right)^3 \times \frac{4}{2} = 1.46$
5. (a) Wurtzite has *face centred cubic* structure in which each Zn^{2+} ion is attached to four S^{2-} ions and each S^{2-} ion remains in contact with four Zn^{2+} ions. Hence coordination number of each ion is 4.
6. (c) The increase in pressure results in decrease in size of ions (more in case of anion than cation), the r^+/r^- increases and the coordination number also increase.
7. (a) $\frac{r^+}{r^-}$ ratio is $\frac{60}{136} = 0.441$, Hence LiF has NaCl structure with CN = 6.
8. (a) Diamond is like ZnS. In diamond cubic unit cell, there are eight corner atoms, six face centered atoms and four more atoms inside the structure.
 Number of atoms present in a diamond cubic cell

$$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} + 4 = 8$$
 (corners) (face centered) (inside body)
9. (a) Fraction of unoccupied sites in NaCl crystal

$$= 1 - \frac{2.165 \times 10^3}{2.178 \times 10^3}$$

$$= \frac{2.178 \times 10^3 - 2.165 \times 10^3}{2.178 \times 10^3} = \frac{0.13 \times 10^3}{2.178}$$

$$= \frac{130}{2178} = 5.96 \times 10^{-3}$$
10. (a) MnO_2
11. (c) I_2 (s)
12. (d)
13. (a)
14. (b) It is the property of liquid crystal.
15. (d) To get a *n*-type semiconductor from silicon, it should be doped with a substance with valency 5.
 e.g., silicon is doped with phosphorus to form a *n*-type semiconductor.
16. (d) $n\lambda = 2d \sin \theta$;

$$2 \times 1\text{\AA} = 2 \times d \sin 60^\circ \Rightarrow d = 1.15\text{\AA} \quad \therefore \left(\sin 60^\circ = \frac{\sqrt{3}}{2} \right)$$
17. (d)
18. (d) $\rho = \frac{Z \times M}{N_o \times a^3}$,

$$2.7 = \frac{Z \times 27}{6.02 \times 10^{23} \times (4.05)^3 \times 10^{-24}} \quad \therefore Z = 4$$
 Hence it is face centred cubic unit lattice.
 Again $4r = a\sqrt{2} = 5.727\text{\AA}$
 $\therefore r = 1.432\text{\AA}$
19. (a) $d_{hkl} = \frac{a}{(h^2 + k^2 + l^2)^{1/2}}$
 Hence $d_{100} = 1$, $d_{110} = \frac{1}{\sqrt{2}}$, $d_{111} = \frac{1}{\sqrt{3}}$
20. (c) For cubic geometry the limiting ratio is
 $\frac{r^+}{r^-} > 0.731$ i.e., (c)
21. (a) Statement (a) is correct.
22. (a) For *bcc* lattice body diagonal = $a\sqrt{3}$.
 The distance between the two oppositely charged ions

$$= \frac{a}{2}\sqrt{3}$$

$$= \frac{387 \times 1.732}{2} = 335\text{pm}$$
23. (d) Number of formulas in cube shaped crystals

$$= \frac{1.0}{58.5} \times 6.02 \times 10^{23}$$
 since in NaCl type of structure 4 formula units form 'a' cell

$$\therefore \text{units cells} = \frac{1.0 \times 6.02 \times 10^{23}}{58.5 \times 4} = 2.57 \times 10^{21} \text{ unit cells.}$$
24. (c) As CsCl is body-centred, $d = \sqrt{3}a/2$.
25. (c) AB is just like NaCl. Thus twelve A^+ are at edges and 1 within body of *fcc* i.e. in octahedral voids and six B^- at faces and 8 at corner.
26. (b) Among the given crystals, only silicon exists as a covalent solid. It has diamond like structure.
27. (c) The p.f. for body centred cube = 0.68 (see text).
28. (d) Calculate the $\frac{r^+}{r^-}$ ratio to get the limiting ratio value and consult the table. All are correct.
29. (c)
30. (b) In Na_2O there is antifluorite structure. Here negative ions form the ccp arrangement so that each positive ion is surrounded by 4 negative ions and each negative ion is surrounded by 8 positive ions. So coordination no. of Na^+ is 4.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC16

- (b) According to Le-chateliers principle, for an exothermic reaction ($\Delta H < 0$) increase in temperature decreases the solubility.
- (b) $P_{\text{total}} = P_A^{\circ} \times x_A + P_B^{\circ} \times x_B$
 $= 80.0 \times 0.4 + 120.0 \times 0.6 = 104 \text{ mm Hg}$
 The observed P_{total} is 100 mm Hg which is less than 104 mm Hg. Hence the solution shows negative deviation.
- (c) $P_A = P_A^{\circ} \times x_A = \text{Total pressure} \times y_A$
 $P_B = P_B^{\circ} \times x_B = \text{Total pressure} \times y_B$
 Where x and y represent mole fraction in liquid and vapour phase respectively.

$$\therefore \frac{P_B^{\circ} x_B}{P_A^{\circ} x_A} = \frac{y_B}{y_A}$$

$$\frac{P_B^{\circ} (1 - x_A)}{P_A^{\circ} x_A} = \frac{1 - y_A}{y_A}$$

$$\frac{1}{x_A} - 1 = \frac{P_A^{\circ}}{P_B^{\circ}} \left(\frac{1}{y_A} - 1 \right)$$

$$= \frac{P_A^{\circ}}{P_B^{\circ}} \cdot \frac{1}{y_A} - \frac{P_A^{\circ}}{P_B^{\circ}}$$

or, $\frac{1}{x_A} = \frac{P_A^{\circ}}{P_B^{\circ}} \cdot \frac{1}{y_A} + \left(1 - \frac{P_A^{\circ}}{P_B^{\circ}} \right)$

$$= \frac{P_A^{\circ}}{P_B^{\circ}} \cdot \frac{1}{y_A} + \frac{P_B^{\circ} - P_A^{\circ}}{P_B^{\circ}}$$

This is equation of straight line.

Slope = $\frac{P_A^{\circ}}{P_B^{\circ}}$, Intercept = $\frac{P_B^{\circ} - P_A^{\circ}}{P_B^{\circ}}$
- (b) $\Delta T_f = 0.3^{\circ} \text{C}$

$$\Delta T_f = 0.3^{\circ} \text{C} = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$$

$$0.3 = \frac{1.86 \times W_B \times 1000}{62 \times 5000}$$

$$\therefore W_B = 50 \text{ g}$$

The amount used should be more than 50 g.
- (b) $\pi V = nRT = \frac{m}{M} RT$ or $M_2 = \left(\frac{m_2}{V} \right) \frac{RT}{\pi}$

Where M_2 = molecular mass of solute and m_2 = mass of solute
- (d) According to Henry's law,
 $m = k \times p$
 given $K_H = 1.4 \times 10^{-3}$
 $P_{O_2} = 0.5$ or
 $P_{O_2} = K_H \times x_{O_2}$
 $\therefore x_{O_2} = \frac{0.5}{1.4 \times 10^{-3}}$
 No. of moles; $n = \frac{m}{M}$
 $0.7 \times 10^{-4} = \frac{m}{32}$
 $m = 22.4 \times 10^{-4} \text{ g} = 2.24 \text{ mg}$
- (c) Let α be the give of dissociation then

$$\text{NaCl} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$$

$$\begin{matrix} 1-\alpha & \alpha & \alpha \end{matrix}$$

Van't Hoff's factor $i = \frac{1 - \alpha + \alpha + \alpha}{1} = 1 + \alpha$

Again Van't Hoff's factor

$$= \frac{\text{Normal mol. wt}}{\text{Observed mol. wt}} = \frac{58.5}{31.8} = 1.83$$

Equating to both values of i, $\therefore 1 + \alpha = 1.83$
 $\therefore \alpha = 0.83$
- (c) These two components A and B follows the condition of Raoult's law if the force of attraction between A and B is equal to the force of attraction between A and A or B and B.
- (c) There is no change in vapour pressure.
- (b) $P_s = x_A (P_A^{\circ} - P_B^{\circ}) + P_B^{\circ}$
 $P_s = P_A^{\circ} \times x_A - P_B^{\circ} \times x_A + P_B^{\circ}$
 $P_s = P_A^{\circ} \times x_A - P_B^{\circ} (1 - x_B) + P_B^{\circ}$
 $\therefore P_s = P_A^{\circ} \times x_A + P_B^{\circ} \times x_B$
 This is condition for ideal solution.
- (d) $\Delta T_f = 0 - (-0.00732^{\circ}) = 0.00732$
 $\Delta T_f = i \times K_f \times m$
 $i = \frac{\Delta T_f}{K_f \times m} = \frac{0.00732}{1.86 \times 0.002} = 2$
- (b) As $\Delta T_f = K_f \cdot m$
 $\Delta T_b = K_b \cdot m$
 Hence, we have $m = \frac{\Delta T_f}{K_f} = \frac{\Delta T_b}{K_b}$
 or $\Delta T_f = \Delta T_b \frac{K_f}{K_b}$

$$\Rightarrow [\Delta T_b = 100.18 - 100 = 0.18^\circ\text{C}]$$

$$= 0.18 \times \frac{1.86}{0.512} = 0.654^\circ\text{C}$$

As the Freezing Point of pure water is 0°C ,

$$\begin{aligned}\Delta T_f &= 0 - T_f \\ 0.654 &= 0 - T_f \\ \therefore T_f &= -0.654\end{aligned}$$

Thus the freezing point of solution will be -0.654°C .

13. (b) $[\Delta H_{\text{mix}} < 0]$
 14. (d) $\text{C}_2\text{H}_5\text{I}$ and $\text{C}_2\text{H}_5\text{OH}$ form non-ideal solution.
 15. (b) For this solution intermolecular interactions between *n*-heptane and ethanol are weaker than *n*-heptane-*n*-heptane & ethanol-ethanol interactions hence the solution of *n*-heptane and ethanol is non-ideal and shows positive deviation from Raoult's law.
 16. (c)
 17. (a) Liquid solvent and solid solvent are in equilibrium.
 18. (d)
 19. (b) The molality involves weights of the solute and the solvent. Since weight does not change with the temperature, therefore molality does not depend upon the temperature.

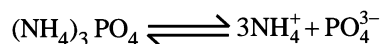
20. (d) Molarity (M) = $\frac{\text{wt} \times 1000}{\text{mol. wt.} \times \text{vol (ml)}}$

$$2 = \frac{\text{wt.}}{63} \times \frac{1000}{250}$$

$$\text{wt.} = \frac{63}{2} \text{ gm}$$

$$\text{wt. of 70\% acid} = \frac{100}{70} \times 31.5 = 45 \text{ gm}$$

21. (c) Vant Hoff factor $i = 4$ in case of $(\text{NH}_4)_3\text{PO}_4$,



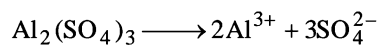
22. (d) $\Delta T_b = iK_b m$
 Given, $(\Delta T_b)_X > (\Delta T_b)_Y$
 $\therefore i_X K_b m > i_Y K_b m$
 (K_b is same for same solvent)

$$i_X > i_Y$$

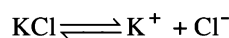
So, x is undergoing dissociation in water. $1 = u/c$

23. (a) Depression in F.P. $\propto$ No. of particles.

$\text{Al}_2(\text{SO}_4)_3$ provides five ions on ionisation



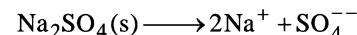
while KCl provides two ions



$\text{C}_6\text{H}_{12}\text{O}_6$ and $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ are not ionised so they have single particle.

Hence, $\text{Al}_2(\text{SO}_4)_3$ have maximum value of depression in F.P or lowest F.P

24. (b) Sodium sulphate dissociates as

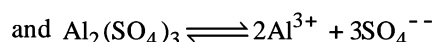
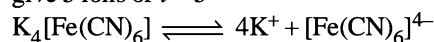


hence van't hof factor $i = 3$

$$\text{Now } \Delta T_f = i k_f .m$$

$$= 3 \times 1.86 \times 0.01 = 0.0558 \text{ K}$$

25. (a) $\text{K}_4[\text{Fe}(\text{CN})_6]$ and $\text{Al}_2(\text{SO}_4)_3$ both dissociates to give 5 ions or $i = 5$



26. (b) 1 molal solution means 1 mole of solute dissolved in 1000 gm solvent.

$$\therefore n_{\text{solute}} = 1 \quad w_{\text{solvent}} = 1000 \text{ g}$$

$$\therefore n_{\text{solvent}} = \frac{1000}{18} = 55.56$$

$$X_{\text{solute}} = \frac{1}{1 + 55.56} = 0.0177$$

27. (b) Concentration of

$$\text{Na}_2\text{CO}_3 = \frac{25.3}{106} \times \frac{1000}{250} = 0.955 \text{ M}$$

$$[\text{Na}^+] = 2 \times 0.955 = 1.91 \text{ M}$$

$$[\text{CO}_3^{2-}] = 0.955 \text{ M}$$

28. (d) Azeotrope of $\text{HCl} + \text{H}_2\text{O}$ contains 20.2% HCl.

29. (d) $\Delta T_f = i . k_f . m$; $\Delta T_b = i . k_b . m$

$$\frac{\Delta T_f}{\Delta T_b} = \frac{k_f}{k_b}$$

$$\Delta T_f = 0 - (-0.186^\circ\text{C}) = 0.186^\circ\text{C}$$

$$\frac{0.186}{\Delta T_b} = \frac{1.86}{0.52} \Rightarrow \Delta T_b = \frac{0.52 \times 0.186}{1.86} = 0.052$$

30. (a) Mole fraction of any component A

$$x = \frac{\text{No. of moles of A}}{\text{Total No. of moles}}$$

As total no. of moles > no. of moles of A
 thus x can never be equal to one or zero.

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1. (c) Conductivity of an electrolyte depends on the mobility of ions and concentration of ions. The motion of an ionic species in an electric field is retarded by the oppositely charged ions due to their interionic attraction. On dilution, concentration of electrolyte decreases and the retarding influence of oppositely charged ions decreases. Therefore mobility of ions increases.

2. (d) Degree of dissociation,

$$\alpha = \frac{\Lambda}{\Lambda_{\infty}} = \frac{8.0}{400} = 2 \times 10^{-2}$$

$$K_a = \frac{C\alpha^2}{(1-\alpha)} \approx C\alpha^2 = \frac{1}{32} \times (2 \times 10^{-2})^2$$

$$= 1.25 \times 10^{-5}$$

3. (b) HCl completely dissociates to give H^+ and Cl^- ions, hence act as very good electrolyte. While others are non-electrolytes.

4. (a) $\Delta G = -nFE_{cell}^{\circ} = -2 \times 96500 \times 1.1 \text{ J} = 212.3 \text{ kJ}$.

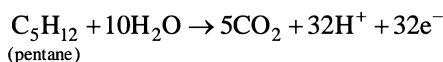
5. (d)

6. (a) $\Lambda_m^{\infty} = 57 + 73 = 130 \text{ Scm}^2 \text{mol}^{-1}$

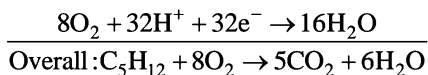
7. (d) Kohlrausch's Law states that at infinite dilution, each ion migrates independently of its co-ion and contributes to the total equivalent conductance of an electrolyte a definite share which depends only on its own nature.

From this definition we can see that option (d) is the correct answer.

8. (c) Writing the equation for pentane-oxygen fuel cell at respective electrodes and overall reaction, we get
At Anode:



At Cathode:



Calculation of ΔG° for the above reaction

$$\Delta G^{\circ} = [5 \times (-394.4) + 6 \times (-237.2)] - [-8.2]$$

$$= -1972.0 - 1423.2 + 8.2 = -3387.0 \text{ kJ}$$

$$= -3387000 \text{ Joules.}$$

From the equation we find $n = 32$

Using the relation, $\Delta G^{\circ} = -nFE_{cell}^{\circ}$ and substituting various values, we get

$$-3387000 = -32 \times 96500 \times E_{cell}^{\circ} \quad (F = 96500C)$$

$$\text{or } E_{cell}^{\circ} = \frac{3387000}{32 \times 96500}$$

$$= \frac{3387000}{3088000} \text{ or } \frac{3387}{3088} \text{ V} = 1.0968 \text{ V}$$

Thus option (c) is correct answer.

9. (d) Standard Gibbs free energy is given as $\Delta G^{\circ} = -nE^{\circ}F$

If $E_{cell}^{\circ} < 0$ i.e., -ve

$$\Delta G^{\circ} > 0$$

Further $\Delta G^{\circ} = -RT \ln K_{eq}$

$$\therefore \Delta G^{\circ} > 0 \text{ and } K_{eq} < 1$$

10. (b) Fe^{+2}/Fe ; $E^{\circ} = -0.44$

$$Fe^{+3}/Fe^{+2} ; E^{\circ} = 0.77$$

The metals having higher negative electrode potential values can displace metals having lower values of negative electrode potential from their salt solutions.

11. (d) $\frac{W_A}{E_A} = \frac{W_B}{E_B}$; $\frac{1.6}{8} = \frac{\text{Wt. of Ag}}{108}$

$$\therefore \text{Wt. of Ag} = 21.6 \text{ g}$$

12. (a) $E_{cell} = E_{right} (\text{cathode}) - E_{left} (\text{anode})$

13. (c) $CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$$\Delta G_r = [\Delta G_f(CO_2, g) + 2\Delta G_f(H_2O, l)] -$$

$$\left[\Delta G_f(CH_3OH, l) + \frac{3}{2}\Delta G_f(O_2, g) \right]$$

$$= -394.4 + 2(-237.2) - (-166.2) - 0$$

$$= -394.4 - 474.4 + 166.2 = -702.6 \text{ kJ}$$

$$\% \text{ efficiency} = \frac{702.6}{726} \times 100 = 97\%$$

14. (b) $\Delta G = -nE^{\circ}F$

$$\text{For concentration cell, } E = \frac{RT}{nF} \ln \frac{C_2}{C_1}$$

In it R, T, n and F are constant

So E is based upon $\ln C_2 / C_1$

$$\text{Now } \Delta G = -nEF = -nF \times \frac{RT}{nF} \ln C_2 / C_1$$

$$= -RT \ln C_2 / C_1$$

At constant temperature ΔG is based upon $\ln (C_2 / C_1)$.

15. (a) 112 ml of H_2 at STP = $\frac{2 \times 112 \text{ g}}{22400}$ (Since 22400 ml at STP

$$= \text{M.wt})$$

$$\text{Amount deposited} = \frac{\text{Eq. wt} \times i \times t}{96500}$$

$$\therefore \frac{2 \times 112}{22400} = \frac{1 \times 965 \times i}{96500};$$

$$i = 1 \text{ amp}$$

16. (b) For $\text{Zn}^{2+} \rightarrow \text{Zn}$

$$E_{\text{Zn}^{2+}/\text{Zn}} = E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} - \frac{2.303RT}{nF} \log \left[\frac{[\text{Zn}]}{[\text{Zn}^{2+}]} \right]$$

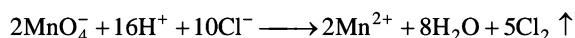
$$= -0.76 - \frac{0.06}{2} \log \frac{1}{[0.1]} = -0.76 - 0.03$$

$$E_{\text{Zn}^{2+}/\text{Zn}} = -0.79 \text{ V}$$

17. (a) The given order of reduction potentials (or tendencies) is $Z > Y > X$. A spontaneous reaction will have the following characteristics
Z reduced and Y oxidised
Z reduced and X oxidised
Y reduced and X oxidised
Hence, Y will oxidise X and not Z.

18. (b) For, $\text{M}^+ + \text{X}^- \longrightarrow \text{M} + \text{X}$, $E_{\text{cell}}^{\circ} = 0.44 - 0.33 = 0.11 \text{ V}$ is positive, hence reaction is spontaneous.

19. (a) MnO_4^- will oxidise Cl^- ion according to the equation



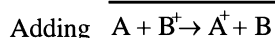
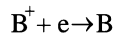
The cell corresponding to this reaction is as follows:



$$E_{\text{cell}}^{\circ} = 1.51 - 1.40 = 0.11 \text{ V}$$

E_{cell}° being +ve, ΔG° will be -ve and hence the above reaction is feasible. MnO_4^- will not only oxidise Fe^{2+} ion but also Cl^- ion simultaneously.

20. (c) The cell reaction is as follows :



21. (b) Conductivity (X) = conductance (c) $\times$ cell constant

$$\therefore \text{Cell constant} = \frac{X}{Y}$$

$$\text{Conductivity of NaOH} = \frac{X}{Y} \cdot Z$$

$$\Delta m(\text{NaOH}) = \frac{X}{Y} \cdot Z \times \frac{1000}{0.1} = \frac{XZ}{Y} 10^4$$

22. (c) From the given data we find Fe^{3+} is strongest oxidising agent. More the positive value of E° , more is the tendency to get oxidized. Thus correct option is (c).
23. (c) At infinite dilution each ion makes a definite contribution towards molar conductance which is given by
 $\Lambda_m^{\circ} = \nu_+ \lambda_+ + \nu_- \lambda_-$
24. (a) In case of equivalent conductance of strong electrolyte there is little increase with dilution.
25. (d) A device that converts energy of combustion of fuels, directly into electrical energy is known as fuel cell.

26. (d) $E = E^{\circ} - \frac{0.0592}{2} \log \frac{[\text{Mn}^{2+}]}{[\text{H}^+]^4}$

$$= E^{\circ} - \frac{0.0592 \times 4}{2} \log \frac{[\text{Mn}^{2+}]^{1/4}}{[\text{H}^+]}$$

$$= E^{\circ} - 0.0592 \times 2 (\log [\text{Mn}^{2+}]^{1/4} + \text{pH})$$

$$\Delta E = E_2 - E_1 = 0.0592 \times 2 (\text{pH}_1 - \text{pH}_2)$$

$$= 0.118 \times 1$$

$$= 0.118 \text{ V}$$

27. (d) Here $n = 4$, and $[\text{H}^+] = 10^{-3}$ (as $\text{pH} = 3$)
Applying Nernst equation

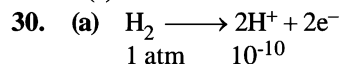
$$E = E^{\circ} - \frac{0.059}{n} \log \frac{[\text{Fe}^{2+}]^2}{[\text{H}^+]^4 (p_{\text{O}_2})}$$

$$= 1.67 - \frac{0.059}{4} \log \frac{(10^{-3})^2}{(10^{-3})^4 \times 0.1}$$

$$= 1.67 - \frac{0.059}{4} \log 10^7 = 1.67 - 0.103 = 1.567 \text{ V}$$

28. (b) In $\text{H}_2 - \text{O}_2$ fuel cell, the combustion of H_2 occurs to create potential difference between the two electrodes

29. (a)



$$E_{\text{H}_2/\text{H}^+} = 0 - \frac{0.059}{2} \log \frac{(10^{-10})^2}{1}$$

$$E_{\text{H}_2/\text{H}^+} = +0.59 \text{ V}$$

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1. (b) $k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303R} \cdot \frac{1}{T}$$

Plot of $\log k$ Vs. $\frac{1}{T}$

$$\text{Straight line Slope} = \frac{-E_a}{2.303R}$$

2. (d) Rate constant $k = 0.6 \times 10^{-3}$ mole per second. (unit $\text{mol L}^{-1}\text{S}^{-1}$ shows zero order reaction)

For a zero order reaction

$$[A] = [A]_0 - kt$$

$$\text{and } [A]_0 - [A] = [B] = kt$$

$$= 0.6 \times 10^{-3} \times 20 \times 60 = 0.72 \text{ M}$$

3. (c) $t_{1/2} \propto \frac{1}{(p)^{n-1}}$, where n is the order of reaction

$$\frac{2}{4} = \left(\frac{50}{100}\right)^{n-1} \quad \text{or} \quad \frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$$

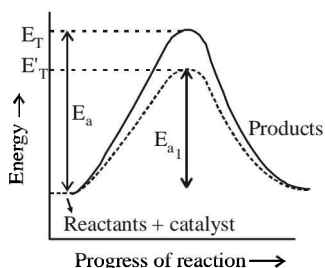
$$\therefore n = 2$$

4. (b) Let x torr of $\text{C}_2\text{H}_4\text{O}$ decompose after 20 min. Then, $80 - x + 2x = 120$; $x = 40$ torr = 50% of initial pressure. Hence $t_{1/2} = 20$ min. For 75% reaction, fraction left

$$= \frac{25}{100} = \frac{1}{4} = \left(\frac{1}{2}\right)^2$$

No. of half lives = 2. Time needed for 75% reaction., $2 \times 20 = 40$ min

5. (b) The presence of enzyme (catalyst) increases the speed of reaction by lowering the energy barrier, i.e. a new path is followed with lower activation energy.



Here E_T is the threshold energy.

E_a and E_{a1} is energy of activation of reaction in absence and presence of catalyst respectively.

6. (c) $\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$; $\frac{120}{240} = \left(\frac{4 \times 10^{-2}}{8 \times 10^{-2}}\right)^{n-1}$; $n = 2$

7. (c) For a first order reaction

$$k = \frac{2.303}{t} \log_{10} \frac{a}{a-x}$$

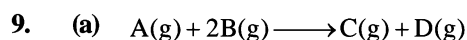
$$\text{when } t = t_{1/2}$$

$$k = \frac{2.303}{t_{1/2}} \log_{10} \frac{a}{a-a/2}$$

$$\text{or } t_{1/2} = \frac{2.303}{k} \log_{10} 2 = \frac{\ln 2}{k}$$

8. (b) Rate = $kC_A^2 C_B^{1/2}$

$$\text{Rate} = k[2C_A]^2 [4C_B]^{1/2} = 4 \times 2 [C_A]^2 [C_B]^{1/2}$$



$$\text{Rate} = k[A][B]^2 = k(0.60)(0.80)^2 \dots\dots\dots(i)$$

when $p_c = 0.20$ atm p_A is reduced to 0.40 and $p_B = 0.40$ (See stoichiometric representation)

$$\text{Rate} = k[0.40][0.40]^2 \dots\dots\dots(ii)$$

$$\therefore (ii) \text{ divide by } (i) = \frac{0.40 \times 0.40 \times 0.40}{0.60 \times 0.80 \times 0.80} = \frac{1}{6}$$

10. (a) From data 1 and 3, it is clear that keeping (B) const, When [A] is doubled, rate remains unaffected. Hence rate is independent of [A]. from 1 and 4, keeping [A] constant, when [B] is doubled, rate become 8 times. Hence

$$\text{rate} \propto [B]^3.$$

11. (a) Since initial velocity is ten times the permissible value

$$\therefore A_0 = 10A$$

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{30}$$

$$t_{1/2} = \frac{2.303}{\lambda} \log \frac{A_0}{A} = \frac{2.303}{0.693/30} \log \frac{10A}{A} = \frac{2.303 \times 30}{0.693} \times \log 10 = 100 \text{ days.}$$

12. (a) Presence of catalyst does not affect enthalpy change of reaction $\Delta H_R = E_f - E_b = 180 - 200 = -20 \text{ kJ/mol}$

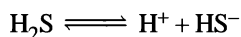
13. (d) Since the slow step is the rate determining step hence if we consider option (A) we find

$$\text{Rate} = k[\text{Cl}_2][\text{H}_2\text{S}]$$

Now if we consider option (B) we find

$$\text{Rate} = k[\text{Cl}_2][\text{HS}^-] \quad \dots(i)$$

For equation,



$$K = \frac{[\text{H}^+][\text{HS}^-]}{[\text{H}_2\text{S}]}$$

$$\text{or } [\text{HS}^-] = \frac{K[\text{H}_2\text{S}]}{[\text{H}^+]}$$

Substituting this value in equation (i) we find

$$\text{Rate} = k[\text{Cl}_2]K \frac{[\text{H}_2\text{S}]}{[\text{H}^+]} = k' \frac{[\text{Cl}_2][\text{H}_2\text{S}]}{[\text{H}^+]}$$

Thus slow step should involve 1 molecule of Cl_2 and 1 molecule of H_2S .

hence only, mechanism (A) is consistent with the given rate equation.

14. (d) From 1st and 2nd sets of data - no change in rate is observed with the change in concentration of 'C'. So the order with respect to 'C' is zero.

From 1st and 4th sets of data

Dividing eq. (4) by eq. (1)

$$\frac{1.25 \times 10^{-3}}{5.0 \times 10^{-3}} = \left[\frac{0.005}{0.010} \right]^x$$

$$\text{or } 0.25 = (0.5)^x \text{ or } (0.5)^2 = (0.5)^x$$

$$\therefore x = 2$$

The order with respect to 'A' is 2 from the 1st and 3rd sets of data dividing eq. (1) by eq. (3)

$$\frac{5.0 \times 10^{-3}}{1.0 \times 10^{-2}} = \left[\frac{0.005}{0.010} \right]^y$$

$$\text{or } (0.5)^1 = (0.5)^y \Rightarrow y = 1$$

The order with respect to 'B' is 1

So the order with respect to the reactants A, B and C is 2, 1 and 0.

15. (b) $\Delta H = E_{a(f)} - E_{a(b)}$

Thus energy of activation for reverse reaction depend upon whether reaction is exothermic or endothermic

If reaction is exothermic,

$$\Delta H = -ve \quad E_{a(b)} > E_{a(f)}$$

If reaction is endothermic

$$\Delta H = +ve \quad E_{a(b)} < E_{a(f)}$$

16. (b) For a first order reaction

$$t_{1/2} = \frac{0.693}{k}; \quad k = \frac{0.693}{1386} = 0.5 \times 10^{-3} \text{ s}^{-1}$$

17. (c) $t_{1/4} = \frac{2.303}{k} \log \frac{1}{3/4} = \frac{2.303}{k} \log \frac{4}{3}$
- $$= \frac{2.303}{k} (\log 4 - \log 3) = \frac{2.303}{k} (2 \log 2 - \log 3)$$
- $$= \frac{2.303}{k} (2 \times 0.301 - 0.4771) = \frac{0.29}{k}$$

18. (b,d)

According to Arrhenius equation

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$= - \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{k_1}{k_2} = - \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

19. (d) Overall order = sum of orders w.r.t each reactant.
Let the order be x and y for G and H respectively

Exp.No.	[G]mole litre ⁻¹	[H]mole litre ⁻¹	rate(mole litre ⁻¹ time ⁻¹)
1	a	b	r
2	$2a$	$2b$	$8r$
3	$2a$	b	$2r$

$\therefore$ For (1) and (3), the rate is doubled when conc. of G is doubled keeping that of H constant i.e.,

$$\text{rate} \propto [\text{G}] \quad \therefore x = 1$$

From (2) and (3), $y = 2$

$\therefore$ Overall order is 3.

20. (c) $\text{Rate}_1 = k[\text{A}]^n[\text{B}]^m$; $\text{Rate}_2 = k[2\text{A}]^n[\frac{1}{2}\text{B}]^m$

$$\therefore \frac{\text{Rate}_2}{\text{Rate}_1} = \frac{k[2\text{A}]^n[\frac{1}{2}\text{B}]^m}{k[\text{A}]^n[\text{B}]^m}$$

$$= [2]^n [\frac{1}{2}]^m = 2^n \cdot 2^{-m} = 2^{n-m}$$

21. (d) Given, $k_1 = 10^{16} \cdot e^{-\frac{2000}{T}}$

$$\text{and } k_2 = 10^{15} \cdot e^{-\frac{1000}{T}}$$

when k_1 and k_2 are equal at any temperature T , we have

$$10^{16} \cdot e^{-\frac{2000}{T}} = 10^{15} \cdot e^{-\frac{1000}{T}}$$

$$\text{or } 10 \times 10^{15} \cdot e^{-\frac{2000}{T}} = 10^{15} \cdot e^{-\frac{1000}{T}}$$

$$\text{or } 10 \cdot e^{-\frac{2000}{T}} = e^{-\frac{1000}{T}}$$

$$\text{or } \ln 10 - \frac{2000}{T} = -\frac{1000}{T}$$

$$\text{or } \ln 10 = \frac{2000}{T} - \frac{1000}{T}$$

$$\text{or } 2.303 \log 10 = \frac{1000}{T}$$

$$\text{or } 2.303 \times 1 \times T = 1000 \quad [\because \log 10 = 1]$$

$$\text{or } T = \frac{1000}{2.303} \text{ K}$$

22. (b)

23. (a) Arrhenius equation is given by

$$k = A e^{-E_a/(2.303RT)}$$

Taking log on both sides, we get

$$\log k = \log A - \frac{E_a}{2.303RT}$$

Arrhenius plot a graph between $\log k$ and $\frac{1}{T}$ whose

$$\text{slope is } \frac{-E_a}{2.303R}.$$

24. (a) When concentration of A is doubled, rate is doubled.
Hence order with respect to A is one.

When concentrations of both A and B are doubled, rate increases by 8 times hence order with respect to B is 2.

$$\therefore \text{rate} = k [A]^1 [B]^2$$

25. (b) $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$

From the unit of rate constant it is clear that the reaction follow first order kinetics. Hence

by rate law equation, $r = k [\text{N}_2\text{O}_5]$

$$\text{where } r = 1.02 \times 10^{-4}, k = 3.4 \times 10^{-5}$$

$$1.02 \times 10^{-4} = 3.4 \times 10^{-5} [\text{N}_2\text{O}_5]$$

$$[\text{N}_2\text{O}_5] = 3\text{M}$$

26. (c) Third order

27. (a) If we write rate of reaction in terms of concentration of NH_3 and H_2 , then

$$\text{Rate of reaction} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt}$$

$$\text{So, } \frac{d[\text{NH}_3]}{dt} = -\frac{2}{3} \frac{d[\text{H}_2]}{dt}$$

28. (a) $E_{a(\text{F.R.})} \neq E_{a(\text{B.R.})}$ E_a can be calculated.

29. (b) For reaction $3\text{A} \longrightarrow \text{B} + \text{C}$

If it is zero order reaction $r = k [\text{A}]^0$, i.e. the rate remains same at any concentration of 'A'. i.e. independent upon concentration of A.

$$30. \text{ (d) } k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{24} \log \frac{1}{\frac{1}{8}} = \frac{2.303}{24} \log 8$$

1. (a) Smaller the charge on anion, lesser will be its coagulating power.
 $\therefore$ KBr have Br^- with least charge of -1 on Br thus KBr is least effective in coagulating $\text{Fe}(\text{OH})_3$.

2. (c)

3. (b)

4. (d)

5. (b) According to Freundlich equation.

$$\frac{x}{m} \propto p^{1/n} \text{ or } \frac{x}{m} = kp^{1/n}$$

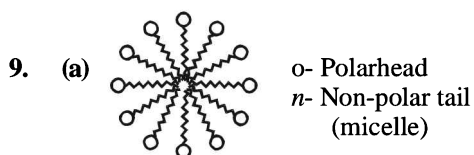
$$\text{or } \log \frac{x}{m} = \log kp^{1/n} \text{ or } \log \frac{x}{m} = \log k + \frac{1}{n} \log p$$

6. (b) 1 micron $\mu = 10^{-6}$ m and 1 millimicron $= 10^{-7}$ m.

7. (d) All are irreversible colloidal systems.

8. (d) Cetyl trimethyl ammonium bromide,

$[\text{C}_{16}\text{H}_{33}(\text{CH}_3)_3\text{N}^+\text{Br}^-]$ is a cationic micelle



10. (c) It is due to impact of molecules of dispersion medium on the colloidal particles.

11. (c)

12. (c) The movement towards anode shows that sol is negative.
For coagulation of negative sol. Cation with higher charge is more effective.

13. (d) $\text{Fe}(\text{OH})_3$ is positive sol. $\text{K}_3[\text{Fe}(\text{CN})_6]$ will provide $[\text{Fe}(\text{CN})_6]^{3-}$ for coagulation.

14. (c) Alum. coagulates mud particles and helps in purifying water.

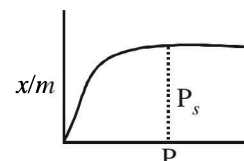
15. (a) On increasing temperature adsorption of a gas on surface of solid decreases. Solid adsorb greater amount of substances at lower temperature.

16. (c) For physisorption the ΔH value is 10 - 40 kJ/mol.

17. (a) On shaking with the dispersion medium, colloids directly form the colloidal sol. Hence they are called intrinsic colloids. i.e., glue.

18. (d) According to Freundlich adsorption isotherm.
At intermediate pressure, extent of adsorption

$$\frac{x}{m} = kP^{1/n} \text{ or } \log \frac{x}{m} = \log k + \frac{1}{n} \log P;$$



plot of $\log \frac{x}{m}$ vs $\log P$ is linear with slope $= \frac{1}{n}$

19. (d) Tannin used in leather industry contains negatively charged colloidal particles.

20. (b) When oppositely charged sols are mixed their charges are neutralised. Both sols may be partially or completely precipitated.

21. (b)

22. (d)

23. (a) According to Langmuir's model of adsorption of a gas on a solid surface the mass of gas adsorbed (x) per gram of the adsorbent (m) is directly proportional to the pressure of the gas (p) at constant temperature i.e.

$$\frac{x}{m} \propto p$$

24. (c) The more the liquefiable nature of a gas, the more is the enthalpy of adsorption. Water is more liquefiable.

25. (a) A catalyst speeds up both forward and backward reaction with the same rate.

So, equilibrium constant is not affected by the presence of a catalyst at any given temperature.

26. (c) As Sb_2S_3 is a negative sol, so $\text{Al}_2(\text{SO}_4)_3$ will be the most effective coagulant due to higher positive charge on Al (Al^{3+}) – **Hardy-Schulze rule**.

27. (d)

28. (d) At high pressure the extent of adsorption follows zero order kinetics.

29. (c) The critical micelle concentration is the lowest concentration at which micelle formation appears. When surfactants are present above that CMC, they can act as emulsifiers that will solubilise a compound which is normally insoluble in the solvent being used.

30. (b) Peptisation is disintegration of colloidal aggregate.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC20

1. (a) The conversion of metal sulphide to metal oxide involves the process of **roasting** (i.e., x is roasting).
The metal oxides can then be converted to impure metal by reduction. Of the given choices in (a) and (b) the reduction process is that of **smelting** (i.e., 'y' is smelting).
The conversion of impure metal to pure metal involves a process of purification. Thus it is electrolysis.
2. (b)
3. (c)
4. (c)
5. (d)
6. (b)
7. (a) Dolomite on calcination gives CaO, MgO which provides basic lining in furnace.
8. (c)
9. (c) Zn being more reactive than Ag and Au, displaces them.

$$4\text{Ag} + 8\text{NaCN} + 2\text{H}_2\text{O} + \text{O}_2 \xrightarrow{\text{Leaching}} 4\text{Na}[\text{Ag}(\text{CN})_2] + 4\text{NaOH}$$

Soluble Sodium dicyanoargentate (I)

Soluble cyanide compound can be treated with Zn to give metal by displacement.

$$2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \xrightarrow{\text{Displacement}} \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}\downarrow$$
10. (d) Decomposition of carbonates and hydrated oxides.
11. (c)
12. (b) Pig iron or cast iron contains 3–5% carbon and varying amounts of Mn, Si, P and S which makes the iron hard and brittle.
13. (b) Cuprite is Cu_2O and Argentite is Ag_2S .
14. (b) Cyanide process is for gold (I-D); floatation process - pine oil (II-B); Electrolytic reduction - Al (III-C); Zone refining - Ge (IV-A).
15. (b) For a spontaneous reaction, ΔG° must be negative and it can be possible only in this case when $x < y$
16. (c) Fused alumina (Al_2O_3) is a bad conductor of electricity. Therefore, cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added to purified alumina which not only make alumina a good conductor of electricity but also reduce the melting point of the mixture to around 1140 K.
17. (d) The sulphide ore is roasted to oxide before reduction because the ΔG_f° of most of the sulphides are greater than those of CS_2 and H_2S , therefore neither C nor H can reduce metal sulphide to metal. Further, the standard free energies of formation of oxide are much less than those of SO_2 . Hence oxidation of metal sulphides to metal oxide is thermodynamically favourable.
18. (a) Cuprous oxide formed during roasting of cuprous sulphide is mixed with few amount of cuprous sulphide and heated in a reverberatory furnace to get metallic copper.

$$2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$$
19. (a) Zone refining is based on the difference in solubility of impurities in molten and solid state of the metal. This method is used for obtaining metals of very high purity.
20. (b) The reactions involved in cyanide extraction process are :

$$\text{Ag}_2\text{S} + 4\text{NaCN} \rightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$$

(argentite)

$$4\text{Na}_2\text{S} + 5\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Na}_2\text{SO}_4 + 4\text{NaOH} + 2\text{S}$$

Oxidizing agent

$$2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}\downarrow$$

(reducing agent)
21. (b) Al_2O_3 cannot be reduced by carbon.
22. (d)
23. (b) The given reaction is the method named as Van Arkel for the purification of titanium
24. (c) Ca and K are strong reducing agents, hence their oxides cannot be reduced with carbon
25. (b) Zr and Ti are purified by van Arkel method.

$$\text{Zr(s)} + 2\text{I}_2(\text{g}) \longrightarrow \text{ZrI}_4(\text{g})$$

$$\text{ZrI}_4(\text{g}) \xrightarrow[\text{filament}]{\text{On the hot}} \text{Zr(s)} + 2\text{I}_2(\text{g})$$
26. (c) (iii) and (iv) reactions occur in the temperature range of 900–1500K in blast furnace.
27. (d) Zincite is ZnO .
28. (a) FeO is capable forming slag with SiO_2

$$\text{SiO}_2 + \text{FeO} \rightarrow \text{FeSiO}_3$$
29. (d)
30. (b) The two equation are:

$$\frac{4}{3}\text{Al(s)} + \text{O}_2(\text{g}) \rightarrow \frac{2}{3}\text{Al}_2\text{O}_3(\text{s}), \Delta_f G^\circ = -827\text{kJ mol}^{-1} \quad \dots(1)$$

$$\frac{4}{3}\text{Cr(s)} + \text{O}_2(\text{g}) \rightarrow \frac{2}{3}\text{Cr}_2\text{O}_3(\text{s}), \Delta_f G^\circ = -540\text{kJ mol}^{-1} \quad \dots(2)$$

Subtracting equation (ii) from equation (i) we have,

$$\frac{4}{3}\text{Al(s)} + \frac{2}{3}\text{Cr}_2\text{O}_3(\text{s}),$$

$$\rightarrow \frac{2}{3}\text{Al}_2\text{O}_3(\text{s}) + \frac{4}{3}\text{Cr(s)}, \Delta_r G^\circ = -287\text{kJ mol}^{-1}$$

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC21

- (a) Hypophosphorous acid is H_3PO_2 in which O.S. of P is +1
- (b) The bond angle decreases on moving down the group due to decrease in bond pair-bond pair repulsion.

NH_3	PH_3	AsH_3	SbH_3	BiH_3
107°	94°	92°	91°	90°

This can also be explained by the fact that as the size of central atom increases sp^3 hybrid orbital becomes more distinct with increasing size of central atom i.e. pure p-orbitals are utilized in M-H bonding.

- (b) $\text{CuSO}_4 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

Blue complex due to $\text{Cu}(\text{NH}_3)_4^{2+}$

- (a)
- (c) The nitroprusside ion is $[\text{Fe}(\text{CN})_6\text{NO}]^{2-}$. The magnetic moment measurements reveal the presence of 4 unpaired electrons in Fe which must be then in Fe^{++} ($3d^6$) and not Fe^{+++} ($3d^5$)
- (c) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ ion is formed
- (d)
- (a) $4\text{HCl} + \text{O}_2 \rightarrow 2\text{Cl}_2 + 2\text{H}_2\text{O}$

air cloud of white fumes

- (a) ICl_7 . The hybridisation is $\frac{1}{2}(7+7+0-0) = 7$ (sp^3d^3)
- (a) Due to large enthalpy of vaporisation SO_2 can be used as refrigerant
- (c) Oxides which are more ionic in nature (salt-like) are known as saline oxides e.g. oxides of alkali metals
- (b) Among the given compounds, the NH_3 is most basic. Hence has highest proton affinity
- (c) Helium is heavier than hydrogen although it is non-inflammable
- (c) It is H_2SO_5 .

- (b) $\text{HO}-\text{P}(\text{OH})_2-\text{H}$ it can form two series of salts by replacement of H attached to oxygen

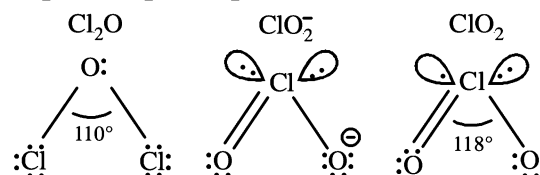
- (b) RCOO^- is not pseudo halide.
- (d) In gaseous state the HCl is covalent in nature while in aqueous solution it ionises to give H^+ and Cl^- ions

- (c) Hypophosphorous acid $\text{H}-\text{O}-\text{P}(\text{H})_2-\text{O}-\text{H}$

Two H-atoms are attached to P atom.

- (b) During disproportionation same compound undergo simultaneous oxidation and reduction.

- (b) $3\text{HOCl} \xrightarrow{\text{Oxidation}} 2\text{HCl} + \text{HClO}_3$ reduction
- (b) XeOF_4 is square pyramidal.
- (c) Yellow ammonium sulphide is $(\text{NH}_4)_2\text{S}_x$
- (c) The correct order of increasing bond angle is $\text{Cl}_2\text{O} < \text{ClO}_2^- < \text{ClO}_2$



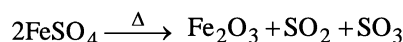
In ClO_2^- there are 2 lone pairs of electrons present on the central chlorine atom. Therefore the bond angle in ClO_2^- is less than 118° which is the bond angle in ClO_2 which has less number of electrons on central chlorine atom.

- (d) The products of the concerned reaction react each other forming back the reactants.
 $\text{XeF}_6 + 3\text{H}_2\text{O} \rightarrow \text{XeO}_3 + 6\text{HF}$
- (b) The hybridization of XeO_3F_2 is sp^3d and its structure is trigonal bipyramidal in which oxygen atoms are situated on the plane and the fluoride atoms are on the top and bottom.

- (d)

- (a) $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$
Tetrathionate

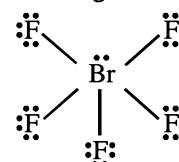
- (c) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \rightarrow \text{FeSO}_4 + 7\text{H}_2\text{O}$



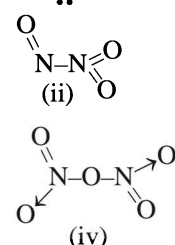
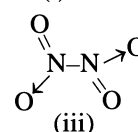
- (d) $\text{HOCl} < \text{HOClO} < \text{HOClO}_2 < \text{HOClO}_3$

In case of oxyacids of similar element as the oxidation number of the central atom increases, strength of acid also increases.

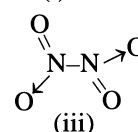
- (b) The molecular geometry of BrF_5 is square pyramidal with asymmetric charge distribution on the central atom.



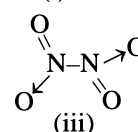
- (d) $\text{N} \equiv \text{N} \rightarrow \text{O}$
(i)



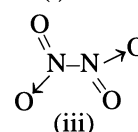
- (d) $\text{N} \equiv \text{N} \rightarrow \text{O}$
(ii)



- (d) $\text{N} \equiv \text{N} \rightarrow \text{O}$
(iii)



- (d) $\text{N} \equiv \text{N} \rightarrow \text{O}$
(iv)



DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC22

- (d) $(n-1)d^5ns^2$ attains the maximum O.S. of +7.
 - (b) $\text{Mn}^{2+} (3d^5)$ is more stable than $\text{Mn}^{3+} (3d^4)$.
 - (c)
 - (d) Super conductors are derived from compounds of transition metals
 - (b) $\text{Mn} - 3d^5 4s^2$

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1

The no. of various oxidation states possible are + 2, +3, +4, +5, +6 and +7.
 - (c) The transition metals and their compounds are used as catalysts. Because of the variable oxidation states they may form intermediate compound with one of the reactants. These intermediate provides a new path with low activation energy. $\text{V}_2\text{O}_5 + \text{SO}_2 \rightarrow \text{V}_2\text{O}_4 + \text{SO}_3$
 $2\text{V}_2\text{O}_4 + \text{O}_2 \rightarrow 2\text{V}_2\text{O}_5$
 - (a) According to their positions in the periods, these values are in the order:
 $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$
At. Nos. 70 61 58 57
This is due to lanthanide contraction.
 - (a) When a solution of potassium chromate is treated with an excess of dilute nitric acid. Potassium dichromate and H_2O are formed.
 $2\text{K}_2\text{CrO}_4 + 2\text{HNO}_3 \rightarrow \text{K}_2\text{Cr}_2\text{O}_7 + 2\text{KNO}_3 + \text{H}_2\text{O}$
Hence $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed.
 - (b) $\text{Mn}^{2+} (d^5)$ is more stable than $\text{Mn}^{3+} (d^4)$, thus
 $E^\circ_{\text{Mn}^{3+}/\text{Mn}^{2+}} = +ve$
 - (c) $\text{Mn}_2\text{O}_3 \xleftarrow{-4e^-} [\text{KMnO}_4] \xrightarrow{-e^-} [\text{MnO}_4]^{-1}$
 $\swarrow -5e^- \quad \searrow -3e^-$
 $\text{Mn}^{2+} \quad \text{MnO}_2$
 - (d) O.N. of Cr in all the species is +6. Configuration of Cr (VI) is $(\text{Ar}) 3d^0$. Hence no $d-d$ electron transition.
 - (c) The number of unpaired electrons in $\text{Ni}^{2+}(\text{aq}) = 2$
Water is weak ligand hence no pairing will take place
spin magnetic moment
 $= \sqrt{n(n+2)} = \sqrt{2(2+2)}$
 $= \sqrt{8} = 2.82$
 - (d) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$
self reduction.
 - (b) Eu^{2+} has electronic configuration $[\text{Xe}]4f^7$ hence stable due to half filled atomic orbitals.
 - (c) $\text{AgNO}_3 \rightarrow \text{Ag} + \text{NO}_2 + \frac{1}{2}\text{O}_2$
 - (a) $2\text{AgNO}_3 + \text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Ag}_2\text{S}_2\text{O}_3 + 2\text{NaNO}_3$
 - (b) Oxidation of water takes place in presence of Mn in biological process
 - (a)
(a) $\text{V} = 3d^3 4s^2$; $\text{V}^{2+} = 3d^3 = 3$ unpaired electrons
 $\text{Cr} = 3d^5 4s^1$; $\text{Cr}^{2+} = 3d^4 = 4$ unpaired electrons
 $\text{Mn} = 3d^5 4s^2$; $\text{Mn}^{2+} = 3d^5 = 5$ unpaired electrons
 $\text{Fe} = 3d^6 4s^2$; $\text{Fe}^{2+} = 3d^6 = 4$ unpaired electrons
Hence the correct order of paramagnetic behaviour
 $\text{V}^{2+} < \text{Cr}^{2+} = \text{Fe}^{2+} < \text{Mn}^{2+}$
 - (b) For the same oxidation state, the ionic radii generally decreases as the atomic number increases in a particular transition series. hence the order is
 $\text{Mn}^{++} > \text{Fe}^{++} > \text{Co}^{++} > \text{Ni}^{++}$
 - (c) In solution, the stability of the compound depends upon electrode potentials, SEP of the transition metal ions are given as
 $\text{Co}^{3+}/\text{Co} = +1.97$, $\text{Fe}^{3+}/\text{Fe} = +0.77$;
 $\text{Cr}^{3+}/\text{Cr}^{2+} = -0.41$, Sc^{3+} is highly stable as it does not show +2 O.S.
 - (d) $\text{Sc} - (+2), (+3)$
 $\text{Ti} - (+2), (+3), (+4)$
 $\text{Cr} - (+1), (+2), (+3), (+4), (+5), (+6)$
 $\text{Mn} - (+2), (+3), (+4), (+5), (+6), (+7)$
i.e. $\text{Sc} < \text{Ti} < \text{Cr} = \text{Mn}$
 - (b) In lanthanides, there is poorer shielding of $5d$ electrons by $4f$ electrons resulting in greater attraction of the nucleus over $5d$ electrons and contraction of the atomic radii.
 - (d)
 - (a)
- | | Eu | La | Gd | Am |
|-----|----|----|----|-------------|
| O.S | +2 | +3 | +3 | +3,+4,+5,+6 |
- (a)
 $2\text{KMnO}_4 + \text{H}_2\text{SO}_4 (\text{Conc}) \rightarrow \text{K}_2\text{SO}_4 + \text{Mn}_2\text{O}_7 + \text{H}_2\text{O}$
Explosive
 - (c)
 $\text{AgCl}(\text{s}) + 2\text{NH}_4\text{OH}(\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
 - (a) Paramagnetic moment is directly proportional to number of unpaired electrons present in the complex.

24. (d) d^5 — strong ligand field

 t_{2g} e_g

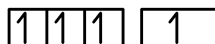
$$\mu = n\sqrt{n+2} = \sqrt{3} = 1.73 \text{ BM}$$

d^3 — in weak as well as in strong field

 t_{2g} e_g

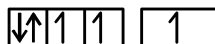
$$\mu = \sqrt{3(5)} = \sqrt{15} = 3.87 \text{ B.M.}$$

d^4 — in weak ligand field

 t_{2g} e_g

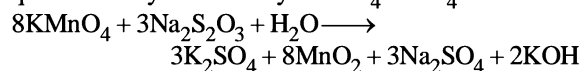
$$\mu = \sqrt{4(8)} = \sqrt{24} = 4.89$$

d^4 — in strong ligand field

 t_{2g} e_g

$$\mu = \sqrt{2(4)} = \sqrt{8} = 2.82$$

25. (b) In neutral or faintly alkaline medium thiosulphate is quantitatively oxidized by KMnO_4 to SO_4^{2-}

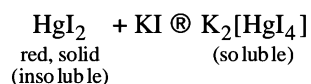


26. (c) Parke's process It is for extraction of silver

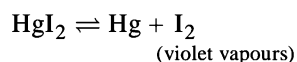
27. (a) Cu, Hg and Ag are attacked by conc acids but gold is not attacked. AuCl_3 forms $\text{H}[\text{AuCl}_4]$ a complex with HCl which is used for toning in photography.

28. (a) $\ddot{\text{N}}\text{H}_3 + \text{H}^+ (\text{acid medium}) \rightleftharpoons \overset{+}{\text{N}}\text{H}_4$

29. (a) When KI is added to mercuric iodide it dissolves in it and forms a complex.



On heating HgI_2 decomposes as



30. (c) $E^\circ_{\text{Cu}^{+2}/\text{Cu}} = 0.34 \text{ V}$

other has –ve $E^\circ_{\text{R.P.}}$

$$E^\circ_{\text{Co}^{++}/\text{Co}} = -0.28 \text{ V}$$

$$E^\circ_{\text{Ni}^{++}/\text{Ni}} = -0.25 \text{ V}$$

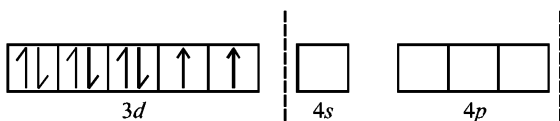
$$E^\circ_{\text{Fe}^{++}/\text{Fe}} = -0.44 \text{ V}$$

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

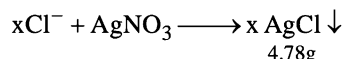
DPP/CC23

- (b) According to spectrochemical series $\text{C}_2\text{O}_4^{2-} > \text{F}^-$.
- (b) CN^- is coordinated to cobalt as the ligand and coordinated compounds have different properties than the individual species.
- (d)
- (d) $[\text{Ni X}_4]^{2-}$, the electronic configuration of Ni^{2+} is



It contains two unpaired electrons and the hybridisation is sp^3 (tetrahedral).

- (b) The crystal field splitting in tetrahedral complexes is lower than that in octahedral complexes, and $\Delta_t = -\frac{4}{9}\Delta_0$.
- (b)
- (a) $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{ClO}_4$. Six monodentate ligands are attached to Co hence C. N. of Co = 6;
O. N. = $x + 5 \times (0) + 1 \times (-2) + 1 \times (-1) = 0 \therefore x = +3$;
electronic configuration of $\text{Co}^{3+}[\text{Ar}] 3d^6 4s^0$ hence number of d electrons is 6. All d electrons are paired due to strong ligand hence unpaired electron is zero.
- (a) EDTA has hexadentate four donor O atoms and 2 donor N atoms and for the formation of octahedral complex one molecule is required
- (c) $[\text{Mn}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are inner orbital complexes and paramagnetic while $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ is diamagnetic in nature.
- (a) $\text{Ti}(\text{C}_2\text{H}_4)_4$ is an organometallic compound due to Ti directly attached to C-atom
- (d) Tetraethyl lead $\text{Pb}(\text{C}_2\text{H}_5)_4$ is not π bonded complex. It is σ bonded organometallic compound
- (a) $\text{CoCl}_3 \cdot 6\text{NH}_3 \longrightarrow x\text{Cl}^-$
2.675g



Number of moles of the complex

$$= \frac{2.675}{267.5} = 0.01 \text{ moles}$$

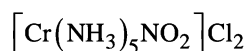
Number of moles of AgCl obtained

$$= \frac{4.78}{143.5} = 0.03 \text{ moles}$$

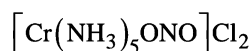
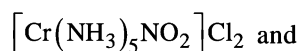
$$\therefore \text{No. of moles of AgCl obtained} = 3 \times \text{No. of moles of complex}$$

$$\therefore n = \frac{0.03}{0.01} = 3$$

- (b) The chemical formula of Pentamminenitrochromium (III) chloride is

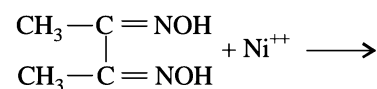


It can exist in following two structures

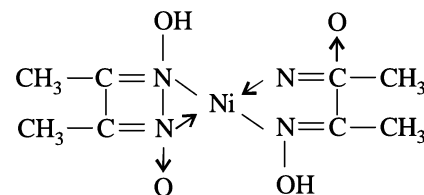


Therefore the type of isomerism found in this compound is linkage isomerism as nitro group is linked through N as $-\text{NO}_2$ or through O as $-\text{ONO}$.

- (c) Nickel ions are frequently detected by the formation of red precipitate of the complex of nickel dimethylglyoxime, when heated with dimethylglyoxime.



Dimethylglyoxime



Nickel dimethylglyoxime

- (d) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} + \text{AgNO}_3 \longrightarrow \text{AgCl} + [\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{NO}_3$

$$\text{Molarity} = \frac{\text{wt}}{\text{mol. mass}} \times \frac{1000}{\text{vol.}}$$

$$\frac{\text{wt}}{\text{mol. mass}} = \text{molarity} \times \frac{\text{vol.}}{1000} = \frac{0.01 \times 100}{1000}$$

$$= 0.001$$

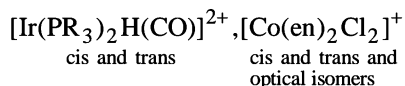
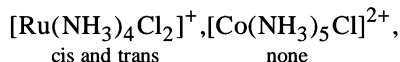
- (c) $\text{Gd} (64) = [\text{Xe}] 4f^7 5d^1 6s^2$
 $\text{Gd}^{3+} = [\text{Xe}] 4f^7 5d^0 6s^0$
i.e. no. of unpaired electrons = 7

$$\mu = \sqrt{n(n+2)} = \sqrt{7(7+2)}$$

$$= \sqrt{63} = 7.93 \text{ BM}$$

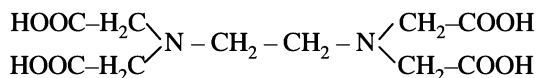
17. (b)

18. (d) isomers



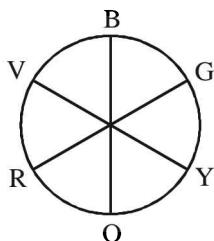
19. (d) Due to some backbonding by sidewise overlapping between d-orbitals of metal and p-orbital of carbon, the F-C bond has σ and π character.

20. (c) The correct structure of EDTA is

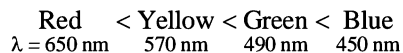


21. (a) $[\text{PdCl}_4]^{2-}$ is dsp^2 hybridized and square planar in shape.

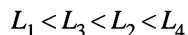
22. (b)



For a given metal ion, weak field ligands create a complex with smaller Δ , which will absorb light of longer λ and thus lower frequency. Conversely, stronger field ligands create a larger Δ , absorb light of shorter λ and thus higher v.i.e. higher energy.

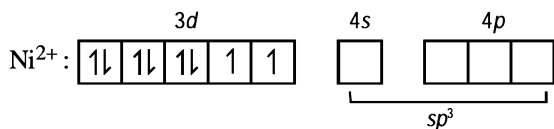


So order of ligand strength is



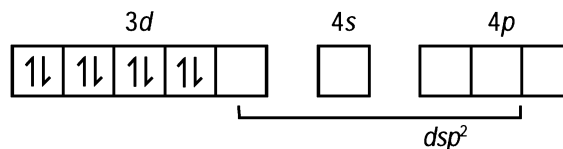
23. (d)

24. (c) In both states (paramagnetic and diamagnetic) of the given complex, Ni exists as Ni^{2+} whose electronic configuration is $[\text{Ar}] 3d^8 4s^0$.

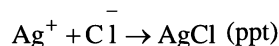
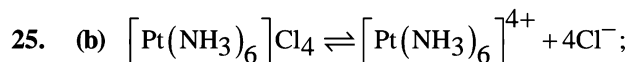


In the above paramagnetic state the geometry of the complex is sp^3 giving tetrahedral geometry.

The diamagnetic state is achieved by pairing of electrons in 3d orbital.

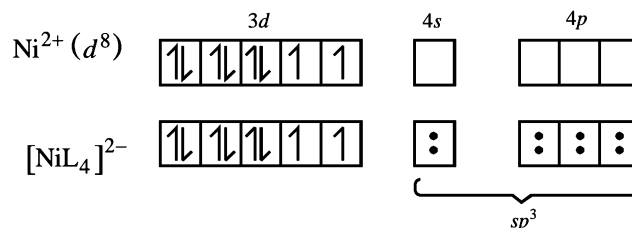


Thus the geometry of the complex will be dsp^2 giving square planar geometry.



26. (b) The correct IUPAC Name of $[\text{Mn}(\text{CN})_5]^{2-}$.
Pentacyanomanganate (III).

27. (a) $[\text{NiL}_4]^{2-}$



i.e, no. of unpaired electron = 2
hybridization – sp^3 .

28. (b)

29. (c) In $[\text{Co}(\text{CO})_5\text{NH}_3]^{2+}$ CO is bonded to cobalt through σ and π bonding.

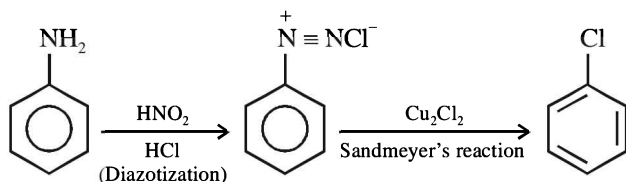
30. (d) $\text{K}_2[\text{Ni}(\text{EDTA})]$. Since EDTA is hexadentate and chelating and coordinates from six sides forming more stable complex.

DAILY PRACTICE PROBLEMS

CHEMISTRY SOLUTIONS

DPP/CC24

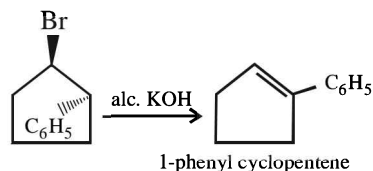
1. (c)



2. (b)

3. (a)

4. (a) The reaction is dehydrohalogenation

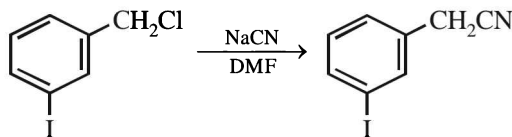


5. (a) Only 1° alkyl halides (i.e. CH_3Br) undergo $\text{S}_{\text{N}}2$ reaction.

6. (a)

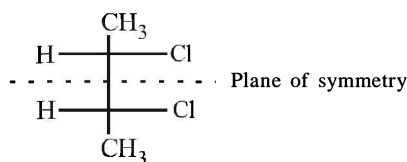
7. (d) $-\text{Cl}$ is *o*, *p*-directing.

8. (b)



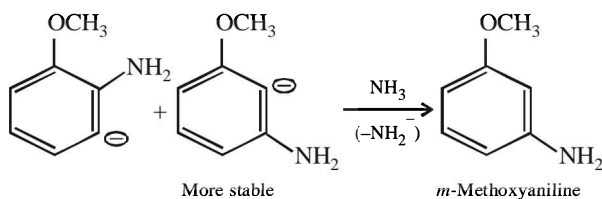
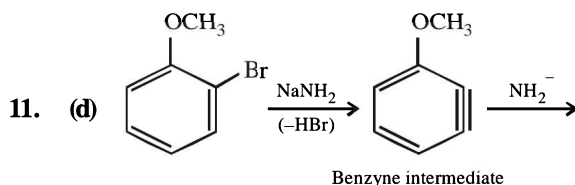
Nuclear substitution will not take place.

9. (b) The compound has two similar asymmetric C-atoms. It has plane of symmetry and exist in meso form.

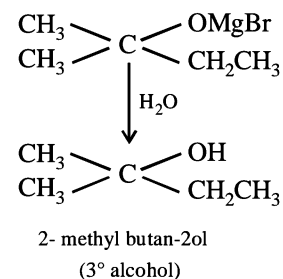
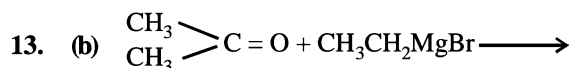


Meso - 2, 3 dichlorobutane

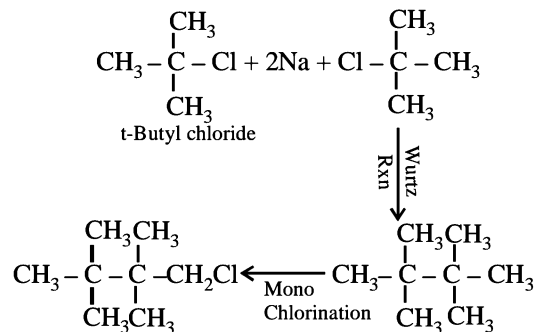
10. (d)



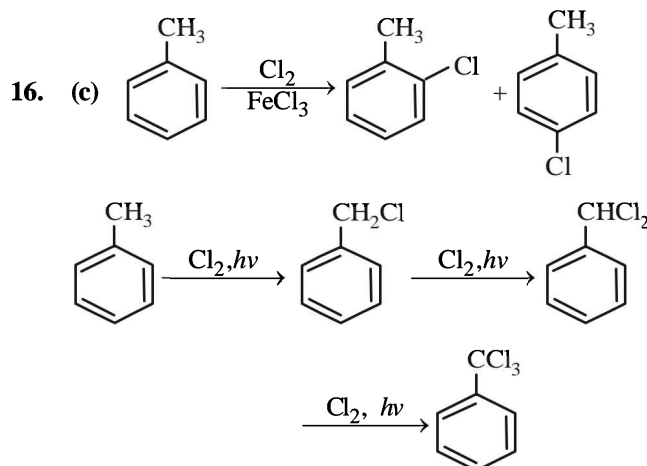
12. (c) CH_3OH does not undergo iodoform reaction, all others do so.



14. (a)

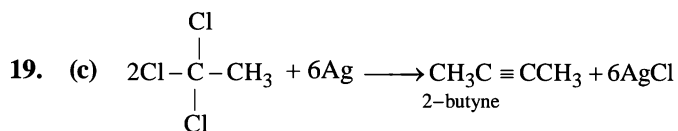
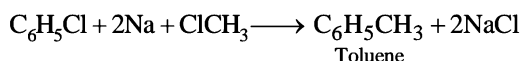


15. (b)



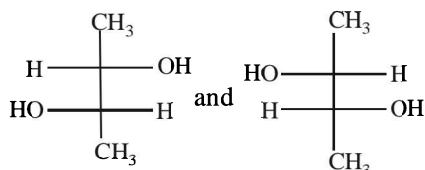
17. (c) The acid character follows the order : $\text{CH}_3\text{COOH} > \text{C}_6\text{H}_5\text{OH} > \text{H}_2\text{O} > \text{CH}_3\text{OH}$
The basic character will follow the order $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{O}^- < \text{CH}_3\text{O}^-$
The stronger the acid, the weaker the conjugate base formed.

18. (b) Reaction between alkyl halide, aryl halide and sodium in presence of ether is known as Wurtz fitting reaction



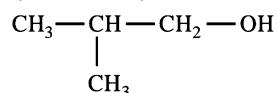
1, 1, 1-trichloroethane

20. (b) Compound which are mirror image of each other and are non superimposable are termed as enantiomers.

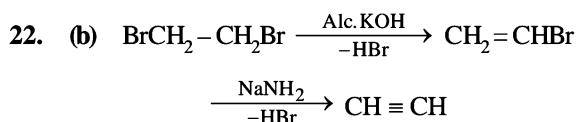


These are enantiomers

21. (d) Iodoform test is given by methyl ketones, acetaldehyde and methyl secondary alcohols.

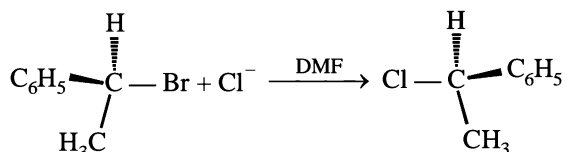


Isobutyl alcohol is a primary alcohol hence doesn't give positive iodoform test.

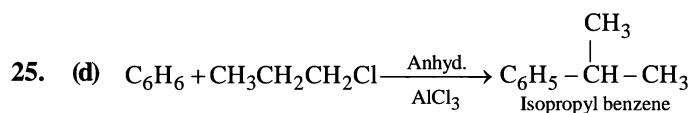
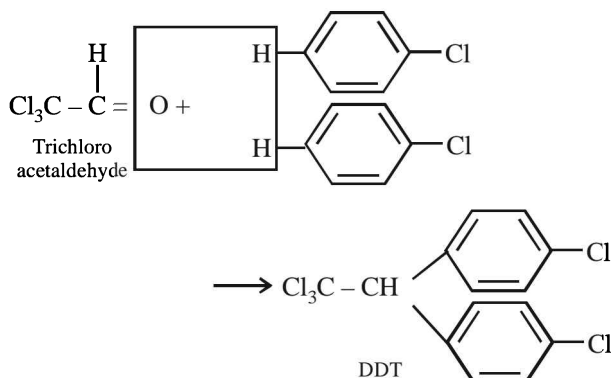


Elimination of HBr from $\text{CH}_2=\text{CHBr}$ requires a stronger base because here, Br acquires partial double bond character due to resonance.

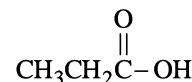
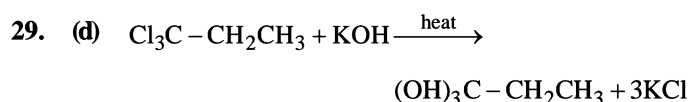
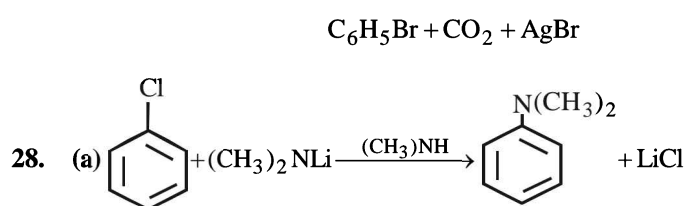
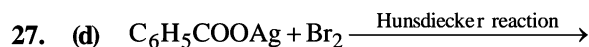
23. (c) $\text{C}_6\text{H}_5\text{CHCH}_3\text{Br}$ being an optically active secondary alkyl bromide undergoes $\text{S}_{\text{N}}2$ nucleophilic substitution reaction. Hence it undergoes complete inversion of configuration.



24. (c) Chloral on reaction with chlorobenzene in the presence of a catalytic amount of sulphuric acid forms DDT (dichlorodiphenyl Trichloro ethane).



26. (d) $-\text{NO}_2$ group withdraws electrons from *o*- and *p*-positions and hence activates the $-\text{Cl}$ present on that position towards nucleophilic substitution.



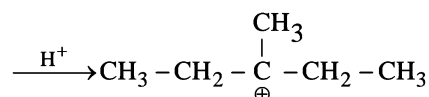
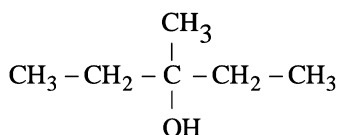
30. (a) A strong nucleophile favours the $\text{S}_{\text{N}}2$ reaction and a weak nucleophile favours the $\text{S}_{\text{N}}1$ reaction. First reaction is $\text{S}_{\text{N}}1$ reaction because $\text{C}_2\text{H}_5\text{OH}$ is used as solvent which is a weak nucleophile. Second reaction is $\text{S}_{\text{N}}2$ reaction because $\text{C}_2\text{H}_5\text{O}^-$ is strong nucleophile.

DAILY PRACTICE PROBLEMS

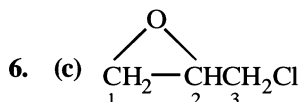
CHEMISTRY SOLUTIONS

DPP/CC25

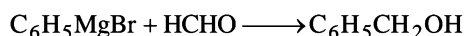
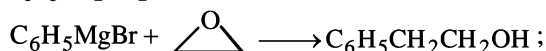
- (a) Lucas reagent is conc. HCl + anhyd. ZnCl_2 .
- (c)
- (a) Methyl alcohol (CH_3OH) is also known as carbinol. Hence vinyl carbinol is $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$
- (a) 3-methyl pentanol-3 will be dehydrated most readily since it produces tertiary carbonium ion as intermediate.



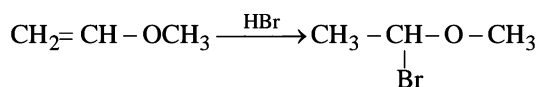
- (c)



- (d) 2-Phenylethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$, is a 1° alcohol which can be prepared from $\text{C}_6\text{H}_5\text{MgBr}$ by treating with ethylene oxide (note that HCHO will introduce only one carbon atom, i.e. it will give $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ and not $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$).

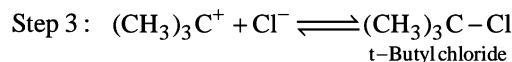
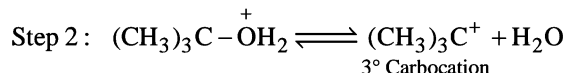
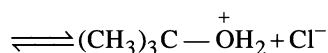
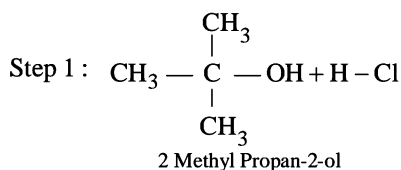


- (b) Methyl vinyl ether under anhydrous condition at room temperature undergoes addition reaction.



- (b) Tertiary alcohols react fastest with conc. HCl and anhydrous ZnCl_2 (lucas reagent) as its mechanism proceeds through the formation of stable tertiary carbocation.

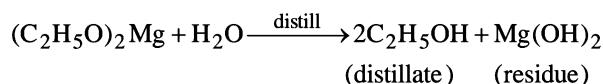
Mechanism



- (d) The two components should be $(\text{CH}_3)_3\text{CONa}$ + $(\text{CH}_3)_3\text{CBr}$. However, tert-alkyl halides tend to undergo elimination reaction rather than substitution leading to the formation of an alkene, $\text{Me}_2\text{C}=\text{CH}_2$

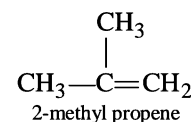
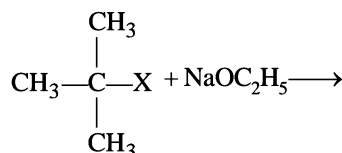
- (d)

- (d)

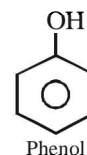
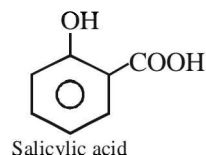
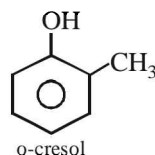


- (a) MnO_2 being a mild oxidising agent stops the oxidation of $-\text{CH}_2\text{OH}$ group at aldehyde stage.

- (b) The tertiary alkyl halide undergo elimination reaction to give alkenes

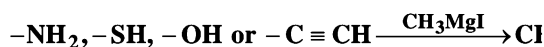


- (d)



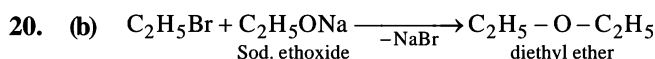
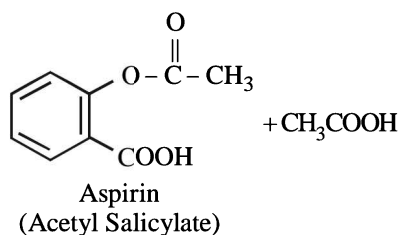
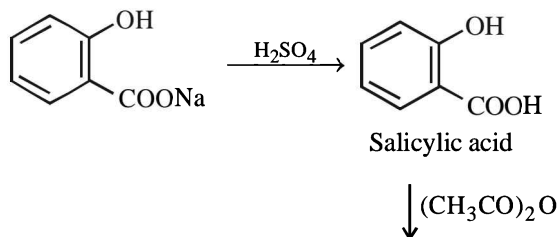
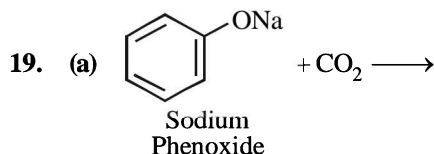
Electron releasing groups ($-\text{CH}_3$, $-\text{OCH}_3$, $-\text{NCH}_3$ etc) intensify the negative charge of phenoxide ion, i.e., destabilises it hence decrease ionization of parent phenol. Therefore decreases acidity while electron donating groups ($-\text{NO}_2$, $-\text{COOH}$, $-\text{CHO}$ etc.) increases acidity.

- (b) Number of active hydrogen in a compound corresponds to the number of moles of CH_4 evolved per mole of the compound.

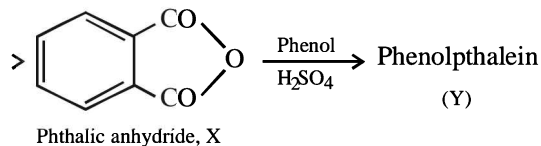
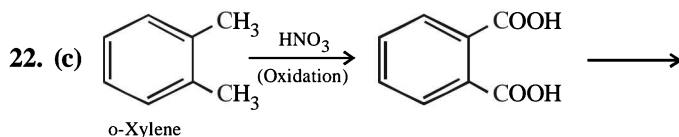


17. (b)

18. (b)

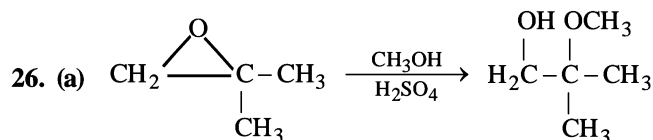
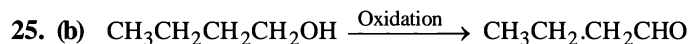


21. (b)



23. (b)

24. (b) The commercial alcohol is made unfit for drinking by mixing in it some copper sulphate (to give it colour) and pyridine (a foul smelling liquid). It is known as denaturation of alcohol.

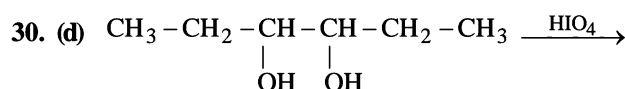


27. (c) Electron withdrawing group stabilises the benzene ring due to delocalisation of charge.

$-\text{CH}_3$ and $-\text{CH}_2\text{OH}$ are electron donating group and hence decrease the stability of benzene ring $-\text{OCH}_3$ is weaker electron withdrawing group than $-\text{COCH}_3$. Hence $-\text{COCH}_3$ group more stabilize the phenoxide ion at *p*-position.

28. (d) This method is suitable for the preparation of a wide variety of unsymmetrical ethers. The nucleophilic substitution of halides with alkoxide leads to desired product.

29. (c) Due to presence of methyl alcohol in liquor.

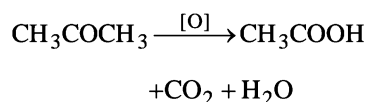


DAILY PRACTICE PROBLEMS

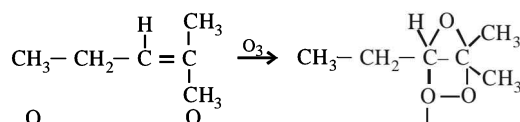
CHEMISTRY SOLUTIONS

DPP/CC26

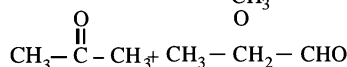
1. (c)
2. (a) Ketones on oxidation give carboxylic acids with lesser number of carbon atoms, i.e.,



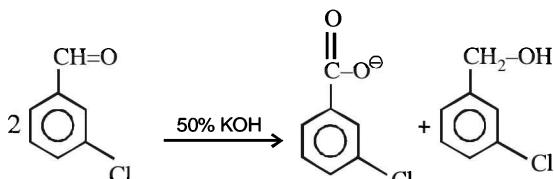
3. (b)



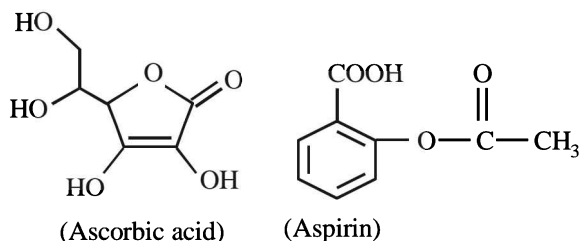
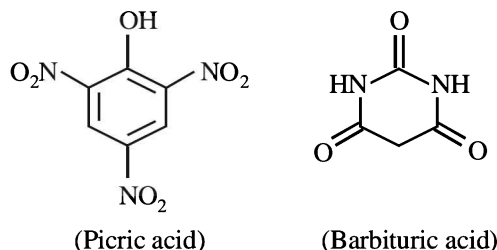
4. (a)



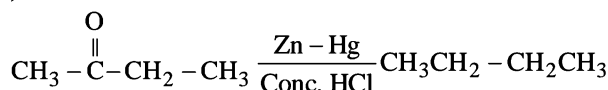
5. (c) Cannizzaro reaction - when an aldehyde containing no α -H undergo reaction in presence of 50% KOH. It disproportionates to form a molecule of carboxylic acid and a molecule of alcohol.



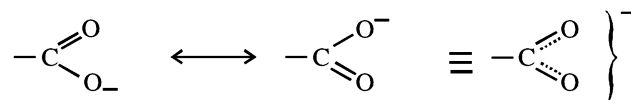
6. (d)



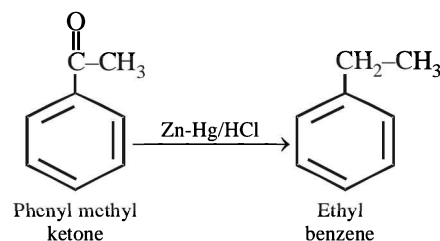
7. (d) It is Clemmensen's reduction



8. (c) Due to resonance in carboxylate ion, the double bond character of $\text{C}=\text{O}$ bond in carboxylic acids is greatly reduced as compared to that in aldehydes and ketones.

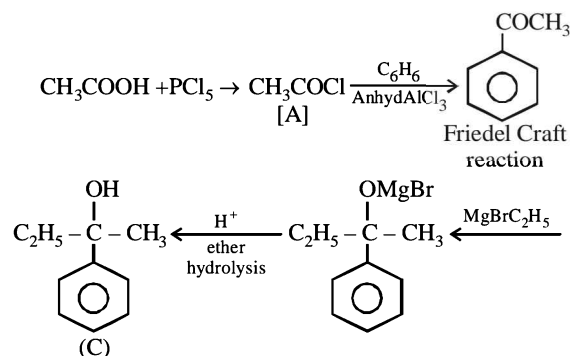


9. (b)

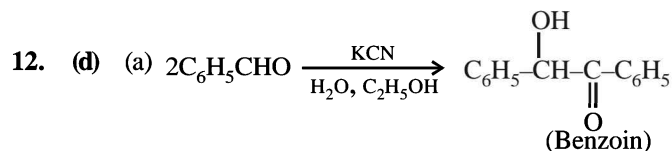
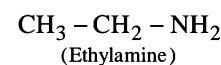
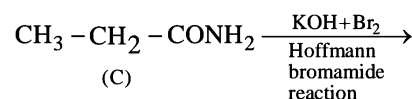
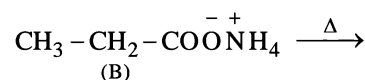
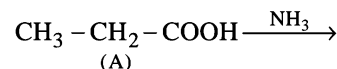


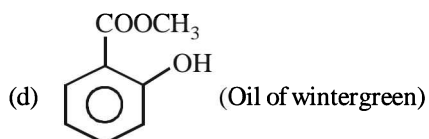
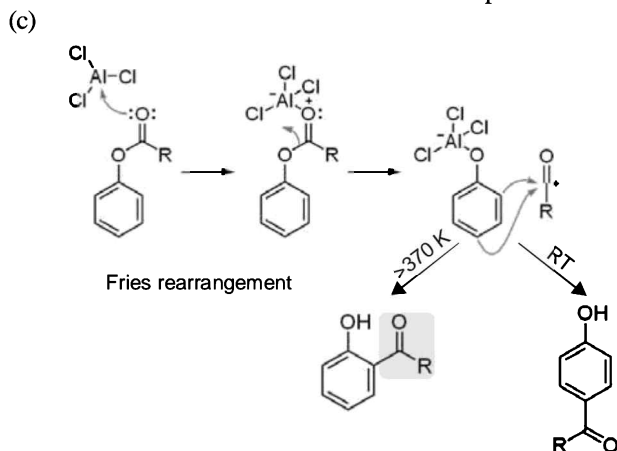
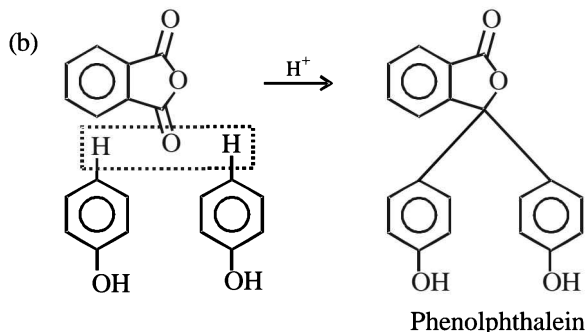
This reaction is known as Clemmensen's reduction.

10. (a)

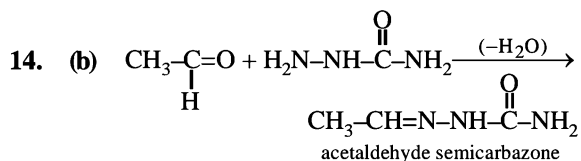
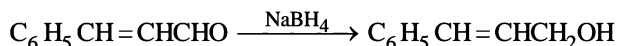


11. (d) Since, C when heated with Br_2 in presence of KOH produces ethylamine, hence it must be propanamide and hence the organic compound (A) will be propanoic acid. The reactions follows.

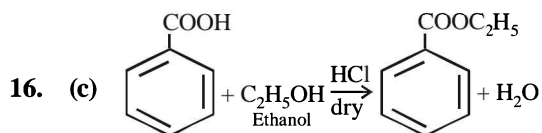




13. (b) NaBH_4 selectively reduces the aldehyde group to alcohol without affecting double bond in a organic compound. So, X is NaBH_4 .



15. (a)

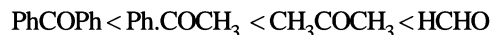


This process is known as esterification.

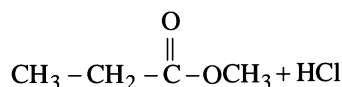
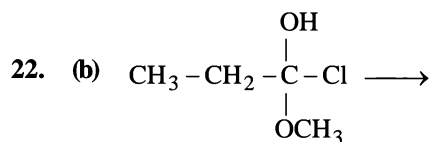
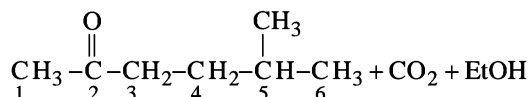
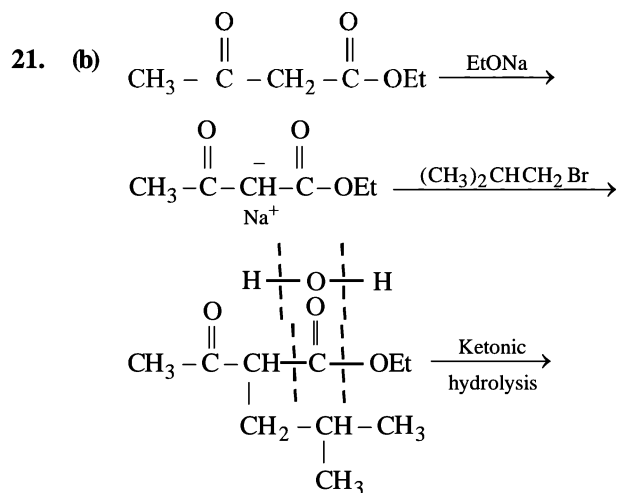
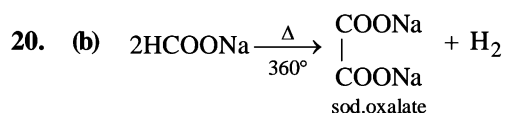
17. (a)

18. (a)

19. (a) Addition of HCN to carbonyl compounds is nucleophilic addition reaction. The order of reactivity of carbonyl compounds is Aldehydes (smaller to higher) Ketones (smaller to higher), Then



The lower reactivity of ketones is due to presence of two alkyl group which shows +I effect. The reactivity of ketones decreases as the size of alkyl group increases.



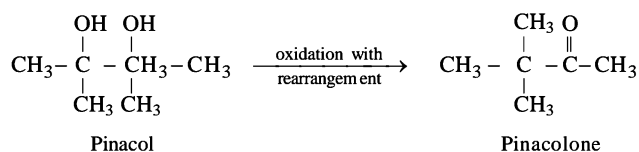
23. (a) The correct order of increasing acid strength
 $(\text{Me})_2\text{CHCOOH} < \text{CH}_3\text{COOH} < \text{MeOCH}_2\text{COOH} < \text{CF}_3\text{COOH}$

Electron withdrawing groups increase the acid strength and electron donating groups decrease the acid strength.

24. (d)

25. (b) Acetic acid does not have $-\text{CHO}$ grouping, while all others (HCOOH , HCHO and CH_3CHO) have $-\text{CHO}$ grouping

26. (b) Pinacolone is oxidation product of pinacol.

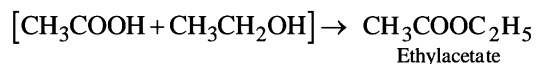
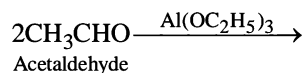


3, 3-dimethyl-2-butanone

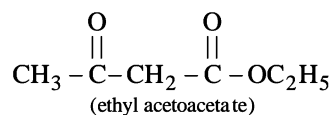
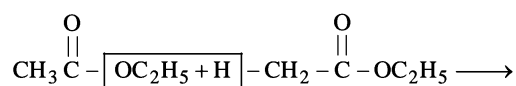
S-58

DPP/ CC26

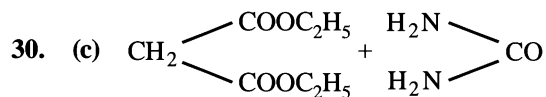
27. (c) Aldehydes having α -H atom, when treated with *aluminium ethoxide* (in place of NaOH or KOH), undergo Cannizzaro type of reaction with a difference that the product isolated is an ester rather than salt of acid or alcohol. Such reaction is called Tischenko reaction.



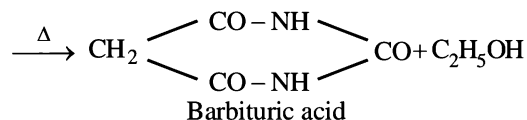
28. (a) It is an example of Claisen condensation. The product is acetoacetic ester.



29. (d) I_2 and Na_2CO_3 react with acetophenone ($\text{C}_6\text{H}_5\text{COCH}_3$) to give yellow ppt. of CHI_3 but benzophenone ($\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$) does not and hence can be used to distinguish between them.



Malonic ester



DAILY PRACTICE PROBLEMS

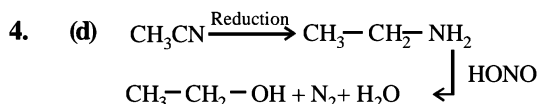
CHEMISTRY SOLUTIONS

DPP/CC27

1. (c) Since the organic amino compound on reaction with nitrous acid at low temperature produces an oily nitrosoamine so the organic amino compound is a secondary aliphatic amines.

2. (b)

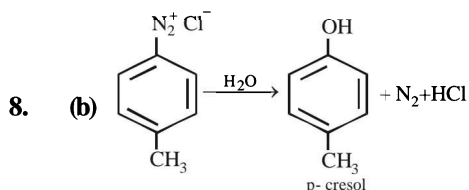
3. (a) The yield in Sandmeyer reaction is found to be better than Gattermann reaction.



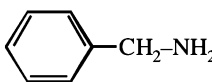
5. (c) Aliphatic amines are more basic than aromatic amines. Resonance decreases the basic character due to delocalisation of shared pair of electrons on nitrogen within benzene nucleus.

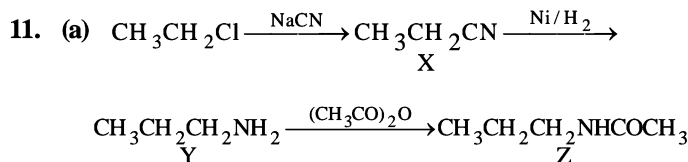
6. (d)

7. (b)

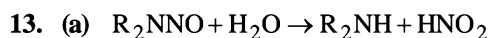


9. (d)

10. (b)  compound is most basic due to localized lone pair of electron on nitrogen atom while other compounds have delocalized lone pair of electron.

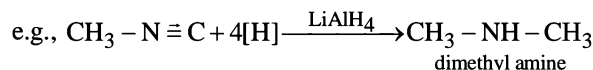
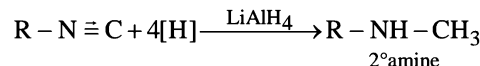


12. (d) (1) $\text{CH}_3\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 (ii) with KOH (alcohol) and CHCl_3 produces bad smell
 (2) $\text{CH}_3\text{C}\equiv\text{CH}$
 (iii) gives white ppt with ammonical AgNO_3
 (3) $\text{CH}_3\text{CH}_2\text{COOCH}_3$
 (i) alkaline hydrolysis
 (4) $\text{CH}_3\text{CHOH}-\text{CH}_3$
 (iv) with Lucas reagent cloudiness appears after 5 minutes



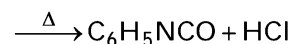
14. (d)

15. (a) Reduction of alkyl isocyanides in presence of LiAlH_4 yields secondary amines containing methyl as one of the alkyl group.

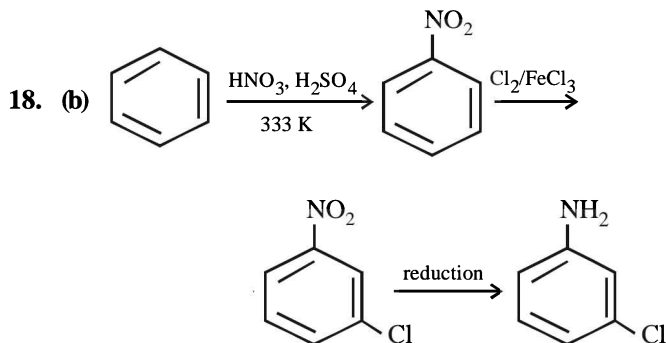
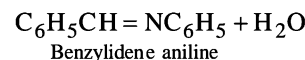
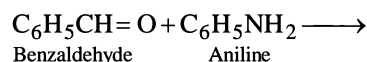


whereas, alkyl cyanides give 1° amine on reduction.

16. (d) $\text{C}_6\text{H}_5\text{NH}_2 + \text{COCl}_2 + \text{KOH} \rightarrow \text{C}_6\text{H}_5\text{NH}\cdot\text{COCl} + \text{HCl}$



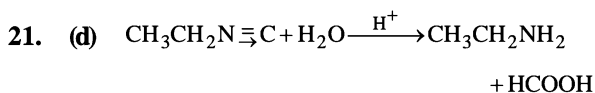
17. (c) Benzaldehyde reacts with primary aromatic amine to form schiff's base

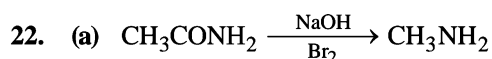


- (a) When chlorination is done earlier than nitration, chlorobenzene formed at first step would introduce $-\text{NO}_2$ group in ortho-position, not in m-position
 (b) Again if $-\text{NO}_2$ group is reduced earlier than the chlorination step, $-\text{NH}_2$ group formed on reduction will again introduce $-\text{Cl}$ in o-position

19. (c)

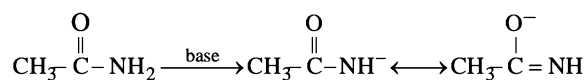
20. (d) Reduction with iron scrap and hydrochloric acid is preferred because FeCl_2 formed gets hydrolysed to release hydrochloric acid during the reaction. Thus, only a small amount of hydrochloric acid is required to initiate the reaction.





(Hofmann bromamide reaction)

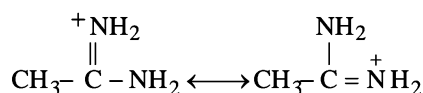
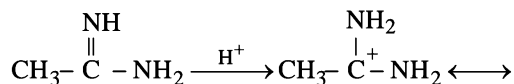
23. (a) Acetamide is basic due to the presence of lone pair of electrons of N it is also acidic because its conjugate base shows resonance.



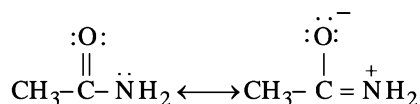
(more stable due to -ve charge on O)

Benzonitrile ($\text{C}_6\text{H}_5\text{C}\equiv\text{N}$) acts as an electrophile (Lewis acid) due to electron deficiency of C of CN as well as nucleophile (Lewis base) in nature due to presence of lone pair electrons on N hence it is neutral Triethylamine and phenol are basic and acidic in nature respectively.

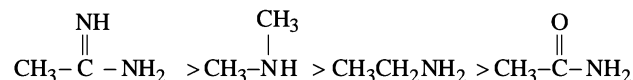
24. (b) Guanidine is most basic because its conjugate acid is stabilized by two equivalent resonance structures,.



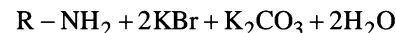
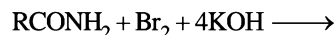
Further 2° amines are more basic than 1°, while amides are least basic due to delocalisation of the lone pair of electrons on N



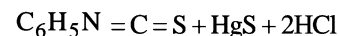
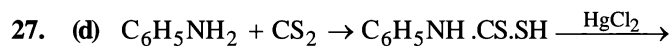
Thus the correct order of basicities is



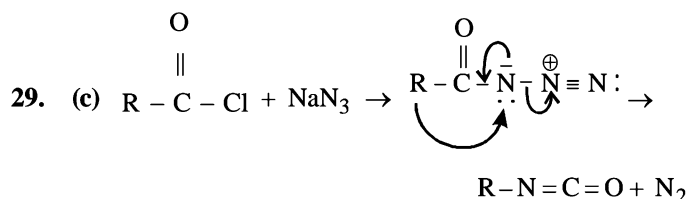
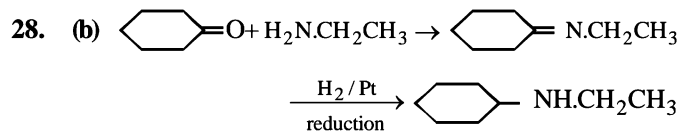
25. (c) CH_3NC (methyl isocyanide) on reduction with LiAlH_4 gives secondary amine.
26. (c) Only 1° amides (i.e. RCONH_2) in the present case undergo Hofmann bromamide reaction.



(Hofmann's bromamide reaction)



The reaction is called mustard oil reaction.



30. (c) Amides are hydrolysed to give acids and NH_3 or amines.

1. (a)
2. (c)
3. (b)
4. (d)
5. (a)
6. (d) Glucose being an aldose responds to Tollen's test while fructose, although a ketose, undergoes rearrangement in presence of basic medium (provided by Tollen's reagent) to form glucose, which then responds to Tollen's test.
7. (d) Since sucrose is dextrorotatory while hydrolysis product of sucrose, having equimolar mixture of glucose and fructose, is laevorotatory. Hence the hydrolysed product of sucrose is known as invert sugar and the hydrolysis of sucrose is known as inversion.
8. (b) All proteins are not found in L-form but they may be present in form of D or L.
9. (a) Butter, being a lipid, dissolves in chloroform.
10. (b) Human body can synthesize 10 of the 20 amino acids obtained by hydrolysis of various proteins
11. (c) Chemically amylose is a long unbranched chain with 200-1000 α -D-(+)-glucose units held by C1-C4 glycosidic linkage.
12. (a)
13. (a) On the basis of structure of guanine and complementary bases present in them, we can say that if the sequence of bases in one strand of DNA is I, then the sequence in the second strand should be II
 A : T : G : C : T : T : G : A : I
 T : A : C : G : A : A : C : T : II
14. (c)

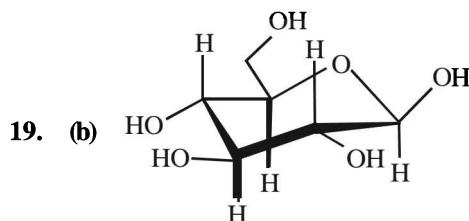
15. (a) Due to resonance, $-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{NH}- \longleftrightarrow -\overset{\overset{\text{O}^-}{\parallel}}{\text{C}}=\text{NH}^+-$, C-N bond acquires some double bond character, hence shorter in length

16. (c) Proline is an acidic amino acid.

17. (a) Isoelectric point (pH)

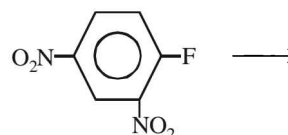
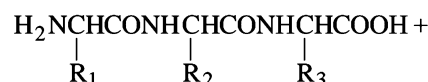
$$= \frac{pK_{a1} + pK_{a2}}{2} = \frac{2.34 + 9.60}{2} = 5.97$$

18. (b)

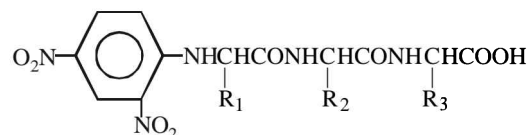


It is a β -pyranose hence it is an aldohexose.

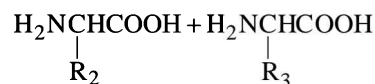
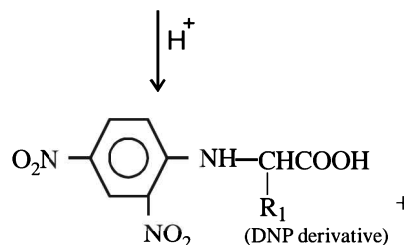
20. (d) 2, 4- Dinitrofluorobenzene, also known as Sanger's reagent, reacts with the H_2N - group of the peptide to form 2, 4- dinitrophenyl (DNP) derivative of the peptide. The DNP derivative of the peptide is hydrolysed to give DNP derivative of the single amino acid



2, 4, Dinitrofluoro benzene

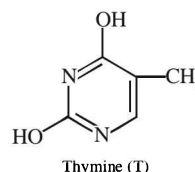


(DNP derivative of polypeptide)



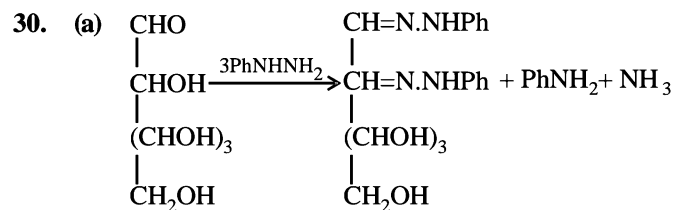
(Amino acid mixture)

21. (d) Zwitter ion contains both +ve and -ve charge. Proton of $-\text{COOH}$ group is transferred to the $-\text{NH}_2$ group. $-\text{NH}_3^+$ group is acidic since it can donate a proton and $-\text{COO}^-$ group is basic since it can accept a proton.
22. (c) Glucose is considered as a typical carbohydrate which contains $-\text{CHO}$ and $-\text{OH}$ group.
23. (a)
24. (a) DNA has double stranded α -helical structure
25. (c)
26. (d) The correct structure of thymine is



Thymine (T)

27. (b) The DNA sequence that codes for a specific protein is called a Gene and thus every protein in a cell has a corresponding gene.
28. (d) Enzymes are very specific biological catalysts possessing well - defined active sites
29. (b) In DNA and RNA heterocyclic base and phosphate ester are at C₁' and C₅' respectively of the sugar molecule.

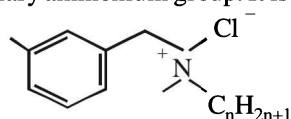


DAILY PRACTICE PROBLEMS

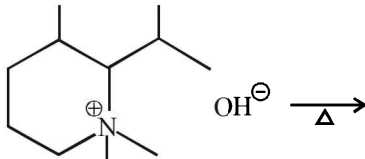
CHEMISTRY SOLUTIONS

DPP/CC30

1. (a) Given drug is used as antacid.
2. (c)
3. (d)
4. (c)
5. (a)
6. (b) It is acetyl salicylic acid i.e., aspirin, analgesic and antipyretic.
7. (a) Whiteware possesses good strength, translucency, and low porosity.
8. (b)
9. (c) Glyceryl dinitrate is not used as fixative in perfumes.
10. (b)
11. (d) Drugs that mimic the natural messenger by switching on the receptor are called agonists. While drugs that binds to the receptor site and inhibit its natural function are called antagonists.
12. (d) The mixture of chloroxylenol and terpenol is dettol which is used as antiseptic.
13. (c) We know that N-acetyl-para-aminophenol (or paracetamol) is an antipyretic which can also be used as an analgesic to relieve pain.
14. (a)
15. (a) Benzalkonium chloride, also known as alkyldimethylbenzylammonium chloride is nitrogenous cationic surface active agent belonging to the quaternary ammonium group. It is used as antiseptic.
16. (a)
17. (d)
18. (d) Bacteriostatic drugs inhibit the growth of organism while bactericidal drugs kill the microorganisms.
19. (c) Narrow spectrum antibiotics are effective against Gram-positive or Gram-negative bacteria. Limited spectrum antibiotics are effective against a single organism or disease.
20. (c) Linear alkylbenzenesulphonates (LAB) carrying phenyl chains at secondary positions are biodegradable
21. (b) The term "antihistamine" refers only to H_1 antagonists, which is also known as H_1 -receptor antagonists and H_1 -antihistamine.
22. (d)
23. (c) Magnesium hydrosilicate forms base of Talcum powder.
24. (b) It is the known structure of Penicillin G
25. (b) Aluminium salts present in deodorants act as antibacterial agents.
26. (a) Pheromones are the chemical substances used by insects for communication.
27. (b) Dilute solutions of boric acid and hydrogen peroxide are weak antiseptics.
28. (d) Sucrose is natural sweetner.
29. (c) Broad spectrum antibiotics act on different antigens.
30. (a)

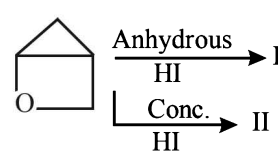

 $n = 8, 10, 12, 14, 16, 18$

Mock Test Full Syllabus Chemistry

- Let ν_1 be the frequency of the series limit of the Lyman series, ν_2 be the frequency of the first line of the Lyman series, and ν_3 be the frequency of the series limit of the Balmer series, then –
 - $\nu_3 = \frac{1}{2} (\nu_1 - \nu_2)$
 - $\nu_2 - \nu_1 = \nu_3$
 - $\nu_1 - \nu_2 = \nu_3$
 - $\nu_1 + \nu_2 = \nu_3$
- 

Number of α -H in alkene which is major product in this reaction is –

 - 2
 - 4
 - 5
 - 7
- When an isolated gaseous cation $X^+(g)$ is converted into $X^-(g)$ anion, the amount of energy released is 16.8 eV. Calculate the electronegativity value of the atom X on Pauling's scale.
 - 1
 - 2
 - 4
 - 3
- Unknown sample (A) on heating swells up first and aqueous solution of (A) gives white ppt. with NaOH which becomes soluble with excess of NaOH addition –
 - Borax
 - Alum
 - Microcosmic salt
 - None
- At 675K, $H_2(g)$ and $CO_2(g)$ react to form $CO(g)$ and $H_2O(g)$, K_p for the reaction is 0.16. If a mixture of 0.25 mole of $H_2(g)$ and 0.25 mol of CO_2 is heated at 675K, mole % of $CO(g)$ in equilibrium mixture is –
 - 7.14
 - 14.28
 - 28.57
 - 33.33
- Choose the correct option for



 - I and II are identical
 - I and II are different
 - Mechanism of formation of I and II are not known
 - None of these
- For the extraction of Pb when impurity content is very high, then during roasting of galena, the formation of which compound (s) is/are prevented by lime –
 - $PbSO_4$
 - $PbSiO_3$
 - Both (a) and (b)
 - $CaSiO_3$
- Polymer obtained by reaction of phenol and formaldehyde is
 - Bakelite
 - Terylene
 - Nylon-66
 - Urea formaldehyde resin
- The conductivity of a saturated solution of Ag_3PO_4 is $9 \times 10^{-6} \text{ Sm}^{-1}$ and its equivalent conductivity is $1.50 \times 10^{-4} \text{ Sm}^2 \text{ equivalent}^{-1}$. The K_{sp} of Ag_3PO_4 is –
 - 4.32×10^{-18}
 - 1.8×10^{-9}
 - 8.64×10^{-13}
 - None of these
- Which statement is incorrect regarding thermite process –
 - 3 part metal oxide + 1 part Al is taken as thermite mixture
 - Ignition mixture (Mg powder + BaO_2) acts as catalyst
 - Metals (Fe, Cr) are obtained in vapour form
 - Reaction is exothermic due to highly +ve enthalpy of formation of Al_2O_3 .
- Which of the following option is having maximum number of unpaired electrons –
 - A tetrahedral d^6 ion
 - $[Co(H_2O)_6]^{3+}$
 - A square planar d^7 ion
 - A co-ordination compound with magnetic moment of 5.92 B.M.
- pH of a saturated solution of silver salt of monobasic acid HA is found to be 9. Find the K_{sp} of sparingly soluble salt $AgA(s)$. Given $K_a(HA) = 10^{-10}$.
 - 1.1×10^{-9}
 - 1.1×10^{-10}
 - 1.1×10^{-11}
 - 10^{-12}
- Which of the following carbide does not release any hydrocarbon on reaction with water.
 - SiC
 - Be_2C
 - CaC_2
 - Mg_2C_3
- Phenol and benzoic acid is separated by –
 - $NaHCO_3$
 - NaOH
 - Na
 - $NaNH_2$

15. For the reaction takes place at certain temperature $\text{NH}_4\text{HS} \rightleftharpoons \text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g})$, if equilibrium pressure is X bar, then ΔG° would be –

(a) $-2RT \ln X$ (b) $-RT \ln (X - \ln 2)$
(c) $-2RT (\ln X - \ln 2)$ (d) None of these

16. Aqueous solution of (M) + $(\text{NH}_4)_2\text{S} \rightarrow$ yellow ppt (B)
 $\xrightarrow{(\text{NH}_4)_2\text{S}_2}$ insoluble.

The cation present in (M) is –

(a) CdS (b) SnS_2
(c) Cd^{2+} (d) Sn^{2+}

17. How many pairs of enantiomers are possible for following complex compound. $[\text{M}(\text{AB})(\text{CD})\text{ef}]^{n\pm}$

(where AB, CD – Unsymmetrical bidentate ligand, e, f – Monodentate ligands)

(a) 20 (b) 5
(c) 10 (d) 8

18. For the reaction $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{D}(\text{g})$; $K_C = 10^{12}$.

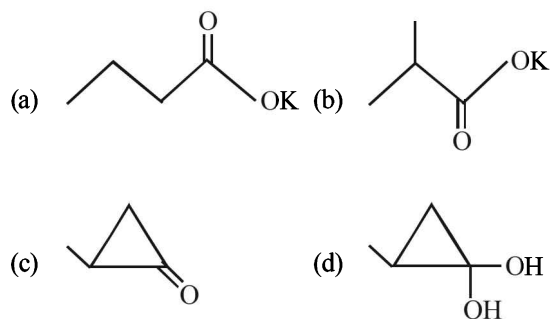
If the initial moles of A, B, C and D are 0.5, 1, 0.5 and 3.5 moles respectively in a one litre vessel. What is the equilibrium concentration of B?

(a) 10^{-4} (b) 2×10^{-4}
(c) 4×10^{-4} (d) 8×10^{-4}

19. H_2S gas is not evolved when SO_3^{2-} ion reacts with following reagents.

(a) Zn + dil. H_2SO_4 (b) Al + Conc. NaOH
(c) Al + dil. HCl (d) None of these

20. $\text{Me}-\text{CH}=\text{CH}_2 + \text{CHCl}_3 \xrightarrow[\text{excess, } \Delta]{\text{aq. KOH}}$ A (Major products) is –



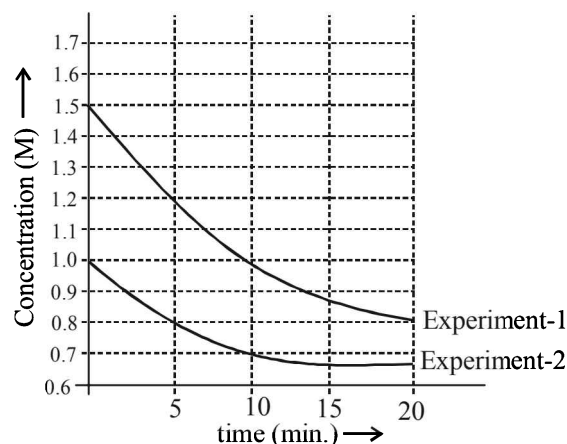
21. This question contains Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement 1 : $[\text{HeH}]^+$ is more stable as compared to H_2^+ .

Statement 2 : Both the above species are having equal bond order.

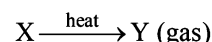
- (a) Statement-1 is false, Statement-2 is true.
(b) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
(c) Statement-1 is true, statement-2 is true and statement-2 is NOT correct explanation for statement-1
(d) Statement-1 is true, Statement-2 is false.

22. The variation of concentration of A with time in two experiments starting with two different initial concentration of A is given in the following graph. The reaction is represented as $\text{A}(\text{aq}) \rightarrow \text{B}(\text{aq})$. What is the rate of reaction (M/min) when concentration of A in aqueous solution was 1.8 M?



- (a) 0.072 M min^{-1} (b) 0.036 M min^{-1}
(c) $0.1296 \text{ M min}^{-1}$ (d) 1 M min^{-1}

23. $\text{NH}_4\text{ClO}_4 + \text{HNO}_3(\text{dilute}) \rightarrow \text{X} + \text{HClO}_4$

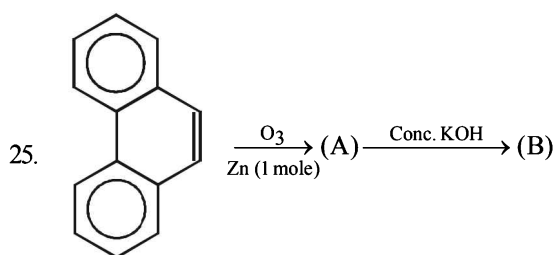


Gas (Y) is –

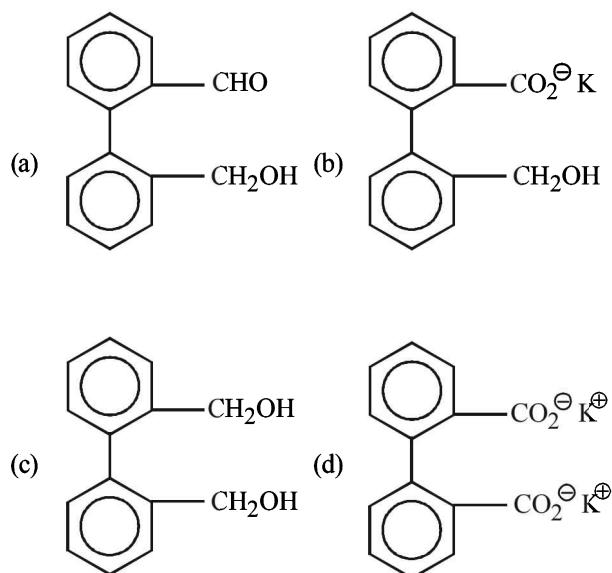
(a) O_2 (b) N_2
(c) NO_2 (d) N_2O

24. Choose the correct order of T (True) and F (False) –

- (1) When the pH of rain water is below 3.6, it is called acid rain.
(2) Ozone hole occurs over Antarctica mainly during September–October and it gets replenished in November–December.
(3) Methylcyclohexane is an ozone-depleting molecule.
(4) COD is always larger than BOD.
(a) TTTF (b) FTFT
(c) FFFF (d) FTTT



End product (B) of above reaction is –



26. Precautions to be taken in the study of reaction rate for the reaction between potassium iodate (KIO_3) and sodium sulphite (Na_2SO_3) using starch solution as indicator at different concentrations and temperature –

- (a) The concentration of sodium thiosulphate solution should always be less than the concentration of the potassium iodide solution.
 (b) Freshly prepared starch solution should be used
 (c) Experiments should be performed with the fresh solutions of H_2O_2 and KI.
 (d) All of these

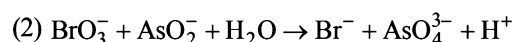
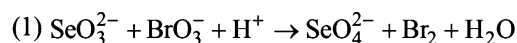
27. Calculate the millimoles of SeO_3^{2-} in solution on the basis of following data :

70 ml of $\frac{\text{M}}{60}$ solution of KBrO_3 was added to SeO_3^{2-} solution.

The bromine evolved was removed by boiling and excess of

KBrO_3 was back titrated with 12.5 mL of $\frac{\text{M}}{25}$ solution of NaAsO_2 .

The reactions are given below.



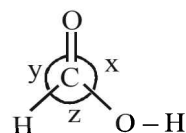
- (a) 1.6×10^{-3} (b) 1.25
 (c) 2.5 (d) None of these

28. 1 mole of gas X is present in a closed adiabatic vessel fitted with a movable frictionless piston. The initial temperature of gas X is 300 K. The vessel is maintained at constant pressure of 1 atm. Keeping the pressure constant at 1 atm the reaction $3\text{X}(\text{g}) \rightarrow 2\text{Y}(\text{g})$; $\Delta H = -30 \text{ kJ/mol}$ is started with the help of negligible amount of electric energy.

If finally 75 mole % of X undergone reaction at constant pressure of 1 atm, find the final temperature (in K) of reaction vessel. Given : $C_{p,m}(\text{X}) = 40 \text{ J/K mole}$, $C_{p,m}(\text{Y}) = 30 \text{ J/K mole}$.

- (a) 600 K (b) 300 K
 (c) 1200 K (d) 1000 K

29. In the given molecule choose the correct order of their angle.



- (a) $x > y > z$ (b) $x < y < z$
 (c) $x = y > z$ (d) Can't be predicted

30. What would be the reduction potential of an electrode at 298 K, which originally contained 1M $\text{K}_2\text{Cr}_2\text{O}_7$ solution in acidic buffer solution of pH = 1.0 and which was treated with 50% of the Sn necessary to reduce all $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} . Assume pH of solution remains constant.

Given : $E^0_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}, \text{H}^+} = 1.33 \text{ V}$, $\log 2 = 0.3$,

$$\frac{2.303RT}{F} = 0.06$$

- (a) 1.285 V (b) 1.193 V
 (c) 1.187 V (d) None of these

SOLUTIONS

1. (c) $v = RcZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

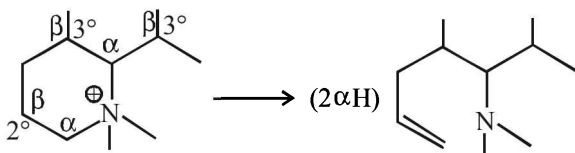
$$v_1 = RcZ^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = RcZ^2$$

$$v_2 = RcZ^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3RcZ^2}{4}$$

$$v_3 = RcZ^2 \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = \frac{RcZ^2}{4}$$

$$\therefore v_1 - v_2 = v_3$$

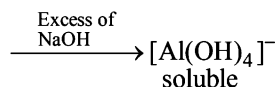
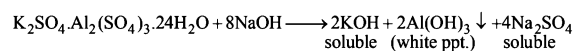
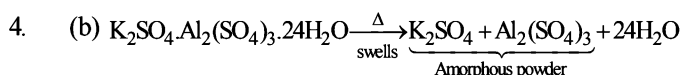
2. (a)



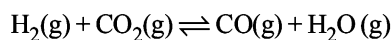
T.S. having carbanion character so Hoffmann alkene is major product.

3. (d) $X^+(g) \longrightarrow X^-(g)$; Energy = IE + EA = 16.8 eV

$$EN_{\text{Pauling}} = \frac{IA + EA}{5.6} = \frac{16.8}{5.6} = 3$$



5. (b)

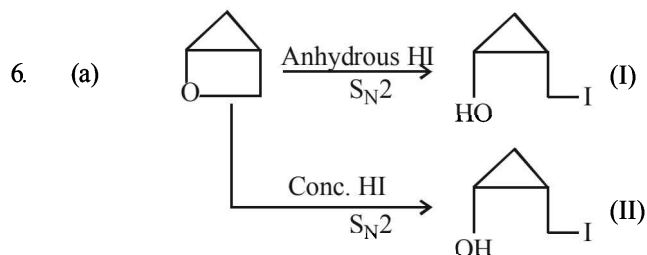


At eq^m 0.25-x 0.25-x x x

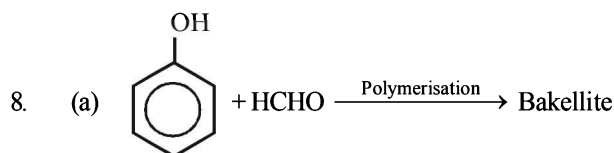
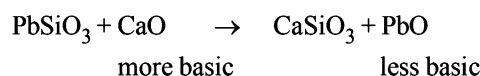
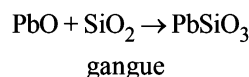
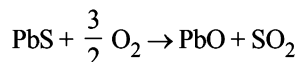
$$K_p = 0.16 = \frac{x^2}{(0.25-x)^2} \Rightarrow 0.4 = \frac{x}{0.25-x} \Rightarrow 0.1 - 0.4x = x$$

$$x = 0.0714$$

$$\text{Mole\% of CO (g)} = \frac{0.0714}{0.50} \times 100 = 14.28$$



7. (c) By intermittently filled lime the blown air in the sinterer gets diffused and PbS particles come under limited supply of air therefore formation of PbSO₄ is prevented.

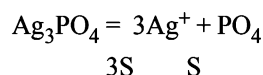


9. (a) $\lambda = K \times \frac{1000}{N}$

$$1.50 \times 10^{-4} \times 10^4 = 9 \times 10^{-6} \times 10^{-2} \times \frac{1000}{N}$$

$$N = 6 \times 10^{-5}$$

$$S = M = \frac{N}{n_f} = \frac{6 \times 10^{-5}}{3} = 2 \times 10^{-5} \text{ mol/L}$$

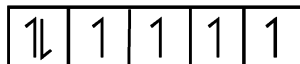


$$K_{sp} = (3S)^3 \cdot S = 27S^4 = 27 \times (2 \times 10^{-5})^4 = 4.32 \times 10^{-18}$$

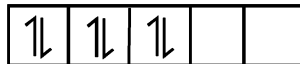
10. (a) (Mg powder + BaO₂) acts as ignition mixture
Fe, Cr are obtained in molten state
Enthalpy of formation of Al₂O₃ is negative.

11. (d)

(a) For tetrahedral d^6 ion,
4 unpaired electrons



(b) For $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$,
0 unpaired electrons



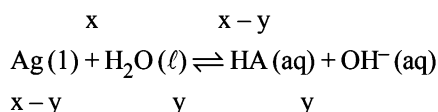
(c) For square planar d^7 ion,
1 unpaired electrons



(d) B.M. = $\sqrt{n(n+2)}$, n = unpaired electrons

$$5.92 \text{ B.M.} = \sqrt{n(n+2)}$$

$n = 5$ unpaired electrons

12. (c) $\text{Ag}(1) \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{A}^-(\text{aq})$ 

$$K_{\text{sp}} = x(x-y)$$

$$K_{\text{h}} = \frac{K_{\text{w}}}{K_{\text{a}}} = \frac{y^2}{(x-y)} ; \frac{10^{-14}}{10^{-10}} = \frac{(10^{-5})^2}{(x-y)}$$

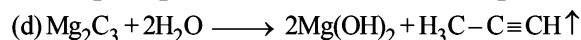
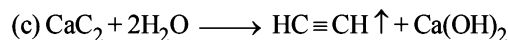
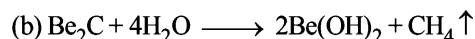
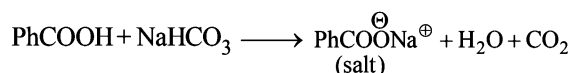
$$x-y = 10^{-6}$$

$$x = 10^{-5} + 10^{-6}$$

$$x = 1.1 \times 10^{-5}$$

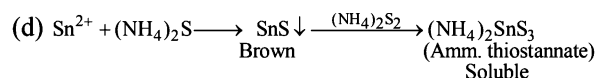
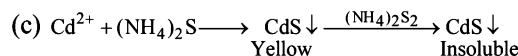
$$K_{\text{sp}} = 1.1 \times 10^{-5} \times 10^{-6}$$

$$K_{\text{sp}} = 1.1 \times 10^{-11}$$

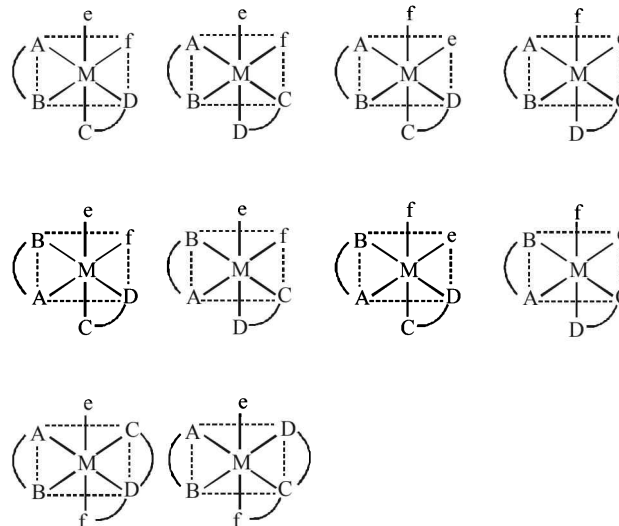
13. (a) (a) $\text{SiC} \Rightarrow$ Covalent carbide14. (a) $\text{PhOH} + \text{NaHCO}_3 \longrightarrow$ No reaction15. (c) $P_{\text{NH}_3} = P_{\text{H}_2\text{S}} = \frac{P}{2} = \frac{X}{2}$

$$\Delta G^\circ = -RT \ln K_p = -RT \ln \left(\frac{X}{2} \right)^2 = -2RT \ln \left(\frac{X}{2} \right)$$

$$= -2RT (\ln X - \ln 2)$$

16. (c)(a) $\text{CdS} \downarrow + (\text{NH}_4)_2\text{S} \longrightarrow \text{CdS} \downarrow \xrightarrow{(\text{NH}_4)_2\text{S}_2} \text{CdS} \downarrow$
Yellow Yellow Insoluble

17. (c)

18. (b) $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{D}(\text{g})$

$$\begin{array}{cccc} \text{At eq}^n & 0.5-x & 1-2x & 0.5+x \quad 3.5+x \\ & \approx y & \approx 2y & \approx 1 \quad \approx 4 \end{array}$$

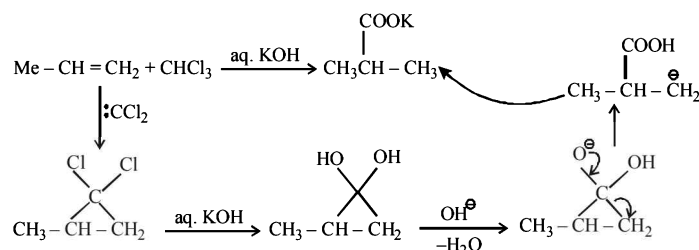
(due to very high value of equilibrium constant)

$$10^{12} = \frac{1 \times 4}{y \times (2y)^2} ; y = 10^{-4}; [\text{B}(\text{g})] = 2y \Rightarrow 2 \times 10^{-4}$$

19. (b) $\text{SO}_3^{2-} + 2\text{Al} + 2\text{OH}^- + 3\text{H}_2\text{O} \longrightarrow \text{S}^{2-} + 2[\text{Al}(\text{OH})_4]^-$

H_2S does not evolve as liberated H_2S is neutralized by NaOH and Na_2S is formed.

20. (b)

21. (d) $[\text{HeH}]^+$, M.O. configuration $\sigma 1s^2$

$$\text{B.O.} = 1.0$$

H_2^+ , M.O. configuration $\sigma 1s^1$

B.O. = 0.5

Higher the B.O. more is the stability.

22. (a) Rate law $r = k[A]^m$

$$r_1 = \left(\frac{0.3}{5}\right) \text{ M/min from graph (exp. 1)}$$

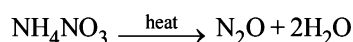
$$r_2 = \left(\frac{0.2}{5}\right) \text{ M/min from graph (exp. 2)}$$

$$\frac{r_1}{r_2} = \frac{3}{2} = \left\{ \frac{[A]_1}{[A]_2} \right\}^m = \left(\frac{1.5}{1} \right)^m \Rightarrow m = 1$$

$$\frac{0.3}{5} = k(1.5) \Rightarrow k = \frac{1}{25} \text{ min}^{-1}$$

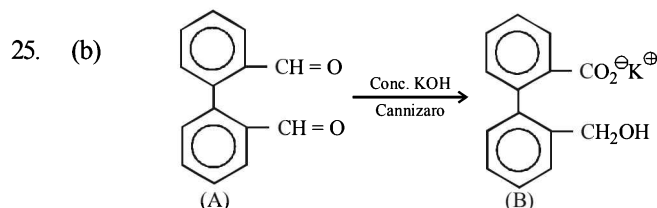
$$r = k \times [A] \Rightarrow \frac{1}{25} \times 1.8 = 0.072 \text{ M min}^{-1}$$

23. (d) $\text{NH}_4\text{ClO}_4 + \text{HNO}_3 \text{ (dilute)} \rightarrow \text{NH}_4\text{NO}_3 + \text{HClO}_4$
(X)



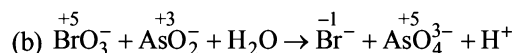
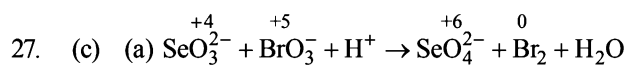
(X) (Y)

24. (b) (1) When the pH of rain water is below 5.6, it is called acid rain.
(2) Ozone hole occurs over Antarctica mainly during September–October and it gets replenished in November–December.
(3) Methylcyclohexane is not an ozone-depleting molecule.
(4) BOD (Biological oxygen demand) is a measure of organic pollutant present in the sample of water. Higher is the value of BOD, higher is the level of organic pollution in water. The amount of oxygen (in mg/L) consumed for oxidising all organic and oxidisable inorganic material in a sample of water is called chemical oxygen demand (COD).
COD is always larger than BOD.



26. (d) (a) The concentration of sodium thiosulphate solution should always be less than the concentration of the potassium iodide solution.

- (b) Freshly prepared starch solution should be used
(c) Experiments should be performed with the fresh solutions of H_2O_2 and KI.



In reaction (2)

gm. eq. of BrO_3^- = gm. eq. of AsO_2^-

$$n_{\text{BrO}_3^-} \times 6 = n_{\text{AsO}_2^-} \times 2 = \frac{12.5}{1000} \times \frac{1}{25} \times 2 = 10^{-3}$$

$$n_{\text{BrO}_3^-} = \frac{10^{-3}}{6}$$

In reaction (1)

$$\text{moles of } \text{BrO}_3^- \text{ consumed} = \frac{70}{1000} \times \frac{1}{60} - \frac{10^{-3}}{6} = 10^{-3}$$

gm eq. of SeO_3^{2-} = gm. eq. of BrO_3^-

$$n_{\text{SeO}_3^{2-}} \times 2 = 10^{-3} \times 5 ; n_{\text{SeO}_3^{2-}} = 2.5 \times 10^{-3}$$

28. (a) Mole of X reacted = 0.75 mole
Heat liberated at constant pressure = $0.75 \times 10 \text{ kJ} = 7.5 \text{ kJ}$

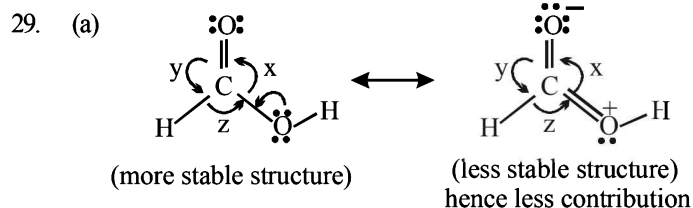
$$\text{Mole of X remaining} = \frac{1}{4} \text{ mole}$$

$$\text{Mole of Y formed} = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$$

$$\left(\frac{1}{4} C_{p,m(X)} + \frac{1}{2} C_{p,m(Y)} \right) (\Delta T) = q = 7500$$

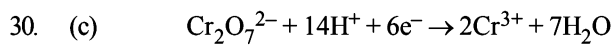
$$\Delta T = \frac{7500}{\frac{40}{4} + \frac{30}{2}} = 300$$

$$\text{Final temperature} = 300 + 300 = 600 \text{ K}$$



Therefore, repulsion between both C–O bonds will be greater than the repulsion between C–H bond and C–OH bond or between C–H and C=O bond.

Repulsion between C – H and C = O bond is greater than that of between C – H and C – OH bond, because double bond character in C – OH bond will always be less than C = O.



Initial conⁿ 1 0.1 0

After reaction 0.5 0.1 M 1M

$$E_{\text{RP}} = E_{\text{RP}}^0 - \frac{0.06}{6} \log \frac{[\text{Cr}^{3+}]^2}{[\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^{14}}$$

$$E_{\text{RP}} = 1.33 - \frac{0.06}{6} \log \frac{1}{(0.5)(0.1)^{14}}$$

$$= 1.33 - \frac{0.06}{6} \log (2 \times 10^{14}) = 1.187\text{V}$$

